

A Structural Approach to Interpretable Deep Learning

From DC Decompositions to Two-Player Games

Thesis in partial fulfillment of the requirements for the degree
Master of Science (M.Sc.)

submitted by
Jakob Zimmermann

Under the scientific supervision of
Prof. Dr. Wojciech Samek

Under the scientific guidance of
Georg Loho

Berlin, November 2026

Technische Universität Berlin
Fakultät IV – Elektrotechnik und Informatik
Fraunhofer Heinrich-Hertz-Institut
Department of Artificial Intelligence

Affidavit

Deutsch

1) Hiermit versichere ich, dass ich die vorliegende Arbeit ohne Hilfe Dritter und ausschließlich unter Verwendung der aufgeführten Quellen und Hilfsmittel angefertigt habe. Alle Stellen, die den benutzten Quellen und Hilfsmitteln unverändert oder inhaltlich entnommen sind, habe ich als solche kenntlich gemacht.
2) Sofern generative KI-Tools verwendet wurden, habe ich Produktnamen, Hersteller, die jeweils verwendete Softwareversion und die jeweiligen Einsatzzwecke (z.B. sprachliche Überprüfung und Verbesserung der Texte, systematische Recherche) benannt. Ich verantworte die Auswahl, die Übernahme und sämtliche Ergebnisse des von mir verwendeten KI-generierten Outputs vollumfänglich selbst.
3) Die Satzung zur Sicherung guter wissenschaftlicher Praxis an der TU Berlin vom 15. Februar 2023[1] habe ich zur Kenntnis genommen.
4) Ich erkläre weiterhin, dass ich die Arbeit in gleicher oder ähnlicher Form noch keiner anderen Prüfungsbehörde vorgelegt habe.

Jakob Zimmermann
Berlin, 10. November 2026

[1] https://www.static.tu.berlin/fileadmin/www/10002457/K3-AMBl/Amtsblatt_2023/Amtliches_Mitteilungsblatt_Nr._16_vom_30.05.2023.pdf

English

1) I hereby declare that I have written this thesis without the help of third parties and exclusively using the sources and aids listed. I have marked as such all passages that are taken from the sources and aids used, either unchanged or in terms of content.
2) Where generative AI tools were used, I have stated the product name, manufacturer, the software version used and the respective purposes (e.g. linguistic review and improvement of texts, systematic research). I am fully responsible for the selection, adoption and all results of the AI-generated output I used.
3) I have taken note of the Statutes for Safeguarding Good Scientific Practice at TU Berlin dated February 15, 2023[2].
4) I further declare that I have not yet submitted the work in the same or a similar form to any other examination authority.

Jakob Zimmermann
Berlin, November 10, 2026

[2] https://www.static.tu.berlin/fileadmin/www/10002457/K3-AMBl/Amtsblatt_2023/Amtliches_Mitteilungsblatt_Nr._16_vom_30.05.2023.pdf

Acknowledgements

I thank Gerrit Holzbach for giving me the opportunity to work on machine learning and gain a practical perspective. I am grateful to Georg Loho for opening my eyes to the theory of deep learning and polyhedral geometry.

I thank Grégoire Montavon for taking the time to discuss the idea of DC decomposition with me and for pointing me toward the Quantus framework and the intricacies of evaluating attribution methods. I am grateful to Wojciech Samek for his Responsible AI lecture, which made me aware of the problems in this area and strategies for addressing them. I also thank him for encouraging me to work on transformers and more modern machine learning models and paradigms, for asking me to study parameter randomization, for our conceptual discussions, and for his organizational support.

I thank Klaus Robert Müller for the interesting discussions of these results. I am also grateful to Shinichi Nakajima for the interesting discussions and for comparing my trajectory distribution view with the Normalized Relevance Measure framework and its signed-measure perspective on layer-wise relevance propagation and concept relevance propagation.

Abstract

Deutsch

Interpretierbarkeit entwickelt sich von der nachträglichen Analyse einzelner Vorhersagen zu einem Werkzeug, mit dem Modelle gezielt verbessert und gesteuert werden. Wenn Attributionen die Korrektur unerwünschter Merkmale, das Training oder die Inferenz beeinflussen, muss deshalb auch ihre Berechnung verstanden und überprüft werden. Diese Arbeit verbindet dafür zwei Perspektiven: die strukturierte Darstellung interner Beiträge und die mathematische Beschreibung ihrer Rückwärtspropagation.

Ein trainiertes neuronales Netz zu verstehen heißt, nachzuvollziehen, wie es eine Eingabe in eine Ausgabe überführt — welche gegenläufigen Einflüsse jede Vorhersage gegeneinander abwägt. Diese Arbeit untersucht Interpretierbarkeit durch *Zerlegung*: Statt ein Netz zur Interpretierbarkeit zu zwingen, legen wir interpretierbare Struktur offen, die ein trainiertes ReLU-Netz bereits enthält. Ein einziger Begriff der *Netzzerlegung* — eine Anhebung der Berechnung in ein Paar koordinierter, nichtnegativer Ströme, deren Differenz das ursprüngliche Netz exakt reproduziert — vereinheitlicht mehrere solcher Strukturen, ohne eine einzige Vorhersage zu verändern.

Die erste ist die Differenz-konvexer-Funktionen-Zerlegung (DC-Zerlegung). Da Monotonie die Interpretierbarkeit fördert, aber die Ausdrucksstärke einschränkt, trainieren wir kein monotones Modell, sondern zerlegen ein trainiertes ReLU-Netz — ein MLP oder ein CNN — in zwei Teilnetze g und h mit $f = g - h$, die in den verdoppelten Eingabekoordinaten konvex und monoton sind und positive und negative Evidenz auf getrennten Pfaden führen. Wir stabilisieren die ausgewertete ReLU-Zerlegung und konstruieren aus der absoluten Krümmung exakte Paare für glatte Aktivierungen (GELU, Swish, Tanh, Sigmoid, Softplus). Explizite Softmax-, Produkt- und LayerNorm-Paare mit positivem ε ergeben zudem eine DC-Zerlegung deterministischer Transformer beliebiger endlicher Tiefe; daraus folgt keine Garantie numerisch stabiler Auswertung. Auf der ReLU-Zerlegung bauen wir die Saliency-Methoden SplitCAM und SplitLRP auf, die auf VGG-16 und ResNet-18 über ImageNet-S konfigurationsabhängige Verbesserungen bei Lokalisierung und Selectivity sowie Zielkonflikte bei Pixel Flipping und Robustheit zeigen. Zudem trainieren wir Differenzen konvexer oder monotoner Netze zur MNIST-Klassifikation und beobachten klassenabhängige Muster in ihren Stromgradienten als qualitative Illustration der Darstellung.

Die zweite ist die Stopp-Zerlegung, die die Gewichte nach Vorzeichen aufteilt, $W = W^+ - W^-$, und exzitatorische von inhibitorischen Beiträgen entlang der Netzpfade trennt. Diese Zerlegung ist die Brücke zu einer spieltheoretischen Deutung des Rückwärtsdurchlaufs. Aufbauend auf dem ReLU-Netz-Spiel von Gaubert und Vlassopoulos fassen wir Rückwärts-Attribution als ein Zwei-Spieler-Nullsummen-Stoppspiel auf einem erweiterten Netzgraphen auf. Gradienten und die vollständige $\alpha\beta$-LRP-Familie ergeben sich als Integrale über Spieltrajektorien unter bestimmten Gleichgewichten, sodass Attributionskarten zu Projektionen einer Verteilung über Trajektorien werden, statt das primäre Objekt zu sein. Erwünschte Eigenschaften einer Interpretation — Lokalisierungsschärfe, Robustheit gegenüber Eingangsrauschen, stabiles Attention-Routing — lassen sich als spieltheoretische Begriffe (Politik-Regularisierung, Risikoaversion, erweiterte Aktionsräume) formulieren, die sich unmittelbar in neue Rückwärtsregeln übersetzen; auf ViT-B/16 übertrifft eine solche Anpassung von $\alpha\beta$-LRP alle betrachteten Vergleichsmethoden, einschließlich transformer-spezifischer Methoden, bei der Attributionslokalisierung. Die revidierte Evaluation trennt Lokalisierung und Faithfulness bei der Validierungsauswahl und berichtet Konfidenzintervalle sowie gepaarte Tests; andere Metriken zeigen Zielkonflikte.

Ein einziger Rahmen macht so mehrere Lesarten derselben trainierten Gewichte zugleich verfügbar. Wir entwickeln zudem die tropisch-geometrische Perspektive, in der die zerlegten Ströme tropische Polynome sind, deren Newton-Polytope — *imaginative Polytope* — die erreichbaren Gradienten als gelernte visuelle Prototypen kodieren, sowie eine Trajektorienverteilungs-Sicht, die den Rückwärtsdurchlauf mit einer Hellinger-Distanz zwischen normalisierten Routing-Verteilungen ausstattet und damit das Routing der Trajektorien vor der modellspezifischen Projektion signierter, diskontierter Belegungsmaße auf Heatmaps vergleicht.

Schlüsselwörter: *DC-Zerlegung, Interpretierbarkeit, Neuronale Netze, Spieltheorie, Tropische Geometrie*

English

Interpretability is moving from post hoc analysis of individual predictions toward improving and controlling models. When attributions guide the correction of spurious features, training, or inference, the calculation producing them must itself be understood and checked. This thesis connects two perspectives on that problem: a structured representation of internal contributions and a mathematical account of their backward propagation.

Understanding a trained neural network means accounting for how it turns an input into an output — for the competing influences that every prediction balances against one another. This thesis studies interpretability through *decomposition*: instead of constraining a network to make it interpretable, we expose interpretable structure that a trained ReLU network already contains. A single notion of a *network decomposition* — a lift of the computation into a pair of coordinated, non-negative streams whose difference reproduces the original network exactly — unifies several such structures, none of which alters a single prediction.

The first is the difference-of-convex (DC) decomposition. Monotonicity is known to aid interpretability but to restrict expressivity, so rather than training a monotone model we split a trained ReLU network — an MLP or a CNN — into two subnetworks g and h with $f = g - h$, convex and monotone in doubled input coordinates and carrying positive and negative evidence on separate pathways. We stabilize the evaluated ReLU split and use absolute curvature to construct exact pairs for smooth activations (GELU, Swish, tanh, sigmoid, softplus). Explicit softmax, product, and positive-epsilon LayerNorm pairs also give a DC decomposition of deterministic Transformers of arbitrary finite depth, without a numerical-stability guarantee. On the ReLU split representation we build the saliency methods SplitCAM and SplitLRP, which show configuration-dependent gains in localization and Selectivity on VGG-16 and ResNet-18 over ImageNet-S, with trade-offs in Pixel Flipping and robustness. We also train differences of convex or monotone networks for MNIST classification and observe class-dependent patterns in their stream gradients, providing a qualitative illustration of the representation.

The second is the stopping decomposition, which splits the weights by sign, $W = W^+ - W^-$, separating excitatory from inhibitory contributions along the paths of the network. This decomposition is the bridge to a game-theoretic reading of the backward pass. Building on Gaubert and Vlassopoulos' ReLU Net Game, we recast backward attribution as a two-player zero-sum stopping game on an extended network graph. Gradients and the full $\alpha\beta$-LRP family arise as integrals over game trajectories under specific equilibria, so that attribution maps become projections of a distribution over trajectories rather than the primary object. Desired properties of an interpretation — localization focus, robustness to input noise, stable attention routing — can be stated as game-theoretic concepts (policy regularization, risk aversion, extended action sets) that translate directly into new backward rules; on ViT-B/16, one such adaptation of $\alpha\beta$-LRP outperforms all evaluated baselines, including transformer-specific methods,

in attribution localization. The revised evaluation selects localization and faithfulness configurations independently on validation data and reports confidence intervals and paired tests; other metrics expose trade-offs.

One framework thus makes several readings of the same trained weights available at once. We develop the tropical-geometric perspective, in which the split streams are tropical polynomials whose Newton polytopes — *imaginative polytopes* — encode the attainable gradients as learned visual prototypes; and a trajectory-distribution view that equips the backward pass with a Hellinger distance between normalized routing laws, comparing trajectory routing before the model-dependent projection of signed discounted occupation measures onto heatmaps.

Keywords: *DC Decomposition, Interpretability, Neural Networks, Game Theory, Tropical Geometry*

Contents

List of Figures i

List of Tables ii

List of Symbols iii

List of Abbreviations v

1 Introduction 1
- 1.1 Motivation and Problem Statement . . . 1
- 1.2 Contributions of This Thesis . . . 3
- 1.3 Positioning within the Literature . . . 5

2 Mathematical Foundations 7
- 2.1 Neural Network Notation and PWL Mappings . . . 7
- 2.2 Convexity and Monotonicity in Neural Networks . . . 9
- 2.3 Game Theory and Markov Decision Processes . . . 11
- 2.4 Tropical Geometry and Polyhedra . . . 16

3 Network Decompositions and DC Formulations 25
- 3.1 Generalized Network Decomposition . . . 25
- 3.2 Convex and Monotone DC Decompositions . . . 27
- 3.3 The Mixing Decomposition family . . . 39
- 3.4 The Stopping Decomposition . . . 41
- 3.5 Regularized Decompositions . . . 43
- 3.6 Numerical Stabilization of Split Passes . . . 46
- 3.7 Generalization to Convolutional Neural Networks (CNNs) . . . 50

4 Game-Theoretic Framework 54
- 4.1 Related Work . . . 54
- 4.2 Backward Attribution . . . 55
- 4.3 Attribution as a Game . . . 56
- 4.4 Game-Theoretic Modes . . . 58
- 4.5 Signed affine output logits . . . 58

5 Extensions: Attention and Transformers 61
- 5.1 Related Work: Interpretability in Transformers . . . 61
- 5.2 Decomposing Multiplicative Operations . . . 62
- 5.3 A Constructive DC Decomposition of Transformers . . . 64
- 5.4 Attention via a detached-logit oracle . . . 68
- 5.5 Extension: Vision Transformer Attention . . . 69

6 Saliency and Attribution Methods 73
- 6.1 Split-Based and Difference-of-Convex Attribution . . . 73
- 6.2 Game-Based Attribution: Modes and Technicalities . . . 84

7 Experiments and Evaluation 86
- 7.1 Experimental Setup . . . 86

7.2 Split Methods: VGG-16 . . . 86
7.3 Split Methods: ResNet-18 . . . 87
7.4 Split-Method Stabilization Ablations . . . 88
7.5 Game-Based Case Study . . . 90

8 The Trajectory Distribution View **93**
8.1 Comparison with Normalized Relevance Measures . . . 93
8.2 Trajectory Sanity Check . . . 94
8.3 Hellinger Distance and Disagreement Map . . . 95
8.4 Randomization Sanity Checks . . . 99

9 Conclusion and Future Work **106**
9.1 Summary of Contributions . . . 106
9.2 Limitations . . . 107
9.3 Broader Impact . . . 108
9.4 Future Research Directions . . . 109

A Experimental Protocol and Reproducibility **111**
A.1 Split-Method Evaluation . . . 111
A.2 Evaluation Protocol . . . 111
A.3 Formal Verification and Mathematical Audit . . . 114

B Publication Provenance and Tool Use **115**
B.1 Author's Contribution . . . 115
B.2 Disclosure in the Revised Game Paper . . . 115
B.3 AI Assistance in the Thesis Revision . . . 116

List of References **117**

List of Figures

1.1 The backward pass as a two-player game 2

2.1 Imaginative Polytope of the VGG-16 dragonfly neuron 23

3.1 The network splitting procedure 25
3.2 Monotone, convex split streams and a standard ReLU layer 31

6.1 Stream gradients of the inverted-input DIC model 83
6.2 Second example of inverted-input DIC stream gradients 83
6.3 Input-gradient visualizations for ten MNIST digits 84

7.1 VGG-16 saliency map comparison 87

8.1 Hellinger trajectory distance under parameter randomization 94
8.2 Input-noise Hellinger trajectory distance 98

List of Tables

3.1 Closed scalar profiles for smooth activations 33
3.2 Linear CNN chunks 51

4.1 Modes of the games 59

7.1 Selected interpretability evaluation configurations: VGG-16 88
7.2 Selected interpretability evaluation configurations: ResNet-18 89
7.3 Alpha comparison for SplitGrad and SplitCAM 89
7.4 ViT-B/16 on the ImageNet-S test split 91

8.1 Reference baselines for the input-noise sweep 99
8.2 Parameter randomization sanity check 102
8.3 Hellinger trajectory distance between random initializations 103
8.4 Per-image Spearman rank correlations 104

List of Symbols

Greek Symbols

Symbol	Description	Unit
α	Backward pass shift parameter	
δ	Network gradient (sensitivity)	
δ^g, δ^h	Split stream gradients	
γ	Discount factor / LRP emphasis parameter	
π, σ	Policies of Max and Min players	
τ	Temperature / entropy regularization parameter	
ε	Numerical stability constant	

Indices

Symbol	Description
c	Channel index
i,j	Neuron indices
l	Layer index

Latin Symbols

Symbol	Description	Unit
$a^{(l)}$	Post-activation at layer l	
$a^{(l,+)}, a^{(l,-)}$	Positive and negative activations of split network	
$d^{(l)}$	Width of layer l	
f	Network function	
f_P	Support function of polytope P	
g	Positive split stream (monotone, convex)	
h	Negative split stream (monotone, convex)	
L	Number of hidden layers	
$P^l_{i\to j}$	Transition probability in game	
$Q[l]$	Auxiliary network with min and absolute weights from layer l	
Q^l	Q-function (action-value function)	
$R^{(l)}$	LRP relevance score at layer l	
$V^l_{i,\pm}$	Game value function at state (l,i,$\pm$)	
$W^{(l)}$	Weight matrix of layer l	
$W^{(l,+)}, W^{(l,-)}$	Positive and negative parts of weight matrix	
x	Network input	
$z^{(l)}$	Pre-activation at layer l	

Operators

Symbol	Description	Definition
∇	Gradient operator	$\left[\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_n}\right]^T$
$\odot$	Hadamard (element-wise) product	
ReLU	Rectified linear unit	$\max\{0, x\}$

List of Abbreviations

Methods

ADF	Assumed Density Filtering
AL	Attribution Localization
CAM	**C**lass **A**ctivation **M**apping
CPWL	**C**ontinuous **P**iece**w**ise **L**inear
DC	**D**ifference of **C**onvex
GAE	Generic Attention Explainability
ICNN	**I**nput-**C**onvex **N**eural **N**etwork
LRP	**L**ayer-wise **R**elevance **P**ropagation
MDP	**M**arkov **D**ecision **P**rocess
NRM	**N**ormalized **R**elevance **M**easure
PG	Pointing Game
RG	**R**outing **G**ame
Sel	Selectivity
SG	**S**topping **G**ame
TIBAV	Transformer Interpretability Beyond Attention Visualization

Architectures

CNN	**C**onvolutional **N**eural **N**etwork
GELU	Gaussian Error Linear Unit
LLM	**L**arge **L**anguage **M**odel
MLP	Multi**l**ayer **P**erceptron
ReLU	**R**ectified **L**inear **U**nit
ViT	**Vi**sion **T**ransformer

1 Introduction

1.1 Motivation and Problem Statement

When we ask a machine what it sees, seek its help, or entrust it with a task, we assume some shared understanding of the image. A robot asked to grasp a cup should recognize its shape and position. A system assisting a radiologist should attend to signs of disease, rather than incidental clues about where the image was acquired—a shortcut documented in medical imaging (Zech et al., 2018). We therefore seek vision models aligned with human perception: models whose decisions rest on objects, their features and relations, in ways we can understand and question. Attribution offers a means of investigating this shared visual grounding, which matters for communication, justified trust, and safety.

Heatmaps make this question concrete: is the prediction supported by the object, or by an incidental mark or background? Attribution has exposed such "Clever Hans" strategies (Lapuschkin et al., 2019), which can undermine generalization when familiar correlations change (Geirhos et al., 2020). In vision systems that guide physical actions, these failures pose safety concerns. Deliberate visual manipulations also create security risks, as physical attacks on road-sign classifiers demonstrate (Eykholt et al., 2018). Heatmaps can help identify suspicious dependencies for further testing. Such inspection also extends to "Clever Hans" behavior in anomaly detection and explanations of one-class models (Kauffmann et al., 2020a,b).

The central motivation of this thesis is to support this scrutiny: to expose the visual dependencies behind predictions and help assess whether they match the intended task. Diagnosed dependencies can inform targeted corrections: constraints on input gradients encode which inputs should be irrelevant (Ross et al., 2017) or encourage adversarial robustness (Ross and Doshi-Velez, 2018); Class Artifact Compensation suppresses identified artifact directions (Anders et al., 2022), and Reveal to Revise combines detection and correction (Pahde et al., 2023). Such interventions require evaluating both the attribution and the resulting model behavior.

This role requires a precise account of what interpretability promises. Lipton (2018) distinguish the different objectives often grouped under the term, and Doshi-Velez & Kim (2017) argue that evaluation must match the intended application and audience. The framework of Murdoch et al. (2019) separates predictive accuracy, descriptive accuracy, and relevance to the user. In this thesis, the immediate object of study is the computation of a trained network and the backward rules used to attribute its output. Faithfulness to a specified computation, numerical stability, localization, and sensitivity to learned parameters can be investigated separately; a convincing heatmap alone does not establish that a model relies on the intended evidence.

Two obstacles connect the papers on which this thesis builds. The first concerns the *representation* being interpreted. Weight signs and activation gates combine opposing contributions into a single activation, obscuring how those contributions accumulate. Monotonicity has long provided useful structure for this problem (Amos et al., 2017; Sill, 1997): with non-negative weights, increasing an input cannot decrease downstream evidence. Networks that enforce non-negative weights represent parts of the learned structure without cancellation between positive and negative weighted contributions within a stream (Ayinde and Zurada, 2018; Lee and Seung, 1999). Monotonicity constraints have also been used to express domain-specific fairness requirements (Wang and Gupta, 2020); the constraint encodes the chosen requirement rather than guaranteeing fairness on its own. Inherently interpretable models are an important option, particularly in the high-stakes settings discussed by Rudin (2019).

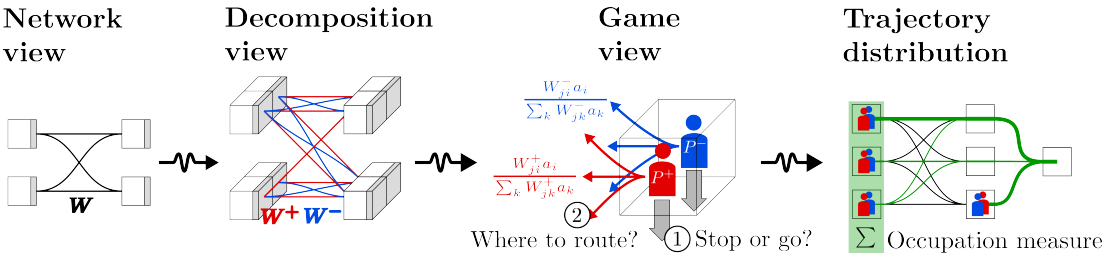


Fig. 1.1: The backward pass as a two-player game. The Stopping Decomposition splits every weight by sign, $W = W^+ - W^-$; at each state of the extended graph the players decide (1) stop or continue and (2) where to route. Their equilibrium policies induce a distribution over backward trajectories, and its signed discounted occupation measures at the input states recover the attribution map.

For an already trained, unconstrained network, however, changing its weights to impose monotonicity generally changes its predictions. Hidden Monotonicity instead splits a trained ReLU network into two monotone, convex stream functions g and h such that $f = g - h$. The key contribution is the numerical stabilization of the forward and backward passes through this representation, making its internal signals accessible without retraining or altering the represented function (Zimmermann and Loho, 2026).

The ***Difference of Convex*** (DC) representation $f = g - h$ suggests a *debate*: one stream accumulates contributions for a class, the other contributions against it, and their difference determines the class score. This motivates inspecting both streams and asking whether the same structure is useful when trained directly. The MNIST difference-model experiments in Section 6.1.4 train two convex or monotone networks jointly; when the h-stream receives the complemented input $\mathbf{1} - x$, the displayed gradients show class-dependent sensitivities on present strokes and candidate missing-stroke locations (Figure 6.1). These qualitative patterns motivate a factual/counterfactual interpretation to be tested through interventions on the complete model.

The second obstacle concerns the *backward calculation* applied to that representation. Gradients, ***Layer-wise Relevance Propagation*** (LRP), Deep Taylor decomposition, and transformer-specific rules are commonly specified through different local recipes and stabilizers. Playing the Network Backward seeks a common account of those rules as weighted trajectories through the computational graph. Building on the ReLU Net Game of Gaubert & Vlassopoulos (2025), its stopping and routing games represent gradients and, at the unit-temperature routing anchor, the full $\alpha\beta$-LRP family through discounted occupation measures. Stop decisions, sign switches, and routing policies then become explicit mathematical objects that can be compared and modified (Figure 1.1).

The thesis joins these perspectives through a single notion of *network decomposition*: lift a signed computation into two coordinated streams whose difference recovers the original network. The DC and stopping decompositions use the same split of weights by sign but retain different internal information. The DC split preserves pre-activation contributions that an inactive ReLU would discard; the stopping split retains the original gate pattern and supports the trajectory representation of the backward pass. A family of mixing and regularized decompositions makes their relationship explicit. This is the common foundation for the split attribution methods, game-based modifications, tropical geometry, and trajectory diagnostics developed in the subsequent chapters.

Oversight depends on the reliability of the evidence it uses. Parameter insensitivity, input dependence, perturbation fragility, and misleading modified-gradient structure (Adebayo et al., 2018; Ghorbani et al., 2019; Kindermans et al., 2018; Nie et al., 2018) can give false assurance about a model's visual dependencies. The two papers address complementary parts of this problem: the split representation

exposes internal contributions and their derivative identities, while the game representation makes routing choices and distributions over calculations explicit. Numerical stabilization enables the former at depth; policy regularization, risk aversion, and extended action sets provide design choices for the latter. Hellinger diagnostics compare the resulting trajectory laws, complementing comparisons of their projected attribution maps. The experiments assess properties of these attribution methods; whether their use improves oversight or supports safer deployment remains a further empirical question.

Three research questions organize this combined account. *What structure can be exposed without changing the network's function?* The decomposition framework connects monotone convex streams, sign-separated paths, and the polyhedral geometry of attainable gradients. *What does a backward attribution actually calculate?* Derivative identities and game occupation measures relate local propagation rules to explicit global objects. *How can these calculations support reliable inspection of vision models?* Stable implementations, controlled modifications of backward rules, and complementary evaluation criteria provide a basis for assessing what an attribution reveals and where its limitations lie.

Publication basis. This thesis brings together and extends two research works on which the thesis author is first author:

- *Hidden Monotonicity: Explaining Deep Neural Networks via their DC Decomposition*, coauthored with Georg Loho, was published at **CVPR 2026 and selected as a Highlight** (Zimmermann and Loho, 2026).[1]
- *Playing the Network Backward: A Game-Theoretic Attribution Framework*, coauthored with Jim Berend, Georg Loho, Sebastian Lapuschkin, and Wojciech Samek, is a **preprint**; this thesis uses arXiv:2605.06212v2, revised for AISTATS (Zimmermann et al., 2026).

Their results are organized by the shared chain of questions: representation, backward semantics, method design, and evaluation. The experiments from each work remain separately identifiable. The thesis reuses their model-evaluation outputs, describes the implemented protocols, and recomputes the VOC uncertainty estimates with images as the resampling unit. It adds a common exposition of the decompositions, further mathematical connections, and a partial Lean formalization; Appendix B records publication provenance and tool use.

Author's contribution. My principal contributions to the two underlying papers were developing the theory, developing and conducting the experiments, and writing all appendices and most of the paper text outside the introductions. These contributions were developed in collaboration with the coauthors listed above. Appendix B.1 distinguishes this work from the subsequent integration and formalization in the thesis.

1.2 Contributions of This Thesis

The contributions of this thesis are the following.

1) **A general framework of network decompositions.** Chapter 3 introduces a *network decomposition* and its *validity*: a lift of a network into a pair of non-negative streams, propagated by non-negative maps $W^{(l,+)}, W^{(l,-)}$ and pairwise activations, whose difference recovers the original computation at every layer. Within it the thesis constructs a whole family — the mixing family f^{η}_{mix} interpolating between maximum- and minimum-based pair activations, the stopping

[1] Official CVPR listing: https://cvpr.thecvf.com/virtual/2026/poster/36564.

decomposition f_{stop}, and regularized decompositions f^{Ω}_{reg} whose entropy-regularized member f^{θ}_{soft} yields sigmoid-gated soft splits. The classical DC split $f = g - h$ is the $\eta = 1$ endpoint of the mixing family rather than an isolated trick, and a splitting theorem for activations with integrable curvature extends the construction beyond ***Re**ctified **L**inear **U**nit* (ReLU) to GELU, Swish, tanh, sigmoid, and softplus networks.

2) **A numerically stable DC decomposition of ReLU networks and *Convolutional Neural Network*s (CNNs).** A naive split can grow rapidly with depth while its difference stays bounded, so floating-point cancellation destroys the signal. Chapter 3 develops stabilization through transformations of the split activations and propagated signals. The exact convex representation extends to convolutions, convexity-preserving max-pooling, average pooling, fixed-statistic batch normalization, residual connections, and fixed dropout masks. The evaluated implementation additionally uses input-dependent recentering, cache correction, and optional winner-takes-it-all pooling. These procedures target forward reconstruction and usable backward signals; the individual stabilized streams do not inherit all convexity and monotonicity guarantees of the original representation.
3) **A game-theoretic reformulation of the backward pass.** Chapter 4 reinterprets the stopping decomposition as a two-player zero-sum stopping game on an extended network graph, adapting the ReLU Net Game of Gaubert & Vlassopoulos (2025): layers are rounds, neurons states, sign switches hand the turn to the opponent, and the network output is the game value. Backward attribution is represented by *signed discounted occupation measures over game trajectories*: propagation rules are equilibrium policies of the players P^+ and P^-, and attribution maps discounted occupation measures of that distribution. Two games anchor the theory — the ***S**topping **G**ame* (SG), whose occupation measures recover the Jacobian, and the entropy-regularized ***R**outing **G**ame* (RG), recovering the full $\alpha\beta$-LRP family — giving the two methods a shared trajectory-space representation. These are representation theorems: the architecture alone does not determine a unique game or discount. The trajectory-space view of Chapter 8 further yields a Hellinger distance between normalized backward *routing laws*, detecting routing changes that pixel-space sanity checks (Adebayo et al., 2018) may obscure, and analytical *modes* — risk aversion, action-space augmentation, policy regularization — mapping back to principled backpropagation rules.
4) **New attribution methods and benchmark evaluations.** Chapter 6 builds three attribution methods on the split representation: SplitCAM, a LayerCAM-inspired (Jiang et al., 2021) ***C**lass **A**ctivation **M**apping* (CAM) method on the split-streams; SplitLRP, an adaptation of LRP (Bach et al., 2015) to the split-pair; and SplitGrad, a gradient method exploiting the improved signal flow. Evaluated with the Quantus suite (Hedström et al., 2023) on VGG-16 (Simonyan and Zisserman, 2015) and ResNet-18 (He et al., 2016) over ImageNet-S (Mao et al., 2022), they show configuration-dependent gains in localization and Selectivity, with trade-offs in Pixel Flipping and robustness (Chapter 7). On ResNet-18, Guided Grad-CAM retains the highest reported Attribution Localization, and SplitCAM's Selectivity advantage over Guided Backprop is 0.014. The chapter also develops directly trained difference-of-convex input neural networks and evaluates their stream sensitivities on MNIST (Section 6.1.4).
5) **A tropical-geometric interpretation and Imaginative Polytopes.** The tropical foundations (Section 2.4) place the decomposition in the polyhedral-geometry view of ReLU networks (Misiakos and Thrampoulidis, 2022; Zhang et al., 2018b): the split-streams g and h are the two tropical polynomials whose difference the network computes, their Newton polytopes evolving across layers by Minkowski sums and convex hulls. The thesis introduces *Imaginative Polytopes* and proves that their vertices encode the attainable gradients of the split-streams, giving them an interpretation as learned visual prototypes and counterfactual feature profiles.

6) **Extension to attention mechanisms and Transformers.** Chapter 5 gives a recursive DC construction for deterministic Transformers of arbitrary finite depth, including signed attention projections, exact softmax, and LayerNorm with positive epsilon. Its streams are convex and monotone on non-negative doubled inputs and recover the full forward map exactly; numerical stability remains a separate issue. A distinct oracle-player construction treats the realized QK-softmax as a reference policy for conditioned Value routing. On ViT-B/16 (Dosovitskiy et al., 2021), a selected game-derived attribution mode improves localization, with the selected configuration also evaluated on Pascal VOC without re-tuning. Pointing Game, Top-k overlap, faithfulness, and robustness reveal distinct trade-offs; the revised evaluation reports independently validation-selected localization and faithfulness configurations, confidence intervals, and paired tests. These empirical results concern conditioned attribution, not the new full Transformer DC algorithm.

1.3 Positioning within the Literature

This section positions the thesis along its three research questions; it is brief, since dedicated related-work discussions open the chapters where the material is developed — games and propagation rules in Section 4.1 and transformer interpretability in Section 5.1.

What structure is hidden inside a trained network? The idea that sign-constrained networks are the interpretable core of deep learning runs from monotonic networks (Daniels and Velikova, 2010; Sill, 1997) through certified and lattice-based monotone models (Gupta et al., 2016; Liu and Coogan, 2020; Nguyen et al., 2023; Runje and Shankaranarayana, 2023; Sivaraman et al., 2020) to ***I**nput-**C**onvex **N**eural **N**etwork*s (ICNNs) (Amos et al., 2017) and convex network calculus (Bach, 2017; Calafiore et al., 2020; Sivaprasad et al., 2021). DC decompositions bridge this to unconstrained networks: classical DC programming (Sriperumbudur and Lanckriet, 2009; Tao and An, 1997; Thi and Dinh, 2018) and the representation theory of piecewise-linear functions as differences of convex ones (Brandenburg et al., 2025; Griewank and Walther, 2020; Kazda and Li, 2024; Koutschan et al., 2023; Kripfganz and Schulze, 1987; Melzer, 1986; Schlüter and Darup, 2021; Zalgaller, 2000) guarantee the split exists; Awasthi et al. (2023) and Siahkamari et al. (2020) construct essentially the split used here, for training and regression, and CDiNN-type architectures make the difference structure trainable (Chen, Wang, et al., 2025; Sankaranarayanan and Rengaswamy, 2022). None confronts the numerical instability of the split or applies it to explanation, where this thesis departs. The geometric reading rests on the identification of ReLU networks with ***C**ontinuous **P**iece**w**ise **L**inear* (CPWL) maps and tropical rational functions (Balestriero and Baraniuk, 2021; Huchette et al., 2025; Koehler and Risteski, 2018; Magron and Wang, 2022; Misiakos and Thrampoulidis, 2022; Mukherjee et al., 2018; Phuong and Lampert, 2020; Zhang et al., 2018b); the Newton-polytope perspective is developed in Section 2.4.

When can explanations be trusted? The attribution toolbox spans gradient saliency (Simonyan et al., 2014), deconvolution and guided backpropagation (Springenberg et al., 2015; Zeiler and Fergus, 2014), LRP and Deep Taylor decomposition (Bach et al., 2015; Montavon et al., 2017, 2018), CAM variants (Chattopadhay et al., 2018; Jiang et al., 2021; Selvaraju et al., 2017), axiomatic methods (Shrikumar et al., 2017; Sundararajan et al., 2017), noise averaging (Smilkov et al., 2017), and perturbation- or surrogate-based explanations (Fel et al., 2023; Fong and Vedaldi, 2017; Kolek et al., 2023; Lundberg and Lee, 2017; Ribeiro et al., 2016). A wave of critiques sharpened the trust question: sanity checks reveal parameter insensitivity (Adebayo et al., 2018), explanations can be fragile or manipulable (Alvarez-Melis and Jaakkola, 2018; Dombrowski et al., 2019; Ghorbani et al., 2019), modified backward rules

can produce structure unrelated to the model (Nie et al., 2018; Sixt et al., 2020), and input dependence confounds attribution (Kindermans et al., 2018). In response, evaluation frameworks such as Quantus (Hedström et al., 2023) operationalize faithfulness, localization (Zhang et al., 2018a), robustness, and complexity (Bhatt et al., 2020; Yeh et al., 2019) as measurable desiderata — the standard this thesis holds its methods to in Chapter 7. The thesis' answer is structural rather than post hoc: attribution is computed on a representation (the split-pair) in which evidence cannot cancel invisibly, or derived from a game whose trajectory distribution records normalized backward routing and whose signed discounted occupation measures recover attribution, so sanity-check failures can be studied at both the trajectory and pixel levels (Chapter 8); their different responses to randomization do not by themselves establish explanation faithfulness.

How can attribution be made stable? The closest precursor to the game framework is the ReLU Net Game of Gaubert & Vlassopoulos (2025), which identifies forward evaluation with the value of a zero-sum stopping game but connects it neither to the backward pass nor to attribution. Related stochastic views of relevance flow include Excitation Backprop's absorbing Markov chain (Zhang et al., 2018a), which the SG generalizes by adding an opponent, and the ***N**ormalized **R**elevance **M**easure* (NRM) framework (Xiong et al., 2026), a signed-measure calculus over walks compared with our trajectory distributions in Section 8.1. Shapley-value methods (Lundberg and Lee, 2017) play a *different*, cooperative game over input coalitions in a black-box setting, complementary to the white-box backward game here. Stability itself has been approached via curvature reduction with softplus activations (Dombrowski et al., 2022) — recovered here as entropy regularization of the stop decision — and via risk-sensitive control (Fei et al., 2020) motivating the risk-averse game modes. On transformers, attention rollout and flow (Abnar and Zuidema, 2020), transformer-LRP (Achtibat et al., 2024; Ali et al., 2022; Chefer et al., 2021a,b), and artifact-aware constructions (Darcet et al., 2024; Wróbel et al., 2026) define the state of the art Chapter 5 engages with; Section 4.1 and Section 5.1 give the full accounts. For visual inspection and attribution-guided correction (Pahde et al., 2023), instability can obscure unwanted dependencies and misdirect subsequent interventions.

Outline. Chapter 2 assembles the foundations: network notation, CPWL mappings, convexity and monotonicity, ***M**arkov **D**ecision **P**rocess*s (MDPs) and zero-sum games — including the ReLU Net Game of Gaubert and Vlassopoulos — and the tropical geometry of the split with the Imaginative-Polytope picture. Chapter 3 develops the decomposition framework, the smooth-activation splitting theorem, numerical stabilization, and the CNN extension; Chapter 4 the two-player stopping and routing games, with gradients and $\alpha\beta$-LRP recovered as equilibrium occupation measures; Chapter 5 the extension and implementation on multiplicative operations and ***Vi**sion **T**ransformer*s (ViTs); and Chapter 6 the split methods SplitCAM, SplitLRP, and SplitGrad, inherently interpretable difference-of-convex architectures, and the attribution modes. Chapter 7 reports the evaluation on ImageNet-S and Pascal VOC across VGG-16, ResNet, and ViT-B/16; Chapter 8 develops the trajectory-distribution view and its Hellinger diagnostics; and Chapter 9 concludes with future directions from ***L**arge **L**anguage **M**odel*s (LLMs) to hardware-level acceleration.

2 Mathematical Foundations

This chapter fixes the notation, definitions, and background theory used throughout. Section 2.1 fixes the feedforward network notation and recalls the equivalence between ReLU networks and CPWL functions. Section 2.2 introduces the two structured classes—monotone networks and ICNNs—into which Chapter 3 splits a general network, together with the Maxout units generalizing ReLU. Section 2.3 collects the background on MDPs and zero-sum turn-based games and presents the ReLU Net Game of Gaubert and Vlassopoulos in full, so that Chapter 4 can state which parts of the framework it inherits and which are new. Section 2.4 provides the tropical-geometric vocabulary—semirings, tropical polynomials, Newton polytopes, Minkowski sums, and support functions—and, in its later subsections, develops the tropical reading of the DC split and the resulting Imaginative Polytopes (Sections 2.4.1–2.4.3). The activation-level construction from absolute curvature, including smooth activations, is developed with the network decomposition theorem in Section 3.2.2.

2.1 Neural Network Notation and PWL Mappings

The standard feedforward architecture serves as the reference model throughout; convolutional and attention-based architectures are treated in Section 3.7 and Chapter 5 as structured special cases and extensions.

Definition 2.1 (Feedforward ReLU network). Let $L \in \mathbb{N}$ and let $d^{(0)}, d^{(1)}, \dots, d^{(L)} \in \mathbb{N}$ denote the *layer widths*, with input dimension $d^{(0)}$ and output dimension $d^{(L)}$. A *feedforward ReLU network* of depth L is given by weight matrices $W^{(l)} \in \mathbb{R}^{d^{(l)} \times d^{(l-1)}}$ and bias vectors $b^{(l)} \in \mathbb{R}^{d^{(l)}}$ for $l = 1, \dots, L$. On an input $x \in \mathbb{R}^{d^{(0)}}$ it computes the recursion

$$\begin{aligned} a^{(0)} &:= x, \\ z^{(l)} &:= W^{(l)} a^{(l-1)} + b^{(l)}, && l = 1, \dots, L, \\ a^{(l)} &:= \mathrm{ReLU}\big(z^{(l)}\big), && l = 1, \dots, L-1, \end{aligned} \tag{2.1}$$

where $\mathrm{ReLU}(z) := \max\{z, 0\}$ acts componentwise, and the output layer is affine, $a^{(L)} := z^{(L)}$. We call $z^{(l)}$ the *pre-activation* and $a^{(l)}$ the *(post-)activation* of layer l, and denote the *network function* by

$$f \colon \mathbb{R}^{d^{(0)}} \longrightarrow \mathbb{R}^{d^{(L)}}, \qquad f(x) := a^{(L)}. \tag{2.2}$$

Layers $1, \dots, L-1$ are the *hidden layers*. Throughout we fix the boundary convention $\mathrm{ReLU}'(0) := 0$, so that the gate indicator $\mathbf{1}\{z_i^{(l)} > 0\}$ returns 0 at $z_i^{(l)} = 0$.

When a statement concerns the layers $l, \dots, L$ only, we write $f^{(l)} \colon \mathbb{R}^{d^{(l)}} \to \mathbb{R}^{d^{(L)}}$ for the tail map, so that $f = f^{(1)} \circ \big(x \mapsto \mathrm{ReLU}(W^{(1)}x + b^{(1)})\big)$ and $f^{(L)}$ is the affine output map. A bias-free formulation is always available by appending an *intercept input* of constant value 1 and absorbing $b^{(l)}$ into a column of $W^{(l)}$; we use this in some proofs without comment.

The decompositions of Chapter 3 act on the signs of parameters and inputs, fixed by the following notation.

Definition 2.2 (Positive and negative parts). For a scalar $u \in \mathbb{R}$ we write $u^+ := \max\{u, 0\}$ and $u^- := \max\{-u, 0\}$, so that $u = u^+ - u^-$ and $|u| = u^+ + u^-$, with $u^+ u^- = 0$. For a matrix or vector W

and a sign label $\sigma \in \{+,-\}$, the *entrywise part* $[W]^\sigma$ is obtained by applying $u \mapsto u^\sigma$ to every entry. For the layer weights and biases of Definition 2.1 we abbreviate

$$W^{(l,\sigma)} := [W^{(l)}]^\sigma, \qquad b^{(l,\sigma)} := [b^{(l)}]^\sigma, \qquad \sigma \in \{+,-\}, \tag{2.3}$$

so that $W^{(l)} = W^{(l,+)} - W^{(l,-)}$ and $|W^{(l)}| = W^{(l,+)} + W^{(l,-)}$ entrywise, and both $W^{(l,\sigma)}$ are entrywise non-negative.

Remark 2.1 (Pair notation for decomposed activations). Superscript sign labels on *activations*, written $a^{(l,\sigma)}$ for $\sigma \in \{+,-\}$, are reserved for the two non-negative *streams* of a network decomposition and are defined in Section 3.1; they do *not* denote the entrywise parts $[a^{(l)}]^\sigma$ of the original activations. The distinction matters: the decomposed streams carry strictly more information than the sign pattern of $a^{(l)}$—the source of the sub-threshold signals exploited in Chapter 6.

Continuous piecewise linear functions

The function class computed by ReLU networks admits an exact geometric description.

Definition 2.3 (CPWL function). A function $f\colon \mathbb{R}^d \to \mathbb{R}^m$ is ***C**ontinuous **P**iece**w**ise **L**inear* (CPWL) if it is continuous and there exist finitely many closed polyhedra $Q_1, \dots, Q_N \subseteq \mathbb{R}^d$ with $\bigcup_{r=1}^N Q_r = \mathbb{R}^d$ such that the restriction of f to each Q_r is an affine map. The full-dimensional polyhedra on which f is affine, taken maximal with respect to inclusion, are called the *linear regions* of f.

For example, $|\cdot|\colon \mathbb{R} \to \mathbb{R}$ is CPWL with the two regions $(-\infty, 0]$ and $[0, \infty)$, and $x \mapsto \max\{x_1, \dots, x_d\}$ is CPWL with d regions.

Observation 2.1 (`Lean_1`). Every feedforward ReLU network represents a CPWL function.

Proof. Affine maps are CPWL, the componentwise ReLU is CPWL, and the composition of two CPWL maps is again CPWL: if f is affine on the polyhedra $Q_1, \dots, Q_N$ covering $\mathbb{R}^d$ and g is affine on the polyhedra $P_1, \dots, P_M$ covering the intermediate space, then $g \circ f$ is affine on each of the at most NM polyhedra $Q_r \cap f^{-1}(P_s)$ (preimages of polyhedra under affine maps are polyhedra), these cover $\mathbb{R}^d$, and continuity is preserved under composition. The claim follows by induction along the recursion (2.1). □

The converse also holds: *every* CPWL function is exactly representable by a ReLU network of bounded depth. This equivalence, due to Mukherjee et al. (2018), is the entry point for the difference-of-convex analysis of Chapter 3 and the tropical-geometric analysis of Section 2.4.1.

Theorem 2.1 (CPWL–ReLU equivalence, (Mukherjee et al., 2018), `Lean_2`). *A function $f\colon \mathbb{R}^d \to \mathbb{R}$ is CPWL if and only if it is representable by a feedforward ReLU network. Moreover, every CPWL function on $\mathbb{R}^d$ is representable by a ReLU network of depth at most $\lceil \log_2(d+1) \rceil + 1$.*

Proof. The "if" direction is Observation 2.1. For the converse we give the classical construction and refer to Mukherjee et al. (2018) for the depth bookkeeping. First, every CPWL function f admits a *difference-of-convex* representation

$$f = g - h, \qquad g = \max_{r \in [N_g]} \ell_r, \qquad h = \max_{s \in [N_h]} m_s, \tag{2.4}$$

with finitely many affine functions $\ell_r, m_s \colon \mathbb{R}^d \to \mathbb{R}$; this is the piecewise-linear DC theorem of Melzer (1986) and Kripfganz & Schulze (1987), and both g and h are convex as finite maxima of affine functions. Second, the maximum of two scalars is a two-layer ReLU computation,

$$\max\{u, v\} = \frac{u+v}{2} + \frac{|u-v|}{2} = \frac{u+v}{2} + \frac{\mathrm{ReLU}(u-v) + \mathrm{ReLU}(v-u)}{2}, \tag{2.5}$$

where the affine pre- and post-combinations are absorbed into the surrounding weight matrices. Arranging the N_g affine arguments of g in a balanced binary tree of pairwise maxima realizes g by a ReLU network of depth $\lceil \log_2 N_g \rceil + 1$, and likewise for h; a final affine layer forms the difference (2.4). This proves representability. The stated depth bound, which is uniform in the number of affine pieces and depends only on the input dimension d, follows from the refinement of Mukherjee et al. (2018): every CPWL function on $\mathbb{R}^d$ can be written as a signed sum of maxima of at most $d+1$ affine functions each, so the trees in the construction above have depth at most $\lceil \log_2(d+1) \rceil$. □

Remark 2.2 (Expressivity and linear regions). Theorem 2.1 identifies the function *class*, not the *cost* of representation: deep networks can represent CPWL functions with exponentially many linear regions in the number of neurons, an interplay of depth, width, and region counts surveyed by Huchette et al. (2025). The tropical reformulation by Zhang et al. (2018b)—ReLU networks as tropical rational maps—is taken up in Sections 2.4 and 2.4.1. The representation (2.4) also foreshadows the central theme of this thesis: the convex CPWL streams g and h of the DC split can be extracted *constructively and locally* from a trained network, the subject of Section 3.2.3.

2.2 Convexity and Monotonicity in Neural Networks

The decompositions of Chapter 3 split a general network into components from two structured classes with sign-constrained weights. We define both and record the load-bearing facts that they represent monotone and convex functions, respectively.

We order $\mathbb{R}^d$ coordinatewise: $x \le y$ if and only if $x_i \le y_i$ for all $i \in [d]$. A function $f \colon \mathbb{R}^d \to \mathbb{R}^m$ is *monotone* (non-decreasing) if $x \le y$ implies $f(x) \le f(y)$.

Definition 2.4 (Monotone neural network). A feedforward network with recursion (2.1) and monotone activation functions is called a *monotone neural network* if all its weight matrices are entrywise non-negative, $W^{(l)} \ge 0$ for all $l = 1, \dots, L$.

Weight-sign constraints are the oldest device for building monotone models (Daniels and Velikova, 2010; Sill, 1997) and remain the structural core of modern certified-monotone architectures (Liu and Coogan, 2020; Runje and Shankaranarayana, 2023).

Proposition 2.2 (Monotone networks represent monotone functions, `Lean_3`**).** *A monotone neural network represents a monotone function; each coordinate function is non-decreasing in every input variable.*

Proof. By induction over the layers. The input layer $a^{(0)} = x$ is monotone in x. Assume $a^{(l-1)}$ is monotone in x. An affine map with non-negative matrix preserves the coordinatewise order: if $a \le a'$, then $W^{(l)}a + b^{(l)} \le W^{(l)}a' + b^{(l)}$ entrywise, because every entry of $W^{(l)}$ is non-negative. Hence $z^{(l)}$ is monotone in x, and applying the monotone activation componentwise preserves monotonicity, so $a^{(l)}$ is monotone in x. The affine output layer is again order-preserving. □

Definition 2.5 (Input-Convex Neural Network, (Amos et al., 2017)). A feedforward network with recursion (2.1) whose activation functions are convex *and* monotone is called an ***I**nput-**C**onvex **N**eural **N**etwork* (ICNN) if all weight matrices except possibly the first are entrywise non-negative, $W^{(l)} \geq 0$ for all $l = 2, \ldots, L$, while $W^{(1)}$ is unconstrained.

Definition 2.5 covers the plain feedforward core of the architecture of Amos et al. (2017); their sign-unconstrained skip ("passthrough") connections do not affect the statement below, since they enter each pre-activation affinely in x.

Proposition 2.3 (ICNNs represent convex functions, `Lean_4`). *An ICNN represents a convex function; each coordinate of every activation $a^{(l)}$, $l \geq 1$, is a convex function of the input x.*

Proof. We use two closure properties of convex functions: (i) a non-negative linear combination of convex functions plus a constant is convex; (ii) if $\varphi\colon \mathbb{R} \to \mathbb{R}$ is convex and non-decreasing and $u\colon \mathbb{R}^d \to \mathbb{R}$ is convex, then $\varphi \circ u$ is convex, since for $\lambda \in [0,1]$,

$$\varphi\big(u(\lambda x + (1-\lambda)y)\big) \leq \varphi\big(\lambda u(x) + (1-\lambda)u(y)\big) \leq \lambda\,\varphi(u(x)) + (1-\lambda)\,\varphi(u(y)),$$

where the first inequality uses convexity of u and monotonicity of φ, and the second convexity of φ.

Now induct over layers. Each coordinate of $z^{(1)} = W^{(1)}x + b^{(1)}$ is affine, hence convex, in x; applying the convex monotone activation gives convex coordinates of $a^{(1)}$ by (ii). Assume all coordinates of $a^{(l-1)}$ are convex in x. For $l \geq 2$ the weights are non-negative, so each coordinate of $z^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)}$ is a non-negative combination of convex functions plus a constant, hence convex by (i), and applying the activation preserves convexity by (ii). The output coordinates $a^{(L)} = z^{(L)}$ are convex by the same argument with the identity activation. □

Remark 2.3 (Monotonicity and convexity combine). If, in addition, $W^{(1)} \geq 0$ in Definition 2.5, the network is simultaneously monotone and input-convex, hence non-decreasing *and* convex by Propositions 2.2 and 2.3. Both ReLU and Maxout (below) are convex and monotone, so non-negative-weight ReLU networks are of exactly this type. These are the building blocks of the DC splitting of Section 3.2.3: the streams g and h are computed by monotone networks acting on a signed embedding of the input, and are input-convex in the original input. The sign constraint also *limits* expressivity—a weight-constrained ReLU network represents only convex functions, precisely Proposition 2.3, which is why monotone architectures add concave counterparts of the activation (Runje and Shankaranarayana, 2023). The decompositions of this thesis turn the limitation into a feature: the difference of two such networks recovers full generality by Theorem 2.1 and (2.4).

Maxout units

Definition 2.6 (Maxout unit and Maxout network, (Goodfellow et al., 2013)). A *rank-k Maxout unit* with weights $w_1, \ldots, w_k \in \mathbb{R}^d$ and biases $\beta_1, \ldots, \beta_k \in \mathbb{R}$ computes

$$x \longmapsto \operatorname{Maxout}(x) := \max_{j \in [k]} \big(w_j^\top x + \beta_j\big). \tag{2.6}$$

A *rank-k Maxout network* is a feedforward network whose neurons are rank-k Maxout units; equivalently, layer l carries k weight matrices $W_1^{(l)}, \ldots, W_k^{(l)}$ and biases $b_1^{(l)}, \ldots, b_k^{(l)}$, and

$$a_i^{(l)} := \max_{j \in [k]} \big(W_j^{(l)} a^{(l-1)} + b_j^{(l)}\big)_i. \tag{2.7}$$

Observation 2.2 (ReLU and max-pooling as Maxout, `Lean_5`*).* The ReLU activation is the rank-2 Maxout unit with parameters $(w_1, \beta_1) = (w, \beta)$ and $(w_2, \beta_2) = (0, 0)$, since $\mathrm{ReLU}(w^\top x + \beta) = \max\{w^\top x + \beta, 0\}$. Max-pooling over a group J of neurons is the rank-$|J|$ Maxout unit whose j-th argument is the coordinate projection onto the j-th element of J. Every rank-k Maxout unit is convex in its input and non-decreasing in each pre-activation argument $w_j^\top x + \beta_j$; hence Maxout qualifies as an admissible activation in Definitions 2.4 and 2.5.

Proof. The first two claims are immediate from (2.6). Convexity holds because a pointwise maximum of affine (hence convex) functions is convex. For monotonicity in the arguments, view the unit as $(u_1, \dots, u_k) \mapsto \max_j u_j$ with $u_j = w_j^\top x + \beta_j$: if $u_j \le u'_j$ for all j, then $\max_j u_j \le \max_j u'_j$. □

Remark 2.4. Maxout networks are the natural piecewise-linear generalization of ReLU: a rank-k unit expresses a maximum of k affine functions in one neuron, matching the building blocks of the DC representation (2.4). The DC splitting of Chapter 3 is developed for ReLU networks and extended to Maxout units—and to the multiplicative interactions needed for attention—in Chapter 5, using the exchange identity $\max_{j\in[k]}(u_j - v_j) = \max_{j\in[k]}\{u_j + \sum_{i\neq j} v_i\} - \sum_{i\in[k]} v_i$ for pairs of non-negative streams.

2.3 Game Theory and Markov Decision Processes

Chapter 4 interprets the backward pass of a decomposed network as a two-player zero-sum game on the network graph. This section fixes the background: MDPs, turn-based zero-sum games, and their optimality equations. We restrict to finite state and action spaces, which suffices throughout.

Definition 2.7 (Markov Decision Process). A (finite) ***M**arkov **D**ecision **P**rocess* (MDP) (Puterman, 1994; Sutton and Barto, 2018) is a tuple $\mathscr{M} = (S, A, \mathbb{P}, R, \gamma)$ consisting of a finite state set S, a finite action set $A(s)$ for each $s \in S$, a *transition kernel* $\mathbb{P}(s' \mid s, a) \ge 0$ with $\sum_{s'} \mathbb{P}(s' \mid s, a) = 1$, a bounded *reward function* $R(s, a) \in \mathbb{R}$, and a *discount factor* $\gamma \in [0, 1)$. A *(stationary) policy* is a map π assigning to each state s a probability distribution $\pi(\cdot \mid s)$ over $A(s)$; π is *pure* (deterministic) if each $\pi(\cdot \mid s)$ is a point mass. A policy induces, for every start state, a Markov chain on S and hence a distribution over trajectories $(s_0, a_0, s_1, a_1, \dots)$; the *value function* of π is the expected discounted return

$$V^\pi(s) := \mathbb{E}^\pi\Big[\textstyle\sum_{t=0}^{\infty} \gamma^t\, R(s_t, a_t) \,\Big|\, s_0 = s\Big], \tag{2.8}$$

and the *optimal value function* is $V^\star(s) := \max_\pi V^\pi(s)$.

Fixing an enumeration of S, a pure policy π turns the kernel into a row-stochastic *transition matrix* $P^\pi \in \mathbb{R}^{S\times S}$ with $P^\pi_{ss'} = \mathbb{P}(s' \mid s, \pi(s))$ and the reward into a vector R^π, and (2.8) becomes the Neumann series $V^\pi = \sum_{t\ge 0} \gamma^t (P^\pi)^t R^\pi$. This matrix picture recurs in Chapter 4, where the (sub)stochastic matrices are built from the normalized weight parts $W^{(l,\sigma)}$ of Definition 2.2.

Proposition 2.4 (Bellman equations, `Lean_6`**).** *Let $\mathscr{M}$ be a finite MDP with $\gamma \in [0,1)$. Then:*

1) V^π is the unique solution of the linear system

$$V(s) = \sum_{a\in A(s)} \pi(a \mid s)\Big[R(s,a) + \gamma \sum_{s'} \mathbb{P}(s' \mid s, a)\, V(s')\Big], \qquad s \in S. \tag{2.9}$$

2) $V^\star$ is the unique solution of the Bellman optimality equation

$$V(s) = \max_{a \in A(s)} \Big\{ R(s,a) + \gamma \sum_{s'} \mathbb{P}(s' \mid s,a)\, V(s') \Big\}, \qquad s \in S, \tag{2.10}$$

and any pure policy $\pi^\star$ that selects, in every state, an action attaining the maximum in (2.10) is optimal: $V^{\pi^\star} = V^\star$.

Proof. Both claims are standard (Puterman, 1994; Sutton and Barto, 2018). The right-hand sides of (2.9) and (2.10) define γ-contractions on $\big(\mathbb{R}^S, \|\cdot\|_\infty\big)$, so each has a unique fixed point by the Banach fixed-point theorem; conditioning (2.8) on the first transition identifies these fixed points as V^π and $V^\star$, and any policy greedy for $V^\star$ is optimal. The acyclic games below make even the contraction argument unnecessary (Remark 2.5). □

Remark 2.5 (State-dependent structural discounts). The games of this thesis depart from Definition 2.7 in one respect: the discount is *state-dependent*, $\gamma(s) \geq 0$, and may exceed 1. Here discounts are not impatience parameters but *structural* normalizers—for neuron i in layer l the absolute weight mass

$$\gamma_i^{(l)} := \sum_j \big(W^{(l,+)} + W^{(l,-)}\big)_{ij} = \sum_j |W_{ij}^{(l)}|, \tag{2.11}$$

which restores the weight mass after row i of $|W^{(l)}|$ is normalized into transition probabilities. Contraction is then unavailable, but also unnecessary: all games are played on *layered acyclic* graphs, so every trajectory is finite and all value functions are well-defined finite sums. Backward induction gives a unique value and the existence of optimal policies; policies need not be unique when actions tie. The two-player statement follows below.

Zero-sum turn-based games

Two-player games enter because a decomposed network carries two interacting streams: a player P^+ who collects positive evidence and a player P^- who collects negative evidence. We use the following conventions throughout: $p, q, r \in \{+,-\}$ denote player, turn, or sign labels; p' denotes the opponent ($+' = -$, $-' = +$); the *sign-XOR* $p \oplus q$ is $+$ if $p = q$ and $-$ otherwise; and $\xi_p \in \{+1, -1\}$ is the numerical sign of the label p, so that $\xi_p\, \xi_q = \xi_{p \oplus q}$.

Definition 2.8 (Zero-sum turn-based game). A (finite) *zero-sum turn-based game* is a tuple $\mathscr{G} = (S_+, S_-, S_{\text{ter}}, A, \mathbb{P}, R, \gamma)$ where the state set $S = S_+ \uplus S_- \uplus S_{\text{ter}}$ is partitioned into states controlled by player P^+, states controlled by player P^-, and terminal states; the action sets $A(s)$, the transition kernel $\mathbb{P}$, and the reward R are as in Definition 2.7 on non-terminal states; $\gamma \colon S \to \mathbb{R}_{\geq 0}$ is a (possibly state-dependent) discount; and each terminal state $s \in S_{\text{ter}}$ carries a fixed one-time payoff $R(s)$. A *policy pair* $\pi = (\pi_+, \pi_-)$ consists of a policy π_p for player P^p on S_p. Play generates a trajectory as in an MDP, with the player in turn choosing the action; the expected total payoff $V^\pi(s)$, defined as in (2.8) with the discount accumulated multiplicatively along the trajectory (γ^t replaced by $\prod_{m=0}^{t-1} \gamma(s_m)$), is paid *by P^- to P^+*: player P^+ maximizes V^π and player P^- minimizes it.

Theorem 2.5 (Shapley–Bellman optimality on layered games, `Lean_7`). *Let $\mathscr{G}$ be a finite zero-sum turn-based game whose transition graph is acyclic. Then $\mathscr{G}$ has a* value*: there is a unique*

function $V^\star\colon S \to \mathbb{R}$ *satisfying the* Shapley–Bellman optimality equations

$$V^\star(s) = \begin{cases} \max\limits_{a \in A(s)} \Big\{ R(s,a) + \gamma(s) \sum\limits_{s'} \mathbb{P}(s' \mid s,a)\, V^\star(s') \Big\}, & s \in S_+, \\ \min\limits_{a \in A(s)} \Big\{ R(s,a) + \gamma(s) \sum\limits_{s'} \mathbb{P}(s' \mid s,a)\, V^\star(s') \Big\}, & s \in S_-, \\ R(s), & s \in S_{\mathrm{ter}}, \end{cases} \tag{2.12}$$

and there exist pure stationary policies $\pi^\star = (\pi_+^\star, \pi_-^\star)$*, obtained by selecting maximizing respectively minimizing actions in* (2.12)*, such that for all* $s \in S$ *and all policies* π_+, π_-*,*

$$V^{(\pi_+^\star, \pi_-)}(s) \;\geq\; V^{\pi^\star}(s) = V^\star(s) \;\geq\; V^{(\pi_+, \pi_-^\star)}(s). \tag{2.13}$$

In particular, $\pi^\star$ *is a Nash equilibrium of every subgame, i.e. a subgame perfect equilibrium.*

Proof. Since the graph is finite and acyclic, its states admit a topological order; we induct backward along it. For terminal states, (2.12) fixes $V^\star(s) = R(s)$ and (2.13) is trivial. Consider a non-terminal state $s \in S_+$ and assume $V^\star$ is uniquely defined and (2.13) holds for all states reachable from s. The bracket in (2.12) is a well-defined finite number for every action, so the maximum exists and is attained by some pure choice $\pi_+^\star(s)$; this defines $V^\star(s)$ uniquely. For the saddle-point property, fix any opponent policy π_- and let P^+ play $\pi_+^\star$. Conditioning on the first transition from s and using the induction hypothesis (the left inequality of (2.13) at the successor states) together with $\gamma(s) \geq 0$,

$$\begin{aligned} V^{(\pi_+^\star, \pi_-)}(s) &= R\big(s, \pi_+^\star(s)\big) + \gamma(s) \sum_{s'} \mathbb{P}\big(s' \mid s, \pi_+^\star(s)\big)\, V^{(\pi_+^\star, \pi_-)}(s') \\ &\geq R\big(s, \pi_+^\star(s)\big) + \gamma(s) \sum_{s'} \mathbb{P}\big(s' \mid s, \pi_+^\star(s)\big)\, V^\star(s') \;=\; V^\star(s), \end{aligned}$$

where the last equality holds because $\pi_+^\star(s)$ attains the maximum in (2.12). Similarly, if P^+ plays an arbitrary π_+ against $\pi_-^\star$, then averaging the bracket over $\pi_+(\cdot \mid s)$ is at most its maximum, so the same conditioning together with the right inequality at the successors yields $V^{(\pi_+, \pi_-^\star)}(s) \leq V^\star(s)$. The argument for $s \in S_-$ is symmetric, with max and min and the inequality directions exchanged. Evaluating both inequalities of (2.13) at $\pi = \pi^\star$ gives $V^{\pi^\star} = V^\star$. Since the argument applies verbatim to the subgame rooted at any state, $\pi^\star$ is subgame perfect. □

Remark 2.6. For general discounted stochastic games with cycles, existence of the value and of optimal stationary policies is Shapley's theorem (Shapley, 1953), proved by the contraction argument of Proposition 2.4 applied to the *Shapley operator*—the min-max analogue of the Bellman operator (2.12). Gaubert & Vlassopoulos (2025) formulate their network games in this operator language. The layered acyclic case of Theorem 2.5 is all we need, and it additionally delivers subgame perfection, on which Chapter 8 relies.

2.3.1 The ReLU Net Game of Gaubert and Vlassopoulos

Gaubert & Vlassopoulos (2025) showed that the forward pass of a ReLU network *is* the value of a zero-sum turn-based stopping game played backward on the network graph. We present their construction in full, in our notation, since Chapter 4 inherits its state space, transition kernel, and stopping structure while changing the payoff bookkeeping. Throughout, fix a network as in Definition 2.1 with scalar

output ($d^{(L)} = 1$), fix an input $x \in \mathbb{R}^{d^{(0)}}$, and assume that no row of any weight matrix vanishes, so that the discounts (2.11) are strictly positive.

Read the identity $a_i^{(l)} = \max\{0, z_i^{(l)}\}$ as a decision: a player at neuron i may *stop*, securing payoff 0, or *continue*, collecting the bias $b_i^{(l)}$ plus the weighted contributions of the previous layer. Continuing propagates the decision backward through the network; a negative weight flips the sign of its contribution, i.e. hands the turn to an adversary facing the mirrored problem. The optimal value at neuron i is exactly the activation $a_i^{(l)}$.

Definition 2.9 (ReLU Net Game, following (Gaubert and Vlassopoulos, 2025)). The *ReLU Net Game* on the network f at input x is the zero-sum turn-based game with the following components.

- **States.** For every hidden layer $l \in \{1, \dots, L-1\}$, every neuron $i \in [d^{(l)}]$, and every turn label $q \in \{+,-\}$ there is an *activation state* $s_{i,q}^{(l)}$, controlled by player P^q, so that $S_q = \{s_{i,q}^{(l)} : l \in [L-1], i \in [d^{(l)}]\}$. For every input coordinate $k \in [d^{(0)}]$ and every $q \in \{+,-\}$ there is a *terminal state* $s_{k,q}^{(0)}$, and there is one absorbing *cemetery state* $\perp$ with zero payoff. Finally, there is a single *output state* $s^{(L)}$ at the affine output neuron, at which no decision is taken.
- **Actions.** At every activation state the player in turn chooses an action from $\{\text{stop}, \text{cont}\}$. The output state has the single action cont.
- **Transitions.** Stopping moves deterministically to the cemetery, $\mathbb{P}(\perp \mid s_{i,q}^{(l)}, \text{stop}) = 1$. Continuation at $s_{i,q}^{(l)}$ moves one layer backward, to neuron j of layer $l-1$ with probability proportional to the absolute weight, keeping the turn on positive weights and passing it to the opponent on negative weights:

$$\mathbb{P}\big(s_{j,r}^{(l-1)} \mid s_{i,q}^{(l)}, \text{cont}\big) := \frac{W_{ij}^{(l,q\oplus r)}}{\gamma_i^{(l)}}, \qquad \gamma_i^{(l)} = \sum_j \big(W^{(l,+)} + W^{(l,-)}\big)_{ij}, \tag{2.14}$$

  where for $l = 1$ the successor $s_{j,r}^{(0)}$ is the corresponding terminal state. The output state transitions by the same rule (2.14) with $l = L$ and turn label $q = +$.
- **Payoffs and discounts.** Continuation at $s_{i,q}^{(l)}$ pays the signed bias $\xi_q\, b_i^{(l)}$; stopping pays 0. A terminal state pays the signed input coordinate once, $R\big(s_{k,q}^{(0)}\big) := \xi_q\, x_k$, and the game ends. The output state pays $b^{(L)}$ upon its forced continuation. The discount at every activation state and at the output state is the structural discount $\gamma\big(s_{i,q}^{(l)}\big) := \gamma_i^{(l)}$ of (2.11).

Player P^+ maximizes and P^- minimizes the expected discounted total payoff, as in Definition 2.8.

The game graph is layered and acyclic—every transition strictly decreases the layer index or ends in the cemetery—so Theorem 2.5 applies despite the discounts $\gamma_i^{(l)}$ being arbitrary positive numbers. Normalization and discounting cancel: the product of the transition probability (2.14) and the discount $\gamma_i^{(l)}$ is exactly the absolute weight $|W_{ij}^{(l)}|$, so the Shapley–Bellman recursion reproduces the affine algebra of the forward pass.

Theorem 2.6 (Forward pass as game value, (Gaubert and Vlassopoulos, 2025), `Lean_8`). *The ReLU Net Game of Definition 2.9 admits the subgame perfect equilibrium $\pi^\star = (\pi_+^\star, \pi_-^\star)$ in which both players use the same pure stopping rule,*

$$\pi_q^\star\big(s_{i,q}^{(l)}\big) = \text{cont} \quad \Longleftrightarrow \quad z_i^{(l)} > 0, \tag{2.15}$$

and the value function of the game evaluates the network:

$$V^\star\big(s_{i,q}^{(l)}\big) = \xi_q\, a_i^{(l)} \quad \text{for } l \in \{0,1,\dots,L-1\}, \qquad V^\star\big(s^{(L)}\big) = f(x), \tag{2.16}$$

where for $l = 0$ we read $a_k^{(0)} = x_k$. The equilibrium is unique up to tie-breaking at neurons with $z_i^{(l)} = 0$, where both actions are optimal; the convention (2.15) stops at ties, matching $ReLU'(0) = 0$.

Proof. By Theorem 2.5 it suffices to verify that the function defined by (2.16) satisfies the Shapley–Bellman equations (2.12), with the maximizers and minimizers given by (2.15). We induct on the layer index.

Base case $l = 0$. Terminal states carry the fixed payoff $R\big(s_{k,q}^{(0)}\big) = \xi_q\, x_k = \xi_q\, a_k^{(0)}$, which is (2.16).

Inductive step. Let $l \geq 1$ and assume (2.16) at layer $l-1$. Consider first a state $s_{i,+}^{(l)} \in S_+$. The stop action yields $0 + V^\star(\bot) = 0$. The continuation value is, by (2.14) and the induction hypothesis,

$$\begin{aligned} R\big(s_{i,+}^{(l)}, \mathrm{cont}\big) + \gamma_i^{(l)} \sum_{j,r} \frac{W_{ij}^{(l,+\oplus r)}}{\gamma_i^{(l)}}\, \xi_r\, a_j^{(l-1)} &= b_i^{(l)} + \sum_j \Big(W_{ij}^{(l,+)}\, a_j^{(l-1)} - W_{ij}^{(l,-)}\, a_j^{(l-1)} \Big) \\ &= b_i^{(l)} + \sum_j W_{ij}^{(l)}\, a_j^{(l-1)} \;=\; z_i^{(l)}, \end{aligned} \tag{2.17}$$

using $+ \oplus r = r$, $\xi_+ = 1$, $\xi_- = -1$, and $W^{(l)} = W^{(l,+)} - W^{(l,-)}$. The maximizing player P^+ therefore obtains

$$V^\star\big(s_{i,+}^{(l)}\big) = \max\big\{0, z_i^{(l)}\big\} = \mathrm{ReLU}\big(z_i^{(l)}\big) = a_i^{(l)} = \xi_+\, a_i^{(l)},$$

continuing exactly when $z_i^{(l)} > 0$. At the mirrored state $s_{i,-}^{(l)} \in S_-$ the continuation payoff is $\xi_- b_i^{(l)} = -b_i^{(l)}$, and the kernel labels satisfy $- \oplus r = r'$, so the computation (2.17) returns the continuation value

$$-b_i^{(l)} + \sum_j \Big(W_{ij}^{(l,-)}\, a_j^{(l-1)} - W_{ij}^{(l,+)}\, a_j^{(l-1)} \Big) = -z_i^{(l)}.$$

The minimizing player P^- obtains $V^\star\big(s_{i,-}^{(l)}\big) = \min\{0, -z_i^{(l)}\} = -\mathrm{ReLU}\big(z_i^{(l)}\big) = \xi_-\, a_i^{(l)}$ and prefers to continue exactly when $-z_i^{(l)} < 0$, i.e. under the same gate (2.15). At $z_i^{(l)} = 0$ both actions give value 0 for either player, which proves the uniqueness-up-to-ties claim. Finally, the output state has the single action cont, and the computation (2.17) with $l = L$ yields $V^\star\big(s^{(L)}\big) = z^{(L)} = a^{(L)} = f(x)$. □

Remark 2.7 (Softplus networks from entropy regularization). Gaubert & Vlassopoulos (2025) also treat softplus networks, by *regularizing* the stop/continue decision. Replace the hard choice at an activation state by a mixed policy $\nu \in \Delta(\{\mathrm{stop}, \mathrm{cont}\})$ rewarded with a Shannon-entropy bonus at temperature $\theta > 0$. By the Legendre–Fenchel identity for the entropy (Boyd and Vandenberghe, 2004),

$$\max_{\nu \in \Delta_n} \Big\{ \textstyle\sum_{i=1}^n \nu_i r_i + \theta\, \mathscr{H}(\nu) \Big\} = \theta \log \textstyle\sum_{i=1}^n e^{r_i/\theta}, \qquad \mathscr{H}(\nu) = -\textstyle\sum_i \nu_i \log \nu_i, \tag{2.18}$$

with the Gibbs distribution $\nu_i^\star \propto e^{r_i/\theta}$ as unique maximizer, the regularized value at a state with continuation value $z_i^{(l)}$ (computed as in (2.17)) becomes $\theta \log\big(1 + e^{z_i^{(l)}/\theta}\big)$—the softplus activation at temperature θ—and the equilibrium policy continues with the sigmoid probability $\big(1 + e^{-z_i^{(l)}/\theta}\big)^{-1}$. The hard game of Definition 2.9 is recovered in the limit $\theta \to 0$. Chapter 4 re-derives this mechanism

from the regularized decompositions of Section 3.5 and reuses it at the routing states of the RG, with the two temperatures θ (activation gates) and τ (routing) kept distinct.

Remark 2.8 (What Chapter 4 inherits and what it changes). The SG of Chapter 4 keeps the structural skeleton of Definition 2.9: per-neuron $\pm$ states, stop-versus-continue actions, the bias as continuation payoff and the input as final payoff, the transition kernel (2.14) driven by the weight parts $W^{(l,\sigma)}$, the cemetery state, and the structural discounts $\gamma_i^{(l)}$. Two changes are load-bearing for attribution. First, the original tracks only the single signed value (2.16); Chapter 4 additionally maintains a *non-negative payoff per player*, identifying the pair of player-specific values $V_p(s)$, $p \in \{+,-\}$, with the two streams of the Stopping Decomposition of Section 3.4 and recovering (2.16) as their difference. Second, the signed terminal payoffs $\xi_q\, x_k$ become the non-negative pair (x_k^+, x_k^-) of Definition 2.2, keeping every player-specific quantity non-negative and exposing the parity structure of backward trajectories. Chapter 4 further extends the graph to skip connections and max-pooling and adds risk aversion and regularized policies, none of which appear in the original game.

2.4 Tropical Geometry and Polyhedra

Tropical geometry replaces $(+, \times)$ by $(\max, +)$, turning polynomials into convex CPWL functions and polynomial multiplication into Minkowski addition of polytopes. This is the algebra governing the convex streams g and h of the DC decomposition, which Sections 2.4.1–2.4.3 exploit to read learned structure off the associated polytopes. We collect the definitions and calculus rules used there.

Definition 2.10 (Max-plus and min-plus semirings). The *max-plus semiring* is the set $\mathbb{R}_{\max} := \mathbb{R} \cup \{-\infty\}$ equipped with *tropical addition* and *tropical multiplication*,

$$u \oplus v := \max\{u, v\}, \qquad u \odot v := u + v, \tag{2.19}$$

with additive neutral element $-\infty$ and multiplicative neutral element 0. Both operations are associative and commutative, and $\odot$ distributes over $\oplus$; additive inverses do not exist, so $\mathbb{R}_{\max}$ is a semiring but not a ring. The *min-plus semiring* $\mathbb{R}_{\min} := \mathbb{R} \cup \{+\infty\}$ is defined dually with $u \oplus' v := \min\{u,v\}$; the map $u \mapsto -u$ is an isomorphism between the two, so every statement below has a min-plus counterpart.

For example, $2 \oplus 3 = 3$ and $2 \odot 3 = 5$, and the tropical square is $u^{\odot 2} = u \odot u = 2u$. More generally, tropical exponentiation is scalar multiplication, $u^{\odot \alpha} = \alpha u$, which is meaningful for any real exponent α. (The symbol $\oplus$ doubles as the sign-XOR of Section 2.3; the operands—real numbers here, sign labels there—always disambiguate.)

Definition 2.11 (Tropical polynomials and rational functions, `Lean_9`). A *tropical monomial* in d variables with exponent $\alpha \in \mathbb{R}^d$ and coefficient $c \in \mathbb{R}$ is the affine function $c \odot x_1^{\odot \alpha_1} \odot \cdots \odot x_d^{\odot \alpha_d} = c + \langle \alpha, x \rangle$. A *tropical polynomial* with finite support $\mathscr{A} \subset \mathbb{R}^d$ and coefficients $(c_\alpha)_{\alpha \in \mathscr{A}}$, $c_\alpha \in \mathbb{R}$, is the tropical sum of monomials,

$$p(x) \;=\; \bigoplus_{\alpha \in \mathscr{A}} c_\alpha \odot x^{\odot \alpha} \;=\; \max_{\alpha \in \mathscr{A}} \big(c_\alpha + \langle \alpha, x \rangle \big). \tag{2.20}$$

A *tropical rational function* is a tropical quotient of two tropical polynomials, $p \oslash q := p - q$. In classical tropical geometry (Maclagan and Sturmfels, 2015) the exponents are restricted to $\mathscr{A} \subset \mathbb{N}^d$ or $\mathbb{Z}^d$; for neural networks, real exponents (network weights) are the natural setting, and all statements below hold verbatim in this generality.

A one-dimensional example: $p(x) = 0 \oplus (0 \odot x) \oplus \big((-1) \odot x^{\odot 2}\big) = \max\{0, x, 2x-1\}$, a convex piecewise-linear function with breakpoints at $x = 0$ and $x = 1$.

Observation 2.3 (Tropical polynomials are the convex CPWL functions, `Lean_10`*).* A function $p\colon \mathbb{R}^d \to \mathbb{R}$ is a tropical polynomial (2.20) if and only if it is convex and CPWL.

Proof. A tropical polynomial is a finite maximum of affine functions, hence convex and CPWL. Conversely, let p be convex and CPWL with affine pieces $\ell_1, \dots, \ell_N$ on the linear regions $Q_1, \dots, Q_N$ of Definition 2.3. We claim $p = \max_r \ell_r$. Pointwise $p \le \max_r \ell_r$, since every x lies in some region Q_r where $p(x) = \ell_r(x)$. For the reverse inequality, fix r and a point x_0 in the interior of the full-dimensional region Q_r. Then ℓ_r agrees with p on a neighborhood of x_0, and it is a supporting affine minorant of the convex function p at x_0: for any x and sufficiently small $t \in (0,1)$, convexity gives $p\big(x_0 + t(x - x_0)\big) \le (1-t)\,p(x_0) + t\,p(x)$, while the left-hand side equals $\ell_r\big(x_0 + t(x-x_0)\big) = (1-t)\,\ell_r(x_0) + t\,\ell_r(x)$; comparing and using $p(x_0) = \ell_r(x_0)$ yields $\ell_r(x) \le p(x)$. Hence $\max_r \ell_r \le p$, and equality holds. □

Combining Observation 2.3 with the DC representation (2.4) and Theorem 2.1 identifies ReLU network functions with tropical rational functions; this is the theorem of Zhang et al. (2018b), restated and connected to the split streams g and h in Section 2.4.1. The expressivity of tropical polynomials as computed by networks is studied by Misiakos et al. (2022) and, in the real-exponent setting, by Brandenburg et al. (2024).

Newton polytopes, Minkowski sums, and support functions

The combinatorics of a tropical polynomial is captured by a polytope—the convex hull of its exponents. We refer to Ziegler (1995) for general polytope theory and write conv for the convex hull.

Definition 2.12 (Newton polytope, `Lean_11`**).** The *Newton polytope* of a tropical polynomial p with support $\mathscr{A}$ as in (2.20) is the convex hull of its exponent vectors,

$$\operatorname{Newt}(p) := \operatorname{conv}\big\{\alpha : \alpha \in \mathscr{A}\big\} \subset \mathbb{R}^d. \tag{2.21}$$

The *lifted Newton polytope* additionally records the coefficients, $\widehat{\operatorname{Newt}}(p) := \operatorname{conv}\big\{(\alpha, c_\alpha) : \alpha \in \mathscr{A}\big\} \subset \mathbb{R}^{d+1}$.

For the example $p(x) = \max\{0, x, 2x-1\}$ above, $\mathscr{A} = \{0, 1, 2\}$ and $\operatorname{Newt}(p) = [0, 2] \subset \mathbb{R}$, while $\widehat{\operatorname{Newt}}(p) = \operatorname{conv}\{(0,0), (1,0), (2,-1)\}$ is a triangle in $\mathbb{R}^2$.

Definition 2.13 (Minkowski sum). The *Minkowski sum* of two sets $P, Q \subseteq \mathbb{R}^d$ is $P + Q := \{u + v : u \in P, v \in Q\}$. For polytopes P and Q the Minkowski sum is again a polytope, and every vertex of $P + Q$ decomposes as the sum of a vertex of P and a vertex of Q (Ziegler, 1995).

For example, the Minkowski sum of two segments in general position is a parallelogram, and the Minkowski sum of k segments is a *zonotope*—the polytope class that Zhang et al. (2018b) showed to govern one-hidden-layer ReLU networks. The analytic tool linking polytopes to convex functions is the support function.

Definition 2.14 (Support function, `Lean_12`**).** The *support function* of a nonempty compact convex set $P \subset \mathbb{R}^d$ is

$$h_P\colon \mathbb{R}^d \to \mathbb{R}, \qquad h_P(x) := \max_{u \in P} \langle u, x \rangle. \tag{2.22}$$

Lemma 2.7 (Properties of support functions, Lean_13). *Let $P, Q \subset \mathbb{R}^d$ be nonempty compact convex sets. Then:*

1) *h_P is convex and positively homogeneous, i.e. $h_P(\lambda x) = \lambda h_P(x)$ for $\lambda \geq 0$; if P is a polytope with vertex set $\operatorname{vert}(P)$, then $h_P(x) = \max_{u \in \operatorname{vert}(P)} \langle u, x \rangle$ is a tropical polynomial with zero coefficients and Newton polytope P.*
2) *For any nonempty finite set $A \subset \mathbb{R}^d$ and $x \in \mathbb{R}^d$, $\max_{u \in \operatorname{conv}(A)} \langle u, x \rangle = \max_{u \in A} \langle u, x \rangle$; in particular $h_{P+Q} = h_P + h_Q$ and $h_{\operatorname{conv}(P \cup Q)} = \max\{h_P, h_Q\} = h_P \oplus h_Q$.*
3) *$P \subseteq Q$ if and only if $h_P \leq h_Q$ pointwise; in particular $h_P = h_Q$ implies $P = Q$, so a compact convex set is uniquely determined by its support function.*

Proof. 1: As a pointwise maximum of the linear functions $x \mapsto \langle u, x \rangle$, $u \in P$, the support function is convex, and homogeneity is clear from (2.22). For a polytope, the maximum of the linear objective $\langle \cdot\,, x \rangle$ over P is attained at a vertex, giving the finite-max representation; its exponent set is $\operatorname{vert}(P)$, whose convex hull is P.

2: Maximizing a linear function over a convex hull adds nothing: any convex combination satisfies $\big\langle \sum_i \lambda_i u_i, x \big\rangle = \sum_i \lambda_i \langle u_i, x \rangle \leq \max_i \langle u_i, x \rangle$, and $A \subseteq \operatorname{conv}(A)$ gives the reverse bound. For the Minkowski sum, $\max_{w \in P+Q} \langle w, x \rangle = \max_{u \in P,\, v \in Q} \big(\langle u, x \rangle + \langle v, x \rangle \big) = h_P(x) + h_Q(x)$, since the two maximizations decouple. For the union, the objective attains the same maximum over $\operatorname{conv}(P \cup Q)$ as over $P \cup Q$ by the first claim (applied with suprema over compact sets in place of finite maxima, with the same one-line proof), and $\max_{u \in P \cup Q} \langle u, x \rangle = \max\{h_P(x), h_Q(x)\}$.

3: If $P \subseteq Q$, the maximum in (2.22) over the larger set is at least as big, so $h_P \leq h_Q$. Conversely, suppose $u_0 \in P \setminus Q$. By the separating-hyperplane theorem for a point and a compact convex set (Boyd and Vandenberghe, 2004), there is $x \in \mathbb{R}^d$ with $\langle u_0, x \rangle > \max_{v \in Q} \langle v, x \rangle = h_Q(x)$, whence $h_P(x) \geq \langle u_0, x \rangle > h_Q(x)$, contradicting $h_P \leq h_Q$. The uniqueness statement follows by applying the equivalence in both directions. □

Support functions translate the tropical operations on polynomials into polytope operations on their Newton polytopes.

Proposition 2.8 (Newton polytope calculus, cf. (Zhang et al., 2018b), Lean_14). *Let p and q be tropical polynomials on $\mathbb{R}^d$. Then*

$$\operatorname{Newt}(p \odot q) = \operatorname{Newt}(p) + \operatorname{Newt}(q), \qquad \operatorname{Newt}(p \oplus q) = \operatorname{conv}\big(\operatorname{Newt}(p) \cup \operatorname{Newt}(q)\big). \tag{2.23}$$

Moreover, the lifted Newton polytope determines the polynomial function:

$$p(x) = h_{\widehat{\operatorname{Newt}(p)}}(x, 1) \qquad \text{for all } x \in \mathbb{R}^d, \tag{2.24}$$

and if all coefficients vanish, $c_\alpha \equiv 0$, then $p = h_{\operatorname{Newt}(p)}$.

Proof. We first prove (2.24). By Lemma 2.7.2 applied to the finite point set $\{(\alpha, c_\alpha) : \alpha \in \mathscr{A}\}$, maximizing the linear functional $\langle \cdot\,, (x,1) \rangle$ over its convex hull $\widehat{\operatorname{Newt}}(p)$ equals maximizing it over the points themselves:

$$h_{\widehat{\operatorname{Newt}(p)}}(x, 1) = \max_{\alpha \in \mathscr{A}} \big\langle (\alpha, c_\alpha), (x, 1) \big\rangle = \max_{\alpha \in \mathscr{A}} \big(\langle \alpha, x \rangle + c_\alpha \big) = p(x),$$

and the zero-coefficient case is the same computation one dimension lower.

For the tropical sum $p \oplus q = \max\{p, q\}$: its support is the union $\mathscr{A}_p \cup \mathscr{A}_q$ (an exponent occurring in both summands keeps the larger coefficient, which does not change the support), so

$$\mathrm{Newt}(p \oplus q) = \mathrm{conv}(\mathscr{A}_p \cup \mathscr{A}_q) = \mathrm{conv}\big(\mathrm{conv}(\mathscr{A}_p) \cup \mathrm{conv}(\mathscr{A}_q)\big) = \mathrm{conv}\big(\mathrm{Newt}(p) \cup \mathrm{Newt}(q)\big),$$

where the middle equality holds because both sides are the smallest convex set containing $\mathscr{A}_p \cup \mathscr{A}_q$.

For the tropical product $p \odot q = p + q$: expanding the product of (2.20) termwise shows that the support of $p \odot q$ is contained in the pointwise sum $\mathscr{A}_p + \mathscr{A}_q$ (tropical addition of coefficients on coinciding exponent sums never removes an exponent, in contrast to classical cancellation), so

$$\mathrm{Newt}(p \odot q) \subseteq \mathrm{conv}(\mathscr{A}_p + \mathscr{A}_q) = \mathrm{conv}(\mathscr{A}_p) + \mathrm{conv}(\mathscr{A}_q) = \mathrm{Newt}(p) + \mathrm{Newt}(q),$$

using the identity $\mathrm{conv}(A + B) = \mathrm{conv}(A) + \mathrm{conv}(B)$: the right-hand side is a convex set containing $A + B$, and conversely any sum of convex combinations $\sum_i \lambda_i u_i + \sum_j \mu_j v_j = \sum_{i,j} \lambda_i \mu_j (u_i + v_j)$ is a convex combination of points of $A + B$. For the reverse inclusion, take a vertex w of $\mathrm{Newt}(p) + \mathrm{Newt}(q)$ and a linear functional $\langle \cdot\,, x^* \rangle$ uniquely maximized over the sum at w; by Definition 2.13 we may write $w = u + v$ with $u \in \mathscr{A}_p$ maximizing $\langle \cdot\,, x^* \rangle$ over $\mathrm{Newt}(p)$ and $v \in \mathscr{A}_q$ maximizing it over $\mathrm{Newt}(q)$ (a vertex of a Newton polytope belongs to the support). Then $w = u + v$ lies in the support of $p \odot q$, so every vertex of $\mathrm{Newt}(p) + \mathrm{Newt}(q)$ lies in $\mathrm{Newt}(p \odot q)$, and taking convex hulls gives the reverse inclusion. □

Remark 2.9 (Layer transitions as polytope operations). Proposition 2.8 is the dictionary used in the remaining subsections: an affine layer with non-negative weights acts on the tropical polynomials computed by the previous layer through tropical products and powers—Minkowski sums and scalings of Newton polytopes—while ReLU and Maxout gates act through tropical sums—convex hulls of unions. The convex streams g and h of the DC decomposition—developed in full in Chapter 3—are, by Observation 2.3, tropical polynomials, so their (lifted) Newton polytopes are computable layer by layer. Section 2.4.3 studies these polytopes: by Lemma 2.7.3 they encode the split streams faithfully, and their vertices are exactly the attainable gradients of g and h—the geometric objects behind the *imaginative polytopes* of this thesis. For the broader landscape connecting polyhedral and tropical geometry with network expressivity, see (Brandenburg et al., 2024; Huchette et al., 2025; Zhang et al., 2018b).

2.4.1 Tropical Rational Maps and the DC Split

The tropical dictionary above reads the DC decomposition geometrically. Zhang et al. (2018b) showed that feedforward ReLU networks with real weights are exactly the *tropical rational maps*—componentwise differences of max-plus polynomials—so that counting linear regions becomes counting polytope vertices; the converse, that every CPWL function arises this way, is Mukherjee et al. (2018).[1] We use this correspondence in the opposite direction: rather than bounding a worst-case network, we take a *trained* network, compute its split streams g, h explicitly—the construction developed in full in Chapter 3—and read the vertices of their Newton polytopes as learned prototypes for interpreting trained networks (Zimmermann and Loho, 2026).

This subsection anticipates the difference-of-convex split developed in full in Chapter 3, presenting its geometry here because it belongs with the tropical and polyhedral foundations of this chapter. The

[1] Related max-affine perspectives include Maxout networks (Goodfellow et al., 2013) and the expressivity program on Newton polytopes (Brandenburg et al., 2024; Haase et al., 2023; Hertrich and Loho, 2025; Huchette et al., 2025; Misiakos et al., 2022).

split forward pass—formalized in Chapter 3—acts on a bias-free network as follows: the input $x \in \mathbb{R}^d$ is replaced by a pair $(a^{(0,+)}, a^{(0,-)})$ with $x = a^{(0,+)} - a^{(0,-)}$, the weights are split as $W^{(l)} = W^{(l,+)} - W^{(l,-)}$ with $W^{(l,\pm)} \geq 0$, and the hidden layers $1 \leq l < L$ iterate entrywise

$$z^{(l,+)} = W^{(l,+)}a^{(l-1,+)} + W^{(l,-)}a^{(l-1,-)}, \qquad z^{(l,-)} = W^{(l,-)}a^{(l-1,+)} + W^{(l,+)}a^{(l-1,-)}, \tag{2.25}$$

$$a^{(l,+)} = \max\{z^{(l,+)}, z^{(l,-)}\}, \qquad a^{(l,-)} = z^{(l,-)}, \tag{2.26}$$

At the final affine readout, use the same pre-activation formulas and set $a^{(L,\pm)} = z^{(L,\pm)}$ without a maximum. Thus for a scalar output the split streams $g, h\colon \mathbb{R}^{2d} \to \mathbb{R}$ map the doubled input $(a^{(0,+)}, a^{(0,-)})$ to the two final components and satisfy $f(a^{(0,+)} - a^{(0,-)}) = g - h$. The following theorem identifies this construction as a tropical rational decomposition whose components carry additional monotone structure.

Theorem 2.9 (The DC split is a monotone tropical rational decomposition, `Lean_15`). *Let $f\colon \mathbb{R}^d \to \mathbb{R}$ be a bias-free ReLU **M**ulti**l**ayer **P**erceptron (MLP) with scalar output and split streams $g, h\colon \mathbb{R}^{2d} \to \mathbb{R}$ defined by (2.25)–(2.26). Then:*

1) *every coordinate of $a^{(l,+)}$ and $a^{(l,-)}$, and in particular g and h, is a positively homogeneous* monotone *tropical polynomial in the doubled variable $(x^+, x^-) \in \mathbb{R}^{2d}$;*
2) *for every splitting $x = x^+ - x^-$,*

$$f(x) \;=\; g(x^+, x^-) \;-\; h(x^+, x^-), \tag{2.27}$$

i.e. f, precomposed with the difference map, is the tropical rational function $g \oslash h$.

Proof. (1). Induct forward. At $l = 0$ the coordinates of $a^{(0,+)} = x^+$ and $a^{(0,-)} = x^-$ are coordinate projections, i.e. monomials $\langle e_j, \cdot \rangle$ with a single non-negative exponent and zero coefficient. For the step, each coordinate of $z^{(l,\pm)}$ in (2.25) is a non-negative combination of layer-$(l-1)$ coordinates (as $W^{(l,\pm)} \geq 0$), and each coordinate of $a^{(l,+)}$ is a pointwise maximum of two of these while $a^{(l,-)} = z^{(l,-)}$. By the closure of positively homogeneous monotone tropical polynomials under non-negative combinations and maxima (Observation 2.3), the property propagates to g and h. *(2).* The identity (2.27) is not a new proof obligation here: the correctness of the split $f = g - h$ is established in Chapter 3, and the present part only restates it in tropical language—by part (1) both g and h are tropical polynomials, so their difference $g \oslash h = g - h$ is a tropical rational function. □

Monotonicity of both components—not guaranteed by generic tropical rational representations—is what confines the Newton polytopes below to the non-negative orthant, via the following elementary fact.

Theorem 2.10 (Monotone CPWL functions have non-negative gradients, `Lean_16`). *Let $u\colon \mathbb{R}^n \to \mathbb{R}$ be monotone and CPWL. Then every gradient attained by u at a point of differentiability is entrywise non-negative.*

Proof. If u is differentiable at x with $(\nabla u(x))_j < 0$, then u is affine on an ε-ball around x, and $\tilde{x} := x - \frac{\varepsilon}{2}e_j \leq x$ satisfies $u(\tilde{x}) = u(x) - \frac{\varepsilon}{2}(\nabla u(x))_j > u(x)$, contradicting monotonicity. □

2.4.2 Newton Polytopes of the Split Streams

By Observation 2.3 and Lemma 2.7, every positively homogeneous convex CPWL function u is the support function $u = h_P$ of a unique polytope $P = \mathrm{Newt}(u)$, recovered as the halfspace intersection

$$P = \big\{y \in \mathbb{R}^n \,:\, \langle y, x\rangle \le h_P(x) \text{ for all } x \in \mathbb{R}^n\big\}, \tag{2.28}$$

and the tropical operations act on P through the Minkowski calculus of Proposition 2.8. Applied to the split streams—positively homogeneous monotone tropical polynomials by Theorem 2.9—this turns the split recursion into a layer-wise propagation of Newton polytopes. Writing $P_j^{(l,\sigma)} \coloneqq \mathrm{Newt}\big(a_j^{(l,\sigma)}\big) \subseteq \mathbb{R}^{2d}$ for the jth split activation of layer l:

Corollary 2.11 (Minkowski-sum propagation across layers, `Lean_17`). *For a bias-free ReLU MLP with split streams (2.25)–(2.26), the Newton polytopes of the split activations satisfy*

$$\begin{aligned} &P_j^{(0,+)} = \{e_j\}, \qquad P_j^{(0,-)} = \{e_{d+j}\}, \qquad j \in [d], \\ &Q_i^{(l,+)} \coloneqq \bigoplus_j \Big(W_{ij}^{(l,+)} P_j^{(l-1,+)} \oplus W_{ij}^{(l,-)} P_j^{(l-1,-)}\Big), \quad Q_i^{(l,-)} \coloneqq \bigoplus_j \Big(W_{ij}^{(l,-)} P_j^{(l-1,+)} \oplus W_{ij}^{(l,+)} P_j^{(l-1,-)}\Big), \end{aligned} \tag{2.29}$$

$$P_i^{(l,+)} = \mathrm{conv}\Big(Q_i^{(l,+)} \cup Q_i^{(l,-)}\Big), \qquad P_i^{(l,-)} = Q_i^{(l,-)}, \tag{2.30}$$

where the activation update (2.30) applies only to $l < L$; at the affine readout set $P_i^{(L,\pm)} = Q_i^{(L,\pm)}$. Here $e_1, \dots, e_{2d}$ are the standard basis of $\mathbb{R}^{2d}$ and $\oplus$ the Minkowski sum. The Newton polytopes $P_g \coloneqq \mathrm{Newt}(g)$ and $P_h \coloneqq \mathrm{Newt}(h)$ of a scalar output neuron are obtained after L steps.

Proof. The base case restates that $a_j^{(0,\pm)}$ are the coordinate functionals $\langle e_j, \cdot\rangle$, $\langle e_{d+j}, \cdot\rangle$. For the step, the ith coordinate of (2.25) is a non-negative combination whose Newton polytope is, by the Minkowski-sum and scaling identities of Proposition 2.8, the weighted sum $Q_i^{(l,\pm)}$; the entrywise maximum in (2.26) corresponds to the convex hull of the union by the same calculus, while $a^{(l,-)}$ passes its polytope through unchanged. □

For a single hidden layer this produces Minkowski sums of segments and points, i.e. *zonotopes*, recovering the shallow-network picture of Zhang et al. (2018b); in deeper networks the alternation of sums and hulls makes vertex counts grow with depth. The recursion is a structural tool, not an algorithm: the split forward pass evaluates these support functions without enumerating vertices. The link between polytope geometry and the differential behavior of its support function is the subdifferential correspondence, the engine behind Section 2.4.3.

Proposition 2.12 (Subdifferential of a support function, `Lean_18`). *Let $P \subseteq \mathbb{R}^n$ be a polytope and $x \in \mathbb{R}^n$. Then $\partial h_P(x) = F_P(x) = \{p \in P : \langle p, x\rangle = h_P(x)\}$, the maximizing face. In particular, h_P is differentiable at x if and only if $F_P(x)$ is a singleton, in which case $\nabla h_P(x) \in P$.*

Proof. "$\supseteq$": For $p \in F_P(x)$ and any y, $h_P(y) \ge \langle p, y\rangle = h_P(x) + \langle p, y - x\rangle$, the subgradient inequality. "$\subseteq$": Let $s \in \partial h_P(x)$. Testing the subgradient inequality with $y = 2x$ and $y = 0$ and using positive homogeneity gives $\langle s, x\rangle = h_P(x)$; substituting back yields $h_P(y) \ge \langle s, y\rangle$ for all y, so $s \in P$ by (2.28), hence $s \in F_P(x)$. Differentiability of a convex function is equivalent to a singleton subdifferential, whose element is the gradient. □

2.4.3 Imaginative Polytopes

We now specialize to the split streams of a trained network. Their Newton polytopes are the central geometric objects of the interpretability perspective of this thesis.

Definition 2.15 (Imaginative Polytopes). Let $g, h\colon \mathbb{R}^{2d} \to \mathbb{R}$ be the split streams of an output neuron of a bias-free ReLU network. The *Imaginative Polytopes* of that neuron are the Newton polytopes $P_g := \mathrm{Newt}(g)$ and $P_h := \mathrm{Newt}(h)$, i.e. the unique polytopes in $\mathbb{R}^{2d}$ whose support functions are g and h (Lemma 2.7).

They exist and are unique by Theorem 2.9 and the support-function bijection, and Corollary 2.11 expresses them as an alternation of weighted Minkowski sums and convex hulls. Monotonicity constrains their location.

Corollary 2.13 (Imaginative Polytopes live in the non-negative orthant, `Lean_19`**).** $P_g, P_h \subseteq \mathbb{R}^{2d}_{\geq 0}$.

Proof. By Theorem 2.9, $g = \max_i \langle a_i, \cdot \rangle$ with all $a_i \geq 0$, and $P_g = \mathrm{conv}\{a_i\}$ is a convex hull of non-negative vectors; likewise for P_h. (Equivalently, every vertex is an attainable gradient of the monotone g by Theorems 2.14 and 2.10.) □

Theorem 2.14 (Vertices encode the attainable split-stream gradients, `Lean_20`**).** *Let $u\colon \mathbb{R}^n \to \mathbb{R}$ be a positively homogeneous convex CPWL function with Newton polytope $P = \mathrm{Newt}(u)$. Then*

$$\mathrm{vert}(P) \;=\; \big\{\nabla u(\xi) \,:\, \xi \in \mathbb{R}^n,\ u \text{ differentiable at } \xi\big\}.$$

In particular, the attainable gradients of the split streams g and h are exactly the vertex sets $\mathrm{vert}(P_g)$ and $\mathrm{vert}(P_h)$ of the Imaginative Polytopes.

Proof. Here $u = h_P$, and $\partial u(\xi) = F_P(\xi)$ by Proposition 2.12. "$\supseteq$": if u is differentiable at ξ, then $F_P(\xi) = \{\nabla u(\xi)\}$ is a singleton face; a single-point face is a vertex, since $p = \frac{1}{2}(p_1 + p_2)$ with $p_i \in P$ would force $p_1, p_2 \in F_P(\xi) = \{p\}$. "$\subseteq$": let $v \in \mathrm{vert}(P)$ and $C := \mathrm{conv}(\mathrm{vert}(P) \setminus \{v\})$; then $v \notin C$, so strict separation yields ξ with $\langle v, \xi\rangle > \max_{v_i \neq v} \langle v_i, \xi\rangle$, and this persists on a ball B. By the vertex representation of the support function (Lemma 2.7), $u(\xi') = \langle v, \xi'\rangle$ on B, so u is differentiable at ξ with $\nabla u(\xi) = v$. The split-stream statement is the case $u = g, h$, admissible by Theorem 2.9. For networks with biases, homogenization gives an ambient support function, but fixing the added coordinates restricts the exposing directions. The input gradients of the restricted function are projections of vertices exposed by directions on that affine slice; not every ambient vertex need be visible there. For example, (1,0) is a vertex of $P = \mathrm{conv}\{(0,1),(2,1),(1,0)\}$, but its value x is strictly below $h_P(x,1) = \max\{1, 2x+1\}$ for every x. Thus the equality with all vertices applies in the homogeneous ambient space, while the biased network uses only the slice-visible vertices. □

Theorem 2.14 turns the support-function representation into an interpretable computational model. Writing $\xi = (x^+, x^-)$ for the doubled input,

$$g(\xi) = \max_{v \in \mathrm{vert}(P_g)} \langle v, \xi\rangle, \qquad h(\xi) = \max_{w \in \mathrm{vert}(P_h)} \langle w, \xi\rangle, \tag{2.31}$$

so each split stream scans a finite catalog of non-negative linear templates (Corollary 2.13)—for image classifiers, pairs of non-negative images—and reports the strongest response; each vertex corresponds

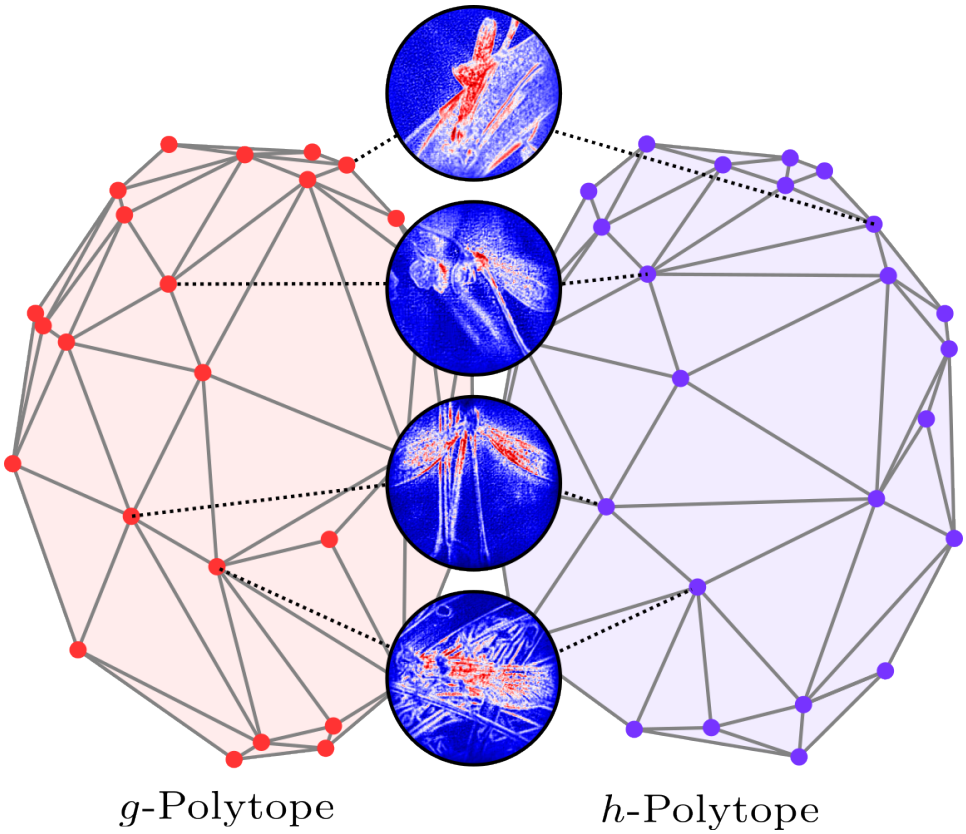


Fig. 2.1: Concept visualization of the Imaginative Polytopes associated with the "dragonfly" output neuron of VGG-16 (Simonyan and Zisserman, 2015) on four ImageNet test images (Deng et al., 2009). Shown are g-stream gradients with respect to the class logit, computed with the stabilized backward pass of Section 3.6; by Theorem 2.14, each unstabilized such gradient is a vertex of P_g, i.e. a learned visual prototype of the class. The h-stream gradients look very similar, since the two streams differ only by the gradient of the original network. Figure adapted from Zimmermann & Loho (2026).

to one gate configuration of (2.26), the linear region on which that stream has the given gradient. We therefore read $\mathrm{vert}(P_g)$ and $\mathrm{vert}(P_h)$ as the neuron's *learned prototypes*, extremal non-redundant feature detectors of which the network applies the best-matching one per input, with the logit the difference of the two strongest responses (2.27); P_g collects excitatory, P_h inhibitory evidence profiles. The same vertices double as *counterfactual profiles*: for an input with active prototype v, the non-negative gaps $\langle v - v', \xi \rangle$ over edge-neighbors v' of v measure how far ξ is from being re-imagined under another prototype, and $v - v'$ shows which evidence would have to move—answering, within the network's own geometry and without any surrogate, what minimally different perception would flip the stream's assessment. Their non-negativity also explains the stability and localization of the split-stream saliency maps studied in Chapters 6 and 7: responses aggregate non-negative evidence without the sign cancellation that erodes ∇f.

The split pair (g, h) is in general *not* minimal—the recursion accumulates common monomials in both streams—but it is computed by the network's own architecture, both components are monotone, and the split activations stay in one-to-one correspondence with the original neurons, properties that Chapters 6 and 4 exploit.[2]

[2]That every CPWL function is a difference of convex CPWL functions is classical (Kripfganz and Schulze, 1987; Melzer, 1986; Zalgaller, 2000); algorithmic and minimality aspects, including the polyhedral structure of the set of all DC

The tropical representation depends on piecewise linearity. The convex and monotone stream construction itself is more general: Section 3.2 characterizes suitable activation pairs, constructs them from absolute curvature, and recovers ReLU as the maximum/identity special case. It also gives explicit formulas for smooth activations.

decompositions of a fixed function, were developed by Griewank & Walther (2020) and in (Brandenburg et al., 2025; Koutschan et al., 2023).

3 Network Decompositions and DC Formulations

The structural approach of this thesis starts with a common operation: separate the positive and negative contributions of a trained network while preserving their difference. This chapter develops that operation into a general framework of network decompositions. The DC instance exposes monotone, convex stream functions for the attribution methods of Chapter 6; the stopping instance preserves the network's gates and supplies the state structure for the games of Chapter 4. Mixing and regularized decompositions connect these choices. Their shared forward invariant lets us compare what each representation retains and how its backward calculation changes, while numerical stabilization makes the constructions usable in deep networks.

3.1 Generalized Network Decomposition

An ordinary network activation is a single signed scalar that collapses two opposing flows—the contributions that *support* the prediction and those that *oppose* it—into one number, hiding their magnitudes. The decompositions of this chapter lay them bare: at every neuron and layer, the signed activation is replaced by a non-negative pair that exposes how positive and negative contributions accumulate through the network, while still recovering the original activation as their difference. Figure 3.1 illustrates the idea for the DC instance of Section 3.2.3.

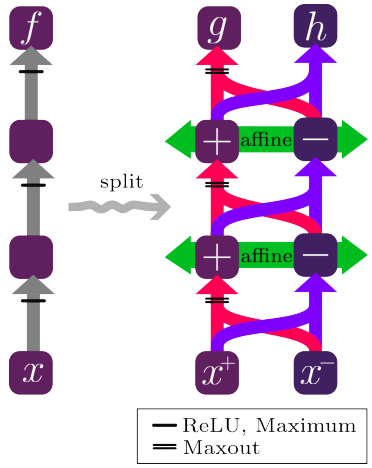


Fig. 3.1: The network splitting procedure: a ReLU network f is decomposed into two monotone, convex streams g and h with $f = g - h$ (Section 3.2.3).

Throughout this chapter we consider a feedforward network of L layers with weights $W^{(l)}$, biases $b^{(l)}$, and an entrywise activation $\varrho\colon \mathbb{R} \to \mathbb{R}$ (e.g. ReLU or softplus), computing the recursion

$$a^{(0)} = x, \qquad z^{(l)} = W^{(l)} a^{(l-1)} + b^{(l)}, \qquad a^{(l)} = \varrho_l\big(z^{(l)}\big), \qquad l = 1, \dots, L, \tag{3.1}$$

where $\varrho_l = \varrho$ for $1 \le l < L$ and $\varrho_L = \mathrm{id}$ is the ungated affine readout, with scalar output $f(x) = a^{(L)} \in \mathbb{R}$; for multi-class networks we fix the target logit and treat the rest as a single-output network. We follow the notation of Section 2.1; g and h are reserved for the split streams of Section 3.2.3.

A network decomposition replaces every activation $a_i^{(l)}$ by a pair of non-negative scalars, propagated through non-negative linear maps and *pairwise activation functions* that act on neuron pairs. We first

record this structural setup, then identify the decompositions that recover the original network as the difference of the decomposed activations.

Definition 3.1 (Decomposition). A *decomposition* of a feedforward network (3.1) with L layers is a calculation scheme that propagates non-negative activation pairs $\left(a_i^{(l,+)}, a_i^{(l,-)}\right) \in \mathbb{R}^2_{\geq 0}$ from input to output using only

1) **non-negative linear maps**—matrices with non-negative entries together with non-negative bias vectors—and
2) **pairwise activation functions** $\phi\colon \mathbb{R}^2_{\geq 0} \to \mathbb{R}^2_{\geq 0}$, $(z^+, z^-) \mapsto (a^+, a^-) = \phi(z^+, z^-)$, applied at every hidden neuron; the final affine readout uses the identity on the pair.

Activations are indexed by $l \in \{0, \ldots, L\}$, affine parameters by $l \in \{1, \ldots, L\}$. The layer-0 pair is any non-negative split of the input,

$$\left(a_k^{(0,+)}, a_k^{(0,-)}\right) := \left(x_k^+, x_k^-\right), \qquad x_k^+ - x_k^- = x_k, \qquad x_k^+, x_k^- \geq 0, \tag{3.2}$$

and at each affine layer l a non-negative linear map sends the layer-$(l-1)$ activation pair to the layer-l pre-activation pair $\left(z_i^{(l,+)}, z_i^{(l,-)}\right) \in \mathbb{R}^2_{\geq 0}$, to which the pairwise activation is applied for $l < L$. At $l = L$ we set $a^{(L,\pm)} = z^{(L,\pm)}$, with no activation or maximum. We call $\left(a_i^{(l,+)}, a_i^{(l,-)}\right)$ the *decomposed activation* and $\left(z_i^{(l,+)}, z_i^{(l,-)}\right)$ the *decomposed pre-activation*.

Canonical non-negative linear map. Throughout the thesis we instantiate the non-negative linear map at layer l by the entrywise positive/negative decomposition of the original network's weights and biases,

$$W^{(l,+)} := [W^{(l)}]^+, \qquad W^{(l,-)} := [W^{(l)}]^-, \qquad b^{(l,+)} := [b^{(l)}]^+, \qquad b^{(l,-)} := [b^{(l)}]^-, \tag{3.3}$$

where $[\cdot]^\pm$ are the entrywise positive and negative parts, so $W^{(l)} = W^{(l,+)} - W^{(l,-)}$ with $W^{(l,\pm)} \geq 0$. The resulting pre-activation pair at neuron i is

$$z_i^{(l,+)} := b_i^{(l,+)} + \sum_j W_{ij}^{(l,+)}\, a_j^{(l-1,+)} + \sum_j W_{ij}^{(l,-)}\, a_j^{(l-1,-)}, \tag{3.4}$$

$$z_i^{(l,-)} := b_i^{(l,-)} + \sum_j W_{ij}^{(l,+)}\, a_j^{(l-1,-)} + \sum_j W_{ij}^{(l,-)}\, a_j^{(l-1,+)}. \tag{3.5}$$

Positive weights preserve stream membership, negative weights swap it: an excitatory activation through a negative weight becomes an inhibitory contribution, and vice versa. Definition 3.1 imposes structure only; the following definition ties the pair back to the original network.

Definition 3.2 (Valid decomposition). A decomposition (Definition 3.1) is *valid* for the network (3.1) with activation function ϱ if the propagated activation pairs satisfy

$$a_i^{(l,+)} - a_i^{(l,-)} = \varrho_l\left(z_i^{(l)}\right) = a_i^{(l)} \tag{3.6}$$

for every layer $l \in \{0, \ldots, L\}$ and every neuron i, where at $l = 0$ condition (3.6) is read as $a_k^{(0,+)} - a_k^{(0,-)} = x_k$.

Validity is a layer-wise condition, but it propagates through the canonical linear map and forces the decomposition to recover the network output at the final layer.

Proposition 3.1 (Validity recovers the network, `Lean_21`). *Let a decomposition use the canonical non-negative linear map (3.4)–(3.5). If the validity condition (3.6) holds at layer $l-1$, then the decomposed pre-activations at layer l satisfy*

$$z_i^{(l,+)} - z_i^{(l,-)} = z_i^{(l)} \qquad \text{for every neuron } i. \tag{3.7}$$

Consequently, every valid decomposition recovers the original network output as the difference of the final decomposed activations,

$$f(x) = a^{(L,+)} - a^{(L,-)}. \tag{3.8}$$

Proof. Subtracting (3.5) from (3.4) and regrouping by predecessor j,

$$z_i^{(l,+)} - z_i^{(l,-)} = b_i^{(l,+)} - b_i^{(l,-)} + \sum_j \big(W_{ij}^{(l,+)} - W_{ij}^{(l,-)}\big)\big(a_j^{(l-1,+)} - a_j^{(l-1,-)}\big) = b_i^{(l)} + \sum_j W_{ij}^{(l)}\, a_j^{(l-1)} = z_i^{(l)},$$

where the second equality uses $W^{(l)} = W^{(l,+)} - W^{(l,-)}$, $b^{(l)} = b^{(l,+)} - b^{(l,-)}$, and the validity condition at layer $l-1$. This proves (3.7). For (3.8), validity at the final layer L gives $a^{(L,+)} - a^{(L,-)} = \varrho_L\big(z^{(L)}\big) = z^{(L)} = f(x)$. □

By Proposition 3.1, the pre-activation pair always carries the original pre-activation in its difference. A valid pairwise activation must therefore recover $\varrho(z^+ - z^-)$ by subtraction and preserve non-negativity on the input quadrant. We first identify the additional conditions that make both streams convex and coordinatewise non-decreasing, construct such pairs from absolute curvature, and recover the ReLU maximum/identity pair as a special case (Section 3.2). Closed formulas then extend the same network construction to smooth activations. The mixing family (Section 3.3), stopping decomposition (Section 3.4), and regularized decompositions (Section 3.5) retain the difference invariant with different structural properties.

3.2 Convex and Monotone DC Decompositions

The required activation-level object is a pair of functions $F_+, F_- : \mathbb{R}^2 \to \mathbb{R}$ satisfying

$$\begin{gathered} \varrho(x-y) = F_+(x,y) - F_-(x,y), \\ F_+, F_- \text{ convex and coordinatewise non-decreasing.} \end{gathered} \tag{3.9}$$

Convexity alone is insufficient for recursive composition: a convex function of convex inputs need not be convex unless the outer function is non-decreasing in those inputs. Conversely, coordinatewise monotonicity does not mean that a function has non-negative values everywhere. We require the pair to be non-negative only on $\mathbb{R}^2_{\geq 0}$, the domain visited by the split network. The global domain $\mathbb{R}^2$ in (3.9) makes the analytic requirements precise and permits signed inputs to the stream functions.

3.2.1 The profile criterion and network composition

A scalar DC decomposition $\varrho = p - q$ becomes a suitable pair after adding the same affine correction to both components. In this activation-level construction, L denotes a slope bound, independently of

the network depth. The following criterion states exactly when this construction works, including at the kink of ReLU.

Proposition 3.2 (Criterion for a convex, monotone pair, Lean_22). *Let $L \geq 0$ and $C\colon \mathbb{R} \to \mathbb{R}$, and define*

$$T_{C,L}(x,y) := C(x-y) + Ly. \tag{3.10}$$

Then $T_{C,L}$ is convex on $\mathbb{R}^2$ if and only if C is convex. It is coordinatewise non-decreasing on $\mathbb{R}^2$ if and only if

$$0 \leq C(v) - C(u) \leq L(v-u) \qquad (u \leq v). \tag{3.11}$$

If C is differentiable, condition (3.11) is equivalent to $0 \leq C' \leq L$. Consequently, given $p - q = \varrho$, the pair

$$F_+(x,y) = p(x-y) + Ly, \qquad F_-(x,y) = q(x-y) + Ly \tag{3.12}$$

is convex and coordinatewise non-decreasing exactly when both profiles are convex and satisfy (3.11). Under these conditions it maps $\mathbb{R}^2_{\geq 0}$ to itself exactly when $p(0),q(0) \geq 0$.

Proof. Affine precomposition and addition of an affine function preserve convexity; conversely, restriction to $y = 0$ recovers C. Increasing x gives the lower bound in (3.11). Increasing y by $v - u$ changes the profile argument from v to u and gives the upper bound. These two inequalities also suffice for monotonicity in the respective coordinates. For a differentiable profile, the equivalence with the derivative bounds follows by taking difference quotients and by the mean value theorem. The common term Ly cancels in $F_+ - F_-$. Finally, monotonicity gives $F_\pm(x,y) \geq F_\pm(0,0)$ on the non-negative quadrant, and evaluation at the origin proves necessity of the two normalization inequalities. □

Theorem 3.3 (Network DC decomposition from activation pairs, Lean_23). *Consider the network (3.1) with hidden activation ϱ. Choose a pair F_+,F_- satisfying (3.9) on all of $\mathbb{R}^2$ and $F_\pm(0,0) \geq 0$. Use this pair entrywise after the canonical affine split (3.4)–(3.5), and use the identity pair at the final affine readout. Then every decomposed activation, and each of the two output streams G,H, is convex and coordinatewise non-decreasing as a function of the independent doubled input coordinates (x^+,x^-). Moreover,*

$$G(x^+,x^-) - H(x^+,x^-) = f(x^+ - x^-), \tag{3.13}$$

and both streams are non-negative when $x^+,x^- \geq 0$. These conclusions also hold for every tail network starting at an intermediate layer.

If each component of an input split $x \mapsto (x^+(x),x^-(x))$ is convex, then $x \mapsto G(x^+(x),x^-(x))$ and $x \mapsto H(x^+(x),x^-(x))$ are convex. In particular, the usual split $x^\pm = [x]^\pm$ gives a DC representation on the original input space.

Proof. The initial doubled coordinates are affine and non-decreasing. At each affine layer, non-negative coefficients preserve convexity and coordinatewise monotonicity. At a hidden layer, a convex, coordinatewise non-decreasing pair component composed with these convex inputs is again convex; composition also preserves monotonicity. At the final layer there is no activation: its two affine combinations retain both properties. This proves the structural assertions by induction, equally for a network or any of its tails.

For recovery, subtraction of the affine pair yields the original affine map by Proposition 3.1. A hidden pair then gives $F_+(z^+,z^-) - F_-(z^+,z^-) = \varrho(z^+ - z^-)$. At $l = L$ the identity pair instead gives the ungated pre-activation difference. Induction proves (3.13), including negative output logits. The origin

normalization and monotonicity preserve non-negativity on the quadrant at every hidden step, and the canonical affine maps do so as well. Finally, the composition rule for a convex non-decreasing function with convex component functions proves the assertion for a convex input split (Amos et al., 2017). □

Coordinatewise monotonicity in this theorem refers to the independent pair coordinates. It generally does *not* survive as monotonicity in the original signed input x: for example, $x \mapsto x^- = \max\{-x,0\}$ decreases as x increases. Convexity does survive this input split. Thus the theorem yields monotone networks on doubled input space and a difference of convex functions on the original input space, with these two domains kept distinct.

3.2.2 Construction from absolute second derivatives

A finite amount of positive and negative curvature provides the required bounded profile slopes. The construction below uses $|f''|$ directly; here f denotes a scalar activation, not the whole network.

Theorem 3.4 (Absolute-curvature construction, `Lean_24`**).** *Let $f\colon \mathbb{R} \to \mathbb{R}$ be differentiable, let f' be locally absolutely continuous, and assume $f'' \in L^1(\mathbb{R})$. Set*

$$\begin{aligned} s_- &\coloneqq \lim_{t\to-\infty} f'(t), \qquad V(t) \coloneqq \int_{-\infty}^{t} |f''(u)|\ du, \\ D(z) &\coloneqq |f(0)| + |s_-|z + \int_0^z V(t)\ dt, \\ p &\coloneqq \frac{D+f}{2}, \qquad q \coloneqq \frac{D-f}{2}. \end{aligned} \tag{3.14}$$

The limits s_- and $s_+ \coloneqq \lim_{t\to+\infty} f'(t)$ exist. With

$$L \coloneqq \frac{\|f''\|_{L^1(\mathbb{R})} + |s_-| + |s_+|}{2}, \tag{3.15}$$

the profiles p,q are convex, satisfy $p - q = f$ and $0 \le p',q' \le L$, and have $p(0) = \max\{f(0),0\}$, $q(0) = \max\{-f(0),0\}$. Therefore (3.12) is a convex, coordinatewise non-decreasing pair on $\mathbb{R}^2$, is non-negative on $\mathbb{R}^2_{\ge 0}$, and recovers $f(x-y)$ exactly.

Proof. Local absolute continuity gives $|f'(v) - f'(u)| \le \int_u^v |f''(t)|\ \mathrm{d}t$ for $u \le v$. Integrability makes f' Cauchy at both ends of the real line, so its finite limits exist. Write

$$P = \frac{|f''| + f''}{2}, \qquad N = \frac{|f''| - f''}{2}. \tag{3.16}$$

Differentiating (3.14) yields

$$p'(z) = \max\{s_-,0\} + \int_{-\infty}^{z} P(u)\ \mathrm{d}u, \qquad q'(z) = \max\{-s_-,0\} + \int_{-\infty}^{z} N(u)\ \mathrm{d}u. \tag{3.17}$$

Both derivatives are non-negative, non-decreasing, and bounded, and $p'' = P$, $q'' = N$ almost everywhere. Their upper limits are $A = \max\{s_-,0\} + \int_{\mathbb{R}} P$ and $B = \max\{-s_-,0\} + \int_{\mathbb{R}} N$. Since

$A + B = |s_-| + \|f''\|_1$ and $A - B = s_+$, their maximum is precisely (3.15). The difference and origin values follow directly from the definition. Proposition 3.2 now proves all pair properties. □

The functions P and N are the positive and negative parts of curvature. They are minimal: any twice weakly differentiable convex profiles $\widetilde{p},\widetilde{q}$ with $\widetilde{p} - \widetilde{q} = f$ satisfy $\widetilde{p}'' \geq P$ and $\widetilde{q}'' \geq N$ almost everywhere. Within the affine lift (3.12), the coefficient in (3.15) is also the least possible common slope bound for globally convex non-decreasing profiles. Indeed, their non-negative limiting slopes at $-\infty$ differ by s_-, so they are at least $\max\{s_-,0\}$ and $\max\{-s_-,0\}$; their curvature measures dominate $P\,\mathrm{d}z$ and $N\,\mathrm{d}z$. Both upper slope bounds are therefore at least A and B from the proof. This is a minimality statement for this profile-and-affine-lift construction, not for arbitrary bivariate decompositions or for the sizes of the resulting network streams.

Remark 3.1 (Why the global domain matters). Under the regularity assumptions on f' above, integrability of f'' is also necessary for the existence of *finite-valued* convex, coordinatewise non-decreasing F_+,F_- on all of $\mathbb{R}^2$ with difference $f(x-y)$. To see this, restrict each component to $(t,0)$ for $t \leq 0$. The resulting convex non-decreasing function has slopes bounded between 0 and a finite one-sided slope at the origin, hence finite curvature mass on the negative half-line. Next restrict each component to $(0,-t)$ for $t \geq 0$. These restrictions are convex and non-increasing, so their slopes lie between a finite slope at the origin and 0, giving finite curvature mass on the positive half-line. In both restrictions the difference is $f(t)$; the total variation of its curvature is bounded by the sum of the two positive curvature measures. Thus $\int_{\mathbb{R}} |f''| < \infty$. This necessity argument concerns the global domain and does not apply if the pair is defined only on the non-negative quadrant. For example,

$$(x-y)^2 = \left(2x^2 + 2y^2\right) - (x+y)^2$$

is a difference of two convex, coordinatewise non-decreasing functions on that quadrant, although the curvature of $f(z) = z^2$ is not integrable on $\mathbb{R}$.

Corollary 3.5 (Exact splitting on a bounded activation interval). *Let $a < b$ and let $f \in C^1([a,b])$ have absolutely continuous derivative. There exist finite-valued convex, coordinatewise non-decreasing functions $F_+,F_-\colon \mathbb{R}^2 \to \mathbb{R}$, non-negative on $\mathbb{R}^2_{\geq 0}$, such that $F_+(x,y) - F_-(x,y) = f(x-y)$ whenever $x - y \in [a,b]$.*

Proof. Extend f by its tangent affine functions at a and b. The extension is continuously differentiable, its derivative is locally absolutely continuous, and its curvature vanishes outside $[a,b]$. Apply Theorem 3.4 to this extension. □

This local statement applies, for example, to exponentiation on a fixed bounded score interval. It does not assert a global monotone convex split for e^z on all of $\mathbb{R}$, whose curvature is not integrable.

3.2.3 ReLU: the maximum and identity pair

For $f(z) = \mathrm{ReLU}(z)$ choose $p(z) = \mathrm{ReLU}(z)$, $q(z) = 0$, and $L = 1$. Both profiles are convex and satisfy (3.11), even though p is not differentiable at the origin. The profile lift gives the exact pair

$$\begin{gathered} F_+(a,b) = \mathrm{ReLU}(a-b) + b = \max\{a,b\}, \qquad F_-(a,b) = b, \\ \mathrm{ReLU}(a-b) = \max\{a,b\} - b. \end{gathered} \tag{3.18}$$

Thus the ReLU construction is a special case of the general criterion: its positive component is a Maxout unit (Goodfellow et al., 2013) and its negative component is the identity in the second coordinate. The same pair is obtained from the absolute-curvature construction by replacing $f''\,\mathrm{d}z$ with the curvature measure δ_0 (Remark 3.5). It recovers the classical DC splitting of ReLU networks (Awasthi et al., 2023; Siahkamari et al., 2020; Zhang et al., 2018b; Zimmermann and Loho, 2026), and underlies the stabilized passes, convolutional generalization, and saliency methods developed later in this thesis.

The hidden-layer update is therefore

$$a^{(l,+)} = \max\{z^{(l,+)}, z^{(l,-)}\}, \qquad a^{(l,-)} = z^{(l,-)}, \qquad 1 \le l < L, \tag{3.19}$$

with $a^{(L,\pm)} = z^{(L,\pm)}$ at the final affine readout. Theorem 3.3 immediately gives validity and convexity of this decomposition. Section 3.3 will identify it as the $\eta = 1$ endpoint of a larger family.

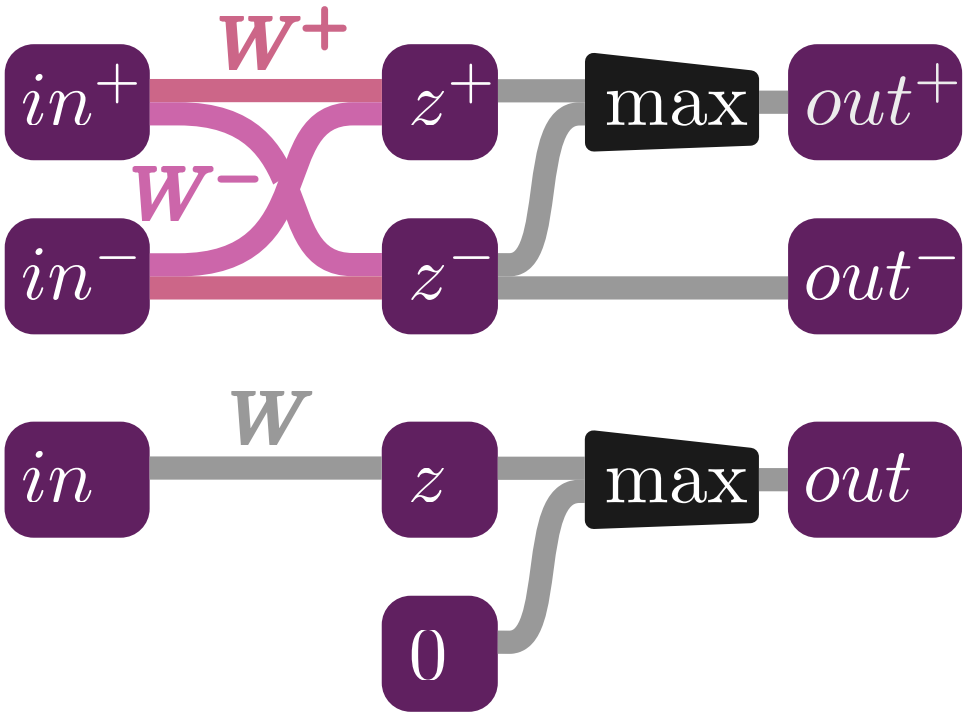


Fig. 3.2: The monotone, convex split streams (top) versus a standard ReLU layer (bottom). The negative weight part transitions positive input to negative pre-activation and vice versa, while the positive part maps positive to positive and negative to negative.

The stream functions g and h. The DC split replaces each layer by a *split pair* of two monotone *split streams*, non-negative on the input quadrant (Figure 3.2). In analogy to the tail networks $f^{(l)}\colon \mathbb{R}^{d^{(l)}} \to \mathbb{R}$ restricting the original network to layers $l+1, \dots, L$, we denote the split streams restricted to those layers by

$$g^{(l)}\colon \mathbb{R}^{d^{(l)}} \times \mathbb{R}^{d^{(l)}} \to \mathbb{R}, \qquad h^{(l)}\colon \mathbb{R}^{d^{(l)}} \times \mathbb{R}^{d^{(l)}} \to \mathbb{R}, \tag{3.20}$$

mapping an activation pair at layer l to the $+$ and $-$ components of the decomposed output; we abbreviate $g := g^{(0)}$, $h := h^{(0)}$. On the output layer $g^{(L)}(x^+, x^-) = x^+$, $h^{(L)}(x^+, x^-) = x^-$, and for $0 \le l < L$,

$$g^{(l)}(x^+, x^-) = g^{(l+1)}\big(a^{(l+1,+)}, a^{(l+1,-)}\big), \qquad h^{(l)}(x^+, x^-) = h^{(l+1)}\big(a^{(l+1,+)}, a^{(l+1,-)}\big), \tag{3.21}$$

where the pair $\big(a^{(l+1,+)}, a^{(l+1,-)}\big)$ is computed from (x^+, x^-) by (3.4)–(3.5) and (3.19). These stream functions—not the individual layer activations—are the central objects of the DC sections of this chapter and of the saliency derivations of Chapter 6: the network is exhibited as $f = g - h$ with structured subnetworks g and h. Concretely, the split forward pass for an MLP initialises $\big(a^{(0,+)}, a^{(0,-)}\big) = (x^+, x^-)$ and, for $l = 1, \dots, L-1$, computes the pre-activation pair (3.4)–(3.5) followed by the entrywise update $a^{(l,+)} = \max\{z^{(l,+)}, z^{(l,-)}\}$, $a^{(l,-)} = z^{(l,-)}$ of (3.19). The final layer uses only the affine pair (3.4)–(3.5), so the two heads may reconstruct a negative logit.

The split streams are affine maps with non-negative coefficients followed by entrywise Maxout gates and projections, i.e. monotone networks with convex, monotone activations; by the composition principles of Section 2.2 (Amos et al., 2017) this yields their two defining properties.

Proposition 3.6 (Monotonicity and convexity of the split streams, `Lean_25`). *For every $0 \le l \le L$, the stream functions $g^{(l)}$ and $h^{(l)}$ of (3.20) are convex and non-decreasing in each coordinate of the pair $(x^+, x^-) \in \mathbb{R}^{d^{(l)}} \times \mathbb{R}^{d^{(l)}}$.*

Proof. Apply Theorem 3.3 to each tail network using the pair $(\max\{a,b\},b)$ from (3.18). The final pair is the identity, so the same argument includes the ungated affine head. □

The correctness statement asserts that the difference of the two split-stream outputs equals the original tail network evaluated at the difference of their inputs—at *every* layer, not only at the input.

Theorem 3.7 (Correctness of the split forward pass, `Lean_26`). *For every $0 \le l \le L$ and all $x^+, x^- \in \mathbb{R}^{d^{(l)}}_{\ge 0}$,*

$$f^{(l)}\big(x^+ - x^-\big) = \big(g^{(l)} - h^{(l)}\big)(x^+, x^-). \tag{3.22}$$

Proof. The recovery part of Theorem 3.3 applies to every tail network. At each hidden layer, subtraction of the maximum/identity pair gives $\mathrm{ReLU}(z^+ - z^-)$ by (3.18); at the final layer, subtraction gives $z^+ - z^-$ directly, because $\varrho_L = \mathrm{id}$. The base tail at $l = L$ is the identity, and the recursion (3.21) gives (3.22) at every preceding layer. □

Remark 3.2 (Preservation of sub-threshold signals). The correctness theorem hides an information-theoretic asymmetry between the original and the split network. When a neuron is inactive, $z_i^{(l)} \le 0$, the original ReLU outputs $a_i^{(l)} = 0$ and discards the *magnitude* of the losing pre-activation: downstream computation is oblivious to how strongly the neuron was suppressed. The split pair reads $\big(a_i^{(l,+)}, a_i^{(l,-)}\big) = \big(z_i^{(l,-)}, z_i^{(l,-)}\big)$: the difference is zero, as validity demands, but both components retain the accumulated magnitude $z_i^{(l,-)}$ of the cancelled mass and propagate it forward. The split streams thus carry strictly more information than the original activations—sub-threshold contributions survive the gate—which the attribution methods of Chapter 6 exploit. The same effect drives the numerical growth of Section 3.6: without cancellation by subtraction, both streams accumulate the absolute mass $z_i^{(l,+)} + z_i^{(l,-)}$ instead of the signed residual.

Remark 3.3 (Flexibility of the input split). Theorem 3.7 holds for *any* non-negative representation $x = x^+ - x^-$: the entrywise split $x^\pm = [x]^\pm$ or the symmetric split $x^\pm = [x]^\pm + c$ with $c \ge 0$. This degree of freedom is exploited both for numerical stabilization and for the baseline choices of the attribution methods in Chapter 6. The half–half split $x^\pm = \pm x/2$ of Section 3.6 is *not* such a non-negative input split—its negative component is signed—but a chain-rule device for the backward pass, used only to differentiate the split streams along the unified input direction.

3.2.4 Closed formulas for smooth activations

The absolute-curvature construction can be integrated explicitly for the activations below. There is also a useful choice: a smooth curvature majorant may give a shorter formula than the minimal positive/negative curvature split. We provide both, distinguishing exact recovery from curvature minimality. Write $\sigma(z) = (1+e^{-z})^{-1}$, let φ,Φ be the standard Gaussian density and distribution function, and set

$$s_\beta(z) := \frac{1}{\beta}\log(1+e^{\beta z}), \qquad \beta > 0.$$

Corollary 3.8 (Explicit convex, monotone activation pairs, Lean_27). *For $f(z) = z\Phi(z)$ (GELU), $f(z) = z\sigma(\beta z)$ (Swish), $f = \sigma$ (sigmoid), and $f = s_{1/\theta}$ (softplus), choose the corresponding row of Table 3.1. Then the functions $F_+(x,y) = p(x-y) + Ly$ and $F_-(x,y) = q(x-y) + Ly$ are convex and coordinatewise non-decreasing on $\mathbb{R}^2$, map $\mathbb{R}^2_{\geq 0}$ to itself, and satisfy $F_+ - F_- = f(x-y)$.*

Activation	$p(z)$	$q(z)$	L
GELU	$2z\Phi(z) + 2\varphi(z)$	$z\Phi(z) + 2\varphi(z)$	2
Swish	$2s_\beta(z)$	$2s_\beta(z) - z\sigma(\beta z)$	2
Sigmoid	$\frac{1}{2}(s_1(z) + \sigma(z))$	$\frac{1}{2}(s_1(z) - \sigma(z))$	$\frac{1}{2}$
Softplus	$s_{1/\theta}(z)$	0	1

Tab. 3.1: Closed scalar profiles for the lift (3.12), with β,$\theta > 0$. These smooth profiles prioritize a short expression; the GELU, Swish, and sigmoid rows use more curvature than the canonical split. GELU means the exact Gaussian activation, not its tanh approximation; SiLU is the case $\beta = 1$ of Swish.

Proof. In every row $p - q = f$. For GELU (Hendrycks and Gimpel, 2016), direct differentiation gives

$$p' = 2\Phi, \quad p'' = 2\varphi, \qquad q' = \Phi - z\varphi, \quad q'' = z^2\varphi.$$

Hence both profiles are convex; p' increases from 0 to 2 and q' from 0 to 1. For Swish (Ramachandran et al., 2017), writing $u = \beta z$ gives

$$p' = 2\sigma(u), \quad p'' = 2\beta\sigma'(u), \qquad q' = \sigma(u) - u\sigma'(u), \quad q'' = \beta u\big(2\sigma(u) - 1\big)\sigma'(u) \geq 0.$$

The last inequality uses that u and $2\sigma(u) - 1$ have the same sign. The slopes again have limits 0,2 for p' and 0,1 for q', so they lie in [0,2]. For the sigmoid profiles,

$$p' = \tfrac{1}{2}\sigma(2-\sigma), \quad q' = \tfrac{1}{2}\sigma^2, \qquad p'' = \sigma'(1-\sigma) \geq 0, \quad q'' = \sigma'\sigma \geq 0,$$

with slopes in [0,1/2]. Softplus has $p' = \sigma(z/\theta) \in [0,1]$ and $p'' = \sigma'(z/\theta)/\theta \geq 0$, while $q = 0$. All profile values at zero are non-negative (for sigmoid, $s_1(0) = \log 2 > 1/2$). Proposition 3.2 proves the claim. □

These formulas require no numerical integration or auxiliary root solve. For example, the softplus row simplifies directly to

$$F_+(x,y) = \theta\log\big(e^{x/\theta} + e^{y/\theta}\big), \qquad F_-(x,y) = y, \tag{3.23}$$

whose limit as $\theta \downarrow 0$ is exactly $(\max\{x,y\},y)$. The sigmoid row gives

$$F_\pm(x,y) = \frac{\log(e^x + e^y) \pm \sigma(x-y)}{2}. \tag{3.24}$$

Writing $d = x - y$, the GELU and Swish pairs are explicitly

$$\begin{aligned} F_+^{\mathrm{GELU}}(x,y) &= 2\big(d\Phi(d) + \varphi(d) + y\big), \\ F_-^{\mathrm{GELU}}(x,y) &= d\Phi(d) + 2\varphi(d) + 2y, \end{aligned} \tag{3.25}$$

and

$$\begin{aligned} F_+^{\mathrm{Swish}}(x,y) &= \frac{2}{\beta}\log\big(e^{\beta x} + e^{\beta y}\big), \\ F_-^{\mathrm{Swish}}(x,y) &= F_+^{\mathrm{Swish}}(x,y) - d\sigma(\beta d). \end{aligned} \tag{3.26}$$

Stable implementations of the exponential formulas should evaluate log-sum-exp through its usual shifted form.

Closed forms for the canonical absolute-curvature split. The smaller coefficients in Theorem 3.4 can also be attained explicitly. For sigmoid,

$$p(z) = \begin{cases} \sigma(z), & z \le 0, \\ \frac{1}{2} + \frac{1}{4}z, & z \ge 0, \end{cases} \qquad q(z) = p(z) - \sigma(z), \qquad L = \tfrac{1}{4}. \tag{3.27}$$

For both GELU and Swish, let $a > 0$ be the positive zero of f'', and define $m = f'(a) - 1$, $c = af'(a) - f(a)$ and $L = 1 + 2m$. Then

$$p(z) = \begin{cases} -c, & z \le -a, \\ f(z) + mz, & -a \le z \le a, \\ Lz - c, & z \ge a, \end{cases} \qquad q(z) = p(z) - f(z). \tag{3.28}$$

The constants are

$$\begin{aligned} &\text{GELU:} \quad a = \sqrt{2}, \quad m = \Phi(\sqrt{2}) + \sqrt{2}\varphi(\sqrt{2}) - 1, \quad c = 2\varphi(\sqrt{2}), \\ &\text{Swish:} \quad a = r/\beta, \quad m = (r-2)/4, \quad c = (r^2-4)/(4\beta), \quad L = r/2, \end{aligned} \tag{3.29}$$

where r is the unique positive solution of $r\tanh(r/2) = 2$ ($r \approx 2.39936$). The left-hand side is strictly increasing from 0 to infinity on $r > 0$, so this fixed constant is unambiguous. Once it is fixed, the Swish pair has a pointwise closed expression with no quadrature. For softplus the canonical profiles are already the last row of Table 3.1.

To verify (3.28), note that both activations satisfy $f(-z) = f(z) - z$, have $f'(-\infty) = 0$, $f'(+\infty) = 1$, and have positive curvature precisely on $(-a,a)$. Differentiating the displayed pieces gives $p'' = \max\{f'',0\}$ and $q'' = \max\{-f'',0\}$; the values and first derivatives match at $\pm a$. At zero $p(0) = q(0) = 0$, so these are exactly the normalized profiles of Theorem 3.4. For sigmoid the curvature changes sign at zero and the same verification gives (3.27). The canonical GELU and Swish profiles can take negative values on the negative half-line; their lifted pair is nevertheless non-negative on the non-negative quadrant by the origin normalization and coordinatewise monotonicity. No global non-negativity claim for the scalar profiles is needed.

Corollary 3.9 (Modern activation functions, `Lean_28`). *Theorems 3.4 and 3.13 apply to the following activation functions, where φ and Φ denote the standard Gaussian density and distribution function and $\sigma(x) = (1 + e^{-x})^{-1}$ the logistic sigmoid:*

1) GELU, $f(x) = x\,\Phi(x)$, *with* $f''(x) = (2 - x^2)\,\varphi(x)$*; the curvature is positive on* $(-\sqrt{2}, \sqrt{2})$ *and negative outside.*
2) Swish/SiLU, $f(x) = x\,\sigma(\beta x)$ *with* $\beta > 0$, *with* $f''(x) = \beta\,\sigma'(\beta x)\big(2 + \beta x\,(1 - 2\sigma(\beta x))\big)$*; the curvature is positive on a bounded interval around the origin and negative in both tails.*
3) Mish, $f(x) = x\tanh\big(\log(1 + e^x)\big)$.
4) Hyperbolic tangent, $f(x) = \tanh(x)$, *with* $f''(x) = -2\tanh(x)\operatorname{sech}^2(x)$, *so* P *is supported on* $(-\infty, 0]$ *and* N *on* $[0, \infty)$.
5) Logistic sigmoid, $f = \sigma$, *with* $f'' = \sigma'\,(1 - 2\sigma)$.
6) Softplus at temperature $\theta > 0$, $f(x) = \theta\log(1 + e^{x/\theta})$, *with* $f''(x) = \sigma'(x/\theta)/\theta \geq 0$*; here* $N = 0$, $C_2 = 0$, *and the decomposition degenerates to the statement that softplus itself is convex.*

In each case $f(0) \geq 0$, *and both constructions preserve non-negativity on the input quadrant (Remark 3.6).*

Proof. All listed functions are smooth with the stated second derivatives, obtained by direct differentiation. In each case f'' is continuous and decays at least exponentially as $|x| \to \infty$ (with Gaussian decay for GELU): for Swish and Mish the polynomial factor x is dominated by the exponential decay of $\sigma'(\beta x)$ and of the derivatives of $\tanh \circ \operatorname{softplus}$, respectively. Hence $f'' \in L^1(\mathbb{R})$ and f' is locally absolutely continuous, so both constructions apply. The values $f(0) \in \{0, \frac{1}{2}, \theta\log 2\}$ are non-negative in every case. □

Corollary 3.10 (Networks with smooth activations, `Lean_29`). *Every finite feedforward network with GELU, Swish/SiLU, sigmoid, or softplus hidden activations, arbitrary signed weights and biases, and an affine output admits an exact decomposition into two convex, coordinatewise non-decreasing streams on doubled input space. Either the explicit profiles of Corollary 3.8 or the canonical profiles (3.27)–(3.28) and (3.23) may be used at every hidden neuron. With the usual non-negative input split, the streams remain non-negative and their pullbacks are convex functions of the original signed input. The same existence result holds for Mish and tanh by Theorem 3.4.*

Proof. The activation pairs have the properties required by Theorem 3.3, by Corollaries 3.8 and 3.9; apply that theorem with the identity pair at the output. □

The same layer-wise induction also permits different qualifying activations or profile choices at different hidden layers, because each step uses only the properties of that layer's pair. The network theorem above states the uniform-activation case used in the formal network interface.

Remark 3.4 (Scope of the smooth extension). The extension above is an exact structural theorem about the scalar activations and their composition into feedforward networks. It does not establish empirical attribution quality for these smooth networks, nor does it by itself extend the ReLU-specific tropical polytope representation, hard-gate games, or numerical stabilization formulas. The experiments later in the thesis evaluate the architectures and methods described in their protocols; the new smooth formulas are not additional experimental results.

3.2.5 Multivariate activations and softmax

The same lifting principle applies to activations that couple several coordinates. Convexity of a vector-valued pair means convexity of every scalar output component. The following differentiable version suffices for softmax.

Proposition 3.11 (Multivariate profile lift, Lean_30). *Let $C \in C^1(\mathbb{R}^n)$ and $\ell \in \mathbb{R}^n_{\geq 0}$. Define*

$$T_{C,\ell}(x,y) = C(x-y) + \langle \ell, y \rangle, \qquad x,y \in \mathbb{R}^n. \tag{3.30}$$

Then $T_{C,\ell}$ is convex on $\mathbb{R}^{2n}$ if and only if C is convex, and is coordinatewise non-decreasing if and only if $0 \leq \partial_j C(z) \leq \ell_j$ for all z and j. Consequently, if $C_+ - C_- = f$, both profiles are convex, and both satisfy these derivative bounds, the common lift gives a convex, coordinatewise non-decreasing pair with difference $f(x-y)$. If $C_+(0), C_-(0) \geq 0$, the lifted pair is non-negative on $\mathbb{R}^{2n}_{\geq 0}$.

Proof. Affine precomposition preserves convexity, and restriction to $y = 0$ proves the converse. The partial derivatives of the lift are $\partial_{x_j} T = \partial_j C$ and $\partial_{y_j} T = \ell_j - \partial_j C$, which proves the monotonicity criterion. The common affine term cancels in the difference, and monotonicity compares every point of the non-negative orthant with the origin. □

Softmax admits a particularly short pair because its curvature is dominated by the Hessian of log-sum-exp. This is a multivariate curvature comparison, rather than a componentwise application of the scalar integrability theorem.

Theorem 3.12 (Explicit convex, monotone softmax pair, Lean_31). *Let $n \geq 1$, and define*

$$H(z) = \log\left(\sum_{j=1}^n e^{z_j}\right), \qquad \pi_i(z) = \frac{e^{z_i}}{\sum_{j=1}^n e^{z_j}}, \qquad c_n = \frac{\log n - 1/n}{2}.$$

For each output coordinate i, set

$$F_{i,\pm}(x,y) = \frac{H(x-y) \pm \pi_i(x-y) + \sum_{j=1}^n y_j}{2} - c_n. \tag{3.31}$$

Both components are finite, convex, and coordinatewise non-decreasing on all of $\mathbb{R}^{2n}$. They are non-negative on $\mathbb{R}^{2n}_{\geq 0}$ and satisfy $F_{i,+}(x,y) - F_{i,-}(x,y) = \pi_i(x-y)$.

Proof. Fix z and a direction $v \in \mathbb{R}^n$, and abbreviate $\mu = \sum_j \pi_j(z) v_j$ and $V = \sum_j \pi_j(z)(v_j - \mu)^2$. Differentiation along $z + tv$ gives

$$\begin{aligned} D^2 H(z)[v,v] &= V, \\ D^2 \pi_i(z)[v,v] &= \sum_j \pi_j(z)\big(\mathbf{1}[j=i] - \pi_i(z)\big)(v_j - \mu)^2. \end{aligned} \tag{3.32}$$

Since $|\mathbf{1}[j=i] - \pi_i| \leq 1$, it follows that $|D^2 \pi_i(z)[v,v]| \leq V$. Thus $H + \pi_i$ and $H - \pi_i$ have positive semidefinite Hessians and are convex. For the profiles $C_{i,\pm} = (H \pm \pi_i)/2 - c_n$, their partial derivatives are

	$\partial_j C_{i,+}$	$\partial_j C_{i,-}$
$j = i$	$\pi_i - \pi_i^2/2$	$\pi_i^2/2$
$j \neq i$	$\pi_j(1-\pi_i)/2$	$\pi_j(1+\pi_i)/2.$

(3.33)

Every entry lies in $[0,1/2]$; for the last entry, use $\pi_j \leq 1 - \pi_i$. Proposition 3.11 with $\ell_j = 1/2$ gives global convexity and monotonicity. At the origin, $F_{i,+}(0,0) = 1/n$ and $F_{i,-}(0,0) = 0$, proving non-negativity on the orthant. Subtraction gives exact recovery. □

For a fixed masked-softmax row, apply the theorem to its nonempty set of allowed coordinates and set the masked output pairs to (0,0). A fully masked row requires an additional output convention and is outside this statement. Fixed positive temperatures and fixed additive score biases can be absorbed into the preceding affine map. Section 5.3 combines this exact softmax pair with convex product and normalization pairs to obtain a DC decomposition of the full Transformer computation.

3.2.6 The symmetric-tilt variant and curvature minimality

For comparison, the following alternative uses the same positive/negative curvature split but adds the common symmetric term $M(a+b)$. We retain it because its fixed-profile minimality statement is useful independently of the asymmetric lift (3.12). The two constructions have the same exact difference but generally different individual streams; in particular, only the asymmetric normalization used above reproduces the literal ReLU pair $(\max\{a,b\},b)$.

Theorem 3.13 (DC–monotone splitting under integrable curvature, `Lean_32`**).** *Let $f:\mathbb{R}\to\mathbb{R}$ be differentiable with locally absolutely continuous derivative f' and $f''\in L^1(\mathbb{R})$. Then the limit $s_-:=\lim_{u\to-\infty}f'(u)$ exists and f' is bounded. Write $P:=\max\{f'',0\}$ and $N:=\max\{-f'',0\}$, so that $f''=P-N$ with $P,N\ge 0$, and define*

$$C_1(x):=f(0)+\int_0^x\Big(s_-+\int_{-\infty}^t P(u)\;du\Big)\,dt,\qquad C_2(x):=\int_0^x\int_{-\infty}^t N(u)\;du\;dt, \tag{3.34}$$

as well as $M:=\max\big\{\sup_{x\in\mathbb{R}}|C_1'(x)|,\sup_{x\in\mathbb{R}}|C_2'(x)|\big\}<\infty$. Then the functions

$$g(a,b):=C_1(a-b)+M(a+b),\qquad h(a,b):=C_2(a-b)+M(a+b) \tag{3.35}$$

are convex on $\mathbb{R}^2$, non-decreasing in each argument, and satisfy

$$f(a-b)=g(a,b)-h(a,b)\qquad \text{for all } (a,b)\in\mathbb{R}^2. \tag{3.36}$$

Proof. Well-definedness. For $x\le y$ we have $|f'(y)-f'(x)|\le\int_x^y|f''(u)|\,\mathrm{d}u$, which tends to 0 as $x,y\to-\infty$ because $f''\in L^1(\mathbb{R})$; by the Cauchy criterion the limit s_- exists. Consequently $|f'(x)|\le|s_-|+\|f''\|_{L^1(\mathbb{R})}$ for all x, so the boundedness of f' is automatic and need not be assumed. Since $0\le P,N\le|f''|$, both P and N lie in $L^1(\mathbb{R})$, the inner integrals in (3.34) are well-defined, non-decreasing in t, and bounded by $\|P\|_{L^1(\mathbb{R})}$ and $\|N\|_{L^1(\mathbb{R})}$ respectively. Hence C_1,C_2 are well-defined with locally absolutely continuous derivatives

$$C_1'(x)=s_-+\int_{-\infty}^x P(u)\;\mathrm{d}u,\qquad C_2'(x)=\int_{-\infty}^x N(u)\;\mathrm{d}u,$$

and $|C_1'|\le|s_-|+\|P\|_{L^1(\mathbb{R})}$, $|C_2'|\le\|N\|_{L^1(\mathbb{R})}$, so $M<\infty$.

Exactness. For almost every x,

$$C_1'(x)-C_2'(x)=s_-+\int_{-\infty}^x\big(P(u)-N(u)\big)\,\mathrm{d}u=s_-+\int_{-\infty}^x f''(u)\;\mathrm{d}u=f'(x),$$

and $C_1(0) - C_2(0) = f(0)$, so integrating from 0 yields $C_1 - C_2 = f$ on all of $\mathbb{R}$. The linear tilt cancels upon subtraction, and (3.36) follows by evaluating at $x = a - b$.

Convexity. C_1' and C_2' are non-decreasing, being (shifted) antiderivatives of the non-negative functions P and N, so C_1 and C_2 are convex on $\mathbb{R}$. Precomposition with the linear map $(a,b) \mapsto a - b$ preserves convexity, and so does adding the linear function $M(a + b)$; hence g and h are convex on $\mathbb{R}^2$. (When f'' is continuous, this is also visible directly: the Hessian of $(a,b) \mapsto C_1(a - b)$ has eigenvalues 0 and $2C_1''(a - b) = 2P(a - b) \geq 0$.)

Monotonicity. By the choice of M,

$$\frac{\partial g}{\partial a}(a,b) = C_1'(a - b) + M \geq 0, \qquad \frac{\partial g}{\partial b}(a,b) = -C_1'(a - b) + M \geq 0,$$

and analogously for h with C_2' in place of C_1'. Hence g and h are non-decreasing in each argument. □

The curvature split is canonical. For the fixed profiles C_1,C_2 in (3.34), the coefficient M is minimal among symmetric tilts $\lambda(a + b)$; this claim does not compare it with asymmetric corrections or other profile normalizations.

Proposition 3.14 (Minimality of the curvature split, `Lean_33`). *Let f be as in Theorem 3.13 and suppose $f = c_1 - c_2$ for convex functions $c_1, c_2 \colon \mathbb{R} \to \mathbb{R}$ whose second distributional derivatives are locally integrable functions. Then $c_1'' \geq P$ and $c_2'' \geq N$ almost everywhere. In particular, (C_1, C_2) is, up to a common affine summand, the unique pair of convex functions with $C_1 - C_2 = f$ of pointwise minimal curvature, and among all tilts of the form $(a,b) \mapsto \lambda(a + b)$, $\lambda \geq 0$, the constant M in Theorem 3.13 is the smallest one rendering both components of (3.35) non-decreasing in each argument.*

Proof. From $c_1'' = f'' + c_2''$ and $c_2'' \geq 0$ we get $c_1'' \geq f''$, and convexity of c_1 gives $c_1'' \geq 0$; together $c_1'' \geq \max\{f'', 0\} = P$ almost everywhere, and symmetrically $c_2'' \geq N$. If equality holds a.e. in both bounds, then $c_1 - C_1$ and $c_2 - C_2$ are affine and, since $c_1 - c_2 = f = C_1 - C_2$, they coincide. For the tilt, non-decreasingness of g in both arguments is equivalent to $\lambda \geq C_1'(x)$ and $\lambda \geq -C_1'(x)$ for all x, i.e. $\lambda \geq \sup_x |C_1'(x)|$, and analogously for h; the smallest admissible λ is exactly M. □

Remark 3.5 (Bounded variation and the piecewise-linear case). The curvature construction extends to locally absolutely continuous f whose weak derivative has a bounded-variation representative: its distributional second derivative is a finite signed Radon measure μ. Replacing $|f''(u)|\,\mathrm{d}u$ by $|\mu|$ in Theorem 3.4, or replacing $P(u)\,\mathrm{d}u$ and $N(u)\,\mathrm{d}u$ by the Jordan parts μ_+,μ_-, gives convex profiles with bounded slopes; the increment criterion (3.11) replaces pointwise derivative conditions at kinks. For ReLU, $\mu = \delta_0$, $s_- = 0$, and the canonical profiles are $p(z) = \max\{z,0\}$, $q = 0$, $L = 1$. Thus the asymmetric lift gives exactly $(\max\{a,b\},b)$, not merely a pair with the same difference. The symmetric variant instead gives $(\mathrm{ReLU}(a - b) + a + b, a + b)$. Leaky ReLU with slope α has curvature measure $(1 - \alpha)\delta_0$; the standard ELU with unit negative-branch scale has curvature $e^z \mathbb{1}_{z<0}\,\mathrm{d}z$. Both fit the finite-measure construction. Piecewise-linear and smooth activations are therefore covered by the same curvature principle, with differentiability replaced by bounded variation where needed.

Remark 3.6 (Compatibility with valid decompositions). Both constructions fit Definition 3.1. For the canonical asymmetric pair, $F_+(0,0) = \max\{f(0),0\}$ and $F_-(0,0) = \max\{-f(0),0\}$ are non-negative; coordinatewise monotonicity then ensures non-negative values throughout $\mathbb{R}^2_{\geq 0}$. For the symmetric pair (3.35), the origin values are $g(0,0) = f(0)$ and $h(0,0) = 0$, so the same argument applies whenever $f(0) \geq 0$, as for every activation in Corollary 3.9. If $f(0) < 0$, adding $|f(0)|$ to both symmetric

components restores non-negativity without affecting their difference. Exactness is precisely the activation-level validity condition; Theorem 3.3 supplies network recovery, convexity, and monotonicity.

3.3 The Mixing Decomposition family

We now enlarge the ReLU pair into a family of pairwise activation functions for ReLU networks. The family exploits the fact that ReLU networks compute CPWL functions (Mukherjee et al., 2018), which admit a DC decomposition (Kripfganz and Schulze, 1987; Melzer, 1986; Zimmermann and Loho, 2026), and organizes the decompositions into a one-parameter family interpolating between two arithmetic extremes.

Definition 3.3 (Mixing decomposition function, `Lean_34`). For $\eta \in [0,1]$, the *mixing decomposition function* is

$$f^{\eta}_{\mathrm{mix}}(z^+, z^-) := \big(\, \eta \max\{z^+, z^-\} + (1-\eta)\, z^+,\;\; \eta\, z^- + (1-\eta) \min\{z^+, z^-\} \,\big). \tag{3.37}$$

The decomposition obtained by applying f^{η}_{mix} at every hidden neuron and retaining the identity pair at the affine output on top of the canonical non-negative linear map (3.4)–(3.5), with the input convention (3.2), is called the *Mixing Decomposition*; when comparing decompositions we tag its activations as $a^{(l,\pm,\eta)}_{\mathrm{mix}}$.

Both components of f^{η}_{mix} are non-negative on $\mathbb{R}^2_{\geq 0}$, being convex combinations and maxima/minima of the non-negative inputs; hence f^{η}_{mix} is a pairwise activation function (Definition 3.1). At $\eta = 1$ the pair reads $\big(\max\{z^+, z^-\}, z^-\big)$, at $\eta = 0$ it reads $\big(z^+, \min\{z^+, z^-\}\big)$, and intermediate η mixes the two linearly. The family is valid for ReLU networks for *every* mixing parameter.

Theorem 3.15 (Layer-wise recovery of the Mixing Decomposition, `Lean_35`). *Let the activation pairs be propagated by the canonical non-negative linear map (3.4)–(3.5), initialised by (3.2), and updated by f^{η}_{mix} for some $\eta \in [0,1]$. Then for every layer l and neuron i,*

$$a^{(l,+,\eta)}_{\mathrm{mix},i} - a^{(l,-,\eta)}_{\mathrm{mix},i} = \varrho_l\big(z^{(l)}_i\big) = a^{(l)}_i, \tag{3.38}$$

i.e. the Mixing Decomposition is a valid decomposition (Definition 3.2) of the ReLU network for every $\eta \in [0,1]$.

Proof. We argue by induction on the layer index l.

Base case $l = 0$. By the input convention (3.2), $a^{(0,+,\eta)}_{\mathrm{mix},k} - a^{(0,-,\eta)}_{\mathrm{mix},k} = x^+_k - x^-_k = x_k = a^{(0)}_k$.

Inductive step. At the final affine readout, recovery is directly Proposition 3.1. For a hidden layer, suppose (3.38) holds at layer $l-1$. By Proposition 3.1, the decomposed pre-activations at layer l satisfy $z^{(l,+)}_i - z^{(l,-)}_i = z^{(l)}_i$. Writing $u = z^{(l,+)}_i$ and $v = z^{(l,-)}_i$, the definition (3.37) gives

$$a^{(l,+,\eta)}_{\mathrm{mix},i} - a^{(l,-,\eta)}_{\mathrm{mix},i} = \eta\big(\max\{u,v\} - v\big) + (1-\eta)\big(u - \min\{u,v\}\big) = \mathrm{ReLU}(u-v) = \mathrm{ReLU}\big(z^{(l)}_i\big) = a^{(l)}_i,$$

using the identities $\max\{u,v\} - v = \max\{u-v, 0\}$ and $u - \min\{u,v\} = \max\{u-v, 0\}$. □

The proof reveals the structural difference within the family: at $\eta = 1$ the non-linearity is carried by the $+$ stream through a per-stream maximum, at $\eta = 0$ by the $-$ stream through a per-stream minimum, both realising the same difference $\mathrm{ReLU}(u-v)$. The endpoints are not symmetric in their

analytic properties, however: the $\eta = 1$ extreme produces *convex* decomposed activations at every layer.

Theorem 3.16 ($\eta = 1$ convex characterization of the Mixing Decomposition, Lean_36). *Under the input convention (3.2) with split components $x_k^{\pm}$ that are convex functions of x (e.g. the entrywise split $x^{\pm} = [x]^{\pm}$, or any affine split), the $\eta = 1$ mixing decomposed activations*

$$a_{\mathrm{mix},i}^{(l,+,1)}(x) = \max\{z_i^{(l,+)}(x), z_i^{(l,-)}(x)\} \qquad \text{and} \qquad a_{\mathrm{mix},i}^{(l,-,1)}(x) = z_i^{(l,-)}(x)$$

are convex in x at every hidden layer l and neuron i; at the final layer, both streams are non-negative affine combinations of the previous streams and remain convex. Consequently $a_i^{(l)} = a_{\mathrm{mix},i}^{(l,+,1)} - a_{\mathrm{mix},i}^{(l,-,1)}$ is a difference of convex functions, the standard DC representation of the CPWL function computed by the network (Mukherjee et al., 2018; Zimmermann and Loho, 2026).

Proof. For $\eta = 1$, the activation pair is $(\max\{z^+,z^-\},z^-)$, so Theorem 3.3 applies. Its final assertion gives convexity after any componentwise convex input split, and its identity output pair includes the final affine readout. □

Remark 3.7 (On the $\eta = 0$ extreme). The dual DC representation writes the output as $u(x) - v(x)$ with v *concave*, and f_{mix}^0 places $a_{\mathrm{mix},i}^{(l,-,0)} = \min\{z_i^{(l,+)}, z_i^{(l,-)}\}$ in the role of v. This is concave only when both decomposed pre-activations are concave (or affine): the minimum of two concave functions is concave, but of two convex functions is in general neither. Under the input convention (3.2), the first pre-activations are convex in x, but later $\eta = 0$ streams and pre-activations need not remain convex or concave. A canonical scalar example is $W^{(1)} = 1$, $b^{(1)} = 1$, giving the negative stream $v(x) = \min\{\max(x,0) + 1, \max(-x,0)\}$. It satisfies $v(-2) = v(-1) = 1$, $v(0) = v(1) = 0$: the midpoint tests at -1 and 0 respectively exclude convexity and concavity. By Theorem 3.15, f_{mix}^0 still yields a valid arithmetic decomposition, and the network-level concave representation is recoverable with an unsplit affine input convention, at the cost of the $\pm$ structure.

The mixing family thus interpolates between a convexity-carrying extreme at $\eta = 1$ and its arithmetic dual at $\eta = 0$; only the former inherits per-layer convexity. The $\eta = 1$ endpoint is the maximum/identity DC split already constructed in Section 3.2.3.

Corollary 3.17 (DC split as the $\eta = 1$ mixing endpoint, Lean_37). *Set $\eta = 1$ in Definition 3.3 and drop the decomposition tag, writing $a^{(l,\pm)} \coloneqq a_{\mathrm{mix}}^{(l,\pm,1)}$. The resulting DC split applies the update (3.19) entrywise at hidden layers $l = 1, \ldots, L-1$, and sets $a^{(L,\pm)} = z^{(L,\pm)}$ at the final affine readout, with the decomposed pre-activations (3.4)–(3.5) induced by the weight split $W^{(l)} = W^{(l,+)} - W^{(l,-)}$ and the input split $x = x^+ - x^-$. It is a valid decomposition of the ReLU network, and under the input convention of Theorem 3.16 its decomposed activations are convex functions of the input.*

Proof. At $\eta = 1$, the pair (3.37) is exactly the maximum/identity pair (3.18). Its properties and network recovery follow from Proposition 3.2 and Theorem 3.3; this identifies the already constructed DC split with the mixing endpoint. □

3.4 The Stopping Decomposition

The mixing family realises the ReLU through per-stream maxima and minima: each stream applies its own non-linearity. A structurally different mechanism applies one *shared* decision to both streams—the hard gate of the original network.

Definition 3.4 (Stopping decomposition function, Lean_38). The *stopping decomposition function* is

$$f_{\text{stop}}(z^+, z^-) := \big(\mathbf{1}[z^+ - z^- > 0]\, z^+,\ \ \mathbf{1}[z^+ - z^- > 0]\, z^-\big). \tag{3.39}$$

The decomposition obtained by applying f_{stop} at every hidden neuron and retaining the identity pair at the affine output on top of the canonical non-negative linear map (3.4)–(3.5), with the input convention (3.2), is called the *Stopping Decomposition*; when comparing decompositions we tag its activations as $a^{(l,\pm)}_{\text{stop}}$.

Both components of f_{stop} are products of an indicator and a non-negative input, so f_{stop} maps $\mathbb{R}^2_{\geq 0}$ to $\mathbb{R}^2_{\geq 0}$ and is a pairwise activation function (Definition 3.1). Its name anticipates the reinterpretation of the gate as a stop/continue decision (Chapter 4).

Theorem 3.18 (Validity of the Stopping Decomposition, Lean_39). *Let the activation pairs be propagated by the canonical non-negative linear map (3.4)–(3.5), initialised by (3.2), and updated by f_{stop}. Then for every layer l and neuron i,*

$$a^{(l,+)}_{\text{stop},i} - a^{(l,-)}_{\text{stop},i} = \varrho_l\big(z^{(l)}_i\big) = a^{(l)}_i, \tag{3.40}$$

i.e. the Stopping Decomposition is a valid decomposition (Definition 3.2) of the ReLU network.

Proof. Induction on l. The base case $l = 0$ is the input convention, and the final affine step follows directly from Proposition 3.1. For a hidden-layer step, assume (3.40) at $l-1$; Proposition 3.1 gives $z^{(l,+)}_i - z^{(l,-)}_i = z^{(l)}_i$. Since both components of (3.39) carry the *same* gate $\mathbf{1}[z^{(l)}_i > 0]$,

$$a^{(l,+)}_{\text{stop},i} - a^{(l,-)}_{\text{stop},i} = \mathbf{1}[z^{(l)}_i > 0]\big(z^{(l,+)}_i - z^{(l,-)}_i\big) = \mathbf{1}[z^{(l)}_i > 0]\, z^{(l)}_i = \mathrm{ReLU}\big(z^{(l)}_i\big) = a^{(l)}_i. \qquad \square$$

Remark 3.8 (Shared gate versus per-stream maxima). The contrast with the mixing family is sharpest at an inactive neuron, $z^{(l)}_i \leq 0$. The DC split of Corollary 3.17 outputs $\big(z^{(l,-)}_i, z^{(l,-)}_i\big)$, preserving the sub-threshold magnitude (Remark 3.2) at the price of letting cancelled mass flow onward. The Stopping Decomposition outputs $(0, 0)$: the shared hard gate annihilates *both* streams exactly where the original ReLU annihilates its activation, so decomposed mass flows only along neurons active in the original network. The mixing streams are convex (at $\eta = 1$) but blend contributions of dead neurons into every downstream pair; the stopping streams sacrifice global convexity—the gate is discontinuous—for exact support agreement with the original forward pass, while remaining as well-behaved as possible on each linear region.

Lemma 3.19 (Monotonicity of the Stopping Decomposition on linear regions, Lean_40). *On each linear region of the ReLU network—a maximal subset of input space on which all gate indicators $\mathbf{1}[z^{(l)}_j > 0]$ are constant—each stopping decomposed activation $a^{(l,\pm)}_{\text{stop},i}$ is a non-negative-linear (i.e. affine with non-negative coefficients) function of the input pair $(x^+, x^-) \in \mathbb{R}^{2d^{(0)}}_{\geq 0}$, hence monotone non-decreasing in each coordinate x^+_k and x^-_k.*

Proof. Induction on l. At $l = 0$, $a^{(0,\pm)}_{\text{stop},k} = x^{\pm}_k$ is a coordinate of (x^+, x^-). At $l \geq 1$, the map (3.4)–(3.5) has $W^{(l,\pm)}, b^{(l,\pm)} \geq 0$, so $z^{(l,\pm)}_i$ is non-negative-linear in the layer-$(l-1)$ activations and, by the hypothesis, in (x^+, x^-). On the linear region the gate $\mathbf{1}\big[z^{(l)}_i > 0\big]$ is constant, so $a^{(l,\pm)}_{\text{stop},i} = \mathbf{1}\big[z^{(l)}_i > 0\big]\, z^{(l,\pm)}_i$ stays non-negative-linear. □

Excitatory and inhibitory path isolation. Because the shared gate never mixes the two streams, the Stopping Decomposition admits an exact accounting of *which input mass reaches which stream*: each decomposed pre-activation is a sum over backward paths, and whether a path feeds the excitatory (+) or inhibitory (−) stream is determined by the parity of the number of negative weights it traverses.

Lemma 3.20 (Parity path view on the Stopping Decomposition, `Lean_41`). *Fix a ReLU network with weights $W^{(l)}$ and, for clarity, vanishing biases. For a neuron i in layer $l \geq 1$, a* backward path *to the input is a list $P = (P_l, \ldots, P_0)$ with $P_l = i$ and P_t a neuron in layer t, using weights $W^{(t)}_{P_t,P_{t-1}}$. Define the absolute weight product, parity sign, and gate factor*

$$|w|(P) \coloneqq \prod_{t=1}^{l} |W^{(t)}_{P_t,P_{t-1}}|,$$
$$\operatorname{sgn}(P) = + \iff \#\big\{t : W^{(t)}_{P_t,P_{t-1}} < 0\big\} \text{ is even},$$
$$G_{<l}(P) \coloneqq \prod_{t=1}^{l-1} \mathbf{1}\big[z^{(t)}_{P_t} > 0\big].$$

Then the decomposed pre-activations of the Stopping Decomposition admit the path-sum representation

$$z^{(l,+)}_i = \sum_{P:\, P_l = i} G_{<l}(P)\, |w|(P)\, x^{\operatorname{sgn}(P)}_{P_0}, \qquad z^{(l,-)}_i = \sum_{P:\, P_l = i} G_{<l}(P)\, |w|(P)\, x^{\operatorname{sgn}(P)'}_{P_0}, \tag{3.41}$$

where $\operatorname{sgn}(P)'$ denotes the opposite sign of $\operatorname{sgn}(P)$.

Proof. Induction on l.

Base case $l = 1$. By (3.4), $z^{(1,+)}_i = \sum_k \big(W^{(1,+)}_{ik} x^+_k + W^{(1,-)}_{ik} x^-_k\big)$. Paths with $P_1 = i$, $P_0 = k$ biject with predecessors k, each with $|w|(P) = \big|W^{(1)}_{ik}\big|$, $G_{<1}(P) = 1$, and $\operatorname{sgn}(P) = +$ iff $W^{(1)}_{ik} \geq 0$. If $W^{(1)}_{ik} > 0$, only $W^{(1,+)}_{ik}$ contributes, giving $|w|(P)\, x^+_k$ with $\operatorname{sgn}(P) = +$; if $W^{(1)}_{ik} < 0$, only $W^{(1,-)}_{ik}$ contributes, giving $|w|(P)\, x^-_k$ with $\operatorname{sgn}(P) = -$. A zero-weight edge has zero contribution regardless of its parity label. Both nonzero cases match (3.41); $z^{(1,-)}_i$ is identical with the parity swapped.

Inductive step $l \geq 2$. The recursion (3.4) takes stopping activations from layer $l-1$ as inputs,

$$z^{(l,+)}_i = \sum_j \Big(W^{(l,+)}_{ij}\, a^{(l-1,+)}_{\text{stop},j} + W^{(l,-)}_{ij}\, a^{(l-1,-)}_{\text{stop},j}\Big), \qquad a^{(l-1,\pm)}_{\text{stop},j} = \mathbf{1}\big[z^{(l-1)}_j > 0\big]\, z^{(l-1,\pm)}_j.$$

At each predecessor j, the gate indicator upgrades the gate factor $G_{<l-1}(P')$ of every sub-path P' ending at j to $G_{\leq l-1}(P')$. Every backward path P with $P_l = i$, $P_{l-1} = j$ is uniquely obtained by prepending i to such a P'; the prepended edge contributes $\big|W^{(l)}_{ij}\big|$ to $|w|(P)$, and $G_{<l}(P) = G_{\leq l-1}(P')$. If $W^{(l)}_{ij} > 0$, only $W^{(l,+)}_{ij}\, a^{(l-1,+)}_{\text{stop},j}$ contributes to $z^{(l,+)}_i$ through j and the edge preserves parity, $\operatorname{sgn}(P) = \operatorname{sgn}(P')$; the hypothesis for $z^{(l-1,+)}_j$ gives the terms of (3.41) with $x^{\operatorname{sgn}(P)}_{P_0}$. If $W^{(l)}_{ij} < 0$, only $W^{(l,-)}_{ij}\, a^{(l-1,-)}_{\text{stop},j}$

contributes and the edge flips parity; the hypothesis for $z_j^{(l-1,-)}$, whose terms carry $x_{P_0'}^{\mathrm{sgn}(P')'} = x_{P_0}^{\mathrm{sgn}(P)}$, again gives the required terms. If $W_{ij}^{(l)} = 0$, no path through j exists. Summing over j yields $z_i^{(l,+)}$; the representation of $z_i^{(l,-)}$ follows with the streams swapped. Biases are accommodated by a bias stub path terminating at each intermediate layer, subject to the same parity rule. □

Subtracting the two path sums and using $x_k^+ - x_k^- = x_k$ recovers the standard active-path-sum representation $z_i^{(l)} = \sum_P G_{<l}(P)\, w(P)\, x_{P_0}$ with signed weight products $w(P)$: the Stopping Decomposition refines it by sorting every active path into the excitatory or inhibitory stream according to its parity. Chapter 4 reinterprets this decomposition as a two-player zero-sum stopping game, in which the gate of (3.39) becomes the players' optimal stop/continue decision.

3.5 Regularized Decompositions

The hard gate of the Stopping Decomposition solves an optimization problem: for any $\Delta \in \mathbb{R}$,

$$\mathrm{ReLU}(\Delta) = \max_{\pi\in[0,1]} \pi\Delta, \qquad \mathbf{1}[\Delta > 0] \in \mathrm{argmax}_{\pi\in[0,1]}\, \pi\Delta, \tag{3.42}$$

so the indicator in (3.39) is the *unregularized* argmax over a stop ($\pi = 0$) versus continue ($\pi = 1$) decision applied to $\Delta = z^+ - z^-$. This suggests a family of decompositions: penalizing the decision variable by a convex regularizer can smooth the gate and changes the recovered activation function from ReLU to the Fenchel conjugate of the regularizer.

Definition 3.5 (Regularized decomposition function, `Lean_42`). Let $\Omega\colon [0,1] \to \mathbb{R}_{\le 0}$ be a proper, closed, convex regularizer with $\Omega \le 0$. Define the *regularized gate* and the *conjugate activation*

$$\pi_\Omega(\Delta) \in \mathrm{argmax}_{\pi\in[0,1]}\big\{\pi\Delta - \Omega(\pi)\big\}, \qquad \Omega^*(\Delta) \coloneqq \max_{\pi\in[0,1]}\big\{\pi\Delta - \Omega(\pi)\big\}, \tag{3.43}$$

where an arbitrary maximizer is fixed if several exist. The *regularized decomposition function* is

$$f_{\mathrm{reg}}^{\Omega}(z^+, z^-) \coloneqq \Big(\pi_\Omega(\Delta)\, z^+ - \Omega\big(\pi_\Omega(\Delta)\big),\ \ \pi_\Omega(\Delta)\, z^-\Big), \qquad \Delta \coloneqq z^+ - z^-. \tag{3.44}$$

The maximum in (3.43) is attained ([0,1] compact, objective upper semicontinuous), and Ω^* is the Fenchel conjugate of Ω (extended by $+\infty$ outside [0,1]). Both components of (3.44) are non-negative on $\mathbb{R}^2_{\ge 0}$, since $\pi_\Omega(\Delta) \in [0,1]$, $z^\pm \ge 0$, $-\Omega \ge 0$; hence $f_{\mathrm{reg}}^{\Omega}$ is a pairwise activation function (Definition 3.1). Every maximizer gives the same reconstructed difference $\Omega^*(\Delta)$, although the individual components may depend on the selected maximizer. For example, with $\Omega = 0$ and $z^+ = z^- = 1$, the choices $\pi = 0$ and $\pi = 1$ produce (0,0) and (1,1) respectively. We therefore retain the fixed selection in Definition 3.5. Throughout this section the untagged pair $\big(a_i^{(l,+)}, a_i^{(l,-)}\big)$ denotes the regularized-family activations produced by $f_{\mathrm{reg}}^{\Omega}$—not the DC ($\eta = 1$) activations to which Corollary 3.17 binds the same untagged symbol.

Theorem 3.21 (Validity of regularized decompositions, `Lean_43`). *Let Ω be as in Definition 3.5, and let the activation pairs be propagated by the canonical non-negative linear map (3.4)–(3.5), initialised by (3.2), and updated by $f_{\mathrm{reg}}^{\Omega}$ at hidden layers, with the identity pair at the final affine readout. Then for every hidden layer l and neuron i,*

$$a_i^{(l,+)} - a_i^{(l,-)} = \Omega^*\big(z_i^{(l)}\big), \tag{3.45}$$

At the final affine layer, the difference is $z^{(L)} = a^{(L)}$. *Thus* f^{Ω}_{reg} *is a valid decomposition of the network* (3.1) *with activation function* $\varrho = \Omega^*$. *Moreover,* Ω^* *is convex, non-decreasing, and* 1*-Lipschitz, and wherever* Ω^* *is differentiable the gate is its derivative,* $\pi_\Omega(\Delta) = (\Omega^*)'(\Delta) \in [0,1]$.

Proof. Recovery. Induction on l. The base case is the input convention. Assume (3.45) at $l-1$ for the network with activation Ω^*; Proposition 3.1—whose proof uses only the difference identity—gives $\Delta_i \coloneqq z_i^{(l,+)} - z_i^{(l,-)} = z_i^{(l)}$. Subtracting the components of (3.44),

$$a_i^{(l,+)} - a_i^{(l,-)} = \pi_\Omega(\Delta_i)\,\big(z_i^{(l,+)} - z_i^{(l,-)}\big) - \Omega\big(\pi_\Omega(\Delta_i)\big) = \pi_\Omega(\Delta_i)\,\Delta_i - \Omega\big(\pi_\Omega(\Delta_i)\big) = \Omega^*\big(z_i^{(l)}\big),$$

where the last step evaluates (3.43) at its maximizer (Fenchel–Young).

Properties of Ω^*. As a supremum of affine functions $\Delta \mapsto \pi\Delta - \Omega(\pi)$ with slopes $\pi \in [0,1]$, Ω^* is convex with subgradients in $[0,1]$, hence non-decreasing and 1-Lipschitz. At a point of differentiability the subdifferential is $\{(\Omega^*)'(\Delta)\}$, and by Fenchel–Young every maximizer $\pi_\Omega(\Delta)$ is a subgradient; thus $\pi_\Omega(\Delta) = (\Omega^*)'(\Delta)$. □

Corollary 3.22 (Unregularized case, `Lean_44`). *For* $\Omega \equiv 0$ *on* $[0,1]$, *Definition 3.5 with the maximizer* $\pi_\Omega(\Delta) = \mathbf{1}[\Delta > 0]$ *recovers the stopping decomposition function,* $f^0_{\text{reg}} = f_{\text{stop}}$, *and Theorem 3.21 specialises to Theorem 3.18 with* $\Omega^*(\Delta) = \max_{\pi\in[0,1]} \pi\Delta = ReLU(\Delta)$.

Proof. For $\Delta \neq 0$ the objective $\pi\Delta$ has the unique maximizer $\mathbf{1}[\Delta > 0]$ over $[0,1]$, and at $\Delta = 0$ every π is a maximizer, including $\mathbf{1}[0 > 0] = 0$; in all cases (3.44) with $\Omega = 0$ coincides with (3.39) and the optimal value is $\mathrm{ReLU}(\Delta)$. □

3.5.1 The Softplus Decomposition as the entropic subcase

The canonical smooth regularizer is the (negative) binary Shannon entropy at temperature $\theta > 0$,

$$\Omega_\theta(\pi) \coloneqq -\theta H(\pi), \qquad H(\pi) \coloneqq -\pi\log\pi - (1-\pi)\log(1-\pi), \tag{3.46}$$

which is strictly convex on $[0,1]$ and satisfies $\Omega_\theta \leq 0$.

Corollary 3.23 (Softplus Decomposition, `Lean_45`). *For* $\Omega = \Omega_\theta$ *as in* (3.46), *the regularized gate and conjugate activation are*

$$\pi_{\Omega_\theta}(\Delta) = \sigma'_\theta(\Delta) \coloneqq \frac{1}{1+e^{-\Delta/\theta}}, \qquad \Omega^*_\theta(\Delta) = \sigma_\theta(\Delta) \coloneqq \theta\log\big(1+e^{\Delta/\theta}\big), \tag{3.47}$$

the sigmoid gate and the softplus function at temperature θ. *The resulting decomposition function is the* softplus decomposition function

$$f^{\theta}_{\text{soft}}(z^+, z^-) = \Big(\sigma'_\theta(\Delta)\,z^+ + \theta\,H\big(\sigma'_\theta(\Delta)\big),\ \ \sigma'_\theta(\Delta)\,z^-\Big), \qquad \Delta = z^+ - z^-, \tag{3.48}$$

and by Theorem 3.21 it yields a valid decomposition—the Softplus Decomposition*—of the softplus network with parameter* $\beta = 1/\theta$. *In the limit* $\theta \to 0$, $f^{\theta}_{\text{soft}} \to f_{\text{stop}}$ *pointwise on* $\{\Delta \neq 0\}$.

Proof. The objective $\pi\Delta + \theta H(\pi)$ is strictly concave on (0,1) with derivative $\Delta - \theta \log \frac{\pi}{1-\pi}$; the first-order condition gives the unique maximizer $\pi = \sigma'_\theta(\Delta)$. Substituting back, with $\log \sigma'_\theta(\Delta) = \Delta/\theta - \log(1 + e^{\Delta/\theta})$ and $\log(1 - \sigma'_\theta(\Delta)) = -\log(1 + e^{\Delta/\theta})$, yields the optimal value

$$\sigma'_\theta(\Delta)\,\Delta + \theta H\big(\sigma'_\theta(\Delta)\big) = \theta \log\big(1 + e^{\Delta/\theta}\big) = \sigma_\theta(\Delta), \tag{3.49}$$

the Legendre–Fenchel form of the binary log-sum-exp at temperature θ over $\{0, \Delta\}$. Then (3.48) is (3.44) with $-\Omega_\theta(\pi_{\Omega_\theta}(\Delta)) = \theta H(\sigma'_\theta(\Delta))$, and validity is Theorem 3.21 with $\Omega^*_\theta = \sigma_\theta$. For the limit, $\sigma'_\theta(\Delta) \to \mathbf{1}[\Delta > 0]$ and $\theta H(\sigma'_\theta(\Delta)) = \sigma_\theta(\Delta) - \sigma'_\theta(\Delta)\Delta \to 0$ as $\theta \to 0$ for $\Delta \neq 0$. □

The entropy bonus $\theta H(\sigma'_\theta(\Delta))$ is carried entirely by the + stream in (3.48); this convention keeps the − stream a plainly gated quantity and reappears in Chapter 4 as the active player's surplus from being allowed to mix.

Remark 3.9 (Loss of stream monotonicity). The soft gate does not inherit the well-behavedness of the hard-gated Stopping Decomposition. Consider the negative stream $a^{(l,-)} = \sigma'_\theta(\Delta)\, z^-$ with $\Delta = z^+ - z^-$: holding z^+ fixed and increasing z^- both raises the factor z^- and, through Δ, *lowers* the gate $\sigma'_\theta(\Delta)$. Differentiating, $\partial a^{(l,-)}/\partial z^- = \sigma'_\theta(\Delta)\big(1 - z^-\, \sigma''_\theta(\Delta)/\sigma'_\theta(\Delta)\big)$ turns negative once z^- exceeds $\theta/\big(1 - \sigma'_\theta(\Delta)\big)$, so $a^{(l,-)}_{\text{soft}}$ is *not* monotone non-decreasing in z^-. The per-region monotonicity of the Stopping Decomposition (Lemma 3.19) therefore does not lift to the soft-gated family; the reader should not expect the Softplus Decomposition to inherit the monotone or convex structure of the hard-gated and DC decompositions. Only the difference recovery of Theorem 3.21 survives.

Remark 3.10 (Probit regularizer and GELU). The framework is not tied to the entropy. For the probit regularizer $\Omega_\Phi(\pi) := -\phi\big(\Phi^{-1}(\pi)\big)$, with ϕ, Φ the standard Gaussian density and distribution function, one has $\frac{\mathrm{d}}{\mathrm{d}\pi}\big(-\Omega_\Phi\big)(\pi) = -\Phi^{-1}(\pi)$, so the first-order condition for (3.43) gives the gate $\pi_{\Omega_\Phi}(\Delta) = \Phi(\Delta)$ and the conjugate activation

$$\Omega^*_\Phi(\Delta) = \Delta\,\Phi(\Delta) + \phi(\Delta) = \mathbb{E}_{\varepsilon\sim\mathcal{N}(0,1)}\big[\max(\Delta + \varepsilon, 0)\big],$$

the Gaussian smoothing of the ReLU. Up to $\phi(\Delta)$, which common implementations drop, this is GELU (Hendrycks and Gimpel, 2016), and the gate $\Phi(\Delta) = \mathbb{E}_\varepsilon\big[\mathbf{1}[\Delta + \varepsilon > 0]\big]$ is the hard stopping gate averaged over a Gaussian shift of its threshold—matching the original stochastic-gate motivation for GELU. The distinction matters for exact decomposability: the conjugate activation $\Omega^*_\Phi(\Delta) = \Delta\,\Phi(\Delta) + \phi(\Delta)$ is the object recovered by f^Ω_{reg} and, being a Fenchel conjugate, is convex, non-decreasing, and 1-Lipschitz (Theorem 3.21). The *implemented* GELU $\Delta \mapsto \Delta\,\Phi(\Delta)$ that drops $\phi(\Delta)$ is non-monotone—it dips below zero for $\Delta < 0$—and is therefore *not* itself a conjugate activation Ω^*; the exactly decomposable object is the $+\phi(\Delta)$ variant, and networks using the dropped-term GELU are only approximately captured by this decomposition.

Remark 3.11 (Connection to the integrable-curvature splitting). Regularized gates connect to the convex activations covered by Theorem 3.13. Suppose Ω is strictly convex and the gate π_Ω is locally absolutely continuous on $\mathbb{R}$. Then Ω^* is differentiable with $(\Omega^*)' = \pi_\Omega$, which is non-decreasing with limits 0 and 1 at $-\infty$ and $+\infty$. Hence the weak curvature $(\Omega^*)'' \geq 0$ belongs to $L^1(\mathbb{R})$ and integrates to 1. A sufficient condition is $\Omega \in C^2((0,1))$ with $\Omega'' > 0$ there: the gate is the inverse of Ω' on its range, extended by 0 and 1 beyond finite endpoint slopes. Strict convexity and smoothness alone do not guarantee local absolute continuity of the gate; the distributional curvature can have a singular part. Under the stated regularity, Theorem 3.13 applies with $N = 0$, $C_2 = 0$, and $C_1 = \Omega^*$, as recorded for the softplus in Corollary 3.9(6). Regularizing the *gate* (this section) and the curvature-based smooth

split of Section 3.2.2 thus yield compatible valid decompositions of the same networks under these hypotheses, cf. Remark 3.6.

3.6 Numerical Stabilization of Split Passes

As observed in Remark 3.2, the split pair lacks cancellation by subtraction: both streams accumulate the *absolute* mass $z^{(l,+)}+z^{(l,-)}$ instead of the signed residual. In deep networks the common magnitude of the two streams therefore grows roughly geometrically, at a rate governed by products of entrywise absolute weight matrices, and quickly leaves the representable range even though the difference $a^{(l,+)}-a^{(l,-)}=a^{(l)}$ stays moderate. This section develops the affine stabilization procedures of Zimmermann & Loho (2026) for both passes: shifting the activations in the forward pass, and the local sensitivities in the backward pass. We absorb biases into the weights and write, for the product of absolute weight matrices above layer l,

$$W_{\mathrm{abs}}^{[l,\to]} \coloneqq \prod_{j=L,\dots,l+1} |W^{(j)}| = \prod_{j=L,\dots,l+1} \left(W^{(j,+)}+W^{(j,-)}\right), \tag{3.50}$$

with the empty product at $l=L$ read as the identity.

3.6.1 Shifting the forward pass

The key fact is that adding a *common* shift to both streams at any layer translates both stream outputs by the same amount, so the difference—and by Theorem 3.7 the represented network value—is invariant.

Theorem 3.24 (Shift equivariance of the split streams, `Lean_46`). *For any $0\le l\le L$ and any $c, a^{(+)}, a^{(-)}\in\mathbb{R}^{d^{(l)}}$ with $a^{(\pm)}, a^{(\pm)}+c\ge 0$,*

$$\begin{aligned} g^{(l)}\left(a^{(+)}+c, a^{(-)}+c\right) &= g^{(l)}\left(a^{(+)}, a^{(-)}\right) + W_{\mathrm{abs}}^{[l,\to]}\, c, \\ h^{(l)}\left(a^{(+)}+c, a^{(-)}+c\right) &= h^{(l)}\left(a^{(+)}, a^{(-)}\right) + W_{\mathrm{abs}}^{[l,\to]}\, c. \end{aligned} \tag{3.51}$$

In particular, $\left(g^{(l)}-h^{(l)}\right)\left(a^{(+)}+c, a^{(-)}+c\right) = \left(g^{(l)}-h^{(l)}\right)\left(a^{(+)}, a^{(-)}\right)$.

Proof. We show the claim for $g^{(l)}$ by induction on l; $h^{(l)}$ is analogous and the difference statement follows by subtraction. The base case $l=L$ is trivial, as $W_{\mathrm{abs}}^{[L,\to]}=I$ and $g^{(L)}$ is a projection. For $0\le l<L$, the positive and negative neurons of layer $l+1$ connect to layer l by the same two matrices with $W^{(l+1,+)}$ and $W^{(l+1,-)}$ flipped (Figure 3.2), so a uniform shift of the layer-l pair produces a uniform shift of the layer-$(l+1)$ pair. For a hidden layer $l+1<L$,

$$\begin{aligned} a^{(l+1,+)}\big|_{+c} &= \max\left\{W^{(l+1,+)}a^{(+)}+W^{(l+1,-)}a^{(-)},\, W^{(l+1,-)}a^{(+)}+W^{(l+1,+)}a^{(-)}\right\} + \left|W^{(l+1)}\right| c, \\ a^{(l+1,-)}\big|_{+c} &= W^{(l+1,-)}a^{(+)}+W^{(l+1,+)}a^{(-)} + \left|W^{(l+1)}\right| c, \end{aligned}$$

since $\left(W^{(l+1,+)}+W^{(l+1,-)}\right)c = \left|W^{(l+1)}\right|c$ enters both arguments of the maximum and factors out. At the final affine layer $l+1=L$, both stream activations equal their respective pre-activations, and non-negative linearity gives the same common shift $|W^{(L)}|c$ directly, without a maximum. Applying

the induction hypothesis at layer $l+1$ with the shift $|W^{(l+1)}|c$,

$$\begin{aligned} g^{(l)}\big(a^{(+)}+c, a^{(-)}+c\big) &= g^{(l+1)}\Big(a^{(l+1,+)}+|W^{(l+1)}|c, a^{(l+1,-)}+|W^{(l+1)}|c\Big) \\ &= g^{(l)}\big(a^{(+)}, a^{(-)}\big) + W_{\mathrm{abs}}^{[l,\to]}\, c. \quad \square \end{aligned}$$

Theorem 3.24 licenses *re-centering*: whenever the common magnitude of a pair explodes, we subtract a common shift without altering the represented function. The canonical choice

$$c^{(l)} = -\min\big\{a^{(l,+)}, a^{(l,-)}\big\}$$

(entrywise) makes at least one component of every pair zero. For a fixed shift, the head Jacobians at the shifted and unshifted split coordinates agree, since the Maxout branch comparisons are unchanged. If the shift is computed from the input, it must be detached when propagating these head sensitivities: differentiating through the shift adds the common term $W_{\mathrm{abs}}^{[l,\to]} Dc$ to both head derivatives. Their difference remains unchanged, but their individual derivatives need not. The original paper reports double-precision intermediate computations; the released evaluation command requires an explicit precision setting. Its forward implementation offers several magnitude-reduction modes. A common shift preserves the pair difference by Theorem 3.24, whereas multiplying both streams by a common factor generally changes that difference. When original activations are cached, the implementation subsequently applies the symmetric correction

$$e^{(l)} = a^{(l)} - \big(a^{(l,+)} - a^{(l,-)}\big), \qquad a^{(l,+)} \leftarrow a^{(l,+)} + \tfrac{1}{2}e^{(l)}, \qquad a^{(l,-)} \leftarrow a^{(l,-)} - \tfrac{1}{2}e^{(l)}.$$

This restores the **forward invariant** in exact arithmetic. The correction is detached in the default backward configuration; differentiating through it instead changes the stream sensitivities. The common-shift theorem does not by itself justify the scaling modes, the gradients of these input-dependent corrections, or convexity and monotonicity of the resulting stabilized streams. Floating-point reconstruction error must still be measured: an exact identity for the correction does not guarantee zero roundoff error.

3.6.2 Shifting the backward pass

All classical derivative statements in this subsection are restricted to points with strict gate inequalities in the relevant head. At a gate tie, a fixed branch convention defines a backpropagated selection rather than asserting classical differentiability. The gradients of g and h involve only non-negative matrices, so they blow up like the forward activations. Let

$$\delta^{(l,+,g)} := \frac{\partial g^{(l)}}{\partial a^{(l,+)}}, \qquad \delta^{(l,-,g)} := \frac{\partial g^{(l)}}{\partial a^{(l,-)}}, \qquad \delta^{(l,+,h)} := \frac{\partial h^{(l)}}{\partial a^{(l,+)}}, \qquad \delta^{(l,-,h)} := \frac{\partial h^{(l)}}{\partial a^{(l,-)}} \tag{3.52}$$

denote the *local sensitivities* of the stream heads with respect to the split activations of layer l. Theorem 3.24 pins down their pairwise structure.

Theorem 3.25 (Sensitivity complementarity, `Lean_47`**).** *For every* $0 \le l \le L$,

$$\delta^{(l,+,g)} = W_{\mathrm{abs}}^{[l,\to]} - \delta^{(l,-,g)}, \qquad \delta^{(l,+,h)} = W_{\mathrm{abs}}^{[l,\to]} - \delta^{(l,-,h)}. \tag{3.53}$$

Proof. For a coordinate direction e_j, $j \in \{1, \dots, d^{(l)}\}$, Theorem 3.24 with $c = \epsilon\, e_j$ gives $g^{(l)}\big(a^{(l,+)} + \epsilon e_j, a^{(l,-)}\big) = g^{(l)}\big(a^{(l,+)}, a^{(l,-)} - \epsilon e_j\big) + \epsilon\, W_{\text{abs}}^{[l,\to]} e_j$. Subtracting $g^{(l)}\big(a^{(l,+)}, a^{(l,-)}\big)$, dividing by ϵ, and letting $\epsilon \to 0$ yields $\delta_j^{(l,+,g)} = W_{\text{abs}}^{[l,\to]} e_j - \delta_j^{(l,-,g)}$. The claim for h follows analogously. □

For completeness, the four block Jacobians underlying the backward recursions are obtained by the chain rule. At a hidden layer with a strict gate pattern,

$$\begin{aligned}
\frac{\partial a^{(l+1,+)}}{\partial a^{(l,+)}} &= \frac{\partial a^{(l+1,+)}}{\partial z^{(l+1,+)}} \frac{\partial z^{(l+1,+)}}{\partial a^{(l,+)}} + \frac{\partial a^{(l+1,+)}}{\partial z^{(l+1,-)}} \frac{\partial z^{(l+1,-)}}{\partial a^{(l,+)}} \\
&= M^{(l+1,+)} W^{(l+1,+)} + M^{(l+1,-)} W^{(l+1,-)}, \\
\frac{\partial a^{(l+1,-)}}{\partial a^{(l,+)}} &= 0 \cdot W^{(l+1,+)} + W^{(l+1,-)}, \\
\frac{\partial a^{(l+1,+)}}{\partial a^{(l,-)}} &= \frac{\partial a^{(l+1,+)}}{\partial z^{(l+1,+)}} \frac{\partial z^{(l+1,+)}}{\partial a^{(l,-)}} + \frac{\partial a^{(l+1,+)}}{\partial z^{(l+1,-)}} \frac{\partial z^{(l+1,-)}}{\partial a^{(l,-)}} \\
&= M^{(l+1,+)} W^{(l+1,-)} + M^{(l+1,-)} W^{(l+1,+)}, \\
\frac{\partial a^{(l+1,-)}}{\partial a^{(l,-)}} &= 0 \cdot W^{(l+1,-)} + W^{(l+1,+)}.
\end{aligned} \tag{3.54}$$

Here $M^{(l,+)}$ and $M^{(l,-)}$ select the two complementary Maxout branches, defined below. For the final ungated affine layer, the same formulas hold with $M^{(L,+)} = I$ and $M^{(L,-)} = 0$. Biases contribute constant shifts to the pre-activations, so their derivatives with respect to the preceding activations vanish; fixed biases also preserve convexity, monotonicity, and the piecewise-affine structure.

Equation (3.53) exposes the problem: the two sensitivities of each pair sum to the exploding matrix $W_{\text{abs}}^{[l,\to]}$, so at least one is numerically large in deep networks. The stabilization subtracts at every layer an $\alpha^{(l)}$-multiple of this common bulk. Starting from $\delta_{\text{shift}}^{(L,\pm,g)} := \delta^{(L,\pm,g)} - \alpha^{(L)}\big(\delta^{(L,+,g)} + \delta^{(L,-,g)}\big)$ and analogously for h, the *stabilized backward pass* iterates for $l = L-1, \dots, 0$: first the ordinary gradient recursion on the already shifted quantities,

$$\delta_{\text{int}}^{(l,+,g)} := \delta_{\text{shift}}^{(l+1,+,g)}\big(M^{(l+1,+)} W^{(l+1,+)} + M^{(l+1,-)} W^{(l+1,-)}\big) + \delta_{\text{shift}}^{(l+1,-,g)}\, W^{(l+1,-)}, \tag{3.55}$$

$$\delta_{\text{int}}^{(l,-,g)} := \delta_{\text{shift}}^{(l+1,+,g)}\big(M^{(l+1,+)} W^{(l+1,-)} + M^{(l+1,-)} W^{(l+1,+)}\big) + \delta_{\text{shift}}^{(l+1,-,g)}\, W^{(l+1,+)}, \tag{3.56}$$

and then the uniform shift

$$\delta_{\text{shift}}^{(l,\pm,g)} := \delta_{\text{int}}^{(l,\pm,g)} - \alpha^{(l)}\big(\delta_{\text{int}}^{(l,+,g)} + \delta_{\text{int}}^{(l,-,g)}\big), \tag{3.57}$$

with the same scheme for h. Here $M^{(l,+)} := \operatorname{diag}\big(\mathbf{1}[z^{(l)} \geq 0]\big)$ and $M^{(l,-)} := \operatorname{diag}\big(\mathbf{1}[z^{(l)} < 0]\big)$ are the Maxout gate patterns, and (3.55)–(3.56) are the exact chain-rule recursions of the unshifted sensitivities, obtained by differentiating (3.4)–(3.5) and (3.19): e.g. $\partial a^{(l+1,+)}/\partial a^{(l,+)} = M^{(l+1,+)} W^{(l+1,+)} + M^{(l+1,-)} W^{(l+1,-)}$ and $\partial a^{(l+1,-)}/\partial a^{(l,+)} = W^{(l+1,-)}$. For $\alpha^{(l)} \equiv 0$ the scheme reproduces the plain split-pair sensitivities. In the implementation, the gate patterns $M^{(l,\pm)}$ are taken from the cached original activations rather than the drifted split pre-activations, and cached original gradients give small corrections enforcing the **backward invariant** $\frac{1}{2}\big[(\delta^{(l,+,g)} - \delta^{(l,-,g)}) - (\delta^{(l,+,h)} - \delta^{(l,-,h)})\big] = \partial f^{(l)}/\partial a^{(l)}$.

To state what the stabilized quantities compute, we express the split gradients with respect to a unified input. Define $\tilde{g}^{(l)}(a^{(l)}) := g^{(l)}\big(a^{(l)}/2, -a^{(l)}/2\big)$, the head on the half–half split, and analogously $\tilde{h}^{(l)}$;

by the chain rule

$$\delta^{(l,g)} := \frac{\partial \tilde{g}^{(l)}}{\partial a^{(l)}} = \frac{\delta^{(l,+,g)} - \delta^{(l,-,g)}}{2}, \qquad \delta^{(l,h)} := \frac{\partial \tilde{h}^{(l)}}{\partial a^{(l)}} = \frac{\delta^{(l,+,h)} - \delta^{(l,-,h)}}{2}, \tag{3.58}$$

and write $\delta^{(l,g)}_{\text{shift}} := \big(\delta^{(l,+,g)}_{\text{shift}} - \delta^{(l,-,g)}_{\text{shift}}\big)/2$, $\delta^{(l,h)}_{\text{shift}} := \big(\delta^{(l,+,h)}_{\text{shift}} - \delta^{(l,-,h)}_{\text{shift}}\big)/2$ for their stabilized counterparts, with $\delta^{g}_{\text{shift}} := \delta^{(0,g)}_{\text{shift}}$, $\delta^{h}_{\text{shift}} := \delta^{(0,h)}_{\text{shift}}$. For the final affine readout we use the uniform-notation convention $M^{(L,+)} = I$, $M^{(L,-)} = 0$; these are algebraic placeholders, not an additional gate. Standard backpropagation gives $\delta^{(l)} := \partial f^{(l)}/\partial a^{(l)} = \prod_{j=L,\dots,l+1} W^{(j,x)}$ with gated matrices $W^{(l,x)} := M^{(l,+)} W^{(l)}$.

Theorem 3.26 (Closed formulas for the stabilized split gradients, `Lean_48`). *With $\overline{\alpha}_j := \prod_{j\le i\le L}\big(1-2\alpha^{(i)}\big)$, the stabilized backward pass (3.55)–(3.57) computes, for every $0 \le l \le L$,*

$$\delta^{(l,g)}_{\text{shift}} = \frac{1}{2}\Big[\delta^{(l)} - \sum_{l\le j<L} \overline{\alpha}_{j+1}\, W^{[j+1,\to]}_{\text{abs}} \big(M^{(j+1,-)} W^{(j+1)}\big) \prod_{i=(j-1),\dots,l} W^{(i+1,x)}\Big], \tag{3.59}$$

$$\delta^{(l,h)}_{\text{shift}} = -\frac{1}{2}\Big[\delta^{(l)} + \sum_{l\le j<L} \overline{\alpha}_{j+1}\, W^{[j+1,\to]}_{\text{abs}} \big(M^{(j+1,-)} W^{(j+1)}\big) \prod_{i=(j-1),\dots,l} W^{(i+1,x)}\Big]. \tag{3.60}$$

The $j = L-1$ correction term is zero because $M^{(L,-)} = 0$: no minimum or auxiliary Q source is inserted at the affine readout. In particular, $\delta^{(l,g)}_{\text{shift}} - \delta^{(l,h)}_{\text{shift}} = \delta^{(l)}$ for all l; for $\alpha^{(i)} \equiv 0$ (i.e. $\overline{\alpha}_j = 1$) the formulas describe the plain split-pair gradients, and for $\alpha^{(i)} \equiv 1/2$ all correction terms vanish and the stabilized quantities are $\pm\delta^{(l)}/2$, halves of the original gradient.

Proof. We record two auxiliary identities.

Claim 1 (bulk of the shifted sensitivities). For every l,

$$\delta^{(l,+,g)}_{\text{shift}} + \delta^{(l,-,g)}_{\text{shift}} = \prod_{l\le j\le L} \big(1-2\alpha^{(j)}\big)\, W^{[l,\to]}_{\text{abs}} = \overline{\alpha}_l\, W^{[l,\to]}_{\text{abs}}. \tag{3.61}$$

At $l = L$, $\delta^{(L,+,g)} + \delta^{(L,-,g)} = I = W^{[L,\to]}_{\text{abs}}$, and the shift multiplies the sum by $(1-2\alpha^{(L)})$, since subtracting $\alpha^{(L)}$ times the sum from *each* component subtracts $2\alpha^{(L)}$ times it from the sum. For the step, adding (3.55), (3.56) and using $M^{(l+1,+)} + M^{(l+1,-)} = I$, $W^{(l+1,+)} + W^{(l+1,-)} = \big|W^{(l+1)}\big|$,

$$\delta^{(l,+,g)}_{\text{int}} + \delta^{(l,-,g)}_{\text{int}} = \big(\delta^{(l+1,+,g)}_{\text{shift}} + \delta^{(l+1,-,g)}_{\text{shift}}\big)\big|W^{(l+1)}\big| = \overline{\alpha}_{l+1}\, W^{[l+1,\to]}_{\text{abs}} \big|W^{(l+1)}\big| = \overline{\alpha}_{l+1}\, W^{[l,\to]}_{\text{abs}}$$

by induction, and the shift (3.57) multiplies this by $(1-2\alpha^{(l)})$, giving $\overline{\alpha}_l\, W^{[l,\to]}_{\text{abs}}$.

Claim 2 (recursion of the stabilized difference). Subtracting (3.56) from (3.55)—the shift cancels in the difference—and regrouping with $M^{(l+1,+)} - M^{(l+1,-)}$ applied to $W^{(l+1)} = W^{(l+1,+)} - W^{(l+1,-)}$,

$$\begin{aligned} 2\,\delta^{(l,g)}_{\text{shift}} &= \big(\delta^{(l+1,+,g)}_{\text{shift}} - \delta^{(l+1,-,g)}_{\text{shift}}\big) M^{(l+1,+)} W^{(l+1)} - \big(\delta^{(l+1,+,g)}_{\text{shift}} + \delta^{(l+1,-,g)}_{\text{shift}}\big) M^{(l+1,-)} W^{(l+1)} \\ &= 2\,\delta^{(l+1,g)}_{\text{shift}}\, W^{(l+1,x)} - \overline{\alpha}_{l+1}\, W^{[l+1,\to]}_{\text{abs}} M^{(l+1,-)} W^{(l+1)}, \end{aligned} \tag{3.62}$$

where the first equality uses $M^{(l+1,+)} + M^{(l+1,-)} = I$ to sort the four terms into the two brackets, the second Claim 1 at $l+1$.

Induction. At the base $l = L$ of (3.59), the initial shift cancels in the difference, so $\delta^{(L,g)}_{\text{shift}} = \big(\delta^{(L,+,g)} - \delta^{(L,-,g)}\big)/2 = I/2 = \delta^{(L)}/2$ with empty correction sum. For the step, unfolding (3.62) with the

hypothesis at $l+1$,

$$2\,\delta_{\text{shift}}^{(l,g)} = \Big[\delta^{(l+1)} - \sum_{l+1\le j<L} \overline{\alpha}_{j+1} W_{\text{abs}}^{[j+1,\to]}\big(M^{(j+1,-)}W^{(j+1)}\big) \prod_{i=(j-1),\dots,l+1} W^{(i+1,x)}\Big] W^{(l+1,x)}$$
$$- \overline{\alpha}_{l+1}\, W_{\text{abs}}^{[l+1,\to]} M^{(l+1,-)} W^{(l+1)}$$
$$= \delta^{(l)} - \sum_{l\le j<L} \overline{\alpha}_{j+1}\, W_{\text{abs}}^{[j+1,\to]}\big(M^{(j+1,-)}W^{(j+1)}\big) \prod_{i=(j-1),\dots,l} W^{(i+1,x)},$$

using $\delta^{(l+1)}W^{(l+1,x)} = \delta^{(l)}$ and absorbing the extra term as summand $j = l$. Equation (3.60) is identical up to the sign flip $\delta_{\text{shift}}^{(L,h)} = -I/2$. Subtracting (3.60) from (3.59) cancels the correction sums, giving the backward invariant $\delta_{\text{shift}}^{(l,g)} - \delta_{\text{shift}}^{(l,h)} = \delta^{(l)}$; the cases $\alpha \equiv 0,\, 1/2$ are immediate from $\overline{\alpha}_j$. □

Remark 3.12 (Interpretation of the correction terms). The correction summands in (3.59) are gradients of an auxiliary network: define $Q[j](x) \coloneqq W_{\text{abs}}^{[j,\to]} \min\{z^{(j)}(x),0\}$, with the original prefix through layer j followed by a linear absolute-weight tail without biases or activation gates. In a bias-free tail this agrees in value with subsequent minimum activations, but the explicit definition also handles arbitrary prefix biases. Then the j-th summand equals $\overline{\alpha}_{j+1}\,\partial Q[j+1]/\partial x$ (at $l=0$), capturing the mass that switches streams at layer $j+1$. These gradients grow rapidly toward the input, so the factors $\overline{\alpha}_j$ act as a per-layer lever that both prevents gradient explosion and regulates how strongly each layer's stream-switching enters the stabilized gradients. In practice all $\alpha^{(l)}$ equal a common α; for $\alpha = 1/2$ the sensitivities are exact halves of the original gradient, for $\alpha = 0$ the raw split-pair gradients, and $0 < \alpha < 1/2$ is the interesting regime. These stabilized gradients are the backbone of the split saliency methods in Chapter 6.

3.7 Generalization to Convolutional Neural Networks (CNNs)

Nothing in Definitions 3.1 and 3.2 is specific to dense layers: a decomposition only requires that every component propagate non-negative pairs through non-negative linear maps and pairwise non-linearities. We extend the DC split to standard CNNs (He et al., 2016; Simonyan and Zisserman, 2015), representing every component as a *chunk module* $(u^+,u^-) \mapsto (v^+,v^-)$ and keeping linear layers and ReLU gates as separate chunks, so that architectures with several linear operations between activations remain expressible; a converter maps supported computation graphs to the split form (Paszke et al., 2019). For each chunk we verify the same three properties—non-negative pairs map to non-negative pairs, the difference invariant $v^+ - v^- = \text{op}(u^+ - u^-)$ holds, and (where applicable) monotonicity and convexity of the streams are preserved (Proposition 3.6)—and chaining them yields Theorem 3.30. Two chunks carry the substance, split convolution and gated max-pooling; every other supported chunk is a non-negative linear map applied streamwise, covered by a single lemma.

Convolution. A convolutional layer with kernel K and bias b computes $u \mapsto K * u + b$, with $*$ the (multi-channel) cross-correlation. Splitting the parameters entrywise, $K = [K]^+ - [K]^-$ and $b = [b]^+ - [b]^-$, we define the split chunk

$$v^+ \coloneqq [K]^+ * u^+ + [K]^- * u^- + [b]^+, \qquad v^- \coloneqq [K]^- * u^+ + [K]^+ * u^- + [b]^-. \tag{3.63}$$

Proposition 3.27 (Correctness of split convolution, `Lean_49`). *The chunk (3.63) maps non-negative pairs to non-negative pairs, each output component is a non-negative linear (affine) function*

of (u^+, u^-)*—hence convex and non-decreasing in each coordinate—and*

$$v^+ - v^- = K * (u^+ - u^-) + b.$$

Proof. Cross-correlation with a non-negative kernel is a linear map with non-negative coefficients, so $v^\pm \geq 0$ for $u^\pm \geq 0$ and both components are affine with non-negative coefficients in (u^+, u^-), hence convex and non-decreasing in each coordinate. By linearity of $*$,

$$v^+ - v^- = \big([K]^+ - [K]^-\big) * u^+ - \big([K]^+ - [K]^-\big) * u^- + [b]^+ - [b]^- = K * (u^+ - u^-) + b. \qquad \square$$

Convolution followed by a ReLU chunk—the entrywise Maxout gate (3.19)—is covered by the argument of Theorem 3.7, a convolution being a dense layer with a sparse, weight-shared matrix. The same entrywise weight split handles every linear chunk at once.

Lemma 3.28 (Non-negative linear chunks preserve the invariants, `Lean_50`**).** *Let a chunk compute an affine map* $u \mapsto Au + c$ *with real matrix* $A = [A]^+ - [A]^-$ *and bias* $c = [c]^+ - [c]^-$*, split streamwise as*

$$v^+ \coloneqq [A]^+ u^+ + [A]^- u^- + [c]^+, \qquad v^- \coloneqq [A]^- u^+ + [A]^+ u^- + [c]^-.$$

Then the chunk maps non-negative pairs to non-negative pairs, each component is affine with non-negative coefficients in (u^+, u^-)*—hence convex and non-decreasing in each coordinate—and* $v^+ - v^- = A(u^+ - u^-) + c$.

Proof. Identical to Proposition 3.27: the coefficients $[A]^\pm, [c]^\pm$ are non-negative, and subtracting the two lines gives $([A]^+ - [A]^-)(u^+ - u^-) + [c]^+ - [c]^- = A(u^+ - u^-) + c$. $\square$

Split convolution (3.63) is the instance $A = K * (\cdot)$; the remaining linear chunks are listed in Table 3.2. Residual connections combine the identity chunk ($A = I$) with an already-split branch by streamwise addition $v^\pm = u^\pm + F^\pm(u^+, u^-)$, preserving all three invariants whenever the branch does.

Tab. 3.2: Linear CNN chunks as instances of Lemma 3.28.

Chunk	Affine map	Notes
Average pooling	$A = \frac{1}{q}\sum_{j\in[q]}(\cdot)_j$, $c = 0$	non-negative A, applied streamwise
Batch norm (eval)	$A = \mathrm{diag}(c_{\mathrm{bn}})$, $c = d$	scale c_{bn} may be signed; 1×1 diagonal conv
Dropout (train)	$A = \mathrm{diag}(m)$, $c = 0$	mask $m \in \{0, \frac{1}{1-p}\}^d$ from the original pass
Residual add	$A = I$ plus split branch	streamwise sum with branch (F^+, F^-)

Two instances deserve a comment. Evaluation-mode batch normalization has a per-channel scale $c_{\mathrm{bn}} = \gamma/\sqrt{\sigma^2 + \epsilon}$ that may carry either sign (Guillemot and Heuselé, 2020; Hui and Binder, 2019); in practice we merge it into an adjacent convolution before splitting. Training-mode batch normalization, whose statistics depend on the batch, is genuinely multiplicative and requires the techniques of Chapter 5. Training-mode dropout uses the *same* realised mask m on both streams and on $u = u^+ - u^-$, so the split network drops exactly the units the original does.

Gated max-pooling. Max-pooling over a window $[q]$ computes $\max_{j\in[q]} u_j$ per channel, which is neither linear nor a shared gate on a single pair. It is a CPWL function of the window and admits its own convexity-preserving splitting identity.

Proposition 3.29 (Correctness of gated max-pooling, `Lean_51`). *For a pooling window $[q]$, define the split chunk*

$$v^+ \coloneqq \max_{j\in[q]}\Big\{u_j^+ + \sum_{i\in[q],\, i\neq j} u_i^-\Big\}, \qquad v^- \coloneqq \sum_{i\in[q]} u_i^-. \tag{3.64}$$

Then $v^\pm \geq 0$ for $u^\pm \geq 0$, both components are convex and non-decreasing in each coordinate of (u^+, u^-), and

$$v^+ - v^- = \max_{j\in[q]}\big(u_j^+ - u_j^-\big). \tag{3.65}$$

Proof. Non-negativity is clear, as every candidate and the sum are sums of non-negative inputs; both components are non-negative-coefficient linear combinations and an entrywise maximum, hence convex and non-decreasing. For (3.65), add and subtract u_j^- in each candidate: for every $j \in [q]$,

$$u_j^+ + \sum_{i\neq j} u_i^- = \big(u_j^+ - u_j^-\big) + \sum_{i\in[q]} u_i^-,$$

and since the second summand does not depend on j, taking the maximum over j and subtracting $v^- = \sum_i u_i^-$ yields $\max_j\big(u_j^+ - u_j^-\big)$. □

Identity (3.65) generalizes (3.18), the case $q = 2$ with the pair $\big(u^+ - u^-, 0\big)$. Optionally, (3.64) can be replaced by a *winner-takes-it-all* rule forwarding, in both streams, the entries where the *difference* is maximal; this preserves the difference invariant but sacrifices monotonicity and convexity, mirroring the shared-gate trade-off of Remark 3.8.

Implementation and caches. The pinned implementation uses PyTorch FX to trace a model and organize supported graph operations into decomposition chunks. A forward/backward cache accumulates inflowing activations and gradients per chunk, and a parallel cache mirrors the original model. The latter serves both analysis—layer-wise activation/gradient error—and optional on-the-fly correction, and reuses the original model's positivity patterns in place of drifted split-pair maxima in the backward pass. The current code provides configurable `float16`, `float32`, or `float64` arithmetic, with `float32` as its default; this configuration choice is distinct from the double-precision setup reported in the original paper. The graph conversion applies to the operations supported by its chunk implementations.

Alternative nonlinearities and exactness. The source implementation discussion proposes replacing alternative nonlinearities such as ELU (Clevert et al., 2016) or Swish (Ramachandran et al., 2017) by small ReLU subnetworks before applying the split procedure. Such a replacement is an approximation: a finite ReLU subnetwork is piecewise affine and cannot reproduce these non-piecewise-affine functions exactly. The resulting split preserves the approximating network, while the approximation error relative to the original activation is a separate quantity. The pinned repository contains a separate approximation utility for selected scalar operations, including Swish, but that utility is not invoked by the inspected hybrid conversion path and does not list ELU among its supported operations. The curvature construction of Section 3.2.2 instead provides exact scalar decompositions for activations satisfying its hypotheses, without replacing them by ReLU subnetworks.

The CNN split pass. The split forward pass processes the chunks of a computation graph $\mathscr{G}$ in topological order, applying to each the split rule of its type—Lemma 3.28 for the linear chunks, the Maxout gate (3.19) or max-pool split (3.64) for the non-linear ones—with optional entrywise re-centering $v^{\pm} \leftarrow v^{\pm} - \min\{v^+, v^-\}$ after each chunk. The stabilization of Section 3.6 applies unchanged, since Theorem 3.24 uses only non-negative linearity and the Maxout gate structure.

Theorem 3.30 (Correctness of the CNN split pass, `Lean_52`). *Let $\mathscr{G}$ be a feedforward computation graph composed of the chunk types above, computing the function $x \mapsto f(x)$. Then for every input split $x = x^+ - x^-$ with $x^{\pm} \geq 0$, the split pass produces non-negative pairs satisfying the difference invariant*

$$v_m^+ - v_m^- = v_m \qquad \text{for every chunk } C_m,$$

where v_m is the corresponding original activation; in particular $f(x) = v_M^+ - v_M^-$. If neither winner-takes-it-all pooling nor input-dependent re-centering is used, every $v_m^{\pm}$ is convex in (x^+, x^-) and non-decreasing in each coordinate.

Proof. Induction over the topological order of $\mathscr{G}$. At the input node the invariant is the input convention, non-negativity holds by assumption, and $x_k^{\pm}$ are affine, convex, non-decreasing. For the step, assume all predecessors of C_m satisfy the three properties. Lemma 3.28 (linear chunks), Proposition 3.29 (max-pool), and Corollary 3.17 with (3.18) (the ReLU gate) establish for every type that (i) non-negative pairs map to non-negative pairs, (ii) the output difference equals the original operation on the input difference—by the invariant the activation v_m—and (iii) each output component is convex and non-decreasing, being built from non-negative linear combinations and entrywise maxima whose compositions preserve both properties (Amos et al., 2017). Re-centering subtracts a common shift, preserving the invariant and non-negativity (Theorem 3.24); it is omitted when convexity and coordinatewise monotonicity are required, since these properties need not survive the subtraction. Already re-centering the pair $(\max\{p,n\},n)$ gives $(\mathrm{ReLU}(p-n),0)$, whose first component decreases when n increases with p fixed. The output claim is the invariant at C_M. □

This completes the decomposition toolbox: a general validity framework, a one-parameter family with a convex DC endpoint, the curvature-based smooth-activation splitting of Section 3.2.2, a hard-gated and a regularized soft-gated decomposition, numerically stabilized forward and backward passes, and a chunk calculus for standard convolutional architectures. Chapter 4 builds the game-theoretic layer on the Stopping Decomposition, and Chapter 6 turns the split streams g and h into attribution methods.

4 Game-Theoretic Framework

The sign-separated streams of Chapter 3 expose how opposing contributions accumulate. This chapter formalizes the backward-game picture introduced in Figure 1.1, asking how a backward rule traverses those contributions. The stopping decomposition gives a common state graph for two games whose policies represent gradients and relevance propagation. Their trajectory distributions make the backward calculation available for both method design in Chapter 6 and diagnostics in Chapter 8. Exact recovery results concern the anchor rules; myopic temperature changes and risk-sensitive policies are controlled deformations of those rules. Detailed definitions and proofs are given in the companion supplement, Chapter S.1.

4.1 Related Work

Ancona et al. (2018) unify several gradient-based rules as modified gradients, a map-level unification that leaves out the $\alpha\beta$ rule; we instead seek a signed trajectory-space representation of gradients and the $\alpha\beta$ rule that exposes their routing laws. Combined with the split representation, this distinguishes two choices that attribution methods make: which internal signals to retain and how to propagate them backward.

Game and trajectory views. The closest mathematical precursor is the ReLU Net Game of Gaubert & Vlassopoulos (2025), which identifies the forward evaluation of a ReLU network with the value of a backward Stopping Game but does not connect that game to the backward pass or to attribution. We reformulate the game to draw the connection to a novel Stopping Decomposition, whose inherent weight decomposition is related to the DC decomposition of neural networks—the decomposition Zimmermann & Loho (2026) use directly for attribution. Our SG and RG are different games utilizing this weight decomposition, chosen to recover backward attribution rules rather than the pure DC decomposition. The NRM framework of Xiong et al. (2026) also starts from walks through the network, assigning them normalized signed relevance. Section 8.1 compares its treatment of LRP and concept relevance propagation with our trajectory probabilities and discounted occupation measures.

Relevance propagation and stochastic explanations. LRP, Deep Taylor, and Excitation Backprop constitute the core relevance propagation family (Bach et al., 2015; Montavon et al., 2017; Zhang et al., 2018a), with later variants modifying redistribution for contrast, stability, or signed relevance (Binder et al., 2016; Gu et al., 2019; Kohlbrenner et al., 2020; Nam et al., 2020; Vukadin et al., 2024). We supply a common trajectory-space semantics for this family: SG recovers gradients, RG recovers the generalized $\alpha\beta$ rule, and its analytical modes map back to principled modifications of existing backward rules. Excitation Backprop (Zhang et al., 2018a) is an absorbing Markov chain recovering the $\alpha\beta$-LRP rule for the special case $(\alpha, \beta) = (1,0)$ known as the z^+-rule, which only considers the positive weights in the relevance propagation. Our game extends it with an opponent player (enabling negative-weight influence) and replaces input-dependent transition probabilities with soft Bellman equations. Deep Taylor (Montavon et al., 2017) justifies that same z^+-rule and explicitly leaves the mathematical modeling of the general $\alpha\beta$ rule as an open problem; RG closes this.

Shapley-value attribution methods. SHAP and KernelSHAP frame explanation as a cooperative game over feature coalitions (Lundberg and Lee, 2017) in a *black-box* setting; they are therefore a distinct, complementary game that explains marginal input-feature contributions relative to a baseline, whereas our backward game explains how relevance is routed through network internals in a *white-box* setting.

Robust explanations via Softplus. Dombrowski et al. (2022) propose Softplus replacement as a curvature-based defense against gradient-explanation manipulation. Section S.1.5 shows that entropy-regularizing the activation-state stop/continue choice in SG recovers Softplus-network gradients, situating their construction within our framework.

Interpretability in Transformers and ViTs. Attention Rollout / Flow read attention as a routing graph (Abnar and Zuidema, 2020); transformer-LRP methods propagate through attention and skip connections (Achtibat et al., 2024; Ali et al., 2022; Chefer et al., 2021b); DAVE addresses ViT tokenization artifacts via conditioned operators and equivariant averaging (Wróbel et al., 2026). The conditioned-operator transformer constructions arise naturally inside our game semantics: the QK-softmax is the oracle player's reference policy, while attribution lives on the Value-routing trajectories. Hence, our method is compatible with Reynolds averaging, which in our RG+Eq mode removes the same grid artifact that DAVE targets—however, with more control via the different game modes.

4.2 Backward Attribution

For the game derivations in this and the attention and trajectory chapters, $W_{ij}^{(l)}$ denotes the edge from predecessor i to successor j. Thus the displayed adjacency matrices are the transposes of the output-by-input matrices used in the DC split recurrences; the edge weights and represented network are identical. Stopping streams are tagged by stop to distinguish them from the DC streams. Consider a ReLU network with layers $l = 1,\dots,L$, input $a^{(0)} = x$, pre-activations $z_j^{(l)} = \sum_i a_i^{(l-1)} W_{ij}^{(l)} + b_j^{(l)}$ and activations $a_j^{(l)} = \mathrm{ReLU}(z_j^{(l)})$, where $W_{ij}^{(l)}$ connects neuron i of layer $l-1$ to neuron j of layer l. The initial derivation explains a scalar activation output $f(x) = a_u^{(L)}$. Section 4.5 extends it to an ungated, possibly negative affine output logit. We write $[u]^+ = \max(u,0)$ and $[u]^- = \max(-u,0)$, so $u = [u]^+ - [u]^-$, and $W^{(l,\pm)} = [W^{(l)}]^\pm$ entrywise.

Layer-wise relevance propagation. LRP (Bach et al., 2015) explains $f(x)$ by redistributing it backward, layer by layer: starting from $R_u^{(L)} = f(x)$, every neuron passes its relevance $R_j^{(l)}$ on to its inputs in proportion to their contributions $a_i^{(l-1)} W_{ij}^{(l)}$ to its pre-activation, so that relevance is conserved from layer to layer up to the share absorbed by biases and stabilizers. Rules differ in how the contributions are weighted. The $\alpha\beta$-rule is parameterized by $\alpha,\beta \geq 0$, with $\alpha - \beta = 1$ giving conservation when both participating sign streams are nonempty and all bias, stabilizer, and empty-stream absorption is accounted for, and a stabilization parameter $\varepsilon \geq 0$ (added to each sign denominator, as in common implementations (Anders et al., 2023)): the relevance of neuron j is split into a positive-contribution fraction scaled by α and a negative-contribution fraction scaled by β, each normalized by its own denominator,

$$
\begin{aligned}
R_i^{(l-1)} &= \sum_j \Big(\alpha \, \frac{[a_i^{(l-1)} W_{ij}^{(l)}]^+}{\Sigma_j^{(l,+)} + \varepsilon} - \beta \, \frac{[a_i^{(l-1)} W_{ij}^{(l)}]^-}{\Sigma_j^{(l,-)} + \varepsilon} \Big) R_j^{(l)}, \\
\Sigma_j^{(l,\pm)} &= \sum_k [a_k^{(l-1)} W_{kj}^{(l)}]^\pm + [b_j^{(l)}]^\pm .
\end{aligned}
\tag{4.1}
$$

Relevance assigned to the bias and to ε is absorbed. $(\alpha,\beta) = (1,0)$ gives the z^+-rule of Deep Taylor decomposition (Montavon et al., 2017), which turns out to be the same rule as Excitation Backprop (Zhang et al., 2018a); other rules (LRP-ε, LRP-γ) change the redistribution fractions.

Transformers. In a Vision Transformer (ViT), LayerNorm couples the features of a token through its input-dependent mean and variance, and the attention matrix $A = \mathrm{softmax}(QK^\top/\sqrt{d})$ depends on the input, which breaks conservation (Ali et al., 2022). Transformer-specific rules therefore *condition* on the forward pass: conservative propagation (Ali et al., 2022) and DAVE (Wróbel et al., 2026) hold A and the LayerNorm statistics fixed and propagate relevance through the Value path only; AttnLRP (Achtibat et al., 2024) also propagates through the softmax and the QK product.

All these rules share one structure: a backward recursion that routes mass from each neuron to its inputs. The games below make the routing explicit.

4.3 Attribution as a Game

Setup and common state-space structure. With the notation of Section 4.2, we build a finite-horizon two-player game (Osborne and Rubinstein, 1994) played *backward* from the output: a state records a position in the backward traversal (layer, neuron, plus game-specific bookkeeping for player turn and stream sign), and at each non-terminal state the active player either stops, moving to an absorbing cemetery state †, or continues. Skip connections become random transitions to summand states; max-pooling sends players greedily to maximal-payoff states. Section S.1.1 collects the bookkeeping labels (Table S.1.1). The SG value $V_p(s)$ is the γ-discounted expected payoff difference $r_p - r_{p'}$ of player p and opponent p' along trajectories from s under the policies, expressed for the player p in turn at s by the Bellman recursion

$$V_p(s) = \max_{a \in \mathscr{A}_x(s)} \Big\{ r_p(s,a) - r_{p'}(s,a) + \gamma(s) \textstyle\sum_{s'} \mathbb{P}_x(s' \mid s,a)\, V_p(s') \Big\}, \tag{4.2}$$

whereas the **R**outing **G**ame replaces the hard maximization at routing states by an entropy-regularized maximization over mixed routing decisions, whose value is the soft game value (4.5). Both games start with unit mass at the positive output state. Any player policies induce a backward trajectory measure Π_x on trajectories $\zeta = (\zeta_0, \zeta_1, \dots)$ with a game-specific cumulative discount $d_t(\zeta)$ up to time t. The **discounted occupation measure** (Puterman, 1994)

$$\Gamma_x(v) \coloneqq \mathbb{E}_{\zeta \sim \Pi_x}\big[\textstyle\sum_t d_t(\zeta)\, \mathbf{1}\{\zeta_t = v\}\big] \tag{4.3}$$

quantifies how much weight the trajectories going through a state have.

4.3.1 The Stopping Game

With each neuron j in layer l, we associate two states $s^{(l)}_{j,+}$ and $s^{(l)}_{j,-}$, where the sign encodes which player is in turn. The biases become the immediate payoffs for continuing; under continuation, weight magnitudes drive the environment transitions with discount $\gamma^{(l)}_j = \sum_i |W^{(l)}_{ij}|$ (positive- and negative-weight predecessors follow $\mathbb{P} \propto W^{(l,\pm)}/\gamma^{(l)}_j$, and a negative weight passes the turn to the opponent). If $\gamma^{(l)}_j = 0$, continuation pays the bias and enters the cemetery. We select $\mathrm{ReLU}'(0) = 0$ and the smallest-index maximizer at pooling ties. The players' expected payoffs realize the novel Stopping Decomposition, separating positive and negative paths to each neuron activation (Lemma S.1.2). The gradient is then the difference of two *non-negative* occupation measures:

Theorem 4.1 (Gradient representation, `Lean_53`). *The Stopping Game allows a subgame-perfect Nash equilibrium, unique up to tie-breaking at zero-activation neurons. On strict gate cells (and with*

unique max-pooling winners where applicable), under the stop-at-tie-breaks equilibrium the discounted occupation measure $\Gamma_x(s_{i,p}^{(l)}) \geq 0$ *equals the output derivative with respect to the corresponding* Stopping Decomposition *activation. At ties, the corresponding identities refer to the backward quantities selected by the stated gate and winner conventions. On these cells, the gradient of* f *with respect to the input is the difference* $\Gamma_x(s_{i,+}^{(0)}) - \Gamma_x(s_{i,-}^{(0)})$.

Section S.1.2 proves the theorem, works through an example, gives the full state space with skip connections and max-pooling, and compares the Stopping Decomposition with the DC decomposition.

Modeling $\alpha\beta$-LRP or attention modules via the SG would require activation-dependent transitions. The RG moves activation-dependence into the players' policies, opening a continuous family of deformations.

4.3.2 The Routing Game

In the RG we model every neuron by three game states: an activation state $s_{j,q}^{(l,\mathrm{act})}$ and two linear routing states $s_{j,q,+}^{(l,\mathrm{lin})}$ and $s_{j,q,-}^{(l,\mathrm{lin})}$, the $+$ stream state and the $-$ stream state, modeling the positive part $W^{(l,+)}$ and the negative part $W^{(l,-)}$ of the affine layer respectively. At an activation state the player in turn stops or continues to both streams with probability $\frac{1}{2}$ each; the $-$ stream passes the turn to the opponent, and discounting the game value by 2 compensates the split. At a *linear routing state* $s_{j,q,\kappa}^{(l,\mathrm{lin})}$ for neuron j in layer l on sign stream $\kappa \in \{+,-\}$ with player q in turn, the active player mixes over predecessors i, a bias node and an outside option $\perp_{\mathrm{lin}}$ under a policy $\pi_q(s_{j,q,\kappa}^{(l,\mathrm{lin})})$; the bias node and $\perp_{\mathrm{lin}}$ absorb the trajectory, with log-payoffs $\log[b_j^{(l)}]^\kappa$ and $\log \varepsilon$. Writing the game in log space turns the active player r's advantage of moving to predecessor i into an additive action–value,

$$Q_r\big(s_{j,q,\kappa}^{(l,\mathrm{lin})}, i\big) = \begin{cases} \log W_{ij}^{(l,\kappa)} + V_r(s_{i,q}^{(l-1,\mathrm{act})}), & r{=}q, l{\geq}2, \\ \log\big[x_i W_{ij}^{(1)}\big]^\kappa, & r{=}q, l{=}1, \\ -\infty, & r \neq q, \end{cases} \tag{4.4}$$

where $V_r(\cdot)$ denotes the *game value* at the successor activation state from the perspective of player r. At each linear routing subgame we regularize the players' policies with Shannon entropy (Geist et al., 2019), which yields the following equilibrium value and Gibbs best response:

$$V_q\big(s_{j,q,\kappa}^{(l,\mathrm{lin})}\big) = \tau \log \textstyle\sum_i \exp\big(Q_q(s_{j,q,\kappa}^{(l,\mathrm{lin})}, i)/\tau\big), \tag{4.5}$$

$$\pi_q(s_{j,q,\kappa}^{(l,\mathrm{lin})})(i) \propto \exp\big(Q_q(s_{j,q,\kappa}^{(l,\mathrm{lin})}, i)/\tau\big), \tag{4.6}$$

where the sums include the bias node and $\perp_{\mathrm{lin}}$. At $\tau > 0$ the game admits an SPNE whose occupation measure at $\tau = 1$ is the $\alpha\beta$-LRP attribution:

For the Routing Game, the cumulative discount is the product of 2α when entering a positive linear-routing state, 2β when entering a negative one, and 1 otherwise; these trajectory discounts are distinct from those used in game-value evaluation.

Theorem 4.2 ($\alpha\beta$-LRP representation, `Lean_54`). *For* $\tau > 0$*, the Gibbs policies of Eq. (4.6) and stopping decisions from the corresponding continuation values yield a canonical SPNE of the entropy-regularized RG. We select stopping at every activation log tie and absorption for an empty routing stream, as specified in Remark S.1.2. At the anchor* $\tau = 1$*, the selected gate is* $\mathbf{1}[z > 0]$ *and the* $\alpha\beta$*-LRP attribution map in (4.1) equals* $R_i^{(l)} = f(x)\,\big(\Gamma_x(s_{i,+}^{(l)}) - \Gamma_x(s_{i,-}^{(l)})\big)$.

Section S.1.3 gives the full game definition, the proofs, and a worked example of the local routing dynamics (Figure S.1.2).

In practice we use τ as a *myopic* control: each linear subgame applies the τ-scaled Gibbs policy to its own action values, but inherits game values from the soft Bellman equations at τ=1. We do not re-solve at τ downstream: that would change the forward activations, while we want trajectories that explain the network's actual forward pass. The myopic setting at $\tau \neq 1$ is a controlled temperature deformation of the occupation measure. Low τ concentrates routing on the dominant pathway.

4.4 Game-Theoretic Modes

Game-theoretic concepts like risk aversion and entropy regularization map back to novel modes of the backward rules for probing and tuning explanation properties (Table 4.1). The exact representations of Section 4.3 are the defaults; every mode other than the LRP parameters (α,β,ε) is a deformation that is off at the anchor. We use five: the temperature τ (focus), risk aversion λ at activation states and its attention-logit variant λ_{sm}, the probit gate $\tilde{\Phi}$ (gate smoothing), and equivariant averaging over N input transformations (grid artifacts).

Focus. The myopic temperature of Section 4.3.2 rescales the Gibbs policy at every linear state while keeping the τ=1 game values. Low τ concentrates routing on dominant predecessors; high τ spreads mass more uniformly, exposing broader contextual features as well as architectural priors. *Risk aversion. Assumed Density Filtering* (ADF; (Gast and Roth, 2018)) propagates a Gaussian mean μ and variance σ^2 for every pre-activation z from an input variance σ^2 in one additional moment pass. The risk-averse player replaces the observed z at activation states by the lower-confidence bound $\mu + \lambda\sigma$ with $\lambda < 0$, the pessimistic counterpart of confidence-bound exploration (Auer et al., 2002; Fei et al., 2020), and so attenuates routes through high-variance neurons; linear-state routing is unchanged. On attention blocks, λ_{sm} applies the same shift to the oracle's attention logits before the softmax, downweighting high-variance key tokens. *Gate smoothing.* The risk-averse stop/continue rule is the hard gate $\mathbf{1}[\mu/\sigma+\lambda > 0]$. The probit gate $\tilde{\Phi}$ lets the player instead continue with probability $\Phi(\mu/\sigma+\lambda)$, the marginal of a player who perceives the continuation payoff as $z+\eta$, $\eta \sim \mathcal{N}(0,1)$, and continues iff it is positive. At μ=z, σ=1, λ=0 this probit-regularized choice recovers the Gaussian-smoothed ReLU $\mathbb{E}[\text{ReLU}(z+\eta)]$, just as entropy regularization recovers Softplus (Section S.1.5); on ReLU networks it is a smoothing deformation. Because the gate uses the ADF σ, σ^2 is active whenever $\tilde{\Phi}$ is on. *Exact modes and cost.* (α,β) weight the two streams of the occupation measure, and the LRP stabilizer ε adds an outside option $\perp_{\text{lin}}$ with log-score $\log\varepsilon$ at every linear state, leaking weak-evidence neurons into the cemetery while leaving strong-evidence ones near-unaffected (Theorem S.1.8). The exact modes and τ add no cost; λ, λ_{sm} and $\tilde{\Phi}$ add one ADF pass (0.10 → 0.17 s per ViT-B/16 image); averaging multiplies the cost by N (4.93 s at N=50, 8.66 s with λ_{sm}). Section S.1.4 derives the modes and gives defaults, costs and an order of use.

4.5 Signed affine output logits

The activation-output formulation above extends to the signed logits of a classifier by an *ungated affine readout*. Let $h_j = a_j^{(L)} \geq 0$ be the last hidden activations and

$$f_u(x) = \sum_j c_{uj} h_j(x) + b_u. \tag{4.7}$$

There is no ReLU at this output: the sign of the logit introduces no stopping gate.

Tab. 4.1: Modes of the games. "Exact": the mode stays within a represented rule; "deform.": optional deformation, off (default value) at the anchor. σ^2: input variance of the ADF moment propagation; N: number of transformed inputs averaged by RG+Eq.

Mode	Effect on the game	Default	Property
(α,β)	branch discounts of the two streams (exact)	(2,1)	support / oppose
ε	outside option $\perp_{\text{lin}}$, payoff $\log \varepsilon$ (exact)	0.5	damps low-inflow neurons
τ	entropy regularization of routing (deform.)	1	sharp / diffuse
λ, σ^2	values $\to$ lower-confidence bounds (deform.)	λ=0	noise / fragility
λ_{sm}	attention logits shifted by λ_{sm} times a per-key ADF std (deform.)	0	stable attention
$\tilde{\Phi}$	noise on player observations (deform.)	off	gate smoothing
N	average over input transformations (deform.)	off	grid artifacts

Stopping Game. Add a root with forced continuation, immediate bias payoff difference b_u, and structural discount $\gamma_u = \sum_j |c_{uj}|$. When $\gamma_u > 0$, it moves to hidden neuron j with probability $|c_{uj}|/\gamma_u$, retaining the player for $c_{uj} \geq 0$ and switching players for $c_{uj} < 0$. Its value is therefore $b_u + \sum_j c_{uj} h_j = f_u(x)$, even when this value is negative. If $\gamma_u = 0$, the root pays b_u and moves to the cemetery; the input derivative is zero. On the strict gate cells of Theorem 4.1, let $\Gamma^{\text{SG}}_{j,\pm}(i)$ denote the input-state occupations from a unit source at hidden neuron j with initial player $+$. The added transition gives

$$\partial_i f_u(x) = \sum_j c_{uj} \big(\Gamma^{\text{SG}}_{j,+}(i) - \Gamma^{\text{SG}}_{j,-}(i)\big). \tag{4.8}$$

This is the chain rule with the same non-negative path occupations; at gate or pooling ties, the selected-backward convention remains in force.

Routing Game. For attribution, append an ungated source to the hidden Routing Game. This source continues unconditionally; its positive and negative branch masses, including the bias, are $\Sigma^{\kappa}_u = \sum_j [h_j c_{uj}]^{\kappa} + [b_u]^{\kappa}$. At $\tau = 1$, the sign-oracle probabilities $1/2$, trajectory discounts 2α and 2β, and the branchwise Gibbs policies give the coefficients

$$q^+_j = \alpha \frac{[h_j c_{uj}]^+}{\Sigma^+_u + \varepsilon}, \qquad q^-_j = \beta \frac{[h_j c_{uj}]^-}{\Sigma^-_u + \varepsilon}. \tag{4.9}$$

Bias and outside-option shares are absorbed. An empty branch is completed by a transition to the cemetery, and its ordinary-successor fractions are zero ($0/0 := 0$). For unit-source hidden occupations $\Gamma^{\text{RG}}_{j,\pm}(i)$, define

$$\widehat{\Gamma}_{\pm}(i) = \sum_j \big(q^+_j \Gamma^{\text{RG}}_{j,\pm}(i) + q^-_j \Gamma^{\text{RG}}_{j,\mp}(i)\big). \tag{4.10}$$

Since $h_j = 0$ implies $q^+_j = q^-_j = 0$, the readout assigns nonzero incoming relevance only to positive hidden activations. Subtracting the two equations yields precisely the affine $\alpha\beta$ redistribution, followed by the hidden-layer recursion of Theorem S.1.8. Thus, for any real initial relevance r_u,

$$R_i = r_u \big(\widehat{\Gamma}_+(i) - \widehat{\Gamma}_-(i)\big). \tag{4.11}$$

The trajectory law and both occupations remain non-negative; a signed seed scales their *difference*. No logarithm of the signed logit is taken: logarithms apply only to the non-negative branch masses. Classical output-seeded LRP chooses $r_u = f_u(x)$, whereas the released Routing-Game attribution routine uses $r_u = 1$. These maps differ by the signed factor $f_u(x)$, including a sign reversal when the target logit is negative. The zero-logit case has zero output-seeded relevance, but can have nonzero unit-seeded relevance.

5 Extensions: Attention and Transformers

This chapter extends the decomposition and routing-game constructions to multiplicative layers and Transformers. It distinguishes the sign-sorted monotone split inherited from Zimmermann & Loho (2026), an exact convex split of the full deterministic Transformer developed here, and the conditioned routing-game rules used in the experiments. Each construction has its own domain and preserves a different set of structural properties.

Two features separate a Transformer (Vaswani et al., 2017) from the pointwise-activation networks treated earlier. First, the attention products $QK^\top$ and AV multiply input-dependent quantities. Second, softmax and LayerNorm couple several coordinates. Section 5.2 develops the sign-sorted product rule, which preserves monotonicity on non-negative split inputs. Section 5.3 instead combines polarization, the multivariate softmax pair of Theorem 3.12, and a reciprocal-square-root pair to preserve convexity as well. Section 5.5 then studies attention as a conditioned Value-routing game. Its implementation on ViT-B/16, described in Section 5.5.3, supplies the attribution results evaluated in Chapter 7; the new full Transformer DC construction is an analytical result.

5.1 Related Work: Interpretability in Transformers

Attribution research on Transformers began at the attention weights themselves: since the rows of an attention matrix are distributions over tokens, early analyses of head roles and pruning read them directly as explanations (Voita et al., 2019). Abnar & Zuidema (2020) objected that single-layer weights ignore propagation through depth, and proposed *attention rollout* — multiplying the residual-corrected attention matrices across layers — and *attention flow*, a max-flow computation on the layered attention graph. Both commit to a routing view: attention matrices are transition kernels of a walk on tokens, and an explanation states where that walk deposits its mass. What they omit is everything outside the attention weights — value magnitudes, the MLPs, the sign structure of the projections — so rollout can assign high relevance to tokens whose values contribute nothing.

A second line adapts LRP (Bach et al., 2015) to the Transformer. Chefer et al. (2021a) combined LRP-style relevance with attention gradients to propagate class-specific relevance through self-attention, later generalized to co-attention and encoder–decoder architectures (Chefer et al., 2021b). Ali et al. (2022) made the structural point that shapes this chapter: naive gradient propagation through softmax and layer normalization *violates conservation*, since both route relevance through data-dependent normalizers, whereas propagating through a locally linearized map with those outputs and statistics held constant ("detached") restores it. AttnLRP (Achtibat et al., 2024) systematized this into dedicated rules for the bilinear products and non-linearities, scaling to LLMs, with concept-level (Achtibat et al., 2023) and sign-aware (Vukadin et al., 2024) extensions. On the vision side, DAVE (Wróbel et al., 2026) traced grid artifacts to the 16×16 patch tokenization of ViTs (Dosovitskiy et al., 2021) — related to the artifact tokens of Darcet et al. (2024) — and countered them with *conditioned* (detached) operators and attribution averaging over random spatial transformations. At LLM scale, gradient approximations of activation patching (Kramár et al., 2024; Syed et al., 2024) localize circuits against interventions (Hanna et al., 2024), and attribution increasingly acts as a control signal for head identification and guided decoding (Kahardipraja et al., 2025; Komorowski et al., 2025).

These approaches motivate two separate questions. The structural question asks whether the full forward map, including the attention-generating operations, admits explicit convex streams; Sections 5.2 and 5.3 develop the relevant product and normalization rules. The attribution question asks how relevance travels through the Value routes when the realized attention matrix is fixed. Section 5.5

formulates that conditioned question as a KL-regularized routing game and recovers the conservative attention-block rule of Ali et al. (2022). This gives a precise interpretation of a routing calculation of the kind motivated by Abnar & Zuidema (2020), with the conditioning assumptions stated explicitly.

5.2 Decomposing Multiplicative Operations

Gating layers, squeeze-and-excitation blocks, and attention multiply two activations entrywise or contract activation tensors against each other. We first present the sign-sorted construction of Zimmermann & Loho (2026), which preserves the difference and monotonicity of non-negative stream pairs. Section 5.3 will give a different product identity that also preserves convexity and combines with the multivariate softmax pair from Chapter 3.

The gated product rule. Suppose two scalar quantities are carried in split form, $a = a^+ - a^-$ and $b = b^+ - b^-$ with all four streams non-negative, as the valid decompositions of Section 3.1 guarantee. Their product again splits into non-negative parts by expanding and sorting the four cross terms by sign:

$$(a^+ - a^-)(b^+ - b^-) \;=\; \underbrace{\big(a^+b^+ + a^-b^-\big)}_{=:\,(ab)^+} \;-\; \underbrace{\big(a^+b^- + a^-b^+\big)}_{=:\,(ab)^-}. \tag{5.1}$$

Both bracketed terms are non-negative and non-decreasing in each incoming stream, so multiplication maps monotone stream pairs to monotone stream pairs — the analogue of routing $W^{(l,+)}$ and $W^{(l,-)}$ into separate channels in Section 3.2.3. Applied entrywise, (5.1) splits Hadamard layers; applied to every scalar contraction, it splits matrix products of two split-stream tensors. What is *not* preserved is convexity: products of convex functions need not be convex, so the split streams need not stay convex in the input. For this sign-sorted construction, the resulting class is "maximum of finitely many posynomials", a *posynomial* being a polynomial with non-negative coefficients: posynomials are closed under addition, multiplication, and positive scaling, and are monotone on the non-negative orthant.

Maxout multiplication networks. To state the splitting theorem in its natural generality we enlarge the architecture class. A rank-k Maxout unit (Goodfellow et al., 2013) computes the maximum of k affine forms and reduces to ReLU for $k = 2$ with one form fixed to zero.

Definition 5.1 (Maxout multiplication network). A *rank-k Maxout multiplication network* is a finite directed acyclic graph of *input neurons*, *addition neurons*, and *multiplication neurons*. An addition neuron v carries weight vectors $w^{uv} \in \mathbb{R}^k$ on its incoming arcs and computes the rank-k Maxout activation $a^v = \max_{i\in[k]} \sum_u w_i^{uv} a^u$; a multiplication neuron carries natural exponents $w^{uv} \in \mathbb{N}$ and computes the monomial $a^v = \prod_u (a^u)^{w^{uv}}$. The network is called *monotone* if all weights are non-negative, and *input-convex* if all weights are non-negative except possibly those on arcs leaving input neurons.

Linear and convolutional layers, biases (via a constant input neuron), maxima and max-pooling, residual connections, Hadamard products, and matrix products are all expressible here; restricting exponents to natural numbers avoids divisions and fractional powers of possibly negative quantities. The two structural adjectives of Definition 5.1 pin down what such networks compute.

Proposition 5.1 (Representation by maxima of posynomials, `Lean_72`). *A monotone rank-k Maxout multiplication network with non-negative intermediate quantities represents the maximum of finitely many posynomials in its non-negative input variables, and is monotone on the non-negative*

orthant. This representation does not imply ordinary convexity: the positive-coefficient monomial xy is a counterexample. Allowing signed input-incident affine maps likewise supplies no general input-convexity guarantee in the presence of multiplication.

Proof sketch. Induct along a topological order on the non-negative orthant. Coordinate projections are posynomials. Non-negative scaling, addition and maxima preserve finite maxima of posynomials. Products do so when the factors are non-negative, by distributing the product over the finite maxima. These are precisely the stated network operations. The failure of convexity is witnessed by $F(x,y) = xy$: $F(1,1) = 1$ exceeds $\frac{1}{2}F(2,0) + \frac{1}{2}F(0,2) = 0$. □

The splitting theorem. The main result transfers the split of Section 3.2.3 to this class. Two local identities drive it. Maxout units are split by the maximum-shift identity

$$\max_{j\in[k]}\left(a_j^+ - a_j^-\right) \;=\; \max_{j\in[k]}\Big\{a_j^+ + \sum_{i\neq j} a_i^-\Big\} \;-\; \sum_{i\in[k]} a_i^-, \tag{5.2}$$

which generalizes both the ReLU splitting identity $\mathrm{ReLU}(a-b) = \max\{a,b\} - b$ of Section 3.2.3 and the gated max-pooling rule of Section 3.7; multiplication neurons are split by the multinomial generalization of (5.1), expanding $\prod_u (a^{u,+} - a^{u,-})^{w^{uv}}$ and sorting the resulting monomials by the parity of the number of $(-)$-stream factors.

Theorem 5.2 (Monotone split of multiplicative ReLU networks, `Lean_73`). *Let N be a (residual) ReLU network containing linear, convolutional, maximum, matrix-product, and Hadamard-product components. Then, on non-negative split-input variables, N has two* monotone *rank-2 Maxout multiplication networks N^+ and N^- satisfying*

$$N(x) = N^+(x^+,x^-) - N^-(x^+,x^-), \qquad x = x^+ - x^-, \quad x^\pm \geq 0,$$

Each component is monotone as a function of its non-negative split inputs and is representable there by a finite maximum of posynomials. This gives a difference-of-monotone representation; it does not establish a difference-of-convex representation in the original inputs. If the number of expanded monomials per multiplication neuron is uniformly bounded, the sizes of the split networks are linear in the size of N.

Proof sketch. Process the neurons in topological order, replacing each by a coordinated pair (v^+, v^-) preserving the computed function; the construction extends the extended-formulation argument of Hertrich & Loho (2025). *Input neurons* are supplied as non-negative split variables x^+,x^- (for instance $\mathrm{ReLU}(x)$,$\mathrm{ReLU}(-x)$) with output weights sign-flipped toward the $(-)$-copy; keeping inputs unsplit does not, in general, yield a convex variant once multiplication is present. *Addition (Maxout) neurons* split their weight vectors entrywise into $[w^{uv}]^+ - [w^{uv}]^-$ and apply (5.2): the maximum term defines v^+, the subtracted sum v^-, both with non-negative weights. *Multiplication neurons* apply the sign-sorted multinomial expansion; one auxiliary multiplication neuron per summand feeds collecting addition neurons v^+ (even parity) and v^- (odd parity), all non-negative. Each rewrite is an identity, so correctness telescopes to $N = N^+ - N^-$; all weights are non-negative, hence both components are monotone in the split variables, with the posynomial representation from Proposition 5.1. The multinomial summand count is the source of growth; bounded fan-in alone does not control it if exponents are unbounded. A uniform bound on the expanded monomial count is sufficient. □

Theorem 5.2 extends the algebraic split to multiplicative layers. The two streams retain non-negativity and monotonicity in the split inputs, while their difference preserves the original computation. The sign-sorted product pair does not preserve convexity. Section 5.3 supplies a convex alternative, while the finite-polytope interpretation of ReLU streams remains specific to the piecewise-linear setting.

Remark 5.1 (Smooth normalizers: softmax and sigmoid gates). Theorem 5.2 covers projections, bilinear products, and residual additions, but its Maxout multiplication architecture does not contain softmax. Scalar sigmoid and GELU gates admit the exact pairs of Section 3.2.4; exponentiation on a fixed bounded score interval is covered by Corollary 3.5, with exactness restricted to that interval. These scalar observations alone do not split the coupled softmax denominator. Theorem 3.12 supplies an exact multivariate pair for the whole softmax row, which the next section combines with convex products and LayerNorm.

A separate choice used in conditioned attribution is to hold the realized softmax matrix A fixed, turning $X \mapsto AV(X)$ into a fixed-coefficient affine map (Achtibat et al., 2024; Ali et al., 2022; Wróbel et al., 2026). The routing-game sections describe this conditioned calculation. Its interpretation is distinct from the full forward-map decomposition, which retains the input dependence of A.

5.3 A Constructive DC Decomposition of Transformers

The loss of convexity in (5.1) is a property of that sign-sorted product split. An alternative identity using squares preserves convexity and coordinatewise monotonicity on non-negative split inputs. Together with the exact softmax pair of Theorem 3.12 and an extension of the reciprocal square root, it supplies a constructive DC decomposition for the full deterministic Transformer forward map. The results in this section are analytical extensions of the thesis; they are separate from the conditioned attribution implementation evaluated later in this chapter.

Proposition 5.3 (Convex product pair by polarization, `Lean_74`). *For $A,B,C,D \geq 0$, let*

$$\begin{aligned} P(A,B,C,D) &= \tfrac{1}{2}\big((A+C)^2 + (B+D)^2\big), \\ Q(A,B,C,D) &= \tfrac{1}{2}\big((A+D)^2 + (B+C)^2\big). \end{aligned} \tag{5.3}$$

Then $P - Q = (A-B)(C-D)$. Both functions are convex, non-negative, and coordinatewise non-decreasing on $\mathbb{R}^4_{\geq 0}$. Hence substituting non-negative convex, coordinatewise non-decreasing incoming streams preserves these properties. In particular,

$$(A-B)^2 = 2(A^2+B^2) - (A+B)^2 \tag{5.4}$$

is a valid convex, monotone square pair on $\mathbb{R}^2_{\geq 0}$.

Proof. Expansion proves the difference identity. Each summand is the square of an affine form, hence convex. Its arguments are non-negative on the stated orthant, so its partial derivatives there are non-negative. The composition rule for convex non-decreasing functions with convex components now applies; non-negative values are immediate from the squares. Setting $C = A$ and $D = B$ gives (5.4). □

The orthant restriction is essential for the monotonicity claim: for example, the derivative of $(A+C)^2$ with respect to A is $2(A+C)$ and can be negative outside that domain. Consequently, the Transformer theorem below asserts the properties on non-negative doubled inputs, even though the softmax pair

itself has a larger domain. Matrix products follow by applying (5.3) to each scalar product and adding the corresponding positive and negative components.

5.3.1 Layer normalization with positive epsilon

For $\varepsilon > 0$ and $d \geq 1$, LayerNorm on $u \in \mathbb{R}^d$ uses

$$\bar{u} = \frac{1}{d}\sum_{j=1}^{d} u_j, \qquad c_i = u_i - \bar{u}, \qquad v = \frac{1}{d}\sum_{j=1}^{d} c_j^2 \geq 0, \tag{5.5}$$
$$\mathrm{LN}_\varepsilon(u)_i = \gamma_i c_i (\varepsilon + v)^{-1/2} + \beta_i.$$

The centering operation is signed affine, the variance is a positive sum of squares, and the final learned scale and bias are signed affine. The only new scalar function is the reciprocal square root on non-negative arguments. Extending it to negative arguments permits a global convex pair even when the two streams representing the variance vary independently.

Proposition 5.4 (Reciprocal-square-root pair, `Lean_75`**).** *Let $\varepsilon > 0$, put $M = (2\varepsilon^{3/2})^{-1}$, and define*

$$r_\varepsilon(t) = \begin{cases} \varepsilon^{-1/2} - Mt, & t \leq 0, \\ (\varepsilon + t)^{-1/2}, & t \geq 0. \end{cases} \tag{5.6}$$

Then

$$R_+(x,y) = r_\varepsilon(x - y) + Mx, \qquad R_-(x,y) = Mx \tag{5.7}$$

are convex and coordinatewise non-decreasing on $\mathbb{R}^2$, are non-negative on $\mathbb{R}^2_{\geq 0}$, and have difference $r_\varepsilon(x - y)$. Whenever $x - y \geq 0$, this difference is $(\varepsilon + x - y)^{-1/2}$.

Proof. The extension is continuously differentiable at zero, with matching derivative $-M$. On $t > 0$ its second derivative is $3/(4(\varepsilon + t)^{5/2}) > 0$, whereas it is affine on $t < 0$. Thus r_ε is convex and $-M \leq r'_\varepsilon \leq 0$. Affine precomposition and addition preserve convexity, and $\partial_x R_+ = r'_\varepsilon + M \geq 0$, $\partial_y R_+ = -r'_\varepsilon \geq 0$. The linear component R_- is also monotone. Their origin values are $\varepsilon^{-1/2}$ and 0, giving non-negativity on the orthant; cancellation proves the stated differences. Equivalently, this is the scalar lift of Proposition 3.2 with profiles $p(t) = r_\varepsilon(t) + Mt$, $q(t) = Mt$, and slope bound M. □

To decompose (5.5), first split the affine centering map. Apply (5.4) to each centered coordinate and average the two streams separately to obtain $v = v^+ - v^-$. Apply (5.7) to (v^+, v^-). Because the recovered variance is non-negative, the extension agrees exactly with the actual normalization factor. Finally use the polarized product pair to multiply each centered coordinate by that factor, followed by the signed affine scale and bias. Every intermediate stream is convex, monotone, and non-negative on the original split-input orthant. The hypothesis $\varepsilon > 0$ ensures that the normalization is defined for every input, including constant vectors.

5.3.2 An explicit pre-normalization architecture

To make the arbitrary-depth claim concrete, we specify the ordinary Transformer forward map before defining its split evaluation. Tokens are indexed by $t \in \{1, \ldots, T\}$, with feature vectors in $\mathbb{R}^d$, where $T, d \geq 1$. Every matrix and bias below is a fixed real parameter and may have either sign.

Definition 5.2 (Finite pre-normalization Transformer, Lean_76). A Transformer is specified by a positional bias $P \in \mathbb{R}^{T\times d}$, a finite list of blocks $B_1,\dots,B_K$, an optional final LayerNorm, and a signed affine scalar readout. In block k, first normalize each token,

$$U_t = \mathrm{LN}_{k,1}(X_t).$$

Each head h has positive query/key width $d_{q,h}$ and value width $d_{v,h}$, signed affine projections, a real score scale τ_h, and a fixed nonempty allowed-key set $I_{h,t} \subseteq \{1,\dots,T\}$ for every query token. It computes

$$\begin{aligned} Q_{h,t} &= W_{Q,h}U_t + b_{Q,h}, \qquad K_{h,t} = W_{K,h}U_t + b_{K,h},\\ V_{h,t} &= W_{V,h}U_t + b_{V,h}, \qquad S_{h,ts} = \tau_h\langle Q_{h,t}, K_{h,s}\rangle,\\ A_{h,ts} &= \frac{e^{S_{h,ts}}}{\sum_{r\in I_{h,t}} e^{S_{h,tr}}} \quad (s\in I_{h,t}), \qquad O_{h,t} = \sum_{s\in I_{h,t}} A_{h,ts}V_{h,s}. \end{aligned} \tag{5.8}$$

Masked coefficients outside $I_{h,t}$ are zero. The usual scaled dot product is the choice $\tau_h = 1/\sqrt{d_{q,h}}$. Summing the per-head output projections is equivalent to concatenation followed by one output projection. The residual attention and pointwise feedforward updates are

$$\begin{aligned} Y_t &= X_t + \sum_h W_{O,h}O_{h,t} + b_O,\\ B_k(X)_t &= Y_t + W_2\varrho_k(W_1\,\mathrm{LN}_{k,2}(Y_t) + b_1) + b_2. \end{aligned} \tag{5.9}$$

Both LayerNorms have positive epsilon and arbitrary signed learned scale and bias as in (5.5). The activation ϱ_k is chosen from ReLU, exact Gaussian GELU, Swish/SiLU, sigmoid, and softplus, with the positive parameters required in Chapter 3. Head widths, feedforward widths, masks, and parameters may differ between blocks, while the residual width is d.

For any finite depth $K \geq 0$, define

$$\begin{aligned} X^{(0)} &= X + P, \qquad X^{(k)} = B_k(X^{(k-1)}) \quad (1\leq k\leq K),\\ \widehat{X}^{(K)} &= \begin{cases} \mathrm{LN}_{\mathrm{out}}(X^{(K)}), & \text{if a final norm is present,}\\ X^{(K)}, & \text{otherwise,} \end{cases}\\ N(X) &= b_{\mathrm{out}} + \sum_{t=1}^{T}\sum_{j=1}^{d} w_{tj}^{\mathrm{out}}\,\widehat{X}_{tj}^{(K)}. \end{aligned} \tag{5.10}$$

The optional final norm is applied tokenwise with positive epsilon. The depth-zero case has no blocks and consists of the positional affine map, the optional final norm, and the readout. Choosing zero readout weights outside one token includes a class-token logit. This is a deterministic inference computation; a fixed dropout mask can additionally be represented by fixed coordinate scalings.

5.3.3 The recursive split algorithm

Expand the affine maps, dot products, and matrix contractions in Definition 5.2 into scalar expressions. The expression constructors are inputs, real constants, sums, signed scalar multiples, products, supported scalar activations, softmax coordinates, and LayerNorm coordinates. Their ordinary evaluation uses the usual real functions in (5.8)–(5.10). No convexity certificate is assumed as part of an ordinary expression.

Write $\mathscr{C}(e) = (e^+,e^-)$ for the recursively constructed pair, as a function of independent non-negative input streams (x,y). Pair addition is componentwise, and signed scalar multiplication is

$$\mathrm{Scale}_c(p,q) = \begin{cases} (cp,cq), & c \geq 0, \\ (-cq,-cp), & c < 0. \end{cases} \tag{5.11}$$

The operators Prod, $\mathrm{Softmax}_i^{\pm}$, and $\mathrm{LN}_i^{\pm}$ below are the explicit constructions of (5.3), Theorem 3.12, and the LayerNorm recipe following Proposition 5.4. In particular, the LayerNorm pair computes the centered pairs, their square pairs, their average, the reciprocal-square-root pair, and the final product and affine parameters; it retains the input-dependent statistics.

Algorithm 5.1: Recursive DC evaluation of a finite Transformer expression

Input: A scalar output expression e from Definition 5.2; split inputs $x,y \geq 0$.
Output: A pair $\mathscr{C}(e)(x,y)$ whose difference is the ordinary output at $x - y$.
For an input coordinate i, return $\mathscr{C}(X_i) = (x_i,y_i)$; for a constant c, return $(\max\{c,0\}, \max\{-c,0\})$.
Recursively compute the pair of each child expression before applying its parent operation.
For a sum $\sum_j e_j$, return $\sum_j \mathscr{C}(e_j)$; for ce_1, return $\mathrm{Scale}_c(\mathscr{C}(e_1))$.
For e_1e_2, return $\mathrm{Prod}(\mathscr{C}(e_1),\mathscr{C}(e_2))$ using the four-square formula (5.3).
For a scalar activation $\varrho(e_1)$, apply its explicit pair to $\mathscr{C}(e_1)$.
For a softmax coordinate, collect the compiled allowed-score pairs and apply $\mathrm{Softmax}_i^{\pm}$ to that vector of pairs.
For a LayerNorm coordinate, collect the compiled feature pairs and apply $\mathrm{LN}_i^{\pm}$ with its positive epsilon and signed scale and bias.
Return the pair at the requested affine readout.

This is a recursive algorithm over real arithmetic. It is defined for every finite expression, independently of a chosen network depth. The Transformer-to-expression translation iterates over the finite block list: the empty list leaves its input expression unchanged, and appending a block substitutes the current token expressions into that block's explicit formulas. The ordinary evaluation of the resulting expression is (5.10), by induction over that list. Shared intermediate expressions may be reused in an implementation, but no bound on expression expansion or practical runtime is needed for the correctness claim.

5.3.4 The full Transformer construction

Theorem 5.5 (Exact DC decomposition of a deterministic Transformer, `Lean_77`). *Let $N\colon \mathbb{R}^{T\times d} \to \mathbb{R}$ be the Transformer of Definition 5.2, with any finite list of blocks, including the empty list, and identify its token-feature input with $\mathbb{R}^m$, where $m = Td$. Algorithm 5.1, applied to its output expression, terminates and exactly decomposes this forward map. It constructs functions $N_+,N_-\colon \mathbb{R}^{2m}_{\geq 0} \to \mathbb{R}_{\geq 0}$, convex and coordinatewise non-decreasing on that orthant, such that*

$$N(x-y) = N_+(x,y) - N_-(x,y) \qquad (x,y \in \mathbb{R}^m_{\geq 0}). \tag{5.12}$$

In particular, the functions

$$g(u) = N_+([u]^+,[u]^-), \qquad h(u) = N_-([u]^+,[u]^-) \tag{5.13}$$

are convex on all of $\mathbb{R}^m$, are non-negative, and satisfy $N = g - h$. The functions N_+,N_- are coordinatewise non-decreasing in the independent split inputs; g,h need not be coordinatewise non-decreasing in the original signed variable u.

Proof. The block-list construction yields a finite expression, so recursion on its syntax terminates. We prove by structural induction that every compiled expression has non-negative convex, coordinatewise non-decreasing heads on the input orthant and that their difference equals its ordinary evaluation at $x - y$.

Inputs satisfy the invariant by coordinate projection, and constants by their positive/negative split. Sums preserve every property. Signed scaling either rescales both heads by a non-negative coefficient or swaps them before rescaling, proving the scalar-multiple case. For a product, Proposition 5.3 supplies exact recovery and convex, monotone composition on non-negative incoming streams. The supported activations satisfy the corresponding scalar pair theorem. Softmax uses Theorem 3.12 on the nonempty allowed-key vector. LayerNorm uses its explicit centering, square, reciprocal, product, and affine construction; its recovered variance is non-negative and its epsilon is positive. Thus every expression constructor preserves the invariant.

Induction over the block list identifies ordinary evaluation of the compiled output expression with the independently specified forward recurrence (5.10). The expression invariant therefore proves (5.12) at every finite depth, including zero, with no activation at the signed affine readout. Finally, the maps $u \mapsto [u]^{\pm}$ have convex components and take values in the orthant. Convex monotone composition proves global convexity of g,h, and subtraction recovers $N(u)$. □

The construction retains the full input dependence of Queries, Keys, Values, softmax, and LayerNorm statistics. It gives a mathematical DC representation of the forward map. Its existence does not establish numerical stability of the stream evaluation or attribution quality. The reciprocal-square-root pair already adds a common term with coefficient $M = (2\varepsilon^{3/2})^{-1}$, which grows rapidly as ε decreases. Repeated products and squares can further amplify this common mass, so the recovered output may require cancellation between extremely large streams. The real-arithmetic proof supplies no floating-point error bound or practical runtime guarantee. Section S.4.5 of the companion supplement records a numerical diagnostic using a post-normalization block assembled from the same primitive pairs; its LayerNorm placement differs from Definition 5.2. In that unstabilized float64 diagnostic, 15 of 20 cases exceed 10^{-8} output error and stream magnitudes reach 5.95×10^{111}. The attribution experiments in this thesis use the conditioned routing and attribution rules described in the following sections; they do not evaluate this new convex Transformer construction.

5.4 Attention via a detached-logit oracle

Two ViT subtleties break conservative transport: LayerNorm's input-dependent statistics and the data-dependent QK-softmax. LayerNorm statistics are detached so the layer becomes a pointwise linear map (Ali et al., 2022; Wróbel et al., 2026); the QK-softmax block is replaced by an *oracle game* emitting a detached reference policy, the softmax row $A_{t\cdot}$, with occupation-measure trajectories flowing only through the Value route. Like conservative propagation (Ali et al., 2022), this explains the network conditioned on its realized attention pattern.

Theorem 5.6 (Attention, informal). *With reference policy $A_{t\cdot}$, the KL-regularized Value-routing subgame has an SPNE whose occupation measure at the block input equals $\alpha\beta$-LRP on the conditioned effective map $x \mapsto \sum_k A_{tk}\, x_k W_v + b_v$ (Theorem S.2.3).*

Our implementation applies this rule to the attention product and the Value projection as one composite step; a full gradient fallback through the attention block degrades localization (AL 0.624 vs. 0.711 at the localization-selected configuration). Equivariant Reynolds averaging optionally suppresses tokenization-grid artifacts (Wróbel et al., 2026) (RG+Eq). The target rule, oracle embedding and implementation are in Section 5.5; the full KL formulation and occupation-measure proof are in Chapter S.2 of the companion technical supplement.

5.5 Extension: Vision Transformer Attention

We state the target rule, oracle interpretation and implementation used by the ViT experiments. The full state-space formulation, KL derivation and occupation-measure proof are given in Chapter S.2 of the companion technical supplement; the risk-averse attention variant is in Section S.2.5.

5.5.1 Attention as fixed-reference KL Value routing

Target rule: $\alpha\beta$-LRP with detached logits

We first state the rule that the Routing-Game state-space construction in Section S.2.1 is designed to recover. We focus on a single attention head; the multi-head extension carries an additional head index h and the recovery is per-head. Note that ViT-B/16 implements the Value projection as a single $D \times D$ linear $v = XW_v$ followed by a reshape into n_h heads (the per-head matrix $W_v^{(h)} \in \mathbb{R}^{D\times d_h}$ is the head-h column block of the full W_v), so one input embedding $X_{k,e}$ feeds Value entries of every head and the per-head R^X contributions sum at $X_{k,e}$ to give the total. The per-head description below applies unchanged. We omit the ε stabilizer inside the attention block — the surrounding ReLU recursion of §S.1.3.2 carries its own ε on every linear-routing state and that mechanism transfers to the Av+W_v composite without modification, but tracking it through the tensor indexing here only obscures the structure.

Forward pass. The Value projection is applied per token: for each $k \in \{1,\dots,S\}$,

$$v_{k,d} \;=\; \sum_{e} W_{v,e,d}\, X_{k,e} + b_{v,d}, \qquad d \in \{1,\dots,d_h\}, \tag{5.14}$$

where the same matrix $W_v \in \mathbb{R}^{D\times d_h}$ and bias $b_v \in \mathbb{R}^{d_h}$ are applied independently to every token. The detached attention matrix $A \in \mathbb{R}^{S\times S}_{\geq 0}$ (with $A_{t,k} \geq 0$ and $\sum_k A_{t,k} = 1$ from softmax) then mixes Values across tokens *independently per feature dimension* d: for each output token $t \in \{1,\dots,S\}$ and feature d,

$$O_{t,d} \;=\; \sum_{k} A_{t,k}\, v_{k,d} \;=\; \sum_{k,e} A_{t,k}\, W_{v,e,d}\, X_{k,e} + b_{v,d}. \tag{5.15}$$

Composite linear map and κ-stream split. Holding A, the LayerNorm statistics and the GELU gates at their forward values (§5.5.3) is a modeling assumption about the explanation question—given the realized attention policy, through which Value routes does relevance flow?—and not a theorem about relevance through Q and K. Our attention extension is therefore a Value-only attribution conditioned on the realized attention matrix, resembling the conservative propagation of Ali et al. (2022) (CP-LRP), which holds the attention matrix constant, and DAVE's conditioned operators (Wróbel et al., 2026). It differs from AttnLRP (Achtibat et al., 2024), which also propagates relevance through the

attention-generating operations (the softmax and the bilinear QK product) and so attributes the Query and Key paths. With A so conditioned, the attention block reduces to the affine *conditioned effective map* $X \mapsto O$ with composite weights $W^{\text{eff}}_{(k,e),\,(t,d)} = A_{t,k}\, W_{v,e,d}$ and bias $b_{v,d}$. Since $A_{t,k} \geq 0$, the sign of the composite weight is the sign of $W_{v,e,d}$:

$$\left[\, W^{\text{eff}}_{(k,e),\,(t,d)} \,\right]^{\kappa} \;=\; A_{t,k}\, W^{\kappa}_{v,e,d}, \qquad \kappa \in \{+,-\}. \tag{5.16}$$

For signed inputs X (LayerNorm outputs), the per-token κ-stream split of the Value tensor (as in the fused input layer of the RG, with the bias as a bias node) and the per-output-feature stream sum are

$$\tilde{v}^{\kappa}_{k,d} \;\coloneqq\; \sum_{e} [X_{k,e} W_{v,e,d}]^{\kappa} + [b_{v,d}]^{\kappa}, \qquad Z^{\kappa}_{t,d} \;\coloneqq\; \sum_{k} A_{t,k}\, \tilde{v}^{\kappa}_{k,d} \;=\; \sum_{k,e} A_{t,k}\, [X_{k,e} W_{v,e,d}]^{\kappa} + [b_{v,d}]^{\kappa}. \tag{5.17}$$

Note that $\tilde{v}^{\kappa}_{k,d}$ is the κ-stream split inherited from W_v and the sign of X (not the standard scalar positive part $[v_{k,d}]^{+} = \max(0, v_{k,d})$); the two coincide only when $v_{k,d}$ has a single-signed predecessor decomposition.

Target rule. Let $R^{O}_{t,d}$ denote the relevance arriving at the output entry $O_{t,d}$ from the surrounding LRP recursion. We define the target rule as the $\alpha\beta$-LRP rule (4.1) on the composite map $X \mapsto O$, written in two stages that carry the κ-stream split $\tilde{v}^{\kappa}_{k,d}$ inherited from W_v via (5.16) through the Value entries. It is not sequential (layer-wise) $\alpha\beta$-LRP on $X \to v \to O$, which splits by the sign of the netted Value entry $v_{k,d}$ and applies (α,β) at both stages; the two rules differ whenever a Value entry has mixed-sign predecessors. The sequential variant is the alternative that this paper does not use: the implementation (Section 5.5.3) and every ViT result reported here apply the composite rule stated below, and the numerical identity between the implementation and (5.21) is checked by the tests of Section S.4.3.4.

Zero-mass convention. All ratios of non-negative contributions in this subsection use

$$\rho(u,D) \coloneqq \begin{cases} u/D, & D > 0, \\ 0, & D = 0, \end{cases} \qquad 0 \leq u \leq D. \tag{5.18}$$

Thus every displayed fraction with a zero denominator is zero. A zero-mass routing state sends probability one to the cemetery and zero to its ordinary successors; a zero vector of ordinary successor probabilities is not itself a probability distribution. In particular, this convention applies both to $Z^{\kappa}_{t,d}$ and to the intermediate Value mass $\tilde{v}^{\kappa}_{k,d}$. A branch with $Z^{\kappa}_{t,d} = 0$ has zero contribution $A_{t,k}\tilde{v}^{\kappa}_{k,d}$ for every key. If $\tilde{v}^{\kappa}_{k,d} = 0$, every matching input and bias contribution is zero as well. The ideal formulas below use $\varepsilon = 0$; the released implementation additionally supports positive stabilization, which replaces the composite denominator by $Z^{\kappa}_{t,d} + \varepsilon$ and assigns the remaining share to the outside option.

AV mixing $O \to v$. Applied to $O_{t,d} = \sum_k A_{t,k}\, v_{k,d}$, the rule gives the relevance at the Value entry $v_{k,d}$ as the weighted difference of two per-stream contributions:

$$R^{\text{val}}_{k,d} \;=\; \alpha\, R^{\text{val},+}_{k,d} \;-\; \beta\, R^{\text{val},-}_{k,d}, \qquad R^{\text{val},\kappa}_{k,d} \;\coloneqq\; \sum_{t} \frac{A_{t,k}\, \tilde{v}^{\kappa}_{k,d}}{Z^{\kappa}_{t,d}}\, R^{O}_{t,d}, \quad \kappa \in \{+,-\}. \tag{5.19}$$

Value projection $v \to X$. Applied to $v_{k,d} = \sum_e W_{v,e,d}\, X_{k,e} + b_{v,d}$, the rule routes each κ-stream contribution $R^{\mathrm{val},\kappa}_{k,d}$ to the input dim e via the matching κ-stream ratio $[X_{k,e}W_{v,e,d}]^\kappa / \tilde{v}^\kappa_{k,d}$ (the bias share is absorbed):

$$R^X_{k,e} \;=\; \sum_d \left(\alpha\, \frac{[X_{k,e}W_{v,e,d}]^+}{\tilde{v}^+_{k,d}}\, R^{\mathrm{val},+}_{k,d} \;-\; \beta\, \frac{[X_{k,e}W_{v,e,d}]^-}{\tilde{v}^-_{k,d}}\, R^{\mathrm{val},-}_{k,d} \right). \tag{5.20}$$

Aggregate. Substituting (5.19) into (5.20) and canceling $\tilde{v}^\kappa_{k,d}$ when it is positive expresses $R^X_{k,e}$ directly in terms of $R^O_{t,d}$. When it is zero, both forms vanish by the zero-mass convention; the same holds when $Z^\kappa_{t,d} = 0$. Thus the identity holds in every case:

$$R^X_{k,e} \;=\; \sum_{t,d} \left(\alpha\, \frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^+}{Z^+_{t,d}} \;-\; \beta\, \frac{A_{t,k}\,[X_{k,e}W_{v,e,d}]^-}{Z^-_{t,d}} \right) R^O_{t,d}, \tag{5.21}$$

which coincides with what the standard $\alpha\beta$-LRP rule produces directly on the conditioned effective map $X \mapsto O$ with effective weight $W^{\mathrm{eff}}_{(k,e),(t,d)} = A_{t,k}\, W_{v,e,d}$ and bias $b_{v,d}$.

The state-space construction and the proof of Theorem S.2.3 are given in Chapter S.2 of the companion technical supplement.

5.5.2 The reference as the policy of a QK oracle player

§5.5.1 treated the softmax row $A_{t,\cdot}$ as a fixed reference policy. It is itself the entropy-regularized best response of an auxiliary *oracle* player whose payoff is the detached query–key logit $e_{tk} = Q_t K_k^\top / \sqrt{d_h}$: by (S.1.56) at temperature 1,

$$\begin{aligned} A_{t,\cdot} \;&=\; \mathrm{argmax}_{\pi \in \Delta(\{1,\dots,S\})} \Big\{ \textstyle\sum_k \pi(k)\, e_{tk} + \mathscr{H}(\pi) \Big\}, \\ \max_\pi \Big\{ \textstyle\sum_k \pi(k)\, e_{tk} + \mathscr{H}(\pi) \Big\} \;&=\; \log \textstyle\sum_k \exp(e_{tk}). \end{aligned} \tag{5.22}$$

The oracle acts in the forward pass only and hands $\bar{\pi}_t = A_{t,\cdot}$ to the Value-routing player; its states carry no occupation mass, so no relevance flows through Q or K (§5.5.3).

5.5.3 Implementation details

The forward and backward passes of the RG on ViT-B/16 share four operator-level conventions that together preserve LRP conservation through the non-linearities of the ViT and align the explanatory walk with the state-space construction of §S.2.1. We refer to this collectively as the DAVE-style *conditioned forward pass* (Wróbel et al., 2026), augmented with the QK gradient detachment that the QK oracle of §5.5.2 requires.

LayerNorm detachment. Standard backpropagation through $\mathrm{LN}(x) = w_{\mathrm{LN}} \odot (x - m(x))/s(x) + b_{\mathrm{LN}}$ routes gradients through the data-dependent statistics $m(x), s(x)$, which couple the feature coordinates of each token (LayerNorm acts per token; it does not mix tokens) and violate LRP conservation (Ali et al., 2022). Following the detached-operator construction of Wróbel et al. (2026), we compute m, s from a detached copy $c = x.\mathrm{detach}()$, so the backward pass sees only the pointwise map $x \mapsto w_{\mathrm{LN}} \odot (x - m(c))/s(c) + b_{\mathrm{LN}}$ and conservation is preserved.

GELU gate detachment. GELU gates are conditioned analogously: the gate is implemented as $x \cdot \Phi(c)$ with $c = x.\text{detach}()$ and Φ the standard normal CDF, so the backward pass treats the gate as a pointwise scaling by the (constant) activation $\Phi(c)$.

Softmax detachment. Softmax outputs are fully detached. This matches the QK oracle interpretation of §5.5.2: the cached softmax row $A_{t,\cdot}$ is the equilibrium policy of an auxiliary entropy-regularized problem, supplied as a fixed input to the Value-routing player; no trajectory of the explanatory walk enters Q or K.

QK gradient detachment. The softmax and QK-matmul backward hooks return zero gradients, ensuring that the QK pathway is used *only* for computing the detached kernel $A_{t,\cdot}$ and that no relevance flows through Q or K. The QK oracle therefore supplies the KL reference, and all explanatory mass through the attention block flows through the Value tensor. The implemented Value rule is the composite rule of Theorem S.2.3: each Value entry keeps the stream masses $\tilde{v}^{\kappa}_{k,d}$ of its Value-projection products, the attention step routes stream κ in proportion to $A_{t,k}\tilde{v}^{\kappa}_{k,d}$, and the Value projection routes stream κ only to the contributions $[X_{k,e}W_{v,e,d}]^{\kappa}$ of the same sign, so the discounts (α,β) act once per block, as in (5.21). This composite rule is the only attention–Value rule in the released code (option `av_rule='merged'`); the sequential variant of §5.5.1 was not used for any reported number. The identity tests of Section S.4.3.4 check the implementation against an independent reference of (5.21).

Together the four conventions realize the detached-operator semantics of Wróbel et al. (2026) for the RG backward pass, so that its explanatory mass flows only through the linear effective transformation. They implement the explanation question of §5.5.1 (Value-only attribution conditioned on the realized attention matrix, as in CP-LRP (Ali et al., 2022)) and are not the rules of AttnLRP (Achtibat et al., 2024), which propagates relevance through the softmax and the QK product as well.

5.5.4 Equivariant Spatial Averaging

The 16×16 patch tokenization breaks translation equivariance. Following DAVE's transform-averaging strategy (Wróbel et al., 2026), RG+Eq computes explanations on transformed inputs, maps each explanation back to the original image coordinates, and averages the aligned maps (a Monte-Carlo Reynolds average),

$$\mathscr{M}_N(x) \;=\; \frac{1}{N}\sum_{n=1}^{N} T_n^{-1}\big[\mathscr{M}(T_n x)\big], \tag{5.23}$$

where $\mathscr{M}$ is the RG attribution map and the T_n are i.i.d. random rotations (uniform in $\pm 10°$), translations (up to $\pm 5\%$ of the image extent per axis) and horizontal flips (probability 0.5), with $N = 50$ and no input noise. The conditioned forward pass of §5.5.3 separates operator variation from the effective transformation, in the same spirit as DAVE's conditioned-operator argument (Wróbel et al., 2026). The transforms do not form a group, so the average is an optional smoothing mode without an exactness claim.

6 Saliency and Attribution Methods

This chapter introduces the novel interpretability methods enabled by the split-stream representations. The guiding objects throughout are the stream functions g and h with $f = g - h$ constructed in Section 3.2.3: every method below reads the two streams *separately* instead of only their difference. Section 6.1 develops the *split-based* route — attributing the decisions of a pretrained network through its DC decomposition, yielding SplitCAM, SplitLRP and SplitGrad (Zimmermann and Loho, 2026), together with the complementary by-design route of difference-of-convex input neural networks (DCINNs), in which the difference of two constrained networks is *trained* directly and exhibits class-dependent stream sensitivities in the displayed MNIST examples. Section 6.2 turns to the *game-based* route of Chapter 4: rather than re-deriving the stopping and routing games, it collects the practical *modes* — the tunable knobs that turn the game equilibria into concrete attribution maps — and the *technicalities* of equivariant averaging that raise attribution quality.

6.1 Split-Based and Difference-of-Convex Attribution

This section develops the split-based methods on a shared conceptual basis, then instantiates SplitCAM, SplitLRP and SplitGrad on a pretrained network and closes with the by-design DCINNs. Post-hoc attribution methods assign to every input feature a score quantifying its influence on a fixed network output. The established families include gradient saliency (Simonyan et al., 2014), deconvolution and guided backpropagation (Springenberg et al., 2015; Zeiler and Fergus, 2014), LRP and deep Taylor decomposition (Bach et al., 2015; Montavon et al., 2017), class-activation mapping (Selvaraju et al., 2017; Zhou et al., 2016), axiomatic methods such as Integrated Gradients and DeepLIFT (Shrikumar et al., 2017; Sundararajan et al., 2017), noise-averaged gradients (Smilkov et al., 2017), and perturbation- or surrogate-based explanations (Fel et al., 2023; Fong and Vedaldi, 2017; Kolek et al., 2023; Lundberg and Lee, 2017; Ribeiro et al., 2016). All of them ultimately probe the same computational graph: the original network with its ReLU gates. This section identifies a structural information loss that this graph inflicts on every backward-pass method, and shows that the split-pair representation of Section 3.2.3 removes it.

Throughout this chapter we consider a feedforward ReLU network with layers $1, \dots, L$, weight matrices $W^{(l)}$, pre-activations $z^{(l)} = W^{(l)} a^{(l-1)}$ and hidden activations $a^{(l)} = \mathrm{ReLU}(z^{(l)})$ for $l < L$, with the ungated readout $a^{(L)} = z^{(L)}$, attributing a scalar output (a fixed class logit), so that $f^{(l)} \colon \mathbb{R}^{d^{(l)}} \to \mathbb{R}$ denotes the function computed by layers $l+1, \dots, L$ from the layer-l activations; biases are absorbed as in Section 3.2.3. The split-pair consists of the stream functions $g^{(l)}, h^{(l)} \colon \mathbb{R}^{d^{(l)}} \times \mathbb{R}^{d^{(l)}} \to \mathbb{R}$ operating on the split activations $a^{(l,+)}, a^{(l,-)}$, with the correctness guarantee

$$f^{(l)}\big(a^+ - a^-\big) = g^{(l)}(a^+, a^-) - h^{(l)}(a^+, a^-) \qquad \text{for all } a^+, a^- \in \mathbb{R}^{d^{(l)}}, \tag{6.1}$$

and we write $g \coloneqq g^{(0)}$, $h \coloneqq h^{(0)}$ with the balanced input split $a^{(0,\pm)} = \pm x/2$. All derivatives in this chapter are treated as row vectors (Jacobians of scalar functions), and statements involving derivatives hold at every differentiability point, hence Lebesgue-almost everywhere, since all functions involved are CPWL.

A ReLU gate transmits its pre-activation if it is positive and emits an exact zero otherwise. In the forward pass the *magnitude* of a negative pre-activation is annihilated; in the backward pass the local derivative of the gate is zero, so no gradient traverses the neuron either. We call this the *subthreshold signal loss*: all information carried by inactive neurons is invisible both to the network

output and to every attribution method that propagates through the original graph. The consequence for gradient-based attribution is not merely attenuation but exact regional blindness.

Proposition 6.1 (Regional blindness of the input gradient, Lean_67). *Let f be a ReLU network as above, let x_0 be an input at which no pre-activation is exactly zero, and let i be an input coordinate such that every path from input neuron i to the output neuron passes through at least one neuron that is inactive at x_0. Then there is an open neighborhood $U \ni x_0$ with*

$$\frac{\partial f}{\partial x_i} \equiv 0 \quad \text{on } U,$$

even if f depends on x_i elsewhere.

Proof. Write $M^{(l,+)} := \operatorname{diag}\big(\mathbf{1}\{z^{(l)} > 0\}\big)$ and $M^{(l,-)} := \operatorname{diag}\big(\mathbf{1}\{z^{(l)} \le 0\}\big)$ for the activation masks of the hidden ReLU layers $1 \le l < L$, and $W^{(l,x)} := M^{(l,+)} W^{(l)}$ for the corresponding masked weight matrices. The readout layer L produces the class logit through a purely affine map with *no* ReLU gate — the logit may be negative — so it carries no activation mask, and we set $W^{(L,x)} := W^{(L)}$; this unmasked factor is also the base case $\partial f / \partial a^{(L-1)} = W^{(L)}$ of the chain-rule iteration below. Since no hidden pre-activation vanishes at x_0 and pre-activations depend continuously on the input, all masks are locally constant: there is an open $U \ni x_0$ on which the activation pattern is fixed and f is affine. On U, iterating the chain rule through the layers gives

$$\frac{\partial f}{\partial x} = W^{(L,x)}\, W^{(L-1,x)} \cdots W^{(1,x)}, \tag{6.2}$$

and expanding the matrix product, the i-th entry is the sum, over all paths from input neuron i through the hidden layers to the output, of the product of the traversed weights multiplied by the indicator that every *hidden* neuron on the path is active (the ungated output neuron carries no such indicator). By assumption every such path contains an inactive hidden neuron, so every summand vanishes, and the expression is constant on U. □

The phenomenon is not an artifact of adversarial constructions; it occurs in the smallest possible network.

Remark 6.1 (The canonical counterexample). Following Sundararajan et al. (2017), consider the one-dimensional ReLU network

$$f(x) = 1 - \mathrm{ReLU}(1 - x), \tag{6.3}$$

and compare the input $x = 2$ against the baseline $x' = 0$. The prediction changes, $f(2) - f(0) = 1 - 0 = 1$, so the single feature is plainly relevant; yet f is constant on $x \ge 1$, so the gradient at $x = 2$ — and with it Input×Gradient and every purely local backward rule — assigns exactly zero attribution. The single hidden neuron is inactive at $x = 2$, and Proposition 6.1 applies with $U = (1, \infty)$.

The failure mode above was axiomatized by Sundararajan et al. (2017) as a minimal requirement on attribution methods.

Definition 6.1 (Sensitivity). An attribution method A, mapping an input x and a baseline x' to feature scores $A_i(x, x') \in \mathbb{R}$, satisfies *Sensitivity* if for every pair (x, x') that differs in exactly one feature i and satisfies $f(x) \neq f(x')$, the attribution of that feature is non-zero, $A_i(x, x') \neq 0$.

Plain gradients violate Sensitivity by Remark 6.1. Integrated Gradients restores the axiom by integrating the gradient along the segment from x' to x, at the cost of many forward–backward passes and a baseline choice that the explanation inherits. The split-pair offers a structurally different resolution: instead of repairing the attribution rule, it repairs the *representation* on which the rule operates, so that the subthreshold signal is never destroyed in the first place.

Recall from Section 3.2.3 how the split-pair processes a ReLU gate: the pre-activation pair $z^{(l,+)}, z^{(l,-)}$ is mapped to

$$a^{(l,+)} = \max\{z^{(l,+)}, z^{(l,-)}\}, \qquad a^{(l,-)} = z^{(l,-)}, \tag{6.4}$$

a rank-2 Maxout unit in the g-stream and the identity in the h-stream. The difference reproduces $\mathrm{ReLU}(z^{(l)})$ by the splitting identity, but inactivity does not force the individual streams to vanish: when the original neuron is inactive, both $a^{(l,+)}$ and $a^{(l,-)}$ equal $z^{(l,-)}$ and still record the full magnitude of the suppressed signal. The subthreshold information is thus not annihilated but relocated — from the value of the difference into the common level of the two streams. The same happens in the backward pass, which we now make precise.

Definition 6.2 (Local sensitivities). For $s \in \{g, h\}$, $\sigma \in \{+, -\}$ and $0 \le l \le L$, the *local sensitivities* of the split-pair at layer l are the row vectors

$$\delta^{(l,\sigma,s)} := \frac{\partial s^{(l)}}{\partial a^{(l,\sigma)}} \in \mathbb{R}^{1\times d^{(l)}},$$

the derivatives of the stream outputs with respect to the positive and negative split activations of layer l. We further abbreviate the original-network sensitivity by $\delta^{(l)} := \partial f^{(l)}/\partial a^{(l)}$ and define the *absolute weight product*

$$W_{\mathrm{abs}}^{[l]} := |W^{(L)}|\,|W^{(L-1)}|\cdots|W^{(l+1)}| \in \mathbb{R}^{1\times d^{(l)}}, \qquad W_{\mathrm{abs}}^{[L]} := \mathbf{1}^\top,$$

where $|\cdot|$ is taken entrywise. Since the attributed output is scalar, $W_{\mathrm{abs}}^{[l]}$ is a row vector, and its i-th entry sums the absolute weight products over *all* paths from neuron i of layer l to the output — active or not.

The four local sensitivities obey coupled backward recursions — the split analogue of standard backpropagation and the computational core of every method in this chapter.

Lemma 6.2 (Split backward recursions, `Lean_68`). *With the activation masks $M^{(l,\pm)}$ of Proposition 6.1, extended by $M^{(L,+)} = I$ and $M^{(L,-)} = 0$ for the affine readout, for every $0 \le l < L$ and $s \in \{g, h\}$,*

$$\delta^{(l,+,s)} = \delta^{(l+1,+,s)}\big(M^{(l+1,+)}W^{(l+1,+)} + M^{(l+1,-)}W^{(l+1,-)}\big) + \delta^{(l+1,-,s)}\,W^{(l+1,-)}, \tag{6.5}$$

$$\delta^{(l,-,s)} = \delta^{(l+1,+,s)}\big(M^{(l+1,+)}W^{(l+1,-)} + M^{(l+1,-)}W^{(l+1,+)}\big) + \delta^{(l+1,-,s)}\,W^{(l+1,+)}, \tag{6.6}$$

with base case $\delta^{(L,+,g)} = \delta^{(L,-,h)} = \mathbf{1}^\top$ and $\delta^{(L,-,g)} = \delta^{(L,+,h)} = 0$.

These are the chain-rule recursions of the split forward equations of Section 3.2.3, derived in Section 3.6.2; the mask $M^{(l+1,+)}$ appears because the maximum in (6.4) is attained by $z^{(l+1,+)}$ exactly when the original neuron is active. The recursions immediately yield the central structural fact of this chapter: the *total* sensitivity mass carried by the two streams does not depend on the activation pattern at all.

Proposition 6.3 (Sensitivity mass conservation, Lean_69). *For every $0 \le l \le L$ and both streams $s \in \{g, h\}$,*

$$\delta^{(l,+,s)} + \delta^{(l,-,s)} = W_{\text{abs}}^{[l]}. \tag{6.7}$$

Proof. Induction on $l = L, \dots, 0$. The base case is $\mathbf{1}^\top + 0 = W_{\text{abs}}^{[L]}$. For the step, add (6.5) and (6.6): the mask terms combine via $M^{(l+1,+)} + M^{(l+1,-)} = I$ into

$$\delta^{(l,+,s)} + \delta^{(l,-,s)} = \big(\delta^{(l+1,+,s)} + \delta^{(l+1,-,s)}\big)\big(W^{(l+1,+)} + W^{(l+1,-)}\big) = \big(\delta^{(l+1,+,s)} + \delta^{(l+1,-,s)}\big)|W^{(l+1)}|,$$

since $W^{(l+1,+)} + W^{(l+1,-)} = |W^{(l+1)}|$ for the entrywise weight split. The induction hypothesis turns the right-hand side into $W_{\text{abs}}^{[l+1]}|W^{(l+1)}| = W_{\text{abs}}^{[l]}$. □

The relation between the split sensitivities and the original gradient is a one-line consequence of the correctness guarantee.

Corollary 6.4 (Backward invariant, Lean_70). *For every $0 \le l \le L$,*

$$\delta^{(l,+,g)} - \delta^{(l,+,h)} = \delta^{(l)} = -\big(\delta^{(l,-,g)} - \delta^{(l,-,h)}\big), \quad \textit{hence} \quad \big(\delta^{(l,+,g)} - \delta^{(l,-,g)}\big) - \big(\delta^{(l,+,h)} - \delta^{(l,-,h)}\big) = 2\,\delta^{(l)}. \tag{6.8}$$

Proof. Differentiate the correctness identity (6.1) with respect to a^+ and a^-, respectively, and subtract the two resulting identities. □

Together, Proposition 6.3 and Corollary 6.4 express a conservation law: the difference of the stream sensitivities reproduces the original gradient exactly, while their sum is pinned to the input-independent quantity $W_{\text{abs}}^{[l]}$. Where Proposition 6.1 shows the original gradient collapsing to zero, the two combined stream gradients coincide: writing $\delta^{(l,s)} := \frac{1}{2}\big(\delta^{(l,+,s)} - \delta^{(l,-,s)}\big)$ for the combined gradient of stream s, Corollary 6.4 gives $\delta^{(l,g)} - \delta^{(l,h)} = \delta^{(l)}$, so the vanishing of $\delta^{(l)}$ is precisely the equality $\delta^{(l,g)} = \delta^{(l,h)}$ of the two combined stream gradients — their difference, the network gradient, is zero. The common value may itself be zero. The individual sensitivities $\delta^{(l,\pm,s)}$ need not equal $\frac{1}{2}W_{\text{abs}}^{[l]}$; their deviation from that pattern-independent half-bulk records precisely the routing decisions taken at the ReLU gates, including the inactive ones. For a feature connected to the output by a path of nonzero weights, the sensitivity-mass identity guarantees that at least one of the two non-negative partial sensitivities of each head is positive. It does not guarantee that a difference of partials, a combined gradient, or an attribution map is nonzero. For example, with $f(x) = \mathrm{ReLU}(\mathrm{ReLU}(x) + \mathrm{ReLU}(-x) - 10)$ and the balanced input split, both heads are locally equal to 10 near $x = 1$ and both combined gradients vanish.

Remark 6.2 (The counterexample, revisited). Split the network (6.3) with the balanced input split $a^{(0,\pm)} = \pm x/2$. A short computation along Section 3.2.3 gives, for $x > 1$,

$$g(x) = \frac{x}{2} + 1, \qquad h(x) = \frac{x}{2}, \qquad g(x) - h(x) = 1 = f(x),$$

so at $x = 2$ both stream gradients equal $\frac{1}{2} \neq 0$ while $f'(2) = 0$: the split representation *sees* that the feature matters — consistent with $f(2) \neq f(0)$ — and encodes the vanishing of f' as the cancellation $\frac{1}{2} - \frac{1}{2}$. For this example and the baseline $x' = 0$, the pair of head gradients therefore supplies a nonzero signal even though the output gradient vanishes. This example does not establish the sensitivity axiom of Definition 6.1 for a general attribution rule or baseline. The attribution methods of the following

sections differ in how they turn this pair-valued signal into a scalar map; the shifted combinations of Section 6.1.3 keep the two streams from cancelling exactly, and in this example assign the feature a strictly positive attribution for every shift parameter $\alpha < \frac{1}{2}$.

There is also a conceptual reading of these results. Networks with non-negative weights implement a computation in which every neuron can only *add* evidence, never cancel it — a scheme long associated with interpretability and with biologically plausible processing (Ayinde and Zurada, 2018; Dayan and Abbott, 2001; Lee and Seung, 1999). The split-pair equips an arbitrary pretrained network with exactly this structure, at the price of doubling the width; the remainder of this chapter harvests the improved signal flow.

6.1.1 SplitCAM

CAM methods explain CNN decisions by combining spatial activation maps of a convolutional layer with class-specific weighting coefficients. The original CAM (Zhou et al., 2016) requires a global-average-pooling head; Grad-CAM (Selvaraju et al., 2017) generalizes it to arbitrary architectures by weighting each channel with the spatially pooled gradient of the class logit, and Grad-CAM++ (Chattopadhay et al., 2018) refines the pooling weights. LayerCAM (Jiang et al., 2021) drops the pooling altogether and weights every spatial location individually: for a layer l with channel activation maps $a_c^{(l)}$ and class logit y,

$$\mathrm{LayerCAM}^{(l)} = \mathrm{ReLU}\Big(\sum_c \frac{\partial y}{\partial a_c^{(l)}} \odot a_c^{(l)}\Big), \tag{6.9}$$

where $\odot$ denotes the elementwise product and the sum runs over the channels. This makes LayerCAM applicable to early layers with high spatial resolution, but it inherits both failure modes of Section 6.1: inactive neurons contribute neither activation nor gradient, and the outer ReLU discards all negative evidence.

SplitCAM transplants the LayerCAM formula onto the split-pair, replacing activations and gradients of the original network by their split-stream counterparts. Since the raw local sensitivities of deep split-pairs are numerically enormous — they are sums of *absolute* path products, cf. Proposition 6.3 — SplitCAM operates on the *stabilized* sensitivities of Section 3.6.2.

Definition 6.3 (Shifted local sensitivities, `Lean_71`). Fix shift parameters $\alpha^{(l)} \in [0, \frac{1}{2}]$. For $s \in \{g, h\}$ define, backward from layer L, intermediate sensitivities $\delta_{\mathrm{int}}^{(l,\pm,s)}$ by evaluating the recursions (6.5)–(6.6) (respectively the base case of Lemma 6.2 at $l = L$) with the already-shifted sensitivities of the subsequent layer, and set

$$\delta_{\mathrm{shift}}^{(l,\pm,s)} := \delta_{\mathrm{int}}^{(l,\pm,s)} - \alpha^{(l)}\big(\delta_{\mathrm{int}}^{(l,+,s)} + \delta_{\mathrm{int}}^{(l,-,s)}\big), \qquad \delta_{\mathrm{shift}}^{(l,s)} := \frac{\delta_{\mathrm{shift}}^{(l,+,s)} - \delta_{\mathrm{shift}}^{(l,-,s)}}{2}. \tag{6.10}$$

Both signs receive the *same* shift, so the informative differences are preserved layer by layer, while by Proposition 6.3 the subtracted direction is the pattern-independent bulk $\delta_{\mathrm{int}}^{(l,+,s)} + \delta_{\mathrm{int}}^{(l,-,s)} = \big[\prod_{i=l+1}^{L}(1 - 2\alpha^{(i)})\big] W_{\mathrm{abs}}^{[l]}$ (the empty product at $l = L$ is 1); the factor $\frac{1}{2}$ in the combined sensitivity $\delta_{\mathrm{shift}}^{(l,s)}$ matches the balanced input split. Unless stated otherwise all $\alpha^{(l)} \equiv \alpha$. For $\alpha = 0$ the shifted sensitivities are the raw ones; for $\alpha = \frac{1}{2}$ they collapse onto multiples of the original gradient (Chapter 3, Theorem 3.26), so the informative regime is $0 < \alpha < \frac{1}{2}$, where the split signal survives with a controlled, layer-dependent damping.

The SplitCAM attribution.

Definition 6.4 (SplitCAM). For a layer l of the decomposed model, a stream $s \in \{g, h\}$ and a sign $\sigma \in \{+, -\}$, the SplitCAM attribution maps are

$$\mathrm{SplitCAM}^{(l,\sigma,s)} := \sum_c \delta^{(l,\sigma,s)}_{\mathrm{shift}}\big|_c \odot a_c^{(l,\sigma)}, \tag{6.11}$$

$$\mathrm{SplitCAM}^{(l,s)} := \sum_c \delta^{(l,s)}_{\mathrm{shift}}\big|_c \odot a_c^{(l)}, \tag{6.12}$$

where $a_c^{(l,\sigma)}$ denotes the channel-c activation map of the σ-stream, $a_c^{(l)} = a_c^{(l,+)} - a_c^{(l,-)}$ the reconstructed original activation map, and $\cdot|_c$ the restriction of a sensitivity tensor to channel c.

Two deliberate deviations from (6.9) deserve comment. First, SplitCAM *omits the outer ReLU*: on the split-pair, evidence against the class is carried by a dedicated non-negative stream rather than by sign flips, so clipping would discard exactly the counter-evidence that the decomposition makes legible. Second, the gradients entering the formula are the shifted sensitivities rather than raw derivatives; by (6.10) this changes each map only by multiples of the pattern-independent vector $W^{[l]}_{\mathrm{abs}}$, but keeps all quantities in a numerically meaningful range.

Both passes run under the numerical stabilization of Section 3.6: the forward pass scales the split-activation pairs and re-imposes the forward invariant against the cached original activation, and the backward pass applies the shifting of Definition 6.3 with routing masks read from the cached original activations, so that the backward invariant (6.8) holds against the cached original gradient.

Layer choice and resolution. Like LayerCAM, SplitCAM is defined at every layer, and the resulting map is bilinearly upsampled to the input resolution. Empirically the method peaks in the deep convolutional blocks — for VGG-16 (Simonyan and Zisserman, 2015) at layer 26 — where it concentrates on entire classified objects noticeably better than LayerCAM at the same layer; the full layer-ablation study and the quantitative comparison on the Quantus benchmark (Hedström et al., 2023) for VGG-16 and ResNet-18 (He et al., 2016) are reported in Chapter 7.

6.1.2 SplitLRP

LRP (Bach et al., 2015) explains a prediction by propagating a relevance budget backward through the network, layer by layer, distributing the relevance of each neuron onto its inputs in proportion to their contributions; see Montavon et al. (2018) for an overview and Kohlbrenner et al. (2020) for best practices. Among the many propagation rules, the γ-rule of Montavon et al. (2019) emphasizes positive contributions: for an affine layer $z_i^{(l+1)} = \sum_j W_{ij}^{(l+1)} a_j^{(l)} + b_i^{(l+1)}$, one defines the γ-adapted contributions

$$\tilde{z}_{ij}^{(l+1)} := a_j^{(l)} \Big(W_{ij}^{(l+1)} + \gamma\, [W_{ij}^{(l+1)}]^+ \Big), \qquad \gamma \geq 0, \tag{6.13}$$

and redistributes the relevances $R_i^{(l+1)}$ onto the inputs as

$$R_j^{(l)} = \sum_i \frac{\tilde{z}_{ij}^{(l+1)}}{\sum_{j'} \tilde{z}_{ij'}^{(l+1)} + \varepsilon\, \mathrm{sign}\big(\sum_{j'} \tilde{z}_{ij'}^{(l+1)}\big)}\, R_i^{(l+1)}, \tag{6.14}$$

with a small stabilizer $\varepsilon > 0$. The choice $\gamma = 0$ recovers the classical ε-rule. Activation functions are transparent to LRP: the rule acts on the affine layers only.

Relevance propagation on the split-pair. *SplitLRP* runs LRP on the decomposed model, treating the positive and negative streams of each layer as two groups of neurons within one unified network. Accordingly, each layer carries *two* relevance tensors $R^{(l,+)}$ and $R^{(l,-)}$, propagated jointly through the four weight blocks built from $W^{(l,+)}$ and $W^{(l,-)}$ that couple the streams. The propagation is initialized at the class-specific positive output $a^{(L,+)}$ of the g-stream. Since *all* weights of the decomposed model are non-negative, the γ-rule loses its purpose — there are no negative contributions to de-emphasize — and we use (6.14) with $\gamma = 0$ and $\varepsilon = 10^{-6}$, i.e. the ε-rule, on the split-pair. Because the Maxout gates (6.4) are activation functions, they require no special treatment in the relevance pass.

The following conventions complete the specification.

- *Forward pass.* The split activations are computed with the same scaling stabilization and forward-invariant correction of Section 3.6 as SplitCAM.
- *Normalization.* No backward stabilization is needed: the denominator of (6.14) renormalizes relevance in every layer, so the total relevance carried by $R^{(l,+)}$ and $R^{(l,-)}$ together remains equal — up to the ε-stabilizer, cf. Remark 6.3 — to the initial budget $a^{(L,+)}$ throughout the backward pass, and this conservation replaces the shifting of Definition 6.3.
- *Pooling and normalization layers.* Relevance passes through max-pooling by the winner-takes-it-all scheme, assigning the full relevance to the location attaining the maximum; batch-normalization layers are merged into the adjacent linear or convolutional layers beforehand; residual additions split relevance evenly across branches.

Definition 6.5 (Combined SplitLRP map). The SplitLRP attribution at layer l is the pair $\big(R^{(l,+)}, R^{(l,-)}\big)$; the *combined* map is

$$R^{(l,\mathrm{comb})} := R^{(l,+)} - R^{(l,-)}. \tag{6.15}$$

The combination is motivated by the semantics that the decomposition imposes: relevance arriving in the positive stream traversed an even number of sign inversions and constitutes evidence *supporting* the prediction, while relevance in the negative stream constitutes evidence *against* it, so (6.15) aggregates support and opposition into a single signed saliency map.

Remark 6.3 (A probabilistic reading). On the original network the ε-rule mixes signed contributions, and the resulting backward transport is a *signed* measure on paths — the source of the interpretability caveats discussed in Chapter 8. On the split-pair all contributions $\tilde{z}_{ij}^{(l+1)}$ are products of non-negative activations and non-negative weights, so up to the stabilizer the LRP propagation matrices are genuinely stochastic: SplitLRP is a Markov chain on the doubled computational graph, and the pair $\big(R^{(l,+)}, R^{(l,-)}\big)$ is an honest, unsigned relevance measure. This is the same mechanism — doubling the state space to absorb the weight signs — by which the stopping game of Chapter 4 turns backward relevance flow into probability.

Remark 6.4 (Sensitivity to learned parameters). The parameter-randomization sanity check of Adebayo et al. (2018) differentiates the three split methods. After reinitializing the last three feedforward layers of VGG-16, every reported SplitCAM and SplitGrad configuration has lower SSIM than every listed baseline, whereas the HOG and Spearman comparisons are mixed. This establishes sensitivity to the tested parameter changes, not faithfulness to the learned decision process. SplitLRP retains high HOG and Spearman similarity despite a decrease in SSIM. Attenuation by the ε-LRP rule is a possible explanation, consistent with concerns about conservative propagation rules (Sixt et al., 2020), but this comparison does not identify the mechanism. Section S.4.1.4 reports the quantitative results and the caveats of Binder et al. (2023) on randomization-based checks.

6.1.3 SplitGrad

The third method dispenses with activation weighting and relevance budgets altogether and uses the stabilized stream gradients themselves as the attribution signal. The stream functions consume a *pair* of inputs, so we fix the balanced embedding $a^{(0,\pm)} = \pm x/2$ and write $\tilde{g}^{(l)}(a) \coloneqq g^{(l)}\big(\frac{a}{2}, -\frac{a}{2}\big)$, $\tilde{h}^{(l)}$ analogously. By the chain rule (Section 3.6.2) the unified stream gradient $\partial \tilde{s}^{(l)}/\partial a$ equals the combined sensitivity $\big(\delta^{(l,+,s)} - \delta^{(l,-,s)}\big)/2$, which justifies the factor $\frac{1}{2}$ in (6.10): the combined shifted sensitivities $\delta^{(l,g)}_{\text{shift}}, \delta^{(l,h)}_{\text{shift}}$ are stabilized gradients of the convex functions $\tilde{g}^{(l)}, \tilde{h}^{(l)}$, and their difference always reproduces the exact original gradient, since the common shift in (6.10) cancels sign-wise.

Definition 6.6 (SplitGrad). For a layer l and stream $s \in \{g, h\}$, the SplitGrad attribution map is the channel mean of the combined shifted sensitivity,

$$\text{SplitGrad}^{(l,s)} \coloneqq \frac{1}{C^{(l)}} \sum_{c=1}^{C^{(l)}} \delta^{(l,s)}_{\text{shift}}\big|_c, \tag{6.16}$$

where $C^{(l)}$ is the number of channels of layer l; the default configuration reads the g-stream, with forward and backward stabilization exactly as in Section 6.1.1.

What SplitGrad measures. Chapter 3 computes these stabilized gradients in closed form: by Theorem 3.26, the input-level stream gradients have original-gradient terms $\delta/2$ and $-\delta/2$, respectively, where $\delta \coloneqq \partial f/\partial x$, and share a subtracted, geometrically damped sum of *subthreshold probe-gradients* — one per layer, each carrying the signal routed through the *inactive* neurons whose loss causes the blindness of Proposition 6.1. Both streams contain this probe sum with the same sign, so it cancels in their difference, which reproduces δ exactly. The probe sum can itself vanish; the decomposition makes subthreshold contributions available without guaranteeing a nonzero attribution for every input. SplitGrad therefore recovers the feature the plain gradient misses: in the counterexample of Remark 6.2, $\delta = 0$ while the probe term is non-zero, so the feature is attributed correctly for every $\alpha < \frac{1}{2}$. The layer-wise damping factors $\overline{\alpha}_j$ of Theorem 3.26 both counteract the rapid growth of the probe-gradient norms toward the input — a failure otherwise reminiscent of shattered gradients (Balduzzi et al., 2017) — and let α interpolate between fidelity to the original gradient ($\alpha = \frac{1}{2}$) and full exposure of the split information ($\alpha = 0$); the sweep over α in Chapter 7 supports a metric- and configuration-dependent choice. For the two main-text configurations, localization optima occur at $\alpha = 0.35$ or 0.4, while Maximum Sensitivity is lowest at $\alpha = 0$; the full ablation is reported in Section S.4.2.

The three methods share one pipeline: a stabilized split forward pass caching the original activations, a stabilized backward pass — relevance-based for SplitLRP (initialized at the class-specific $a^{(L,+)}$ and propagated by the ε-rule), sensitivity-based otherwise (the shifted recursions of (6.10)) — and a method-specific readout, with the resulting map bilinearly upsampled to the input resolution. The network can be split once per model. In the reported implementation, SplitCAM takes approximately 14–29 times as long per image as the corresponding ordinary Grad-CAM and LayerCAM configurations; SplitGrad also takes substantially longer. Their timings are of the same order as the row labeled Guided Grad-CAM, but the released VGG-16 configuration dispatches Guided LayerCAM, limiting identification of that baseline. Section S.4.1.5 details this ambiguity and the measurement scope.

6.1.4 Directly Trained Difference-of-Convex Input Neural Networks

The methods of Sections 6.1.1–6.1.3 explain a *pretrained* network by imposing the split structure post hoc. This raises a converse question: if the difference-of-two-streams representation makes explanations legible, what happens when the difference structure is not derived but *trained* — when the hypothesis class itself is the set of differences of two constrained networks? This section studies DC input neural networks (DCINNs) and their monotone relatives. The displayed MNIST examples show class-dependent stream sensitivities on present strokes and candidate missing strokes, at a modest and architecture-dependent cost in accuracy.

Architecture and training. Recall from Section 2.2 that an ICNN (Amos et al., 2017) is a network with convex, monotone activation functions and non-negative weights in all layers except possibly the first, and that such a network represents a convex function; a network with monotone activations and *all* weights non-negative represents a monotone function. Throughout this section we view ICNNs as monotone networks acting on an affine transformation of the input, embodied by the unconstrained first layer.

Definition 6.7 (DCINN and difference-of-monotone models). Fix a base architecture. A *DCINN* (in short: DIC model) is the function

$$f_{\mathrm{DIC}} := g - h,$$

where g and h are two ICNNs of the base architecture with independent parameters, trained *jointly* by standard empirical risk minimization on the difference. The *DM model* is defined analogously with both streams monotone networks. The *inverted-input* variants DICIN and DMIN feed the h-stream the complemented input $\mathbf{1} - x$ (for inputs normalized to $[0,1]^d$) instead of x.

The construction mirrors the split-pair — two non-negative streams whose difference is the model — but with a crucial change of direction: nothing is split; both streams are free to learn any convex (respectively monotone) function, and the difference structure is a training-time inductive bias rather than a post hoc reparameterization. Closely related architectures appeared as convex difference neural networks (CDiNN) in optimal control (Sankaranarayanan and Rengaswamy, 2022), with recurrent extensions for safety-critical model predictive control (Chen, Wang, et al., 2025); the interpretability perspective developed here is complementary to that line of work. The inverted-input variant is motivated by monotonicity itself: a monotone h-stream can only accumulate evidence from the *presence* of input mass, so handing it $\mathbf{1} - x$ lets it accumulate evidence from what is *missing* in the input.

Three implementation choices matter in practice.

- *Activations.* Both streams use rank-2 Maxout units (Goodfellow et al., 2013), which are convex and monotone as required, and which consistently outperformed ReLU streams in the difference setting. This aligns the trained models exactly with the hypothesis class of Section 3.2.3: the split-streams obtained there *are* rank-2 Maxout networks with non-negative weights, so training DIC and DM models searches the very class into which the splitting maps.
- *Non-negativity.* Constrained weights are parameterized as $p(\tilde{W})$ with free parameters $\tilde{W}$ and a positivity transform p; among softplus, exponential and squaring, $p = \mathrm{ReLU}$ with absolute-value-of-Gaussian initialization gave the best test accuracy. The transform is also applied to biases and to the scale and shift parameters of batch-normalization layers.

– *Regularization.* Since non-negative weights preclude cancellation, activations can grow quickly; a small initialization scale and a weight decay of 10^{-3} keep the streams stable during training.

The empirical study uses two MNIST base architectures: `cnn_maxout`, a CNN with four convolutional layers (32, 64, 128 and 128 channels with batch normalization) followed by three fully connected layers, roughly 1.2 million parameters in total, and `mlp_maxout`, a wide four-layer MLP ($784 \to 800 \to 800 \to 800 \to 10$) with roughly 4.2 million parameters — each in eight variants: unconstrained (None), single-stream constrained (IC, Mon), and difference-of-two models (DIC, DM, DICIN, DMIN, and an unconstrained difference baseline).

Stream sensitivities and candidate counterfactual features. The by-design route exposes the *plain input gradients* of the trained streams without shifting, relevance rules, or probe networks. For an inverted-input DIC class logit $f(x) = g(x) - h(\mathbf{1} - x)$, the two gradients describe complementary local sensitivities:

– the g-stream gradient $\nabla g(x)$ describes the sensitivity of the contribution $g(x)$ to changes in the input;
– the h-stream gradient $(\nabla h)(\mathbf{1} - x)$ describes the sensitivity of the contribution $-h(\mathbf{1} - x)$ to increasing the original input: the input complement and the subtraction cancel the two minus signs in the chain rule.

At differentiable inputs, the complete model gradient is therefore

$$\nabla f(x) = \nabla g(x) + (\nabla h)(\mathbf{1} - x).$$

The first affine layer of an ICNN is unconstrained, so convexity of the streams does not make either gradient non-negative in the original input coordinates. A positive entry of one stream gradient can be outweighed by a negative entry of the other. Even a positive entry of the complete gradient only establishes an increase for sufficiently small feasible increases in that coordinate at a differentiable input; it does not establish a necessary feature or a change in the predicted class. Finite counterfactual claims require evaluating the complete model under the proposed intervention. The displayed maps are thus stream-local sensitivities; their concentration on present and candidate missing strokes motivates a factual/counterfactual interpretation to be tested by interventions.

Figure 6.1 overlays both gradients, per class logit, for a test digit "3". For the ground-truth logit the g-gradient traces the strokes most accurately, while for the competing classes "6", "8" and "9" the h-gradients concentrate densely *between the arches* of the "3" — locations where adding pixels would close the loops geometrically. A second correctly classified "3" in Figure 6.2 shows the same qualitative pattern. The *single* constrained model (`cnn_maxout` IC) has gradients that are mostly positive on strokes and negative on the background, but does not show the same class-dependent directional variation in the displayed examples. This comparison suggests a useful role for the difference structure, without establishing a general disentanglement property.

Qualitative comparison with a single input-convex model. Figure 6.3 reproduces the source paper's ten-class input-gradient comparison for the single `cnn_maxout` IC model. Its gradients mostly mark positive local logit sensitivities on digit strokes and negative sensitivities on the background. The maps follow the input structure, but show little class-dependent directional variation. By contrast, the DICIN maps in Figures 6.1 and 6.2 show different stream-sensitivity patterns for competing class logits: the h-stream highlights locations of candidate missing strokes. These visual examples motivate

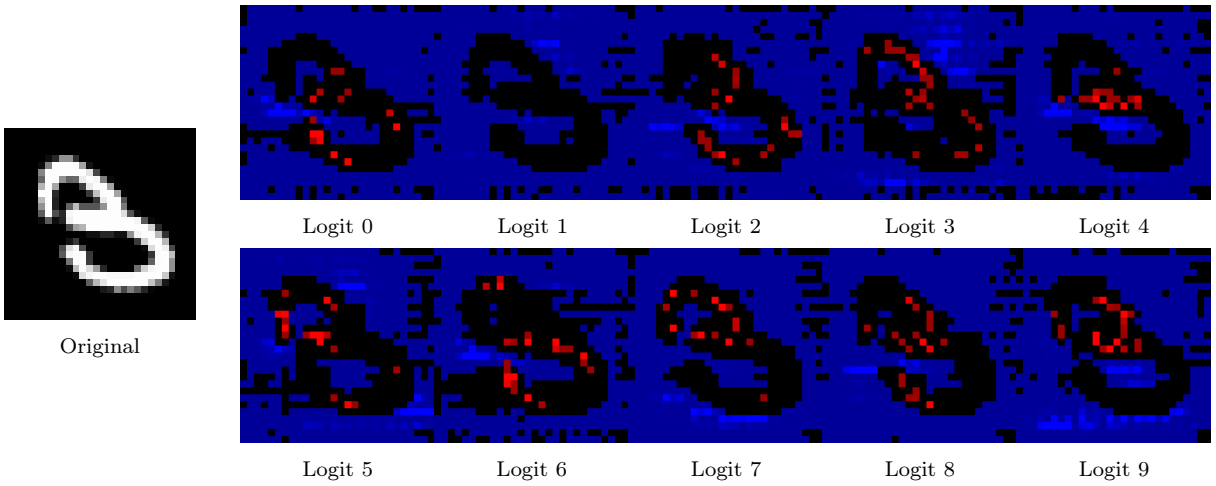


Fig. 6.1: Stream gradients of the inverted-input DIC model (`cnn_maxout` DICIN) on a correctly classified MNIST test digit "3", thresholded and overlaid per class logit. The g-stream gradient (red) highlights sensitivities on present strokes, while the h-stream gradient (blue) highlights sensitivities at candidate missing-stroke locations. These stream-local maps do not by themselves establish the effect of adding pixels on the complete model. Left: the original digit.

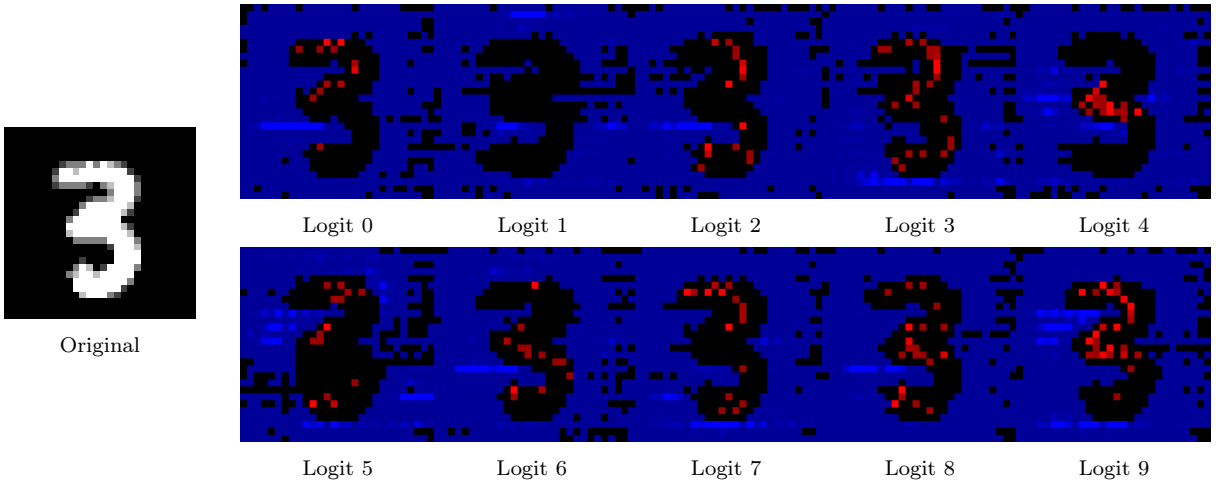


Fig. 6.2: Second example of stream gradients of the inverted-input DIC model (`cnn_maxout` DICIN) on a correctly classified MNIST test digit "3", thresholded and overlaid per class logit. The g-stream gradient (red) highlights sensitivities on present strokes, while the h-stream gradient (blue) highlights sensitivities at candidate missing-stroke locations. These stream-local maps do not by themselves establish the effect of adding pixels on the complete model. Left: the original digit.

the factual/counterfactual reading of the two streams; they establish neither the effect of a finite intervention nor a separation that holds for all inputs or models.

Accuracy trade-offs. Architectural constraints can reduce expressivity (Liu and Coogan, 2020; Sill, 1997), but for the difference-of-two construction the accuracy cost is small on the convolutional base model and even negative on the MLP, where every difference-of-two variant outperforms the unconstrained baseline; the full accuracy table across ten seeds and all eight variants is reported in Chapter 7.

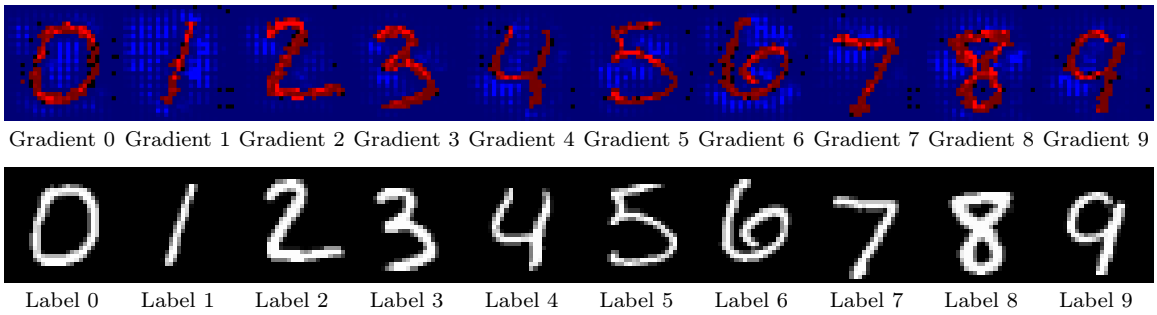


Fig. 6.3: Input-gradient visualizations of the `cnn_maxout` IC model for ten MNIST digits, each with respect to its ground-truth label. Top: saliency maps; bottom: input images. As in the source figure, positive and negative gradient values above threshold 0.2 are colored red and blue, with ranking colorized using exponential decay.

Remark 6.5 (Post-hoc and by-design, reconciled). The two routes of this chapter probe the same structural hypothesis — that difference-of-monotone representations can make network sensitivities more legible — from opposite ends. The MNIST study establishes that differences of constrained networks can be trained directly to competitive accuracy. Its visual examples also exhibit class-dependent stream sensitivities on present and candidate missing strokes. These qualitative observations do not establish faithful counterfactual explanations or a general separation of explanatory roles, either for the trained models or for the split-pairs of the pretrained VGG-16 and ResNet-18 models. Evaluating the proposed interpretation through interventions and testing it on larger architectures remain open directions, taken up again in Chapter 9.

6.2 Game-Based Attribution: Modes and Technicalities

The split-based methods above read a fixed DC decomposition. The complementary route of Chapter 4 attributes through the equilibrium trajectory distribution of the SG and RG: the per-neuron difference of the two players' discounted occupation measures is the attribution map, and at the anchor temperature $\tau = 1$ this difference recovers the $\alpha\beta$-LRP family exactly (Theorem 4.2), just as the plain gradient arises as a difference of occupation measures. Those equivalences — the stopping and routing games, the recovery of gradients and of the $\alpha\beta$-LRP rule — are stated in Chapter 4, with proofs in the companion supplement, and are not restated here. This section collects only what turns those equilibria into concrete, tunable attribution maps: the *modes* that deform the games, and the input-side *technicalities* that raise attribution quality.

6.2.1 Attribution Modes

Chapter 4 deforms the equilibrium trajectory distribution through a small set of orthogonal knobs — the routing and stopping temperatures τ, θ, the policy-entropy bonus, the risk parameter $\lambda \leq 0$, the stream balance (α, β), and the LRP stabilizer $\epsilon \geq 0$ — and establishes an *exact dictionary* (Section 4.4) relating each knob to the backward rule it deforms. We do not restate that dictionary here. What matters for attribution is only the operational role of the knobs: each is a control that turns the abstract game equilibrium into a concrete, tunable attribution map, and together they are the parameters swept in the evaluation of Chapter 7.

Two notational cautions specific to this chapter are worth flagging, since two symbols carry a different meaning here than in the split-based methods above. First, the stream balance (α, β) — the $\alpha\beta$-LRP discount pair, with $\alpha - \beta = 1$ and the theorem-driven anchor $(\alpha, \beta) = (2, 1)$ that reweights support

against opposition — is unrelated to the backward-shift parameter $\alpha \in [0, \frac{1}{2}]$ of Section 6.1; wherever a bare "α" appears in this section it denotes the LRP stream weight. Second, the routing temperature τ (which recovers $\alpha\beta$-LRP exactly only at the anchor $\tau = 1$, Theorem 4.2, and is applied myopically so the recovered forward pass is unchanged) is distinct from the spatial input transforms τ_t introduced in the Technicalities below.

6.2.2 Technicalities

A second, orthogonal lever improves the maps on the input side without touching the game: averaging attributions over input transformations, and the detached-operator (DAVE-style) reading of the resulting gradient decomposition. Both matter most on the ViT, where the 16×16 patch tokenization breaks translation equivariance and introduces grid artifacts (Dosovitskiy et al., 2021; Wróbel et al., 2026).

Equivariant Reynolds averaging. Following DAVE's transform-averaging strategy (Wróbel et al., 2026), an attribution is computed on N inputs transformed by random spatial maps $\tau_1, \ldots, \tau_N$ — small rotations, translations, and horizontal flips (distinct from the routing temperature τ of the games) — each explanation is mapped back to the original image coordinates by the inverse transform τ_t^{-1}, and the aligned maps are averaged. This classical Reynolds-operator construction cancels architecture-specific, transform-dependent noise while preserving the image-dependent structure common to all transforms; the sample count N trades compute for smoothness and is ablated in Chapter 7.

Connection to DAVE-style gradient decomposition. The averaging pairs naturally with a *conditioned forward pass* (Wróbel et al., 2026): LayerNorm statistics, the GELU gate, and the softmax row are detached, so the backward pass sees only the linear effective transformation and LRP conservation is preserved through the non-linearities. The resulting attribution takes the transform-averaged input×gradient form $\frac{1}{N}\sum_t \tau_t^{-1}(W_t) \odot X$, where W_t is the conditioned backward signal on the t-th transformed input. The game-based maps of Chapter 5 inherit exactly these conditioned operators, so equivariant averaging applies to them as a drop-in quality technique.

7 Experiments and Evaluation

This chapter evaluates the two design choices developed throughout the thesis: changing the representation available to attribution through the split streams, and changing the backward routing rule through the game policies. The common questions concern localization, faithfulness, robustness, and the numerical or computational cost of obtaining useful attribution signals. The split-method experiments and the game-based case study provide complementary evidence for these questions. Their original checkpoints, sample exclusions, metric conventions, and selection protocols are retained, and numerical comparisons are made within each campaign. The model-evaluation scores are reused from the papers; the Pascal VOC uncertainty estimates and comparison statistics are recomputed locally with images as the independent sampling units.

7.1 Experimental Setup

The split campaign evaluates pretrained VGG-16 and ResNet-18 on a custom 50-class ImageNet-S subset at 224×224 resolution. The source paper reports a 50-image validation split (one image per class) for selecting layer–formula combinations and a 566-image test split. Its released loader selects from the image–mask pairs available locally without the label-agreement filter used by the game campaign below; the historical sample membership cannot be recovered from these counts alone. The explanations target the model's predicted class, including for misclassified images, while localization is measured against the annotated object mask. Baseline attributions use Captum, with Zennit for γ-LRP and separate Grad-CAM++ and LayerCAM implementations. For each method, localization-selected and Pixel-Flipping-selected configurations are retained. Full settings and results are given in Section S.4.1 of the supplement.

The split tables report Pixel Flipping as area under the score-drop curve ($\mathrm{PF}_5^{\mathrm{drop}}$, $\mathrm{PF}_{20}^{\mathrm{drop}}$; ↑), perturbing 112 scalar input features per step, and Maximum Sensitivity with 25 perturbations. The revised game campaign reports area under the remaining-logit curve (PF_5, PF_{20}; ↓), using 1,568 scalar input features per step, and sensitivity with 50 perturbations on a common 200-image test subset. In both Pixel-Flipping implementations, features index the flattened $C \times H \times W$ input after broadcasting the spatial attribution over color channels; for RGB images at this resolution there are 150,528 such features. Selectivity, Attribution Localization and Pointing Game are measured using Quantus. VGG-16 is shared, while ResNet-18 belongs to the split campaign and ResNet-50 to the game campaign; their scores are not interchangeable. Throughout the tables, AL denotes Attribution Localization, PG Pointing Game, Sel Selectivity, MaxS Maximum Sensitivity, and AvgS Average Sensitivity. Method abbreviations are GCAM (Grad-CAM), GCAM++ (Grad-CAM++), LCAM (LayerCAM), IntGrad (Integrated Gradients), and SmGrad (SmoothGrad); Guided GCAM denotes Guided Grad-CAM. Dark blue shading and bold type mark the best values within the comparison stated in each caption. The notation n/r means that the source paper did not report a result; n/a means that the entry does not apply.

7.2 Split Methods: VGG-16

Qualitative analysis. Figure 7.1 qualitatively compares saliency maps from our split methods with their baseline counterparts at the best-performing layers. Mid-layer SplitCAM shares key visual characteristics with classical γ-LRP from earlier layers. Compared to standard LayerCAM on the same layer it focuses more weight to all object pixels, while still marking the key object features as

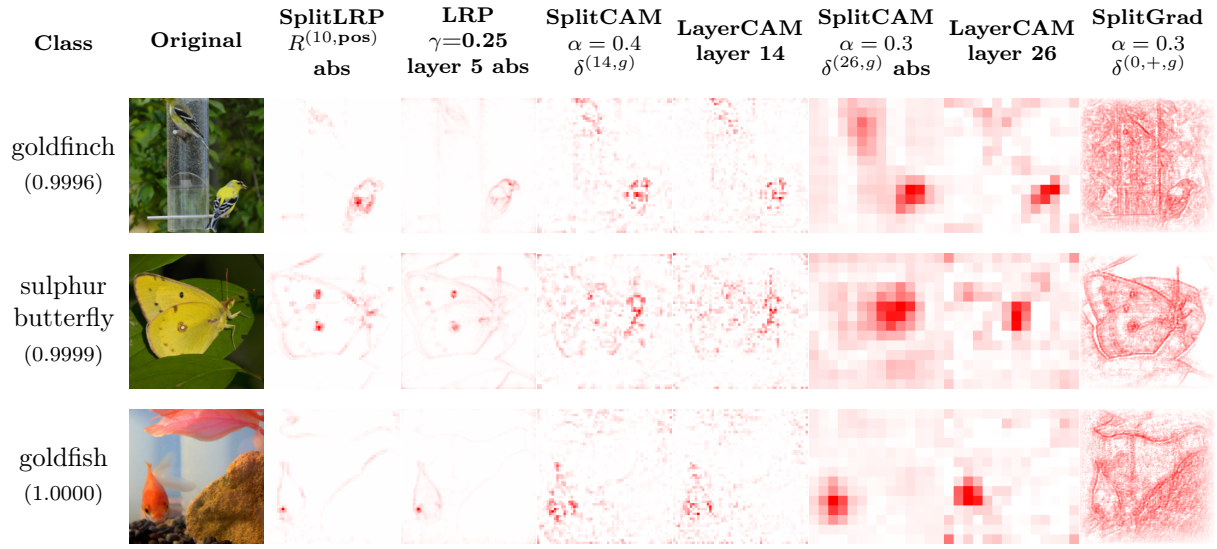


Fig. 7.1: Saliency map comparison for VGG-16 on our custom ImageNet-S test split. Each row shows the predicted class (with confidence), original image, and saliency maps from different methods.

especially important. For $\alpha \ll 0.5$, SplitGrad yields less focused yet finely aligned gradients that closely follow image details. We attribute this to the Maxout activation retaining pre-activation information, unlike ReLU, which discards it in inactive neurons. Overall, mid-layer SplitCAM offers an effective balance between interpretability and focus (low entropy). Further qualitative examples are collected in Section S.4.1 of the supplement.

Quantitative analysis. Table 7.1 and Table 7.2 compare selected attribution configurations on VGG-16 and ResNet-18 across five key Quantus metrics: Selectivity, Attribution Localization, Pointing Game accuracy, Pixel Flipping (score-drop AUC, $\text{PF}_5^{\text{drop}}$ and $\text{PF}_{20}^{\text{drop}}$) and Maximum Sensitivity. The *Abs* column specifies whether absolute saliency values were used prior to evaluation in Quantus. A full table and explanation of both VGG-16 and ResNet-18 evaluation results with all selected layers and other parameter configurations is found in Section S.4.1 of the supplement.

Overall, the considered backpropagation-based attribution methods heavily depend on the selected network layer, also see Section S.4.2 of the supplement for extensive ablation.

On VGG-16, the gains depend on the metric and configuration. In Table 7.1, SplitCAM reaches a Pointing Game score of 0.938 versus 0.887 for Guided Backprop, Attribution Localization of 0.651 versus 0.632 for Guided Backprop, and Selectivity of 2.727 versus 2.792 for LayerCAM. These SplitCAM values come from different layers. LayerCAM retains the highest Pixel Flipping score-drop AUCs, 0.317 and 2.474, compared with SplitCAM's 0.300 and 2.404 in the displayed configurations. SplitLRP has the lowest Maximum Sensitivity, 0.282, compared with 0.375 for Guided Backprop. Its Pointing Game score exceeds those of the displayed classical γ-LRP configurations, while its Attribution Localization and Pixel Flipping scores are lower.

7.3 Split Methods: ResNet-18

Table 7.2 reports selected configurations on ResNet-18 across the same five Quantus metrics, with the *Abs* column again specifying whether absolute saliency values were used prior to evaluation; the full evaluation tables are given in Section S.4.1 of the supplement. SplitCAM has the lowest reported

Tab. 7.1: Selected interpretability evaluation configurations (VGG-16, ImageNet-S test split). The *wta* abbreviation indicates the use of the "winner-takes-it-all" max-pooling implementation; *sc* emphasizes the scaling procedure in the forward pass (Section 3.6). Best values per metric among the displayed rows are bold with dark blue shading. The bracket notation in the layer column follows the upper indices of the split methods as introduced in Chapter 6.

Method	Layer	Abs	Sel↓	AL↑	PG↑	PF_5^{drop}↑	PF_{20}^{drop}↑	MaxS↓
Split Methods								
SplitCAM (sc, $\alpha = 0.4$)	(12, g)	n	4.720	**0.651**	0.912	0.213	1.794	0.389
SplitCAM (sc, $\alpha = 0.4$)	(14, g)	n	**2.727**	0.563	0.836	0.300	2.404	1.012
SplitCAM (sc, $\alpha = 0.4$, wta)	(26, +, g)	y	4.711	0.635	**0.938**	0.165	1.590	0.456
SplitGrad ($\alpha = 0.4$, wta)	(12, +, g)	n	4.735	0.635	0.883	0.186	1.756	0.651
SplitGrad ($\alpha = 0.4$, wta)	(14, +, g)	y	3.287	0.540	0.818	0.249	2.186	0.857
SplitLRP (sc, wta)	(26, pos)	y	5.168	0.588	0.871	0.151	1.473	**0.282**
Baselines (With Layer Optimization)								
Guided Backprop	2	n	3.276	0.632	0.883	0.281	2.302	0.571
Guided Backprop	5	n	3.448	0.614	0.887	0.240	2.091	0.375
LCAM	14	n/a	2.792	0.576	0.798	**0.317**	**2.474**	2.656
LCAM	2	n/a	3.128	0.580	0.855	0.300	2.191	5.508
LRP $\gamma = 0.25$	0	n	3.189	0.615	0.680	0.218	1.918	0.519
LRP $\gamma = 0.25$	14	n	3.593	0.577	0.611	0.279	2.255	0.491
Baselines								
DeepLIFT	0	y	4.447	0.551	0.726	0.281	1.945	1.053
Feature Ablation	0	y	6.029	0.479	0.624	0.123	1.250	0.859
GradientSHAP	0	y	4.808	0.552	0.783	0.261	1.862	1.142
GCAM++	30	n	5.955	0.564	0.827	0.130	1.245	0.401
IntGrad	0	y	4.838	0.554	0.797	0.262	1.851	1.069
Occlusion	0	y	5.214	0.499	0.716	0.130	1.217	0.699
Random Baseline	0	y	3.570	0.425	0.458	0.203	1.585	n/r

Selectivity, 3.503, compared with 3.517 for the strongest baseline, Guided Backprop: a numerical difference of 0.014. SplitCAM and SplitGrad reach a Pointing Game score of 0.972, compared with LayerCAM's 0.971. SplitCAM's Attribution Localization of 0.673 exceeds LayerCAM's 0.664, but the full comparison reports a higher value of 0.742 for Guided Grad-CAM (Table S.4.2). The baselines also retain the best Pixel Flipping scores and Maximum Sensitivity, so the results support metric-specific gains.

7.4 Split-Method Stabilization Ablations

7.4.1 Backward-Pass Stabilization of the Split Methods

Table 7.3 presents the performance of two selected SplitGrad and SplitCAM configurations across different shifting parameters α, evaluated on the custom ImageNet-S test split. As introduced in Section 3.6, the α value determines how strongly the backward pass is stabilized: $\alpha = 0$ results in no stabilization, and $\alpha = 0.5$ results in local sensitivities resembling scaled versions of the original network gradient. Within this ablation, reducing α from 0.5 to 0.4 improves Pointing Game and Attribution Localization for both configurations, while SplitCAM's PF_{20}^{drop} decreases from 1.607 to

Tab. 7.2: Selected interpretability evaluation configurations (ResNet-18, ImageNet-S test split). Highlighting identifies the best values among the displayed rows; the full comparison is in Table S.4.2. The triples $x.y.z$ are a ResNet-18-internal indexing of the convolutional layer at which the attribution maps are calculated. See the caption of Table 7.1 for further abbreviations and table explanation.

Method	Layer	Abs	Sel↓	AL↑	PG↑	PF_5^{drop}↑	PF_{20}^{drop}↑	MaxS↓
SplitCAM Methods								
SplitCAM (sc, $\alpha = 0.3$, wta)	(1.0.2, g)	n	**3.503**	0.487	0.747	0.168	1.395	1.277
SplitCAM (sc, $\alpha = 0.3$, wta)	(4.1.2, g)	n	5.654	**0.673**	**0.972**	0.081	0.880	0.233
SplitGrad ($\alpha = 0.3$, wta)	(1.0.2, g)	y	4.449	0.513	0.777	0.149	1.266	0.813
SplitGrad ($\alpha = 0.3$, wta)	(4.1.2, g)	y	5.663	0.649	0.972	0.082	0.882	0.219
SplitLRP (sc, wta)	(conv1, pos)	y	6.313	0.506	0.587	0.056	0.697	0.755
Baselines (With Layer Optimization)								
GCAM	4.1.2	n	6.070	0.554	0.945	0.076	0.829	**0.131**
LCAM	2.0.2	n	3.690	0.549	0.865	0.144	1.428	0.913
LCAM	4.1.1	n	5.661	0.664	0.971	0.083	0.890	0.227
LRP $\gamma = 0.25$	conv1	y	3.550	0.656	0.852	**0.213**	1.737	0.521
Baselines								
Guided Backprop	conv1	n	3.517	0.621	0.867	0.212	**1.757**	0.459
GradientSHAP	conv1	y	5.269	0.527	0.779	0.166	1.227	1.074
IntGrad	conv1	y	5.265	0.528	0.777	0.167	1.226	0.994
Feature Ablation	conv1	n	4.655	0.136	0.684	0.188	1.388	1.077
Occlusion	conv1	n	5.300	0.179	0.726	0.130	1.030	0.785
Deconvolution	conv1	n	3.934	0.484	0.666	0.147	1.203	0.751
Random Baseline	n/a	y	4.071	0.425	0.440	0.147	1.067	n/r

Tab. 7.3: Comparison of different α parameters with two SplitGrad and SplitCAM base configurations, both in *wta* and *abs* mode, evaluated on the custom ImageNet-S test split.

Method	**Metric**	α .50	α .45	α .40	α .35	α .30	α .00
SplitGrad (26, g)	PG↑	0.898	0.906	0.947	**0.956**	0.889	0.401
	AL↑	0.543	0.548	0.605	**0.617**	0.535	0.449
	MaxS↓	0.616	0.607	0.440	0.238	0.334	**0.028**
	PF_{20}^{drop} ↑	1.527	1.534	**1.581**	1.448	1.317	1.247
SplitCAM (26, +, g)	PG↑	0.868	0.868	**0.938**	0.933	0.613	0.855
	AL↑	0.550	0.554	0.635	**0.663**	0.403	0.513
	MaxS↓	0.691	0.682	0.457	0.265	0.251	**0.120**
	PF_{20}^{drop} ↑	1.607	**1.609**	1.585	1.499	0.853	1.507

1.585. SplitGrad reaches its highest PF_{20}^{drop} at $\alpha = 0.4$ and its highest localization scores at $\alpha = 0.35$; SplitCAM's Pointing Game and Attribution Localization optima occur at 0.4 and 0.35, respectively. Maximum Sensitivity is lowest at $\alpha = 0$ for both configurations. The sweep therefore supports a metric-dependent choice of α and does not isolate the effect of numerical stabilization from the change in attribution rule. The full α grid, including the comparison between the convex and the winner-takes-it-all max-pooling modes, is reported in Section S.4.2 of the supplement (Table S.4.5).

For the same SplitCAM configuration, Tables 7.1 and 7.3 retain conflicting source values: $\mathrm{PF}_{20}^{\mathrm{drop}} = 1.590$ versus 1.585 and MaxS= 0.456 versus 0.457. The unavailable per-image results prevent resolving this discrepancy.

Forward-pass stabilization. In the forward-stabilization experiment of Section 3.6, the split-stream activations of late VGG-16 layers without stabilization grow to the order of 10^{18}, producing numerical errors of order 10^{9}, and SplitLRP produces nearly uniform maps. Scaling gives better values than shifting for all reported metrics in the two early-layer SplitLRP configurations. At later layers, shifting gives higher Attribution Localization and Pixel-Flipping score-drop AUC, scaling gives lower Maximum Sensitivity, and the Pointing Game comparison depends on the configuration (Table S.4.6 in the supplement; the corresponding SplitCAM grids without stabilization and with shift stabilization are Tables S.4.7 and S.4.8).

7.5 Game-Based Case Study

Setup. We evaluate RG and SG against standard backward baselines (Section S.4.3.1) on VGG-16 (Simonyan and Zisserman, 2015), ResNet-50 (He et al., 2016), and ViT-B/16 (Dosovitskiy et al., 2021) with torchvision ImageNet-1K checkpoints (Deng et al., 2009; Paszke et al., 2019; Russakovsky et al., 2015), on a 50-class ImageNet-S subset (Gao et al., 2023) at 224×224 (50-image validation split, 565-image test split), and on Pascal VOC 2012 (Everingham et al., 2009). For the game campaign, the 752 released ImageNet-S50 annotation records are filtered to 616 whose class agrees with the official ILSVRC2012 label; one image per class is selected for validation with seed 42, and one of the remaining 566 test images is excluded because its object mask is outside the center crop. The explanation target is the dataset class, including on misclassified images (Section A.2). ImageNet-S provides the pixel masks required by the localization metrics; ImageNet boxes over-credit mass anywhere inside the box. All metrics are computed through Quantus (Hedström et al., 2023) (definitions in Section A.2.2): localization (*Attribution Localization* (AL), *Pointing Game* (PG), Top-k Intersection (TK); ↑), faithfulness (Pixel Flipping area under the remaining-logit curve over the first 5% and 20% of flips (PF_5, PF_{20}), *Selectivity* (Sel); ↓), robustness (Maximum Sensitivity (MaxS), Average Sensitivity (AvgS); ↓). Baselines run in their reference implementations (lxt, zennit, captum, pytorch-grad-cam, and the authors' code for TIBAV and GAE) at the authors' settings, plus declared grids (Achtibat et al., 2024; Anders et al., 2023; Chefer et al., 2021a,b; Gildenblat and contributors, 2021; Kokhlikyan et al., 2020). RG analytical modes are reported alongside each configuration: temperature τ, risk aversion λ (softmax-shift variant λ_{sm} on ViT attention logits), LRP stabilizer ε, equivariant-Reynolds sample count N.

Protocol. Methods are evaluated under a two-stage protocol that treats baselines and our games alike. Stage 1 runs a grid search on the validation split (one image per class) for every method with tunable parameters, including DAVE (Section S.4.3.2, Table S.4.9; 2,021 configurations in total). Stage 2 picks, for each method, the *loc* configuration with the best localization rank-sum over $\{\mathrm{AL}, \mathrm{PG}, \mathrm{TK}\}$ and, independently, the *faith* configuration with the best faithfulness rank-sum over $\{\mathrm{PF}_5, \mathrm{PF}_{20}, \mathrm{Sel}\}$, and evaluates both once on the test split, with 95% percentile-bootstrap confidence intervals (Efron and Tibshirani, 1993) and Holm-corrected (Holm, 1979) paired tests against the best row of each metric (Section A.2.4); for RG / RG+Eq we additionally show the $\alpha\beta$-LRP anchor $(\alpha,\beta,\varepsilon,\tau)=(2,1,0.5,1)$ as a theorem-driven reference (Theorem 4.2). Localization is a natural target for modes designed to control routing focus, and object annotations provide external ground truth: Kohlbrenner et al. (2020) compare LRP rule composites by the share of relevance inside ground-truth bounding boxes, and we

use pixel masks. The main table reports the loc row of every method; since mask agreement is not faithfulness, the faith rows of all methods are reported in Section S.4.4.

Tab. 7.4: ViT-B/16 on the ImageNet-S test split (n=565; MaxS/AvgS on a fixed random subset of 200 test images). **Bold**, dark shading: best in column; ≈, light shading: not significantly different from the best (paired bootstrap / McNemar, Holm-corrected, $p \geq 0.05$); small numbers: 95% percentile-bootstrap confidence intervals over test images (Section A.2.4). One row per method, selected on the validation split by the localization rank-sum (faithfulness-selected rows and further baselines: Section S.4.4); the RG anchor is the exact $\alpha\beta$-LRP representation (Theorem 4.2). t/img: seconds per image on one GPU. Config: SmoothGrad averages N samples at noise level σ relative to the input range; LRP-γ has stabilizer ε and positive-weight boost γ; AttnLRP has layer-type parameters $\gamma_{\text{conv}}, \gamma_{\text{lin}}, \varepsilon_{\text{attn}}$; KernelSHAP draws N samples for M superpixels; RG modes as in Table 4.1.

Method	Config	AL↑	PG↑	TK↑	PF_5↓	PF_{20}↓	Sel↓	MaxS↓	AvgS↓	t/img
Gradient	fixed	0.411 [0.389,0.432]	0.361 [0.322,0.402]	0.386 [0.359,0.413]	8.45 [8.33,8.56]	8.08 [7.94,8.22]	6.033 [5.905,6.154]	1.290 [1.233,1.351]	0.922 [0.893,0.953]	**0.02**
SmGrad	N=50,σ/range=0.2	0.517 [0.498,0.536]	0.706 [0.667,0.743]	0.675 [0.654,0.694]	8.10 [7.96,8.24]	7.02 [6.83,7.20]	5.029 [4.894,5.163]	0.597 [0.569,0.630]	0.468 [0.451,0.485]	1.12
LRP-γ	ε=0.1,γ=2	0.644 [0.628,0.660]	0.873 [0.844,0.899]	0.816≈ [0.800,0.832]	7.93 [7.78,8.08]	6.64 [6.44,6.83]	**4.451** [4.324,4.572]	0.300 [0.287,0.314]	0.218 [0.210,0.226]	0.08
AttnLRP	γ_{conv}=0.25,γ_{lin}=0.05,$\varepsilon_{\text{attn}}$=0.25	0.644 [0.628,0.660]	0.855 [0.825,0.883]	0.809 [0.794,0.825]	**7.77** [7.61,7.93]	**6.43** [6.23,6.62]	4.582 [4.452,4.706]	0.456 [0.430,0.484]	0.342 [0.324,0.364]	0.10
TIBAV	fixed	0.615 [0.597,0.634]	0.878 [0.851,0.904]	0.782 [0.759,0.805]	8.13 [8.01,8.24]	7.60 [7.46,7.74]	4.906 [4.774,5.033]	1.062 [1.001,1.126]	0.512 [0.486,0.540]	0.15
GAE	fixed	0.533 [0.514,0.551]	0.743 [0.708,0.779]	0.649 [0.623,0.677]	8.13 [8.01,8.25]	7.62 [7.47,7.76]	5.098 [4.961,5.231]	0.525 [0.494,0.558]	0.368 [0.346,0.391]	0.03
KernelSHAP	N=2M+2048, 80 superpixels	0.565 [0.548,0.583]	**0.961** [0.945,0.977]	0.823≈ [0.799,0.848]	8.29 [8.18,8.40]	7.50 [7.35,7.65]	5.140 [5.012,5.261]	0.897 [0.868,0.928]	0.650 [0.632,0.667]	6.71
RG anchor	τ=1	0.509 [0.490,0.528]	0.724 [0.685,0.759]	0.699 [0.678,0.720]	8.19 [8.05,8.31]	6.97 [6.78,7.15]	4.854 [4.732,4.972]	**0.166** [0.161,0.172]	0.120 [0.117,0.124]	0.10
RG	τ=0.4,λ_{sm}=−10,σ^2=5,ε=0.1	**0.711** [0.692,0.730]	0.825 [0.793,0.855]	0.788 [0.768,0.808]	8.27 [8.14,8.39]	7.48 [7.31,7.64]	5.109 [4.987,5.226]	0.729 [0.685,0.777]	0.494 [0.471,0.518]	0.17
RG+Eq anchor	τ=1,N=50	0.520 [0.501,0.539]	0.761 [0.726,0.796]	0.752 [0.729,0.775]	8.41 [8.29,8.52]	7.72 [7.56,7.87]	4.864 [4.742,4.981]	0.170≈ [0.165,0.175]	**0.115** [0.112,0.118]	4.93
RG+Eq	τ=0.5,λ_{sm}=−10,σ^2=2,N=50	0.698 [0.680,0.717]	0.855 [0.827,0.883]	**0.836** [0.815,0.857]	8.39 [8.28,8.50]	7.80 [7.65,7.94]	5.010 [4.889,5.127]	0.441 [0.421,0.463]	0.301 [0.290,0.313]	8.66

ViT-B/16 (Table 7.4). Under the localization-selected protocol, the RG family beats every baseline on attribution localization, including the prior transformer-specific backward methods AttnLRP (Achtibat et al., 2024), DAVE (Wróbel et al., 2026), TIBAV (Chefer et al., 2021b) and GAE (Chefer et al., 2021a): the selected mode reaches AL 0.711 against 0.644 for the best baseline (LRP-γ), and RG+Eq 0.698, both significantly ahead of every baseline row after Holm correction. RG+Eq also has the highest observed TK (0.836), but the Holm-corrected tests do not detect a difference from KernelSHAP at 0.823 or LRP-γ at 0.816. The pointing game is led by KernelSHAP (0.961), whose maps are constant on superpixels: a hit needs only one pixel of the maximal region inside the mask. Scoring a single pixel drawn from that region instead, KernelSHAP falls below RG+Eq, which is then not significantly below the best baseline (LRP-γ; Section S.4.4.6). Among the localization-selected rows, faithfulness favors the LRP rules: AttnLRP is best on PF_5 and PF_{20} and LRP-γ on Sel (PF_{20} 6.43 versus 7.48 for RG), and the faithfulness-selected RG row narrows but does not close the gap (PF_{20} 6.78). Across both selection objectives, faithfulness-selected AttnLRP has the lowest PF_5, PF_{20} and Selectivity, at 7.64, 6.12 and 4.441, respectively. The two anchors are the most robust rows (MaxS 0.166 and 0.170 versus 0.208 for AR), whereas the localization-selected mode reaches MaxS 0.729: sharper routing trades robustness and faithfulness for localization. ADF adds modest overhead (RG: 0.17 s vs. 0.10 s base), still below sampling baselines like SmoothGrad or IntGrad; Reynolds averaging is the only expensive variant (8.66 s), exceeding even the perturbation method KernelSHAP (6.71 s).

Higher τ spreads attribution into context. The localization-optimal modes are the sharpest ones: they place mass on the few patches that align with the segmentation mask. As τ increases, the same

equilibrium spreads routing over more predecessors; in the displayed examples, the object remains in focus while attribution extends into the surrounding context. This visual pattern does not by itself establish which contextual features the network relies on. On ViT-B/16 attribution localization worsens monotonically with τ around the localization-selected setting (AL 0.723 at τ=0.3 down to 0.461 at τ=1), while the faithfulness side is non-monotonic. High τ illustrates how localization scores favor concentrated maps within the same game; no perceptual study was conducted to establish which maps users find more informative.

Cross-dataset transfer (Pascal VOC). A natural worry with validation-tuned modes is that the chosen τ, λ_{sm} and σ^2 are overfitted to the ImageNet-S 50-class subset. We test transfer on a per-image disjoint split of Pascal VOC 2012 trainval (2,743 test (image, foreground-class) samples with pixel-accurate segmentation masks): RG and RG+Eq use the *same* mode parameters as Table 7.4 without any VOC-side re-tuning, while every baseline gets a fresh per-method validation grid on the new pool. This tests the portability of the selected parameters across these datasets for the same pretrained model, without identifying whether the underlying structure is model-specific or dataset-specific. The localization lead carries over (AL 0.459 for RG and 0.456 for RG+Eq versus 0.413 for TIBAV, both significant against every baseline row), while TK and PG are led by TIBAV and the perturbation methods, and faithfulness by the LRP rules (Section S.4.4.4).

VGG-16 and ResNet-50 (Section S.4.4). On the CNNs, where LRP-type rules are already strong, the localization-selected RG rows also lead on AL within the localization-selected comparison, clearly on VGG-16 (0.685 versus 0.613 for LRP-ε) and narrowly on ResNet-50 (0.719 versus 0.711 for LRP-γ). The Holm-corrected paired comparison does not detect a difference between these ResNet-50 AL scores (adjusted $p = 0.420$); the numerical lead therefore does not establish superiority. Across both selection objectives, the strongest VGG-16 baseline AL is 0.639 for faithfulness-selected LRP-ε, still below RG's 0.685. Pixel flipping favors LRP-ε for VGG-16 PF_5 (12.43), LRP-γ for VGG-16 PF_{20} (7.89, compared with 8.17 for LRP-ε), and LRP-γ for both ResNet-50 metrics, while the CAM methods and the perturbation methods lead TK and PG. On VGG-16 the risk-averse SG mode is the most robust method (MaxS 0.278 versus 0.314 for GCAM++).

7.5.1 Game-Method Ablations

The revised one-factor ablations in Section S.4.4.5 vary one mode around each validation-selected configuration and evaluate the resulting rows on the 565-image test split. They are diagnostic results and do not feed back into selection. Section S.4.4.6 controls the pointing-game comparison for map granularity, and Section S.4.4.7 compares the composite Value-routing attention rule with a gradient fallback. These analyses expose trade-offs between mask agreement, perturbation faithfulness and robustness rather than a uniform improvement of all desiderata.

8 The Trajectory Distribution View

The attribution methods of Chapter 6 turn internal signals and backward routing choices into maps, and Chapter 7 evaluates those maps. For the intervention-oriented motivation of this thesis, we also need to examine the calculation producing them. The equilibrium policies of Chapter 4 induce distributions over backward trajectories, making such a comparison possible before projection onto input pixels. This chapter studies their Hellinger distance, its exact per-input-state disagreement map, and the distinction between complete and survival-conditioned laws. It then evaluates these diagnostics under input noise and parameter randomization.

8.1 Comparison with Normalized Relevance Measures

The NRM framework of Xiong et al. (2026), developed by Ping Xiong, Thomas Schnake, Grégoire Montavon, Klaus-Robert Müller, and Shinichi Nakajima, provides a closely related view of the backward calculation. Its basic object is a normalized signed measure on complete neuron walks. Relevance assigned to selected neurons or concepts is obtained by summing over the walks passing through them; conditional relevance is a ratio of joint and marginal relevance when the conditioning relevance is nonzero. With suitable normalized propagation rules, this includes ordinary LRP, concept relevance, and Concept Relevance Propagation (CRP) (Achtibat et al., 2023). In particular, their CRP quantity is the joint relevance of an input neuron, an intermediate concept neuron, and the target output, rather than automatically a conditional probability.

Both views therefore retain paths through the network before reducing them to an attribution map. Their factorization supports sum-product computation and queries involving neurons in several layers. The distinction is what a path carries. In NRM, walk relevance may be negative even though its total is one. Here, the game policies define a non-negative probability law Π on the expanded state graph, including the player label and termination. Gradients and relevance scores are recovered from signed differences of discounted occupation measures of this law (Chapter 4); they are generally not its unweighted marginals. The player expansion therefore separates trajectory probability from the sign and scale of the backward contribution. It does not by itself identify Π with a given NRM measure.

This distinction matters for normalization and comparison. The NRM construction normalizes propagation columns under positive-sum assumptions; its identification with globally normalized original walk relevance additionally requires the constant-sum conditions of Xiong et al. (2026, Section 3.5). Our transition probabilities remain normalized while discounts retain the scale of the backward rule, and bias and stabilizer absorption are accounted for explicitly. Hellinger distance applies directly to the resulting probability laws; it does not apply directly to arbitrary signed relevance measures. Conditioning Π on a concept requires a positive-probability event, whereas signed relevance can have zero total through cancellation of nonzero terms.

The two perspectives suggest a useful extension: query the game trajectories through selected concepts, retaining the signs and discounts when computing their attributions, and compare the resulting trajectory laws across inputs or models. Establishing when those quantities coincide with a particular NRM/CRP construction requires matching the propagation rules, normalization, and absorption conventions. This is a proposed connection, not an equivalence proved in this thesis.

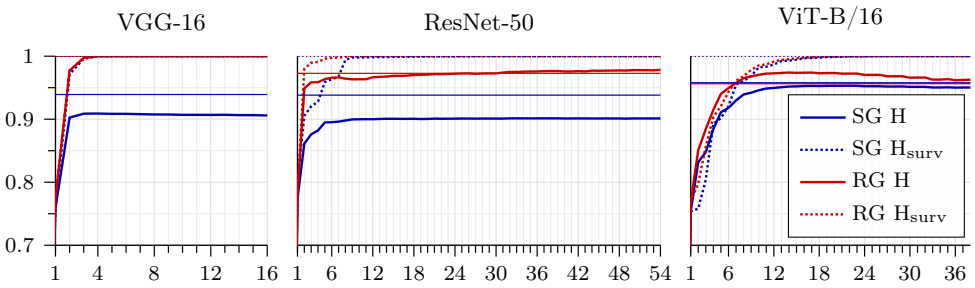


Fig. 8.1: Hellinger trajectory distance under cascading parameter randomization (Adebayo et al., 2018). Hellinger distance H (vertical axis, from 0.7) against the number of cascadingly re-initialized layers, output to input (horizontal axis; $H = 0$ before the first layer is randomized). Solid: plain H; dotted: H_{surv}, conditioned on survival to an ordinary input-layer state. Thin horizontal reference lines mark the fully-random-model values from Table 8.3 (solid: plain H; dotted: H_{surv}), in the color of the corresponding game.

8.2 Trajectory Sanity Check

The cascading-randomization sanity check of Adebayo et al. (2018) asks whether an explanation meaningfully depends on the learned parameters, and answers by comparing attribution maps in pixel space. The object of interest here is the normalized backward routing—a distribution Π over graph trajectories. The attribution map is recovered through a model-dependent signed discounted occupation projection of this law. The **Hellinger distance** (Hellinger, 1909) $H(\Pi_A, \Pi_B) = (1-BC)^{1/2}$, with Bhattacharyya coefficient (Bhattacharyya, 1943) $BC = \sum_\zeta \sqrt{\Pi_A(\zeta)\,\Pi_B(\zeta)}$, measures the routing logic directly rather than its visual projection. This section probes the representation claim: parameter randomization should move Π even when the resulting attribution map remains stable. We remark that $H = 0$ iff $\Pi_A = \Pi_B$, and the metric takes its maximal value $H = 1$ iff their supports are disjoint. Unlike KL-type divergences, H stays finite when randomization opens or closes routes, and since both laws are layered Markov chains on the same state graph, BC factorizes along layers: one backward pass computes H exactly, in time linear in the number of edges, and H^2 decomposes into per-input-state contributions (Section 8.3).

8.2.1 Trajectories change, maps need not

Under cascading randomization (layers cumulatively re-initialized output→input), both SG and RG diverge from the first randomized layer and plateau near the fully-random-model reference of Table 8.3 (Figure 8.1). The reference itself sits below the ceiling $H = 1$ because two fully randomized models still share some short trajectories: both may stop early at structurally dead neurons and route mass to the common cemetery state. Such short paths have a higher chance of coinciding across models than full trajectories reaching the input layer. Conditioning on survival removes this shared cemetery mass and gives the strict ceiling $H_{surv} \approx 1$. Trajectory- and map-level readouts show weak observed per-image rank association (Spearman $|\rho| \leq 0.19$ between H and pixel-level Gradient or $\alpha\beta$-LRP similarity): images with similar attribution maps need not have similar calculations. Per-game values, the conditioning, the floor mechanism and the per-image analysis are in Section 8.4.

What the map hides. Cascading randomization on the maps reproduces Adebayo et al. (2018) and Sixt et al. (2020): some methods retain map structure under full randomization. On ViT-B/16, the RG anchor itself retains Spearman $|\rho| = 0.454$, the highest of all methods, and HOG $r = 0.469$; on VGG-16, LRP-ε retains 0.341 and 0.457 (Table 8.2). The localization-selected RG modes are slightly

more model sensitive than the anchor on all three models. Read against Figure 8.1, the two readouts disagree: H sits at the fully random reference while the attribution map retains structure. The map-level failure stands; H reveals changes in normalized routing that are not apparent in the map. The discounted occupation projection can also change across models, so this discrepancy does not isolate a loss under one fixed projection.

8.3 Hellinger Distance and Disagreement Map

We compare two layered Markov processes on a common state graph and seek both a scalar divergence of the trajectory distributions and an exact per-pixel decomposition of this divergence at the input layer. We use the squared Hellinger distance because the geometric mean $\sqrt{PQ}$ preserves the multiplicative structure of Markov trajectory distributions and yields an exact linear-time dynamic program. Unlike Jeffreys divergence, it remains finite under support mismatch; unlike Jensen–Shannon, it admits an exact recursion on the layered graph. Later, we decompose the squared Hellinger distance as a sum of non-negative per-pixel contributions over all input-layer states, giving a disagreement map at the input layer (Theorem 8.1).

The formal foundations and proofs are given in companion Chapter S.3. We compare two processes on a common finite layered state graph with a shared deterministic starting state s_0. At layer l, $\mathscr{S}_+^{(l)} = \mathscr{S}^{(l)} \cup \{\dagger\}$ contains the ordinary states and an absorbing cemetery state; $T_M^{(l)}$ is the transition kernel from layer l to $l-1$, and $m_M^{(l)}(s)$ is the marginal probability of visiting s. The full trajectory law is $\Pi_M = \Pi_M^{(0)}$. For the RG, full-law diagnostics retain the gate, branch, and routing substeps; only ordinary surviving paths admit exact act-only elimination (Remark S.3.1).

8.3.1 Bhattacharyya Coefficient and the Geometric-Mean Kernel

Definitions. For two probability distributions P, Q on a common finite set, the *Bhattacharyya coefficient* is the geometric overlap

$$\mathrm{BC}(P,Q) \;:=\; \sum_y \sqrt{P(y)\,Q(y)} \;\in\; [0,1], \tag{8.1}$$

and the *squared Hellinger distance* is

$$\mathrm{H}^2(P,Q) \;:=\; \tfrac{1}{2}\sum_y \big(\sqrt{P(y)} - \sqrt{Q(y)}\big)^2 \;=\; 1 - \mathrm{BC}(P,Q) \;\in\; [0,1]. \tag{8.2}$$

The Hellinger distance $\mathrm{H}(P,Q) = \sqrt{1-\mathrm{BC}(P,Q)}$ is a true metric: $0 \le \mathrm{H} \le 1$, with $\mathrm{H} = 0$ iff $P = Q$ and $\mathrm{H} = 1$ iff P and Q have disjoint supports. Crucially, $\sqrt{0 \cdot Q(y)} = 0$, so mismatched supports cause no infinity—only a smaller overlap.

Geometric-mean kernel. For each layer $l \ge 1$ and pair of states $(s, s') \in \mathscr{S}_+^{(l)} \times \mathscr{S}_+^{(l-1)}$, define the *geometric-mean kernel*

$$\widetilde{T}^{(l)}(s,s') \;:=\; \sqrt{T_A^{(l)}(s \to s')\; T_B^{(l)}(s \to s')}. \tag{8.3}$$

This kernel is non-negative but in general sub-stochastic. Its row sum

$$\widetilde{R}^{(l)}(s) \;\coloneqq\; \sum_{s' \in \mathscr{S}_+^{(l-1)}} \widetilde{T}^{(l)}(s, s') \tag{8.4}$$

satisfies $0 \le \widetilde{R}^{(l)}(s) \le 1$ by Cauchy–Schwarz, with equality iff $T_A^{(l)}(s \to \cdot) = T_B^{(l)}(s \to \cdot)$. Cemetery handling is automatic: $T_A^{(l)}(\dagger \to \dagger) = T_B^{(l)}(\dagger \to \dagger) = 1$, so $\widetilde{T}^{(l)}(\dagger, \dagger) = 1$ and $\widetilde{R}^{(l)}(\dagger) = 1$.

8.3.2 Hellinger backward pass

The chain rule (Lemma S.3.1) yields a backward DP that propagates the per-state cumulative $\widetilde{T}$-mass $g^{(l)}(s)$ from the start state s_0 down through the layers. The prefix Bhattacharyya coefficient at each layer is the row sum $\mathrm{BC}^{(l)} = \sum_s g^{(l)}(s)$, the prefix Hellinger distance is $\mathrm{H}^{(l)} = \sqrt{1 - \mathrm{BC}^{(l)}}$, and the per-terminal-state disagreement map is read off at $l = 0$ from (8.5). The map is defined on the full terminal set $\mathscr{S}_+^{(0)} = \mathscr{S}^{(0)} \cup \{\dagger\}$; the visualized pixel attribution map is its restriction to ordinary pixels $\mathscr{S}^{(0)}$, and its sum is therefore $\mathrm{H}^2_{\mathrm{ord}} = \mathrm{H}^2 - \mathrm{H}^2(\dagger)$, not the full H^2 (Theorem 8.1(iii)).

Algorithm 8.1: Hellinger backward pass (with per-terminal-state disagreement map)

Initialize the occupation marginals $m_M^{(L)}$ and overlap mass $g^{(L)}$ at the target output.;
For $l = L, L-1, \ldots, 1$, propagate each model's marginals with its kernel and propagate overlap by $g^{(l-1)}(s') = \sum_s g^{(l)}(s) \sqrt{T_A^{(l)}(s,s') T_B^{(l)}(s,s')}$.;
At each layer compute $\mathrm{BC}^{(l)} = \sum_s g^{(l)}(s)$ and $\mathrm{H}^{(l)} = \sqrt{1 - \mathrm{BC}^{(l)}}$.;
At every terminal state return $\mathrm{H}^2(s) = \frac{1}{2}(m_A^{(0)}(s) + m_B^{(0)}(s)) - g^{(0)}(s)$. The pixel disagreement map restricts this output to ordinary input states.

8.3.3 Per-Pixel Disagreement Map

The squared Hellinger distance H^2 admits an exact non-negative decomposition over input-layer states: each input pixel carries a calibrated share of the total trajectory disagreement (Theorem 8.1), giving a disagreement *map* that is itself an attribution of the trajectory mismatch back to the input. The conditioned-on-survival variant removes the cemetery contribution that otherwise bounds plain H from below on randomized models.

Per-Pixel Decomposition: The Disagreement Map

The squared Hellinger distance at layer 0 decomposes exactly over terminal pixels.

Theorem 8.1 (Per-pixel decomposition, `Lean_85`). *Let* $\mathrm{H}^2 \coloneqq \mathrm{H}^2\big(\Pi_A^{(0)}, \Pi_B^{(0)}\big)$. *For* $s \in \mathscr{S}_+^{(0)}$, *define*

$$\mathrm{H}^2(s) \;\coloneqq\; \tfrac{1}{2}\big(m_A^{(0)}(s) + m_B^{(0)}(s)\big) \;-\; g^{(0)}(s), \tag{8.5}$$

where $g^{(0)}(s)$ *is the cumulative* $\widetilde{T}$*-mass at* s. *Then:*

(i) $\mathrm{H}^2(s) \ge 0$ *for every* $s \in \mathscr{S}_+^{(0)}$.

(ii)

$$\mathrm{H}^2 \;=\; \sum_{s\in\mathscr{S}_+^{(0)}} \mathrm{H}^2(s). \tag{8.6}$$

(iii)

$$\mathrm{H}^2_{\mathrm{ord}} \;\coloneqq\; \sum_{s\in\mathscr{S}^{(0)}} \mathrm{H}^2(s) \;=\; \mathrm{H}^2 - \mathrm{H}^2(\dagger) \tag{8.7}$$

Proofs of the backward pass and the decomposition are given in companion Chapter S.3.

The disagreement map. Theorem 8.1 defines $\mathrm{H}^2(\cdot)$ on the full terminal set $\mathscr{S}_+^{(0)} = \mathscr{S}^{(0)} \cup \{\dagger\}$: this is the *full terminal-state decomposition*, with $\sum_{s\in\mathscr{S}_+^{(0)}} \mathrm{H}^2(s) = \mathrm{H}^2_{(0)}$, and the cemetery contribution $\mathrm{H}^2(\dagger)$ is part of it. The *Hellinger disagreement map* is the ordinary-pixel restriction $s \mapsto \mathrm{H}^2(s)$ on $\mathscr{S}^{(0)}$, which is the quantity we visualize. It is non-negative, sums to $\mathrm{H}^2_{\mathrm{ord}}$ over ordinary pixels, and is computed from $g^{(0)}, m_A^{(0)}, m_B^{(0)}$. The quantity $\mathrm{H}^2_{\mathrm{ord}}$ is an unnormalized ordinary-state subsum; the conditioned-on-survival Hellinger distance that normalizes it is introduced below. Recall from Section S.3.1.2 that layer-0 ordinary states of the two games are *tagged* pixel pairs $s^{(0)}_{j,\pm}$, tagged by the player q for the SG and by the stream sign κ for the RG; the per-pixel attribution displayed in figures aggregates the two tags,

$$\mathrm{H}^2_{\mathrm{pixel}}(j) \;=\; \mathrm{H}^2\big(s^{(0)}_{j,+}\big) + \mathrm{H}^2\big(s^{(0)}_{j,-}\big), \tag{8.8}$$

which remains non-negative and sums to $\mathrm{H}^2_{\mathrm{ord}}$ over pixels. By the Cauchy–Schwarz step in the proof of Theorem 8.1(i), $g^{(0)}(s) \le \sqrt{m_A^{(0)}(s)\, m_B^{(0)}(s)}$, so $\mathrm{H}^2(s)$ is bounded below by the marginal surrogate $\frac{1}{2}\big(m_A^{(0)}(s) + m_B^{(0)}(s)\big) - \sqrt{m_A^{(0)}(s)\, m_B^{(0)}(s)}$, the per-pixel squared Hellinger distance between the layer-0 marginal occupations; the trajectory map is strictly larger whenever the conditional distributions of trajectories reaching pixel s differ between models.

Conditioning on Survival to the Ordinary Input Layer

When the cemetery absorbs a large fraction of trajectory mass, the total Hellinger distance mixes two effects: disagreement about *whether* a trajectory survives and disagreement about *where surviving trajectories go*. To isolate the second effect, we condition each model's trajectory law on the event that the trajectory reaches an ordinary input-layer state. The resulting scalar compares the two *surviving* trajectory distributions directly and removes the common cemetery floor. Because survival to the input layer factorizes along the layered graph, this conditioning remains Markov and is implemented by a per-state Doob h-transform (with $h = \psi_M$, the remaining survival probability).

Let Surv denote this event and assume that both survival masses are positive. With

$$Z_M = \mathbb{P}_M(\mathsf{Surv}) = \sum_{s\in\mathscr{S}^{(0)}} m_M^{(0)}(s) > 0, \qquad \mathrm{BC}^{(0)}_{\mathrm{ord}} = \sum_{s\in\mathscr{S}^{(0)}} g^{(0)}(s),$$

the conditioned-survival Hellinger distance is

$$\mathrm{H}^2_{\mathrm{surv}} = \mathrm{H}^2\big(\Pi_A(\,\cdot \mid \mathsf{Surv}), \Pi_B(\,\cdot \mid \mathsf{Surv})\big) = 1 - \frac{\mathrm{BC}^{(0)}_{\mathrm{ord}}}{\sqrt{Z_A Z_B}}.$$

Theorem S.3.4 gives the conditioned kernels and the proof; Equation (S.3.46) gives the corresponding non-negative map. Post-hoc normalization becomes numerically infeasible for deep networks when either survival probability gets too small; the conditioned-kernel computation avoids this final division.

8.3.4 Experiments

This subsection puts the trajectory-space machinery to two empirical tests, holding the model fixed: how the Hellinger trajectory distance reacts to input perturbations, and whether the resulting distance separates inputs from different ImageNet-S classes from one another and from random noise. Here SG is the Stopping-Game kernel (S.3.4) without risk aversion ($\lambda = 0$; named "ADF" in the code) and RG is the Routing-Game kernel (S.3.5) at $\tau = 1$. The diagnostic Routing-Game kernel omits the bias action and renormalizes the remaining branches, so it can differ from the bias-inclusive attribution game.

Input-Noise Sensitivity

For each architecture and each noise standard deviation $\sigma \in \{0.005, 0.01, 0.02, 0.03, 0.05, 0.1, 0.2, 0.3, 0.5\}$ we draw one standard-Gaussian tensor η per image and compute $\mathrm{H}(\Pi_A(x), \Pi_A(x + \sigma\eta))$ under both game kernels with the argmax target of the clean input. Figure 8.2 reports the mean H and $\mathrm{H}_{\mathrm{surv}}$ curves over the swept σ; both increase with σ for both games on all three architectures. On VGG-16 the RG H and $\mathrm{H}_{\mathrm{surv}}$ are equal, as the kernel routes no trajectories to states belonging to dead neurons. On ResNet-50 and ViT-B/16, skip connections route trajectories to dead-neuron states, and on ViT-B/16 the GELU gate also sends some trajectories to the cemetery at every state.

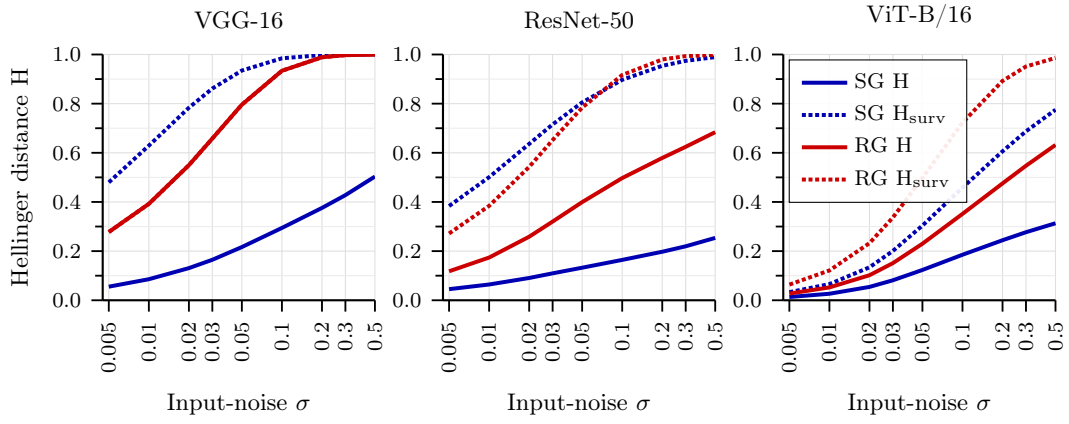


Fig. 8.2: Input-noise Hellinger trajectory distance. Solid: plain H; dotted: $\mathrm{H}_{\mathrm{surv}}$, conditioned on survival to an ordinary input-layer state. x-axis is log-scale input-noise σ; same (A,A) model on both sides, only the input differs.

Class Separability and Reference Baselines

Can the Hellinger distance differentiate between inputs of different classes? Partly; the reference baselines in Table 8.1 make the answer architecture- and kernel-specific:

- On **VGG-16**, the SG H is the kernel that does the work: clean-intraclass H = 0.513 sits below clean-interclass H = 0.607. The RG kernel saturates at H = 1 on every row on VGG-16, so it cannot resolve class structure on this architecture.
- On **ResNet-50**, the RG kernel is the discriminating one: clean-intraclass H = 0.786 vs. clean-interclass H = 0.902. The SG kernel does not separate classes on ResNet-50 (intraclass 0.272 vs. interclass 0.268).
- On **ViT-B/16**, the RG kernel gives intraclass 0.789 vs. interclass 0.882; the SG gap (0.422±0.023 vs. 0.435 ± 0.017) is smaller than the spread over pairs, so the SG does not separate classes there.

Random-vs-random is the lowest row in every column whose rows differ, and clean-vs-random is the highest row of every plain-H column except the SG on ResNet-50 (0.269 vs. intraclass 0.272) and the saturated RG on VGG-16, consistent with the model routing Gaussian inputs differently from in-distribution images. The conditioned variant H_{surv} is at least 0.99 on every row of VGG-16 and ResNet-50 and for the RG on ViT-B/16, so it carries no class-separability signal there. For the SG on ViT-B/16 it separates random-vs-random (0.852) from the clean rows (0.957–0.968), but neither intraclass from interclass pairs (0.963 vs. 0.968) nor clean-vs-random (0.957) from the clean pairs.

		Stopping Game		Routing Game	
Architecture	Baseline	H	H_{surv}	H	H_{surv}
VGG-16	clean-intraclass	0.513 ± 0.071	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
	clean-interclass	0.607 ± 0.032	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
	clean-vs-random	0.663 ± 0.020	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
	random-vs-random	0.320 ± 0.019	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
ResNet-50	clean-intraclass	0.272 ± 0.056	0.999 ± 0.000	0.786 ± 0.044	1.000 ± 0.000
	clean-interclass	0.268 ± 0.045	1.000 ± 0.000	0.902 ± 0.033	1.000 ± 0.000
	clean-vs-random	0.269 ± 0.035	1.000 ± 0.000	0.943 ± 0.010	1.000 ± 0.000
	random-vs-random	0.152 ± 0.008	0.992 ± 0.000	0.739 ± 0.011	0.999 ± 0.000
ViT-B/16	clean-intraclass	0.422 ± 0.023	0.969 ± 0.008	0.789 ± 0.067	1.000 ± 0.000
	clean-interclass	0.435 ± 0.017	0.974 ± 0.005	0.882 ± 0.010	1.000 ± 0.000
	clean-vs-random	0.439 ± 0.014	0.963 ± 0.006	0.891 ± 0.007	1.000 ± 0.000
	random-vs-random	0.377 ± 0.013	0.867 ± 0.013	0.476 ± 0.018	0.997 ± 0.000

Tab. 8.1: Reference baselines for the input-noise sweep, averaged over R pairs drawn from the 565-image ImageNet-S test split (clean rows: $R = 200$ sampled pairs; clean-vs-random: $R = 565$ images paired with a fresh Gaussian each; random-vs-random: $R = 200$ Gaussian pairs); entries are mean ± standard deviation over pairs. **clean-intraclass**: $H(\Pi_A(x_1), \Pi_A(x_2))$ with x_1, x_2 from the same ground-truth class; **clean-interclass**: same, with distinct classes; **clean-vs-random**: one side clean, other Gaussian; **random-vs-random**: both sides Gaussian.

8.4 Randomization Sanity Checks

This section collects the theoretical and empirical material behind the cascading-randomization experiments of Section 8.2. §8.4.1 carries the protocol of Adebayo et al. (2018) over to the trajectory-space Hellinger diagnostic of Section 8.3; §8.4.2 is the pixel-level heatmap-similarity sanity check on the main-paper attribution methods; §8.4.3 gives the fully random reference; §8.4.4 compares trajectory and map diagnostics per image.

8.4.1 Cascading Randomization in Trajectory Space

We apply the Hellinger framework of Section 8.3 to the parameter randomization test of Adebayo et al. (2018). Replacing one weight matrix yields two kernels on the same layered graph, and the Hellinger pass quantifies the resulting trajectory-space change. Notation follows Section 8.3: $\widetilde{T}^{(l)}$ is the geometric-mean kernel and $g^{(l)}(s)$ the overlap mass reaching state s, so $\mathrm{BC}^{(l)} = \sum_{s \in \mathscr{S}_+^{(l)}} g^{(l)}(s)$; $\mathrm{BC}^{(l)}_{\mathrm{ord}}$ denotes the same sum over ordinary states $s \in \mathscr{S}^{(l)}$ only.

The Stopping-Game contraction bound assumes a conditional alive probability bounded away from one; the Routing-Game bound assumes a uniform row-overlap bound strictly below one. The precise hypotheses and proofs are in Theorems S.3.6 and S.3.7; the cemetery-floor analysis is in Remark S.3.2.

The discounted occupation measures at the input layer recover gradient and $\alpha\beta$-LRP maps (Theorems S.1.6 and S.1.8). The undiscounted endpoint marginal $m_M^{(0)}$ is generally different: Hellinger compares the trajectory routing laws and does not include the path discounts. Under their respective contraction hypotheses, Theorems S.3.6–S.3.7 show that Π_M changes drastically under randomization, while Table 8.2 shows that the resulting attribution maps can remain similar. The two diagnostics therefore detect different aspects of parameter sensitivity: the map-level test of Adebayo et al. (2018) asks whether the explanation delivered to the user depends on the learned parameters, the trajectory-level test whether its normalized routing does. A map can fail the first while its routing passes the second; this discrepancy does not isolate a fixed projection loss, since the discounts can also change across models, and it does not make such a map adequate for model debugging (Adebayo et al., 2018; Sixt et al., 2020).

8.4.2 Heatmap similarity at full randomization

We apply the cascading parameter randomization test of Adebayo et al. (2018) to the attribution methods of Table 8.2. Starting from a pretrained network, we cumulatively re-initialize the convolutional and linear layers from the output toward the input (weights Kaiming normal (He et al., 2015), biases $\mathcal{N}(0, 0.01^2)$); BatchNorm (ResNet-50) and LayerNorm, the packed query/key/value projection of the attention, the positional embedding and the class token (ViT-B/16) keep their pretrained values, here and in the Hellinger cascade of Section 8.2.1. At each randomization depth we compute the attribution for every test image under both the original and the (partially) randomized model, then measure the similarity between the two attribution maps. A method that genuinely reflects the learned parameters should produce increasingly dissimilar maps as more layers are randomized; a method that merely responds to low-level input structure (edges, textures) will show high similarity even at full randomization.

Metrics. We report two complementary similarity measures between attribution maps:

- **Spearman** $\rho(|a|)$: The Spearman rank correlation of the absolute attribution values, measuring whether the two maps rank pixels in the same order of importance. We use absolute values because sign conventions may differ between original and randomized attributions; the correlation itself is signed.
- **HOG** r: The Pearson correlation between Histograms of Oriented Gradients (HOG) descriptors (Dalal and Triggs, 2005) extracted from the attribution maps, capturing structural similarity at the level of local gradient orientations.

We intentionally do *not* report SSIM here: Binder et al. (2023) show that randomization-based similarity scores can remain deceptively high because input-dependent structure and activation scales survive top-down randomization. All values are averaged over the 565-image test split of ImageNet-S (50 classes); a map that is constant over the image (LIME and KernelSHAP on the fully randomized ResNet-50, whose output no longer depends on the superpixels) has similarity 0 by convention. Table 8.2 reports the similarity at *full randomization* (all convolutional and linear layers re-initialized).

Discussion. Several patterns emerge. First, Gradient and Grad-CAM stay closest to zero on the CNNs (and LayerCAM on VGG-16), and Gradient, SmoothGrad and GAE on ViT-B/16: they are sensitive to the parameters, though this does not imply that they are faithful. Second, LRP-ε keeps the most rank structure on VGG-16 (0.341); on ResNet-50 the LRP rules (LRP-γ 0.254, LRP-ε 0.250) are close to Integrated Gradients and LayerCAM, which keep the most there. Third, LIME and KernelSHAP keep a HOG correlation of about 0.86 on VGG-16 and ViT-B/16 at low Spearman correlation: their superpixels are computed from the input image, so the maps inherit its edge structure; on ResNet-50 the fully randomized model's output no longer depends on the superpixels and their maps are constant (‡). Fourth, the RG rows lie between Gradient and LRP-ε on VGG-16, slightly below the LRP rules on ResNet-50, and keep the highest Spearman correlation of all methods on ViT-B/16 (0.454 for the anchor, 0.450 for the selected mode). Two maps become anti-correlated rather than unrelated (Grad-CAM++ on ResNet-50, TIBAV on ViT-B/16). The complementary trajectory-space analysis of §8.4.1 measures divergence of the underlying trajectory distributions rather than pixel-level map similarity.

Tab. 8.2: Parameter randomization sanity check (Adebayo et al., 2018). Each cell reports the average similarity between the attribution of the original pretrained model and the attribution of a fully randomized copy (every convolutional and linear layer re-initialized; normalization layers, the attention's query/key/value projection and the ViT embeddings stay pretrained). Values near zero indicate that the method detects the parameter change; the Spearman correlation is signed, so negative values occur. Bold marks the value closest to zero per metric within each architecture, among maps that are not constant; ‡: the map is constant at full randomization (similarity 0 by convention). Rows: the reported configuration of every method (Section 7.5).

Method	Spear. $\rho(\lvert a\rvert)$	HOG r
VGG-16		
Gradient	0.013	**0.009**
SmGrad	0.084	0.101
IntGrad	0.207	0.348
DeepLIFT	0.216	0.340
LRP-ε	0.341	0.457
LRP-γ	0.092	0.370
GCAM	**-0.007**	0.069
GCAM++	0.180	0.129
LCAM	-0.011	0.052
LIME	0.257	0.864
KernelSHAP	0.075	0.868
SG (λ=-1,σ^2=5, gate on)	0.289	0.456
RG anchor	0.299	0.535
RG (τ=0.3,ε=0.1)	0.284	0.287
ResNet-50		
Gradient	0.148	0.134
SmGrad	0.233	0.214
IntGrad	0.277	0.424
DeepLIFT	0.224	0.351
LRP-ε	0.250	0.393
LRP-γ	0.254	0.130
GCAM	**0.020**	**0.072**
GCAM++	-0.502	0.151
LCAM	0.266	0.145
LIME‡	0.000	0.000
KernelSHAP‡	0.000	0.000
RG anchor	0.217	0.344
RG (τ=0.9,ε=1)	0.180	0.316
ViT-B/16		
Gradient	**-0.003**	0.041
SmGrad	0.010	**0.038**
IntGrad	0.272	0.362
LRP-ε	0.165	0.300
LRP-γ	0.027	0.234
AttnLRP	0.124	0.176
DAVE	0.178	0.323
TIBAV	-0.175	0.150
GAE	-0.009	0.130
AR	0.120	0.153
LIME	0.248	0.861
KernelSHAP	0.131	0.867
RG anchor	0.454	0.469
RG (τ=0.4,λ_{sm}=-10,σ^2=5,ε=0.1)	0.450	0.310

Spear. $\rho(\lvert a\rvert)$ = Spearman rank correlation of absolute attribution values; HOG r = Pearson correlation of HOG descriptors. RG anchor = the $\alpha\beta$-LRP representation (τ=1, ε=0.5); the second RG row and the SG row are the localization-selected configurations and list the modes that differ (τ = temperature; λ, λ_{sm} = risk aversion; σ^2 = ADF input variance; ε = LRP stabilizer; $\tilde{\Phi}$ = noisy-observation gate).

8.4.3 Double-Random Empirical Baseline

To calibrate the asymptotic value that Theorems S.3.6–S.3.7 approach under *full* randomization, we compare two *independently random-initialized* copies of each architecture. This endpoint matches the deepest step of the cascading test of Adebayo et al. (2018) while eliminating any residual correlation with the pretrained weights.

Protocol. For each architecture we run $R = 500$ independent rounds. Each round: *(i)* instantiate two copies A_r, B_r of the pretrained checkpoint and re-initialize every `Conv2d`/`Linear` layer of each copy independently as in the cascade, matching its endpoint; the remaining parameters (normalization layers, the attention's packed query/key/value projection, positional embedding and class token) keep their pretrained values in both copies (starting from the checkpoint rather than from `weights=None`, whose ViT classification head is identically zero and would make the two trajectory laws trivially indistinguishable; the same construction is used for all three architectures). *(ii)* Draw one image index with replacement from the 565-image ImageNet-S test split (the 500 draws hit 336 distinct images under the fixed seed we use). *(iii)* Compute the plain Hellinger $\mathrm{H}(\Pi_{A_r}, \Pi_{B_r})$ and conditioned-survival $\mathrm{H}_{\mathrm{surv}}$ for both the Stopping Game and Routing Game kernels using the ground-truth label as target class. Table 8.3 reports the round-to-round mean and standard deviation; the round-to-round variance mixes two sources (image choice and the random-pair realization).

	Stopping Game		Routing Game	
Architecture	H	$\mathrm{H}_{\mathrm{surv}}$	H	$\mathrm{H}_{\mathrm{surv}}$
VGG-16	0.939 ± 0.003	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000
ResNet-50	0.938 ± 0.003	1.000 ± 0.000	0.973 ± 0.002	1.000 ± 0.000
ViT-B/16	0.958 ± 0.002	1.000 ± 0.000	0.956 ± 0.001	1.000 ± 0.000

Tab. 8.3: Hellinger trajectory distance between two independently random-initialized copies of the same architecture, averaged over $R = 500$ rounds (one fresh ImageNet-S test image per round). Plain H concentrates at an architecture-dependent cemetery floor of 0.94–0.97; conditioned-survival $\mathrm{H}_{\mathrm{surv}}$ saturates at unity. The VGG-16 Routing Game cell is the one case in which the plain distance also saturates at 1.000.

Interpretation. Plain H concentrates at a model-dependent floor of 0.94–0.97, matching the cemetery-floor prediction of Remark S.3.2: once both random kernels produce uncorrelated dead masks, the shared dead-in-both coincidence rate bounds $\mathrm{BC}^{(0)}$ below by a strictly positive constant and plain H cannot reach unity. The conditioned-survival distance removes the cemetery floor by renormalizing on survival and correctly reports the two independent random trajectory laws as essentially support-disjoint on ordinary input states ($\mathrm{H}_{\mathrm{surv}} \approx 1$ throughout). The pretrained-vs-randomized Hellinger distance that the cascading sanity check reports should therefore be read against this architecture-specific asymptote rather than against unity: values close to the double-random baseline already indicate that the game-theoretic trajectory law has become fully decoupled from the pretrained one.

8.4.4 Per-Image Association Between Trajectory and Map Diagnostics

Theorems S.3.6–S.3.7 and the double-random baseline of Table 8.3 bound the *aggregate* behavior of the Hellinger trajectory distance and the pixel-level attribution-map similarity. A complementary question, motivated by the argument of §8.4.1, is whether the two diagnostics agree *per image*: at

fixed randomization depth, does a larger displacement of an image's trajectory distribution correspond to a more dissimilar attribution map? Table 8.4 tests this directly.

Protocol. For each architecture we pick a cascading step at the same network depth in the Hellinger and attribution-sanity runs: the layers down to `features.26` on VGG-16 and down to `layer4.1.conv3` on ResNet-50 (identical sets of re-initialized layers), and the last encoder block on ViT-B/16, where the Hellinger cascade re-initializes the MLP and the output projection of the block and the attribution cascade additionally its Q, K and V projections. The interventions therefore match on the CNNs; on ViT-B/16, the comparison also reflects this difference in the randomized operators. The mean H / H_{surv} at these steps can be read off Figure 8.1. For each of the 565 ImageNet-S test images we then record four Hellinger scalars (H plain/surv for the Stopping and Routing Games) and four attribution-map scalars (Spearman $\rho(|a|)$ and HOG, each computed between the pretrained and partially-randomized model, for Gradient and for the $\alpha\beta$-RG anchor at τ=1; these are the attribution methods whose marginal laws line up with the Hellinger game kernels). Table 8.4 reports Spearman rank correlations across the 565 images.

		Within Hellinger				Gradient		RG anchor	
Architecture	Hellinger variant	$H_{RG,pl}$	$H_{RG,surv}$	$H_{SG,pl}$	$H_{SG,surv}$	Spear. $\rho(\lvert a\rvert)$	HOG r	Spear. $\rho(\lvert a\rvert)$	HOG r
VGG-16	$H_{RG,plain}$	1.00	+1.00	−0.33	+0.85	+0.07	+0.10	−0.01	+0.04
	$H_{RG,surv}$	+1.00	1.00	−0.33	+0.85	+0.07	+0.10	−0.01	+0.04
	$H_{SG,plain}$	−0.33	−0.33	1.00	−0.30	+0.12	+0.11	+0.00	−0.02
	$H_{SG,surv}$	+0.85	+0.85	−0.30	1.00	+0.01	+0.05	−0.10	−0.10
ResNet-50	$H_{RG,plain}$	1.00	+0.58	−0.08	+0.70	+0.05	−0.01	−0.08	−0.19
	$H_{RG,surv}$	+0.58	1.00	+0.16	+0.89	+0.15	+0.08	+0.05	−0.05
	$H_{SG,plain}$	−0.08	+0.16	1.00	+0.16	−0.04	+0.07	−0.01	+0.02
	$H_{SG,surv}$	+0.70	+0.89	+0.16	1.00	+0.16	+0.07	−0.01	−0.10
ViT-B/16	$H_{RG,plain}$	1.00	+0.96	+0.29	+0.30	+0.02	+0.00	−0.04	−0.08
	$H_{RG,surv}$	+0.96	1.00	+0.25	+0.28	+0.02	+0.02	−0.07	−0.07
	$H_{SG,plain}$	+0.29	+0.25	1.00	+0.99	−0.12	−0.06	−0.03	−0.05
	$H_{SG,surv}$	+0.30	+0.28	+0.99	1.00	−0.12	−0.06	−0.04	−0.05

Tab. 8.4: Per-image Spearman rank correlations across the 565-image ImageNet-S test split at the selected cascading steps of the Hellinger and attribution-sanity pipelines. The randomized layers match on the CNNs; the ViT-B/16 attribution cascade additionally randomizes Q, K, and V. Left block: within-Hellinger cross-correlations of the plain ($H_{\cdot,pl}$) and survival-conditioned ($H_{\cdot,surv}$) distances under the RG and SG kernels (self-entries shown as −; trivially +1). Right block: Hellinger-vs-map cross-correlations. The Hellinger-vs-map block is uniformly $|\rho| \leq 0.19$: per-image Hellinger and per-image attribution-map similarity have weak observed rank association.

Readings. The Hellinger-vs-map cross-block in Table 8.4 is uniformly near zero ($|\rho| \leq 0.19$ on all three architectures): per-image trajectory displacement and per-image attribution-map drift have weak observed rank association. This does not establish statistical independence or exclude nonlinear dependence. For ViT-B/16, the unequal interventions also limit conclusions about the diagnostics alone.

The within-Hellinger block shows that per-image H carries a stable ordering of the images. Where little trajectory mass reaches the cemetery, the plain and conditioned distances of the same game rank images almost identically: $\rho = +0.96$ (RG) and +0.99 (SG) on ViT-B/16, and identical values for the RG on VGG-16, whose kernel routes no trajectories to states of dead neurons. Under the SG on the ReLU backbones the two normalizations decouple ($\rho = -0.30$ on VGG-16, +0.16 on ResNet-50): the cemetery share of plain H varies with the image's dead masks, and the renormalization by $1/\sqrt{Z_A Z_B}$

removes exactly this share. The conditioned distances of the two games agree strongly on the ReLU backbones ($\rho = +0.85$ on VGG-16, $+0.89$ on ResNet-50) and only weakly on ViT-B/16 ($+0.28$), where the two kernels read partly different per-image structure.

A secondary observation from the within-method pattern: Spearman $\rho(|a|)$ and HOG within a single attribution method agree well ($\rho = 0.65$–0.82, computed from the same per-image joins), but Gradient and $\alpha\beta$-RG similarities barely agree across images (ρ between -0.10 and $+0.30$ depending on architecture and similarity metric). Together with the near-zero Hellinger-vs-map block, this tightens the conclusion of §8.4.1: trajectory-space and pixel-level diagnostics yield weakly associated per-image rankings in these runs, and within pixel-level diagnostics the choice of attribution method changes *which* images a randomization sanity check flags.

9 Conclusion and Future Work

Section 9.1 retraces the preceding chapters; Section 9.2 records the limitations of the theory and evidence; and Section 9.4 collects the directions this work opens.

9.1 Summary of Contributions

The starting point was the growing role of interpretability in improving and controlling models. When attribution guides a correction or an intervention, both the internal representation it reads and the backward calculation it performs matter. This thesis connects those two choices through *network decomposition*: expose structure a trained ReLU network already contains, preserve its predictions, and study how attribution propagates through that structure.

The common mathematical foundation is the decomposition theory of Chapter 3. A *valid decomposition* replaces each layer map by a pair of coordinated split-streams whose difference reproduces the network exactly, and the chapter populated the definition with instances of increasing structure. The mixing family f^{η}_{mix} interpolates between trivial and fully separated splits; its endpoint $\eta = 1$ recovers the classical DC decomposition of a ReLU network into a difference $f = g - h$ of two convex, monotone split-streams, routing the weight parts $W^{(l,+)}, W^{(l,-)}$ into separate channels (Zimmermann and Loho, 2026). Beyond the piecewise-linear case, Theorem 3.13 established a DC–monotone splitting for every activation with integrable curvature — GELU (Hendrycks and Gimpel, 2016), Swish (Ramachandran et al., 2017), Mish, tanh, sigmoid, softplus — with a canonicity statement singling out the split of minimal curvature, and a bounded-variation extension recovering ReLU as the Dirac limit. The stopping decomposition f_{stop} added an early-termination channel, and the regularized family f^{Ω}_{reg} showed that a convex regularizer on the stop/continue decision generates regularized decompositions systematically, the softplus decomposition f^{θ}_{soft} arising as the binary-entropy subcase — placing the softplus replacement of Dombrowski et al. (2022) inside the framework rather than beside it. The chapter closed with the stabilization that makes split passes usable at depth and a full CNN extension: convolutions, gated pooling, batch normalization, and residual connections.

The game semantics of Chapter 4 gives the stopping instance a second, complementary reading. Building on the ReLU-network games of Gaubert & Vlassopoulos (2025), the stopping decomposition was reinterpreted as a two-player zero-sum SG on a doubled state graph: weight signs route between the players P^+ and P^-, gradients and $\alpha\beta$-LRP attributions emerge as discounted occupation measures under equilibrium policies, and the deformation parameters of Chapter 3 — temperature, risk aversion, stabilizers — become principled modifications of the game. Chapter 8 takes the decisive step: the equilibrium policies induce a genuine *probability distribution* Π over complete backward trajectories, and this distribution records normalized backward routing. Recovering attribution additionally requires its signs and model-dependent discounts. The comparison with the NRM framework of Xiong et al. (2026) in Section 8.1 identifies their shared path perspective: NRM assigns normalized signed relevance to neuron walks, while the game separates a probability law on expanded states from the signs and discounts used to recover attribution. A direct identification requires matching the propagation and normalization conventions. The probability law makes trajectory-level statistical comparison possible: the Hellinger distance between two trajectory laws is computable in one linear-time pass, and the parameter-randomization sanity check of Adebayo et al. (2018) lifts from attribution maps to trajectory space, where Theorem S.3.6 shows Bhattacharyya mass decays geometrically in the number of randomized layers — detecting collapse even when heatmaps remain deceptively similar (Sixt et al., 2020).

These representations lead to the attribution methods of Chapter 6. SplitCAM, SplitLRP, and SplitGrad apply established attribution principles to the decomposed streams, exposing separate positive and negative contributions; the difference-of-convex input neural networks (DCINNs) of Section 6.1.4 added a counterpart trained directly in split form, with visually distinct stream sensitivities in the displayed MNIST examples. The tropical foundations (Section 2.4) supply the geometric picture: through the correspondence between ReLU networks and tropical rational maps (Zhang et al., 2018b), the split-streams g and h acquire Newton polytopes, and the *imaginative polytope* of Section 2.4.3 interprets what the decomposition adds to the geometry. Chapter 5 extends the framework to Transformers (Vaswani et al., 2017) in two distinct ways. The constructive DC decomposition of Section 5.3 retains the full input dependence of Queries, Keys, Values, softmax, LayerNorm, and GELU, producing convex streams that are monotone in the independent non-negative split inputs. Separately, the conditioned Value-routing game supplies the attribution implementation evaluated on ViT-B/16 (Dosovitskiy et al., 2021).

The shared evaluation questions in Chapter 7 concern localization, faithfulness, robustness, and the cost of obtaining these signals. The split methods were evaluated on VGG-16 and ResNet-18 on ImageNet-S; the revised game study covers VGG-16, ResNet-50, and ViT-B/16 on ImageNet-S, with a ViT-B/16 transfer study on Pascal VOC. No single method dominates every metric. On ViT-B/16, one adapted $\alpha\beta$-LRP configuration outperforms all evaluated baselines in attribution localization, and this lead transfers to Pascal VOC without re-tuning. The LRP baselines remain stronger on the faithfulness metrics, while the anchor configurations are particularly robust. Independent validation selections for localization and faithfulness, bootstrap confidence intervals, and Holm-corrected paired tests make these trade-offs explicit. Hellinger diagnostics complement pixel-level comparisons: they distinguish trained from randomized calculations even where heatmaps remain similar, increase with Gaussian input noise, and separate intra- from inter-class pairs for at least one game on each tested architecture.

Taken together, the thesis substantiates its central claim: monotone, convex, game-theoretic, and polyhedral structure need not be trained into a network — it is hidden in every trained ReLU network, and a single decomposition framework makes all four readings available at once.

9.2 Limitations

The results above come with qualifications that should be stated plainly.

Computational overhead. A split pass propagates two coordinated streams per channel and, currently, materializes the split network with duplicated weight matrices. In the reported implementation, SplitCAM takes approximately 14–29 times as long per image as the corresponding ordinary Grad-CAM and LayerCAM configurations; Section S.4.1.5 states the measurement scope. The available artifacts do not identify the hardware, numerical precision or exact configuration IDs of the historical timing run. Masked views of the original weights could avoid parameter duplication, but would not remove the need to propagate the two streams. This thesis has not implemented or benchmarked that optimization.

Stabilization hyperparameters. The stabilization of Section 3.6 is essential at depth and introduces choices — thresholds, shifting schemes, and in the game view the temperatures τ, θ and the risk-aversion parameter λ — that the theory constrains but does not determine. The revised experiments use declared validation grids and independent localization and faithfulness selections. Their differing

winners show that there is no universally optimal setting; transfer to Pascal VOC is demonstrated for configurations fixed on ImageNet-S. The same holds for layer selection in SplitCAM and SplitLRP: different layers excel at different metrics.

Transformer structure and numerical stability. The exact constructive DC theorem of Section 5.3 covers fixed finite inference graphs built from the stated operations, with positive LayerNorm ε and at least one allowed key in every masked-softmax row. Its full input dependence is distinct from the evaluated Value-routing game, which conditions on attention, LayerNorm statistics, and GELU gates and does not recover the full derivative through Q and K. That game represents a specified backward rule, without establishing a unique explanatory game. Exact recovery in real arithmetic also does not guarantee stable stream evaluation: the numerical Transformer prototype exhibits severe cancellation, and a practical numerically stable implementation of the full DC construction remains open. The ViT-B/16 attribution results evaluate the conditioned game, not this new construction.

Benchmark scope. The empirical evidence is specific to the architectures (VGG-16, ResNet-18, ResNet-50, ViT-B/16), datasets (ImageNet-S, Pascal VOC), and metric selections of the two underlying studies. Localization against human masks measures spatial agreement, not faithfulness to the model's decision process; sharper attributions can discard network-relevant evidence outside the annotated object. The separate faithfulness metrics assess specified perturbation protocols and do not establish a general causal explanation. All experiments concern vision models; whether the gains transfer to other modalities or into downstream task gains is untested. Finally, the trajectory-level diagnostics were validated on sanity checks and controlled perturbations — dedicated calculation-level benchmarks do not yet exist.

Stream interpretation. The MNIST stream-gradient examples are qualitative. In an inverted-input DCINN, the complete logit gradient is the sum of the two plotted stream sensitivities, so either stream alone can suggest a direction that the other outweighs. Convexity does not prevent this when the first affine layer is unconstrained. The displayed concentration on present and candidate missing strokes therefore motivates an interpretation to be evaluated through complete-model interventions; it does not certify finite counterfactuals or necessary features.

9.3 Broader Impact

The work is methodological: it studies attribution methods for existing image classifiers and does not introduce a deployed decision system, a new dataset of people, or a high-risk generative model. Its intended positive impact is to make interpretability methods more diagnostically accountable. In particular, trajectory-level comparisons can reveal cases where visually similar attribution maps arise from different backward calculations, and can therefore support more careful auditing of interpretability pipelines used in scientific or engineering workflows.

The same capability can also have negative or misleading uses. Sharper or more stable attribution maps may create unwarranted confidence in a model if users treat attributions as causal guarantees rather than as post hoc diagnostics. In safety-critical or fairness-sensitive settings, improved-looking attributions could be used to justify a classifier whose predictions remain brittle, biased, or dataset-dependent. The Hellinger trajectory distance also compares internal calculation paths rather than social validity; small trajectory distance does not imply that the underlying model is fair, robust, or appropriate for deployment.

We therefore view the framework as an analysis tool rather than a deployment certificate. The experiments use public benchmark classifiers and ImageNet-derived evaluation data, and the results should be interpreted as evidence about attribution behavior under these benchmarks, not as evidence that the underlying models are suitable for consequential use.

9.4 Future Research Directions

Four directions appear most promising.

Concept queries on the expanded state graph. Section 8.1 identifies a possible connection with the NRM framework (Xiong et al., 2026). Its queries over sets of neurons suggest corresponding events for game trajectories. Their probabilities and, separately, their signed discounted occupation measures could support concept-level analysis across layers. Establishing a correspondence with a chosen NRM or CRP rule requires matching the propagation factors, normalization, and absorption conventions; discounts do not replace probability normalization. For positive-probability concept events, the conditioned game trajectories could then be compared using the Hellinger machinery. This remains a proposed extension and could supply calculation-level benchmarks of the kind discussed in Section 9.2.

Regularized decompositions for smooth activations. Theorem 3.13 guarantees canonical DC–monotone splits for GELU, Swish, and their relatives, and the regularized family $f_{\mathrm{reg}}^{\Omega}$ of Section 3.5 explains softplus as an entropy-regularized gate — but the two are not yet joined. A direct identification is impossible for GELU and Swish: a conjugate activation

$$\Omega^*(z) = \sup_{\pi \in [0,1]} \{\pi z - \Omega(\pi)\}$$

is always convex, non-decreasing, and 1-Lipschitz, whereas these activations do not have those global properties. The open question is how to combine their exact DC–monotone decompositions with games on the non-negative streams, or to formulate local game interpretations conditional on the realized activation pattern. Such a construction would need its own recovery and backward-occupation theorems before trajectory diagnostics could be transferred to it.

DCINNs at scale. The MNIST experiments of Section 6.1.4 showed class-dependent stream sensitivities on present and candidate missing strokes, but the evidence is qualitative and limited to small models. Establishing whether these patterns support reliable counterfactual explanations requires interventions on the complete model. Scaling DCINNs also raises optimization and theory questions: whether the difference-of-monotone parametrization trains stably at ImageNet scale, whether DC-programming techniques (Thi and Dinh, 2018) or ideas from ICNNs (Amos et al., 2017) accelerate training, and how the accuracy–interpretability trade-off behaves with capacity. Positive results would strengthen the case for using split architectures as a design principle for interpretable models.

Transformers, LLMs, and efficient split passes. The constructive theorem already covers fixed causal masks with nonempty allowed-key sets. Applying it to autoregressive LLMs still requires stable stream evaluation, efficient key–value cache management, and robustness testing; extending trajectory diagnostics additionally requires a specified backward rule and its recovery guarantees. Implementing split passes through masked views rather than materialized duplicates could remove the

weight duplication discussed in Section 9.2, and the nonnegativity of the split weights invites hardware-level exploitation. The observed layer dependence of attribution quality calls for principled layer selection or multi-layer aggregation, for which the trajectory distribution offers the right vocabulary: layers are intermediate marginals of one measure Π.

The unifying lesson is that a trained network is a richer object than the function it computes. One set of weights carries a difference of convex monotone parts, a zero-sum game between cooperating adversaries, a probability distribution over backward trajectories, and a pair of Newton polytopes — none of which requires changing the network. Interpretability, on this view, provides structured access to computations that can be inspected, compared, and modified. The decomposition and game perspectives supply a mathematical basis for that access; evaluating how these tools improve a model through downstream correction, training, or steering is the next step.

Appendix A

Experimental Protocol and Reproducibility

The complete evaluation tables, validation grids, ablations and qualitative examples are collected in the companion supplement, Chapter S.4.

A.1 Split-Method Evaluation

The split-method campaign explains the model's predicted class, including for misclassified images. Its localization metrics compare that attribution with the annotated object mask; when the prediction differs from the annotated class, the explanation target and the mask therefore refer to different classes. The game-method campaign below instead explains the dataset target class. Both retain misclassified images, and their results should be read under these distinct target conventions.

A.1.1 Artifact Availability and Reproduction Scope

The Hidden Monotonicity tables retain their published-source provenance. The available code checkout does not contain the original per-image evaluation outputs, the complete historical evaluation environment, or the six training configurations advertised for the MNIST multi-seed campaign. The runnable notebook-derived example is a separate example and does not reconstruct those missing settings. Consequently, exact reproduction of the historical tables cannot be established from the available artifacts alone.

The current evaluation implementation has been exercised on deterministic CPU fixtures, including double-precision reconstruction, input derivatives, attribution, timing, and Pixel Flipping with an independently evaluated initial target score. Those checks establish the exercised implementation contracts; the original ImageNet campaign has not been rerun, so agreement with its published values is unverified. The historical training runner used the test loader during epoch validation. The current runner instead reserves a deterministic 10% of the training pool for validation and evaluates the test set only after training, recording the partition indices and seed. This current protocol does not establish which checkpoint-selection procedure generated the historical MNIST figures.

A.2 Evaluation Protocol

A.2.1 Data and splits

The game-method campaign uses the released ImageNet-S50 validation annotations (Gao et al., 2023), which contain 752 images from 50 classes. We exclude 136 images whose annotation class differs from the official ILSVRC2012 label, leaving 616 eligible images. With seed 42, one image per class forms the **50-image validation split**, used only for configuration selection; the remaining 566 form the test pool. Images whose object mask lies entirely outside the 224×224 center crop are excluded after the split, because the mask-based metrics are undefined for them. This removes one test image and leaves **565 test images**. The explanation target is the dataset class, including when the model misclassifies the image. This population and target convention belong to the game-method protocol and need not coincide with the split-method campaign. The Pascal VOC 2012 transfer split is described in Section S.4.4.4.

A.2.2 Metric definitions

Let $x \in \mathbb{R}^{C\times H\times W}$ be an image, let y be its target class, and let $a \in \mathbb{R}^{H\times W}$ be the scalar attribution map produced by an explanation method after channel reduction. Write $d \coloneqq HW$ and let $\mathrm{vec}(a) \in \mathbb{R}^d$ denote the flattened map. For any binary segmentation mask $m \in \{0,1\}^{H\times W}$, write $\mathrm{vec}(m) \in \{0,1\}^d$. When a metric is configured with `abs=true` and `normalise=true`, the evaluator applies the max-abs normalization (not min–max)

$$\psi(a) = \frac{|\mathrm{vec}(a)|}{\max_i |\mathrm{vec}(a)_i|}, \tag{A.1}$$

with the convention that the all-zero map remains zero. We denote the preprocessed attribution vector by $\bar{a}$. All metrics below use $\bar{a}$; faithfulness metrics use raw logits (no softmax) to avoid saturation effects, and perturbations use the Quantus per-sample mean replacement baseline.

Attribution localization (AL). The inside-mass ratio of Kohlbrenner et al. (2020), who compute it with ground-truth bounding boxes; we use the ImageNet-S pixel masks:

$$\mathrm{AL}(a,m) = \frac{\sum_{i=1}^{d} \bar{a}_i m_i}{\sum_{i=1}^{d} \bar{a}_i + 10^{-9}}. \tag{A.2}$$

Pointing game (PG). Following the Excitation Backprop protocol (Zhang et al., 2018a), the pointing-game config uses `abs=true` and no normalization. Let $M(a) \coloneqq \{i \in \{1,\dots,d\} : \bar{a}_i = \max_j \bar{a}_j\}$ be the entire set of maximizers. Then

$$\mathrm{PG}(a,m) = \mathbf{1}\big\{M(a) \cap \{i : m_i = 1\} \neq \varnothing\big\}. \tag{A.3}$$

Ties are scored as a hit whenever at least one maximizer lies inside the object mask. PG is binary per image–class record, so its ImageNet-S test mean is a multiple of $1/565$.

Top-k intersection (TK). Following Theiner et al. (2021), with $k = 1000$ and the permutation π that sorts $\bar{a}$ in descending order:

$$\mathrm{TK}(a,m) = \frac{1}{1000} \sum_{r=1}^{1000} m_{\pi_r}. \tag{A.4}$$

Pixel-flipping AUC ($\mathbf{PF}_5$, $\mathbf{PF}_{20}$). Pixel flipping (Bach et al., 2015; Samek et al., 2017) removes the most relevant features first and tracks the class score. The evaluator uses `pixel_flipping_abs` with `features_in_step = 1568` and `perturb_baseline = mean`. Unlike the spatial localization metrics above, pixel flipping operates on $D \coloneqq CHW$ scalar input features. The spatial attribution is broadcast over the color channels; let $\widetilde{\pi}$ sort the flattened broadcast attribution in descending order. For $T \coloneqq \lceil D/1568 \rceil$, the input $x^{(t)}$ is obtained from $x^{(t-1)}$, starting at $x^{(0)} = x$, by replacing features $\widetilde{\pi}_{(t-1)\cdot 1568+1}, \dots, \widetilde{\pi}_{\min\{t\cdot 1568, D\}}$ with the scalar mean of the current flattened input, as implemented by Quantus. The evaluator computes

$$f_t \coloneqq F(x^{(t)})_y, \qquad t = 1,\dots,T, \tag{A.5}$$

and PF_p is the trapezoidal area over the first p% of this flipping curve, i.e. over its first K_p points with the abscissa rescaled to [0,1]: $\mathrm{PF}_p = \mathrm{trapz}\big(\{(t-1)/(K_p-1), f_t\}_{t=1}^{K_p}\big)$, $K_p = \max(2, \lceil pT/100 \rceil)$. For $C = 3$ and $H = W = 224$, $D = 150{,}528$ and $T = 96$ steps, giving $K_5 = 5$ and $K_{20} = 20$. The curve contains only post-perturbation scores; the original input score is not an additional point. Lower is better.

Selectivity (Sel). Following Montavon et al. (2018) and its Quantus implementation (Hedström et al., 2023), $\bar{a}$ is summed over square patches of 39×39 pixels (side $\lfloor\sqrt{1568}\rfloor$); non-overlapping patches are removed (replaced by the per-sample mean) in decreasing order of their summed attribution, and the target logit is recorded after each removal. Sel is the trapezoidal area under this curve, with the fraction of removed patches on the abscissa. Lower is better.

Max and average sensitivity (MaxS, AvgS). Following Yeh et al. (2019), the evaluator regenerates attributions on the same $N_s = 50$ inputs for both metrics, $x + \xi_k$ with uniform noise $\xi_k \sim \mathscr{U}[-r, r]^{C\times H\times W}$, $r = 0.2$, and reports the maximum (MaxS) or mean (AvgS) over k of $\|\psi(a(x+\xi_k)) - \psi(a(x))\|_F / \|\psi(a(x))\|_F$. Lower is better, as it indicates stability of the explanation to input noise. The noise is seeded per image, so every row sees the same perturbed inputs. MaxS and AvgS are computed on a fixed random subset of 200 test images (seed 0) that is identical for all rows and models; all other metrics use the full test split.

A.2.3 Selection

For every tunable method, baselines and ours alike, two configurations are selected on the validation split: `loc` minimizes the rank-sum over {AL, PG, TK} and `faith` the rank-sum over $\{\mathrm{PF}_5, \mathrm{PF}_{20}, \mathrm{Sel}\}$. Ranks are computed within each method's grid, respecting metric directions; ties are broken in favor of the default (for RG, the anchor), then of fewer non-default parameters, then of grid order. A configuration is admissible only if its map is non-zero and every metric is defined on every validation image; inadmissible configurations are excluded before ranking (5 of the 40 SG configurations on VGG-16, all with strong risk aversion and the deterministic gate, whose maps vanish on some images; no other configuration). MaxS and AvgS are not used for selection because each evaluation re-explains every image $2N_s$ times. The `loc` row is the one reported in the main tables, and the `faith` row of every method is reported alongside (Section S.4.4): localization is a natural target because the analytical modes are designed to control routing focus, because mask localization has an external ground truth whereas perturbation metrics evaluate off-distribution inputs, and because comparing LRP rule choices by object localization has precedent, there with bounding boxes (Kohlbrenner et al., 2020); localization is not faithfulness, so both rows are reported and the trade-off stays visible.

A.2.4 Statistics

The game-method test tables report 95% percentile bootstrap confidence intervals (Efron and Tibshirani, 1993) using $B = 10{,}000$ resamples and seed 0. On ImageNet-S, the sampling units are images, with the same resampled image indices used across method rows. MaxS and AvgS intervals resample the fixed 200-image subset of Section A.2.2. Continuous metrics use a two-sided paired bootstrap test of the mean per-image difference: the bootstrap distribution is centered at zero, and p is the fraction of centered resampled differences at least as large in absolute value as the observed difference. Binary PG uses McNemar's test (McNemar, 1947) with continuity correction.

On Pascal VOC, the estimand remains the mean over image–class records, but the independent sampling units are the $G = 2{,}293$ original test images. Each bootstrap replicate samples G images with replacement and retains every class record of each sampled image. Its mean is the sum over these records divided by their replicate-specific count. Thus, classes from the same image remain together, while images with more evaluated classes retain their contribution to the reported record-weighted mean. Paired comparisons use common finite records and common image draws. Every VOC metric, including binary PG, uses the corresponding two-sided, null-centered paired cluster-bootstrap test. For observed mean difference $\widehat{\Delta}$ and bootstrap differences Δ_b^*, its finite-resample p-value is

$$p = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{|\Delta_b^* - \widehat{\Delta}| \geq |\widehat{\Delta}|\}}{B + 1}. \tag{A.6}$$

For the main and full test tables, Holm correction (Holm, 1979) is applied to all unordered pairs of eligible rows within each metric and reported table, before extracting comparisons with the observed best row or declaring a row ahead of every baseline. The observed winner therefore does not determine the comparison family. A difference is called significant at adjusted $p < 0.05$; otherwise, no difference was detected at this threshold. This does not establish practical equivalence. These intervals and tests condition on the fixed checkpoints, selected configurations and realized attribution evaluations; they do not quantify training or configuration-selection variability. The split-method tables report the source point estimates without corresponding confidence intervals.

A.3 Formal Verification and Mathematical Audit

The accompanying thesis repository contains the TeX sources, pinned links to both official code repositories, and a Lean 4 project with mathlib pinned to version 4.35.0. The formalization makes the assumptions of each checked statement explicit. It remains partial: a related lemma does not certify an entire theorem or either paper.

Numbered `Lean_1`, `Lean_2`, ... references appear in the brackets of theorem and lemma headings, following the convention of the planar-features paper. Each links to the checked declaration or components of the corresponding statement; the coverage map distinguishes these cases. The exact assumptions and remaining parts are recorded by source label in `formalization/coverage_map.json`. Selected equation numbers also link to their checked identities or definitions. All links identify a fixed Git commit and the actual declaration line, following the TeX-linking approach of arXiv:2610.01728; subsequent edits to the default branch cannot move these targets.

Further details of the checked developments and mathematical audit are given in the companion supplement, Section S.4.6.

Appendix B

Publication Provenance and Tool Use

The results and experiments integrated in this thesis originate in two coauthored works: *Hidden Monotonicity*, published at CVPR 2026 and selected as a Highlight (Zimmermann and Loho, 2026), and the *Playing the Network Backward* preprint (Zimmermann et al., 2026), in its revised version arXiv:2605.06212v2. The thesis author is first author of both works. The game paper is presented here as a preprint, without a claim of AISTATS acceptance. Their technical content is organized by the shared decomposition framework rather than reproduced as two separate papers. The model-evaluation outputs and reported point estimates are reused. VOC confidence intervals and paired comparisons are recomputed from those outputs by resampling whole images, as specified in Section A.2.4. The thesis adds common definitions, mathematical extensions and connections between the constructions, and Lean formalization.

B.1 Author's Contribution

My principal contributions to *Hidden Monotonicity* and *Playing the Network Backward* were the development of the theory, the development and execution of the experiments, and the writing of all appendices and most of the paper text outside the introductions. The papers are collaborative works with the coauthors identified in Section 1.1; this account specifies my responsibilities within those collaborations.

The thesis brings the two works into a common mathematical exposition, connects their decomposition constructions, and adds the accompanying Lean formalization. The tool-use disclosure below distinguishes the preparation of the original papers from the subsequent thesis integration, formalization, and revision.

B.2 Disclosure in the Revised Game Paper

The revised game paper contains the following AI-use statement, reproduced here as a statement about the preparation of that publication:

> In this work, we used generative AI tools for implementing standard baseline methods and evaluation code, for producing plots and tables of results, for drafting parts of the text, and for editing (grammar, spelling, word choice, and reformatting). We have not used generative AI tools to generate research ideas, to formulate the games, or to produce the theorems and their proofs. All AI-assisted code was reviewed and tested by the authors, including unit tests that check the game-theoretic occupation measures against reference gradient and $\alpha\beta$-LRP implementations; all AI-assisted text was checked by the authors for correctness, and every reference was verified manually. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

B.3 AI Assistance in the Thesis Revision

The November 2026 thesis revision used OpenAI Codex, supplied by OpenAI, for mathematical exploration and proof development, Lean implementation and checking, code review and repair, statistical reaggregation, and drafting and editing the thesis. The mathematical assistance included developing and checking extensions of the convex, monotone stream construction. These activities are distinct from the preparation of the original paper described above; its historical statement about theorem and proof development does not describe this later thesis work.

The formalization repository records reviewed theorem types, proof-checking receipts, and remaining coverage limits. Successful Lean checking certifies the stated formal propositions; source correspondence still requires mathematical review, and the kernel checks do not certify the empirical experiments.

List of References

Abnar, S., W. Zuidema (2020): Quantifying Attention Flow in Transformers. *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Association for Computational Linguistics, 4190–4197. DOI: 10.18653/v1/2020.acl-main.385. URL: http://dx.doi.org/10.18653/v1/2020.acl-main.385 (cit. on pp. 6, 55, 61, 62).

Achtibat, R., M. Dreyer, I. Eisenbraun, S. Bosse, T. Wiegand, W. Samek, S. Lapuschkin (2023): From Attribution Maps to Human-Understandable Explanations through Concept Relevance Propagation. *Nature Machine Intelligence* 5 (9), 1006–1019. DOI: 10.1038/s42256-023-00711-8 (cit. on pp. 61, 93).

Achtibat, R., S. M. Vakilzadeh Hatefi, M. Dreyer, A. Jain, T. Wiegand, S. Lapuschkin, W. Samek (2024): AttnLRP: Attention-Aware Layer-Wise Relevance Propagation for Transformers. *Proceedings of the 41st International Conference on Machine Learning (ICML)*. Vol. 235. Proceedings of Machine Learning Research. PMLR, 135–168 (cit. on pp. 6, 55, 56, 61, 64, 69, 72, 90, 91).

Adebayo, J., J. Gilmer, M. Muelly, I. J. Goodfellow, M. Hardt, B. Kim (2018): Sanity Checks for Saliency Maps. *NeurIPS*, 9525–9536 (cit. on pp. 2, 4, 5, 79, 94, 99, 100, 102, 103, 106).

Ali, A., T. Schnake, O. Eberle, G. Montavon, K.-R. Müller, L. Wolf (2022): XAI for Transformers: Better Explanations through Conservative Propagation. *Proceedings of the 39th International Conference on Machine Learning (ICML)*. Vol. 162. Proceedings of Machine Learning Research. PMLR, 435–451 (cit. on pp. 6, 55, 56, 61, 62, 64, 68, 69, 71, 72).

Alvarez-Melis, D., T. S. Jaakkola (2018): *On the Robustness of Interpretability Methods*. DOI: 10.48550/ARXIV.1806.08049. URL: https://arxiv.org/abs/1806.08049 (cit. on p. 5).

Amos, B., L. Xu, J. Z. Kolter (2017): Input Convex Neural Networks. *Proceedings of the 34th International Conference on Machine Learning*. Ed. by D. Precup, Y. W. Teh. Vol. 70. Proceedings of Machine Learning Research. PMLR, 146–155. URL: https://proceedings.mlr.press/v70/amos17b.html (cit. on pp. 1, 5, 10, 29, 32, 53, 81, 109).

Ancona, M., E. Ceolini, C. Öztireli, M. Gross (2018): *Towards better understanding of gradient-based attribution methods for deep neural networks*. In *International Conference on Learning Representations (ICLR)*, 2018. URL: https://openreview.net/forum?id=Sy21R9JAW (cit. on p. 54).

Anders, C. J., D. Neumann, W. Samek, K.-R. Müller, S. Lapuschkin (2023): *Software for Dataset-wide XAI: From Local Explanations to Global Insights with Zennit, CoRelAy, and ViRelAy*. arXiv: 2106.13200 [cs.LG]. URL: https://arxiv.org/abs/2106.13200 (cit. on pp. 55, 90).

Anders, C. J., L. Weber, D. Neumann, W. Samek, K.-R. Müller, S. Lapuschkin (2022): Finding and removing Clever Hans: Using explanation methods to debug and improve deep models. *Information Fusion* 77, 261–295. ISSN: 1566-2535. DOI: https://doi.org/10.1016/j.inffus.2021.07.015. URL: https://www.sciencedirect.com/science/article/pii/S1566253521001573 (cit. on p. 1).

Auer, P., N. Cesa-Bianchi, P. Fischer (2002): Finite-time Analysis of the Multiarmed Bandit Problem. *Machine Learning* 47 (2-3), 235–256. ISSN: 1573-0565. DOI: 10.1023/a:1013689704352. URL: http://dx.doi.org/10.1023/A:1013689704352 (cit. on p. 58).

Awasthi, P., A. Mao, M. Mohri, Y. Zhong (2023): DC-programming for neural network optimizations. *Journal of Global Optimization*. DOI: 10.1007/s10898-023-01344-2 (cit. on pp. 5, 31).

Ayinde, B. O., J. M. Zurada (2018): Deep Learning of Constrained Autoencoders for Enhanced Understanding of Data. *IEEE Transactions on Neural Networks and Learning Systems* 29 (9), 3969–3979. DOI: 10.1109/TNNLS.2017.2747861 (cit. on pp. 1, 77).

Bach, F. (2017): Breaking the Curse of Dimensionality with Convex Neural Networks. *Journal of Machine Learning Research* 18 (19), 1–53. URL: http://jmlr.org/papers/v18/14-546.html (cit. on p. 5).

Bach, S., A. Binder, G. Montavon, F. Klauschen, K.-R. Müller, W. Samek (2015): On Pixel-Wise Explanations for Non-Linear Classifier Decisions by Layer-Wise Relevance Propagation. *PLOS ONE* 10 (7), 1–46. DOI: 10.1371/journal.pone.0130140. URL: https://doi.org/10.1371/journal.pone.0130140 (cit. on pp. 4, 5, 54, 55, 61, 73, 78, 112).

Balduzzi, D., M. Frean, L. Leary, J. P. Lewis, K. W.-D. Ma, B. McWilliams (2017): The Shattered Gradients Problem: If resnets are the answer, then what is the question?. *Proceedings of the 34th International Conference on Machine Learning*. Ed. by D. Precup, Y. W. Teh. Vol. 70. Proceedings of Machine Learning Research. PMLR, 342–350. URL: https://proceedings.mlr.press/v70/balduzzi17b.html (cit. on p. 80).

Balestriero, R., R. G. Baraniuk (2021): Mad Max: Affine Spline Insights Into Deep Learning. *Proceedings of the IEEE* 109 (5), 704–727. DOI: 10.1109/JPROC.2020.3042100 (cit. on p. 5).

Bhatt, U., A. Xiang, S. Sharma, A. Weller, A. Taly, Y. Jia, J. Ghosh, R. Puri, J. M. F. Moura, P. Eckersley (2020): Explainable machine learning in deployment. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. FAT* '20. Barcelona, Spain: Association for Computing Machinery, 648–657. ISBN: 9781450369367. DOI: 10.1145/3351095.3375624. URL: https://doi.org/10.1145/3351095.3375624 (cit. on p. 6).

Bhattacharyya, A. (1943): *On a measure of divergence between two statistical populations defined by their probability distributions. Bulletin of the Calcutta Mathematical Society*, 35:99–109, 1943 (cit. on p. 94).

Binder, A., G. Montavon, S. Lapuschkin, K.-R. Müller, W. Samek (2016): Layer-wise Relevance Propagation for Neural Networks with Local Renormalization Layers. *International Conference on Artificial Neural Networks (ICANN)*. Vol. 9887. Lecture Notes in Computer Science. Springer, 63–71. DOI: 10.1007/978-3-319-44781-0_8 (cit. on p. 54).

Binder, A., L. Weber, S. Lapuschkin, G. Montavon, K.-R. Müller, W. Samek (2023): Shortcomings of Top-Down Randomization-Based Sanity Checks for Evaluations of Deep Neural Network Explanations. *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 16143–16152. DOI: 10.1109/cvpr52729.2023.01549. URL: http://dx.doi.org/10.1109/CVPR52729.2023.01549 (cit. on pp. 79, 101).

Boyd, S., L. Vandenberghe (2004): *Convex Optimization*. Cambridge University Press. ISBN: 9780511804441. DOI: 10.1017/cbo9780511804441. URL: http://dx.doi.org/10.1017/CBO9780511804441 (cit. on pp. 15, 18).

Brandenburg, M.-C., M. Grillo, C. Hertrich (2025): Decomposition Polyhedra of Piecewise Linear Functions. *International Conference on Representation Learning*. Ed. by Y. Yue, A. Garg, N. Peng, F. Sha, R. Yu. Vol. 2025, 62808–62837. URL: https://proceedings.iclr.cc/paper_files/paper/2025/file/9d66f74820f11ce037fb5f711ab9acd4-Paper-Conference.pdf (cit. on pp. 5, 24).

Brandenburg, M.-C., G. Loho, G. Montufar (2024): The Real Tropical Geometry of Neural Networks for Binary Classification. *Transactions on Machine Learning Research*. ISSN: 2835-8856. URL: https://openreview.net/forum?id=I7JWf8XA2w (cit. on pp. 17, 19).

Calafiore, G. C., S. Gaubert, C. Possieri (2020): A Universal Approximation Result for Difference of Log-Sum-Exp Neural Networks. *IEEE Trans. Neural Networks Learn. Syst.* 31 (12), 5603–5612. DOI: 10.1109/TNNLS.2020.2975051. URL: https://doi.org/10.1109/TNNLS.2020.2975051 (cit. on p. 5).

Chattopadhay, A., A. Sarkar, P. Howlader, V. N. Balasubramanian (2018): Grad-CAM++: Generalized Gradient-Based Visual Explanations for Deep Convolutional Networks. *2018 IEEE Winter Conference on Applications of Computer Vision (WACV)*, 839–847. DOI: 10.1109/WACV.2018.00097 (cit. on pp. 5, 77).

Chefer, H., S. Gur, L. Wolf (2021a): Generic Attention-Model Explainability for Interpreting Bi-Modal and Encoder-Decoder Transformers. *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 397–406 (cit. on pp. 6, 61, 90, 91).

Chefer, H., S. Gur, L. Wolf (2021b): Transformer Interpretability Beyond Attention Visualization. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 782–791 (cit. on pp. 6, 55, 61, 90, 91).

Chen, Z., Y. Wang, et al. (2025): Recurrent Convex Difference Neural Networks for Safety-Critical Model Predictive Control. *IEEE Robotics and Automation Letters*. Early access; bibliographic details may update. URL: https://www.researchgate.net/publication/391585816_Recurrent_Convex_Difference_Neural_Networks_for_Safety-Critical_Model_Predictive_Control (cit. on pp. 5, 81).

Clevert, D., T. Unterthiner, S. Hochreiter (2016): Fast and Accurate Deep Network Learning by Exponential Linear Units (ELUs). *4th International Conference on Learning Representations, ICLR 2016, San Juan, Puerto Rico, May 2-4, 2016, Conference Track Proceedings*. Ed. by Y. Bengio, Y. LeCun. URL: http://arxiv.org/abs/1511.07289 (cit. on p. 52).

Dalal, N., B. Triggs (2005): *Histograms of oriented gradients for human detection*. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 886–893, 2005 (cit. on p. 100).

Daniels, H., M. Velikova (2010): Monotone and partially monotone neural networks. *Trans. Neur. Netw.* 21 (6), 906–917. ISSN: 1045-9227. DOI: 10.1109/TNN.2010.2044803. URL: https://doi.org/10.1109/TNN.2010.2044803 (cit. on pp. 5, 9).

Darcet, T., M. Oquab, J. Mairal, P. Bojanowski (2024): Vision Transformers Need Registers. *ICLR*. OpenReview.net (cit. on pp. 6, 61).

Dayan, P., L. F. Abbott (2001): *Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems*. Cambridge, MA: MIT Press. ISBN: 9780262041997 (cit. on p. 77).

Deng, J., W. Dong, R. Socher,
bibinitperiod Li, K. Li, L. Fei-Fei (2009): ImageNet: A Large-Scale Hierarchical Image Database. English (US). *2009 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2009*. 2009 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2009. Publisher Copyright: © 2009 IEEE.; 2009 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2009 ; Conference date: 20-06-2009 Through 25-06-2009. United States: IEEE Computer Society, 248–255. ISBN: 9781424439911. DOI: 10.1109/CVPR.2009.5206848 (cit. on pp. 23, 90).

Dombrowski, A.-K., M. Alber, C. J. Anders, M. Ackermann, K.-R. Müller, P. Kessel (2019): Explanations Can Be Manipulated and Geometry Is to Blame. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 32 (cit. on p. 5).

Dombrowski, A.-K., C. J. Anders, K.-R. Müller, P. Kessel (2022): Towards robust explanations for deep neural networks. *Pattern Recognition* 121, 108194. ISSN: 0031-3203. DOI: 10.1016/j.patcog.2021.108194. URL: http://dx.doi.org/10.1016/j.patcog.2021.108194 (cit. on pp. 6, 55, 106).

Doshi-Velez, F., B. Kim (2017): *Towards A Rigorous Science of Interpretable Machine Learning*. DOI: 10.48550/arXiv.1702.08608. arXiv: 1702.08608. URL: https://arxiv.org/abs/1702.08608 (cit. on p. 1).

Dosovitskiy, A., L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, N. Houlsby (2021): An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. *ICLR*. OpenReview.net (cit. on pp. 5, 61, 85, 90, 107).

Efron, B., R. J. Tibshirani (1993): An Introduction to the Bootstrap. Chapman & Hall, 1993 (cit. on pp. 90, 113).

Everingham, M., L. Van Gool, C. K. I. Williams, J. Winn, A. Zisserman (2009): The Pascal Visual Object Classes (VOC) Challenge. *International Journal of Computer Vision* 88 (2), 303–338. ISSN: 1573-1405. DOI: 10.1007/s11263-009-0275-4. URL: http://dx.doi.org/10.1007/s11263-009-0275-4 (cit. on p. 90).

Eykholt, K., I. Evtimov, E. Fernandes, B. Li, A. Rahmati, C. Xiao, A. Prakash, T. Kohno, D. Song (2018): Robust Physical-World Attacks on Deep Learning Visual Classification. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 1625–1634. URL: https://openaccess.thecvf.com/content_cvpr_2018/html/Eykholt_Robust_Physical-World_Attacks_CVPR_2018_paper.html (cit. on p. 1).

Fei, Y., Z. Yang, Y. Chen, Z. Wang, Q. Xie (2020): Risk-Sensitive Reinforcement Learning: Near-Optimal Risk-Sample Tradeoff in Regret. *NeurIPS* (cit. on pp. 6, 58).

Fel, T., M. Ducoffe, D. Vigouroux, R. Cadène, M. Capelle, C. Nicodème, T. Serre (2023): Don't Lie to Me! Robust and Efficient Explainability With Verified Perturbation Analysis. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) — Poster*. Method: EVA. URL: https://cvpr.thecvf.com/virtual/2023/poster/22072 (cit. on pp. 5, 73).

Fong, R., A. Vedaldi (2017): Interpretable Explanations of Black Boxes by Meaningful Perturbation. *IEEE International Conference on Computer Vision (ICCV)*, 3429–3437 (cit. on pp. 5, 73).

Gao, S., Z.-Y. Li, M.-H. Yang, M.-M. Cheng, J. Han, P. Torr (2023): Large-Scale Unsupervised Semantic Segmentation. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45 (6), 7457–7476. ISSN: 1939-3539. DOI: 10.1109/tpami.2022.3218275. URL: http://dx.doi.org/10.1109/TPAMI.2022.3218275 (cit. on pp. 90, 111).

Gast, J., S. Roth (2018): Lightweight Probabilistic Deep Networks. *2018 IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE, 3369–3378. DOI: 10.1109/cvpr.2018.00355. URL: http://dx.doi.org/10.1109/cvpr.2018.00355 (cit. on p. 58).

Gaubert, S., Y. Vlassopoulos (2025): *Relu and softplus neural nets as zero-sum turn-based games*. arXiv: 2512.20582 [cs.LG]. URL: https://arxiv.org/abs/2512.20582 (cit. on pp. 2, 4, 6, 13–15, 54, 106).

Geirhos, R., J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, F. A. Wichmann (2020): Shortcut Learning in Deep Neural Networks. *Nature Machine Intelligence* 2. DOI: 10.1038/s42256-020-00257-z. URL: https://arxiv.org/abs/2004.07780 (cit. on p. 1).

Geist, M., B. Scherrer, O. Pietquin (2019): *A theory of regularized Markov decision processes*. In *International Conference on Machine Learning (ICML)*, pages 2160–2169, 2019 (cit. on p. 57).

Ghorbani, A., A. Abid, J. Zou (2019): Interpretation of Neural Networks Is Fragile. *Proceedings of the AAAI Conference on Artificial Intelligence* 33 (01), 3681–3688. ISSN: 2159-5399. DOI: 10.1609/aaai.v33i01.33013681. URL: http://dx.doi.org/10.1609/aaai.v33i01.33013681 (cit. on pp. 2, 5).

Gildenblat, J., contributors (2021): *PyTorch library for CAM methods*. https://github.com/jacobgil/pytorch-grad-cam, 2021. URL: https://github.com/jacobgil/pytorch-grad-cam (cit. on p. 90).

Goodfellow, I., D. Warde-Farley, M. Mirza, A. Courville, Y. Bengio (2013): Maxout Networks. *Proceedings of the 30th International Conference on Machine Learning*. Ed. by S. Dasgupta, D. McAllester. Vol. 28. Proceedings of Machine Learning Research 3. Atlanta, Georgia, USA: PMLR, 1319–1327. URL: https://proceedings.mlr.press/v28/goodfellow13.html (cit. on pp. 10, 19, 31, 62, 81).

Griewank, A., A. Walther (2020): Polyhedral DC Decomposition and DCA Optimization of Piecewise Linear Functions. *Algorithms* 13 (7), 166. DOI: 10.3390/a13070166. URL: https://www.mdpi.com/1999-4893/13/7/166 (cit. on pp. 5, 24).

Gu, J., Y. Yang, V. Tresp (2019): Understanding Individual Decisions of CNNs via Contrastive Backpropagation. *Computer Vision – ACCV 2018, Revised Selected Papers, Part III*. Vol. 11363. Lecture Notes in Computer Science. Springer, 119–134 (cit. on p. 54).

Guillemot, M., C. Heuselé (2020): Breaking Batch Normalization for Better Explainability of Deep Neural Networks through Layer-Wise Relevance Propagation. *arXiv preprint arXiv:2002.11018* (cit. on p. 51).

Gupta, M., A. Cotter, J. Pfeifer, K. Voevodski, K. Canini, A. Mangylov, W. Moczydlowski, A. van Esbroeck (2016): Monotonic Calibrated Interpolated Look-Up Tables. *Journal of Machine Learning Research* 17 (109), 1–47. URL: http://jmlr.org/papers/v17/15-243.html (cit. on p. 5).

Haase, C. A., C. Hertrich, G. Loho (2023): Lower Bounds on the Depth of Integral ReLU Neural Networks via Lattice Polytopes. *The Eleventh International Conference on Learning Representations* (cit. on p. 19).

Hanna, M., S. Pezzelle, Y. Belinkov (2024): Have Faith in Faithfulness: Going Beyond Circuit Overlap When Finding Model Mechanisms. *CoRR* abs/2403.17806 (cit. on p. 61).

He, K., X. Zhang, S. Ren, J. Sun (2015): *Delving deep into rectifiers: Surpassing human-level performance on ImageNet classification*. In *IEEE International Conference on Computer Vision (ICCV)*, pages 1026–1034, 2015 (cit. on p. 100).

He, K., X. Zhang, S. Ren, J. Sun (2016): Deep Residual Learning for Image Recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 770–778. DOI: 10.1109/cvpr.2016.90. URL: http://dx.doi.org/10.1109/CVPR.2016.90 (cit. on pp. 4, 50, 78, 90).

Hedström, A., L. Weber, D. Krakowczyk, D. Bareeva, F. Motzkus, W. Samek, S. Lapuschkin, M. M.-C. Müller (2023): Quantus: An Explainable AI Toolkit for Responsible Evaluation of Neural Network Explanations and Beyond. *Journal of Machine Learning Research* 24 (34), 1–11. URL: http://jmlr.org/papers/v24/22-0142.html (cit. on pp. 4, 6, 78, 90, 113).

Hellinger, E. (1909): *Neue Begründung der Theorie quadratischer Formen von unendlichvielen Veränderlichen*. *Journal für die reine und angewandte Mathematik*, 136:210–271, 1909 (cit. on p. 94).

Hendrycks, D., K. Gimpel (2016): Gaussian Error Linear Units (GELUs). *arXiv preprint arXiv:1606.08415*. URL: https://arxiv.org/abs/1606.08415 (cit. on pp. 33, 45, 106).

Hertrich, C., G. Loho (2025): *Neural Networks and (Virtual) Extended Formulations*. arXiv: 2411.03006 [math.CO]. URL: https://arxiv.org/abs/2411.03006 (cit. on pp. 19, 63).

Holm, S. (1979): *A simple sequentially rejective multiple test procedure*. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979 (cit. on pp. 90, 114).

Huchette, J., G. Muñoz, T. Serra, C. Tsay (2025): *When Deep Learning Meets Polyhedral Theory: A Survey*. arXiv: 2305.00241 [math.OC]. URL: https://arxiv.org/abs/2305.00241 (cit. on pp. 5, 9, 19).

Hui, L. Y. W., A. Binder (2019): BatchNorm Decomposition for Deep Neural Network Interpretation. *Advances in Computational Intelligence – IWANN 2019*. Vol. 11507. Lecture Notes in Computer Science. Springer, 280–291. DOI: 10.1007/978-3-030-20518-8_24 (cit. on p. 51).

Jiang, P.-T., C.-B. Zhang, Q. Hou, M.-M. Cheng, Y. Wei (2021): LayerCAM: Exploring Hierarchical Class Activation Maps for Localization. *IEEE Transactions on Image Processing* 30, 5875–5888. DOI: 10.1109/TIP.2021.3089943 (cit. on pp. 4, 5, 77).

Kahardipraja, P., R. Achtibat, T. Wiegand, W. Samek, S. Lapuschkin (2025): The Atlas of In-Context Learning: How Attention Heads Shape In-Context Retrieval Augmentation. *NeurIPS* (cit. on p. 61).

Kauffmann, J., K.-R. Müller, G. Montavon (2020a): Towards explaining anomalies: A deep Taylor decomposition of one-class models. *Pattern Recognition* 101, 107198. ISSN: 0031-3203. DOI: https://doi.org/10.1016/j.patcog.2020.107198. URL: https://www.sciencedirect.com/science/article/pii/S0031320320300054 (cit. on p. 1).

Kauffmann, J., L. Ruff, G. Montavon, K.-R. Müller (2020b): *The Clever Hans Effect in Anomaly Detection*. arXiv: 2006.10609 [cs.LG]. URL: https://arxiv.org/abs/2006.10609 (cit. on p. 1).

Kazda, K., X. Li (2024): A linear programming approach to difference-of-convex piecewise linear approximation. *European Journal of Operational Research* 312 (2), 493–511. ISSN: 0377-2217. DOI: https://doi.org/10.1016/j.ejor.2023.07.026. URL: https://www.sciencedirect.com/science/article/pii/S0377221723005647 (cit. on p. 5).

Kindermans, P.-J., K. T. Schütt, M. Alber, K.-R. Müller, D. Erhan, B. Kim, S. Dähne (2018): Learning How to Explain Neural Networks: PatternNet and PatternAttribution. *International Conference on Learning Representations (ICLR)*. URL: https://openreview.net/forum?id=Hkn7CBaTW (cit. on pp. 2, 6).

Koehler, F., A. Risteski (2018): Representational Power of ReLU Networks and Polynomial Kernels: Beyond Worst-Case Analysis. *CoRR* abs/1805.11405. arXiv: 1805.11405. URL: http://arxiv.org/abs/1805.11405 (cit. on p. 5).

Kohlbrenner, M., A. Bauer, S. Nakajima, A. Binder, W. Samek, S. Lapuschkin (2020): Towards Best Practice in Explaining Neural Network Decisions with LRP. *IJCNN*. IEEE, 1–7 (cit. on pp. 54, 78, 90, 112, 113).

Kokhlikyan, N., V. Miglani, M. Martin, E. Wang, B. Alsallakh, J. Reynolds, A. Melnikov, N. Kliushkina, C. Araya, S. Yan, O. Reblitz-Richardson (2020): *Captum: A unified and generic model interpretability library for PyTorch*. arXiv: 2009.07896 [cs.LG] (cit. on p. 90).

Kolek, S., R. Windesheim, H. Andrade-Loarca, G. Kutyniok, R. Levie (2023): Explaining Image Classifiers With Multiscale Directional Image Representation. *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) — Poster*. Method: ShearletX. URL: https://cvpr.thecvf.com/virtual/2023/poster/21819 (cit. on pp. 5, 73).

Komorowski, P., E. Golimblevskaia, R. Achtibat, T. Wiegand, S. Lapuschkin, W. Samek (2025): Attribution-Guided Decoding. *CoRR* abs/2509.26307 (cit. on p. 61).

Koutschan, C., B. Moser, A. Ponomarchuk, J. Schicho (2023): Representing piecewise linear functions by functions with small arity: Representing piecewise linear functions by functions with.... *Appl. Algebra Eng., Commun. Comput.* 36 (4), 595–610. ISSN: 0938-1279. DOI: 10.1007/s00200-023-00627-1. URL: https://doi.org/10.1007/s00200-023-00627-1 (cit. on pp. 5, 24).

Kramár, J., T. Lieberum, R. Shah, N. Nanda (2024): *AtP*: An efficient and scalable method for localizing LLM behaviour to components*. arXiv: 2403.00745 [cs.LG]. URL: https://arxiv.org/abs/2403.00745 (cit. on p. 61).

Kripfganz, A., R. Schulze (1987): Piecewise affine functions as a difference of two convex functions. *Optimization* 18 (1), 23–29. DOI: 10.1080/02331938708843210. URL: https://www.tandfonline.com/doi/abs/10.1080/02331938708843210 (cit. on pp. 5, 9, 23, 39).

Lapuschkin, S., S. Wäldchen, A. Binder, G. Montavon, W. Samek, K.-R. Müller (2019): Unmasking Clever Hans predictors and assessing what machines really learn. *Nature Communications* 10 (1), 1096. DOI: 10.1038/s41467-019-08987-4. URL: https://doi.org/10.1038/s41467-019-08987-4 (cit. on p. 1).

Lee, D. D., H. S. Seung (1999): Learning the Parts of Objects by Non-Negative Matrix Factorization. *Nature* 401 (6755), 788–791. DOI: 10.1038/44565. URL: https://doi.org/10.1038/44565 (cit. on pp. 1, 77).

Lipton, Z. C. (2018): The Mythos of Model Interpretability. *Communications of the ACM* 61 (10), 36–43. DOI: 10.1145/3233231. URL: https://arxiv.org/abs/1606.03490 (cit. on p. 1).

Liu, X., S. Coogan (2020): Certified Monotonic Neural Networks. *Advances in Neural Information Processing Systems (NeurIPS)*. URL: https://proceedings.neurips.cc/paper/2020/file/b139aeda1c2914e3b579aafd3ceeb1bd-Paper.pdf (cit. on pp. 5, 9, 83).

Lundberg, S. M., S.-I. Lee (2017): A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*. Ed. by I. Guyon, U. V. Luxburg, S. Bengio, H. Wallach, R. Fergus, S. Vishwanathan, R. Garnett. Vol. 30. Curran Associates, Inc. URL: https://proceedings.neurips.cc/paper_files/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf (cit. on pp. 5, 6, 54, 73).

Maclagan, D., B. Sturmfels (2015): *Introduction to Tropical Geometry*. Vol. 161. Graduate Studies in Mathematics. American Mathematical Society. DOI: 10.1090/gsm/161 (cit. on p. 16).

Magron, V., J. Wang (2022): *Sparse Polynomial Optimization: Theory and Practice*. arXiv: 2208.11158 [math.OC]. URL: https://arxiv.org/abs/2208.11158 (cit. on p. 5).

Mao, X., G. Qi, Y. Chen, X. Li, R. Duan, S. Ye, Y. He, H. Xue (2022): Towards Robust Vision Transformer. *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 12032–12041. DOI: 10.1109/CVPR52688.2022.01173 (cit. on p. 4).

McNemar, Q. (1947): *Note on the sampling error of the difference between correlated proportions or percentages*. *Psychometrika*, 12(2):153–157, 1947 (cit. on p. 113).

Melzer, D. (1986): On the expressibility of piecewise-linear continuous functions as the difference of two piecewise-linear convex functions. In: *Quasidifferential Calculus*. Vol. 29. Mathematical Programming Studies. Berlin, Heidelberg: Springer, 118–134. DOI: 10.1007/BFb0121142. URL: https://link.springer.com/chapter/10.1007/BFb0121142 (cit. on pp. 5, 9, 23, 39).

Misiakos, D., A. Weissenbacher, G. Loho (2022): Neural Network Expressivity and Complexity of Tropical Polynomial Functions. *arXiv preprint arXiv:2202.05093* (cit. on pp. 17, 19).

Misiakos, K., C. Thrampoulidis (2022): On the geometry of ReLU networks with one hidden layer. *arXiv preprint arXiv:2201.09816* (cit. on pp. 4, 5).

Montavon, G., A. Binder, S. Lapuschkin, W. Samek, K.-R. Müller (2019): Layer-Wise Relevance Propagation: An Overview. In: *Explainable AI: Interpreting, Explaining and Visualizing Deep Learning*. Ed. by W. Samek, G. Montavon, A. Vedaldi, L. K. Hansen, K.-R. Müller. Cham: Springer International Publishing, 193–209. ISBN: 978-3-030-28954-6. DOI: 10.1007/978-3-030-28954-6_10. URL: https://doi.org/10.1007/978-3-030-28954-6_10 (cit. on p. 78).

Montavon, G., S. Lapuschkin, A. Binder, W. Samek, K.-R. Müller (2017): Explaining nonlinear classification decisions with deep Taylor decomposition. *Pattern Recognition* 65, 211–222. ISSN: 0031-3203. DOI: https://doi.org/10.1016/j.patcog.2016.11.008. URL: https://www.sciencedirect.com/science/article/pii/S0031320316303582 (cit. on pp. 5, 54, 55, 73).

Montavon, G., W. Samek, K.-R. Müller (2018): Methods for interpreting and understanding deep neural networks. *Digital Signal Processing* 73, 1–15. ISSN: 1051-2004. DOI: https://doi.org/10.1016/j.dsp.2017.10.011. URL: https://www.sciencedirect.com/science/article/pii/S1051200417302385 (cit. on pp. 5, 78, 113).

Mukherjee, A., A. Basu, R. Arora, P. Mianjy (2018): Understanding Deep Neural Networks with Rectified Linear Units. English. International Conference on Learning Representations , ICLR ; Conference date: 30-04-2018 Through 03-05-2018. URL: https://iclr.cc/Conferences/2018 (cit. on pp. 5, 8, 9, 19, 39, 40).

Murdoch, W. J., C. Singh, K. Kumbier, R. Abbasi-Asl, B. Yu (2019): Definitions, methods, and applications in interpretable machine learning. *Proceedings of the National Academy of Sciences* 116 (44), 22071–22080. DOI: 10.1073/pnas.1900654116. URL: https://doi.org/10.1073/pnas.1900654116 (cit. on p. 1).

Nam, W., S. Gur, J. Choi, L. Wolf, S. Lee (2020): Relative Attributing Propagation: Interpreting the Comparative Contributions of Individual Units in Deep Neural Networks. *AAAI*. AAAI Press, 2501–2508 (cit. on p. 54).

Nguyen, A.-P., D. L. Moreno, N. Le-Bel, M. Rodríguez Martínez (2023): MonoNet: enhancing interpretability in neural networks via monotonic features. *Bioinformatics Advances* 3 (1), vbad016. ISSN: 2635-0041. DOI: 10.1093/bioadv/vbad016. eprint: https://academic.oup.com/bioinformaticsadvances/article-pdf/3/1/vbad016/50156829/vbad016.pdf. URL: https://doi.org/10.1093/bioadv/vbad016 (cit. on p. 5).

Nie, W., Y. Zhang, A. Patel (2018): A Theoretical Explanation for Perplexing Behaviors of Backpropagation-based Visualizations. *ICML*. Vol. 80. Proceedings of Machine Learning Research. PMLR, 3806–3815 (cit. on pp. 2, 6).

Osborne, M. J., A. Rubinstein (1994): A Course in Game Theory. MIT Press, 1994 (cit. on p. 56).

Pahde, F., M. Dreyer, W. Samek, S. Lapuschkin (2023): Reveal to Revise: An Explainable AI Life Cycle for Iterative Bias Correction of Deep Models. In: *Medical Image Computing and Computer Assisted Intervention – MICCAI 2023*. Springer Nature Switzerland, 596–606. ISBN: 9783031438950. DOI: 10.1007/978-3-031-43895-0_56. URL: http://dx.doi.org/10.1007/978-3-031-43895-0_56 (cit. on pp. 1, 6).

Paszke, A., S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, S. Chintala (2019): PyTorch: an imperative style, high-performance deep learning library. *Proceedings of the 33rd International Conference on Neural Information Processing Systems*. Red Hook, NY, USA: Curran Associates Inc. (cit. on pp. 50, 90).

Phuong, M., C. H. Lampert (2020): Functional vs. parametric equivalence of ReLU networks. *ICLR*. OpenReview.net (cit. on p. 5).

Puterman, M. L. (1994): *Markov Decision Processes: Discrete Stochastic Dynamic Programming*. John Wiley & Sons. DOI: 10.1002/9780470316887 (cit. on pp. 11, 12, 56).

Ramachandran, P., B. Zoph, Q. V. Le (2017): Searching for Activation Functions. *CoRR* abs/1710.05941. arXiv: 1710.05941. URL: http://arxiv.org/abs/1710.05941 (cit. on pp. 33, 52, 106).

Ribeiro, M. T., S. Singh, C. Guestrin (2016): "Why Should I Trust You?": Explaining the Predictions of Any Classifier. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. KDD '16. San Francisco, California, USA: Association for Computing Machinery, 1135–1144.

ISBN: 9781450342322. DOI: 10.1145/2939672.2939778. URL: https://doi.org/10.1145/2939672.2939778 (cit. on pp. 5, 73).

Ross, A. S., F. Doshi-Velez (2018): Improving the Adversarial Robustness and Interpretability of Deep Neural Networks by Regularizing Their Input Gradients. *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 32. 1. DOI: 10.1609/aaai.v32i1.11504. URL: https://arxiv.org/abs/1711.09404 (cit. on p. 1).

Ross, A. S., M. C. Hughes, F. Doshi-Velez (2017): Right for the Right Reasons: Training Differentiable Models by Constraining their Explanations. *Proceedings of the Twenty-Sixth International Joint Conference on Artificial Intelligence*, 2662–2670. DOI: 10.24963/ijcai.2017/371. URL: https://ijcai.org/proceedings/2017/371 (cit. on p. 1).

Rudin, C. (2019): Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead. *Nature Machine Intelligence* 1 (5), 206–215. DOI: 10.1038/s42256-019-0048-x. URL: https://doi.org/10.1038/s42256-019-0048-x (cit. on p. 1).

Runje, D., S. M. Shankaranarayana (2023): Constrained monotonic neural networks. *Proceedings of the 40th International Conference on Machine Learning*. ICML'23. Honolulu, Hawaii, USA: JMLR.org (cit. on pp. 5, 9, 10).

Russakovsky, O., J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, A. C. Berg, L. Fei-Fei (2015): ImageNet Large Scale Visual Recognition Challenge. *International Journal of Computer Vision* 115 (3), 211–252. ISSN: 1573-1405. DOI: 10.1007/s11263-015-0816-y. URL: http://dx.doi.org/10.1007/s11263-015-0816-y (cit. on p. 90).

Samek, W., A. Binder, G. Montavon, S. Lapuschkin, K.-R. Müller (2017): *Evaluating the visualization of what a deep neural network has learned. IEEE Transactions on Neural Networks and Learning Systems*, 28(11):2660–2673, 2017 (cit. on p. 112).

Sankaranarayanan, P., R. Rengaswamy (2022): CDiNN – Convex difference neural networks. *Neurocomput.* 495 (C), 153–168. ISSN: 0925-2312. DOI: 10.1016/j.neucom.2022.01.024. URL: https://doi.org/10.1016/j.neucom.2022.01.024 (cit. on pp. 5, 81).

Schlüter, N., M. S. Darup (2021): Novel Convex Decomposition of Piecewise Affine Functions. *arXiv preprint arXiv:2108.03950*. URL: https://arxiv.org/abs/2108.03950 (cit. on p. 5).

Selvaraju, R. R., M. Cogswell, A. Das, R. Vedantam, D. Parikh, D. Batra (2017): Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. *2017 IEEE International Conference on Computer Vision (ICCV)*, 618–626. DOI: 10.1109/ICCV.2017.74 (cit. on pp. 5, 73, 77).

Shapley, L. S. (1953): Stochastic Games. *Proceedings of the National Academy of Sciences* 39 (10), 1095–1100. DOI: 10.1073/pnas.39.10.1095 (cit. on p. 13).

Shrikumar, A., P. Greenside, A. Kundaje (2017): Learning Important Features Through Propagating Activation Differences. *Proceedings of the 34th International Conference on Machine Learning*. Ed. by D. Precup, Y. W. Teh. Vol. 70. Proceedings of Machine Learning Research. PMLR, 3145–3153. URL: https://proceedings.mlr.press/v70/shrikumar17a.html (cit. on pp. 5, 73).

Siahkamari, A., A. Gangrade, B. Kulis, V. Saligrama (2020): Piecewise Linear Regression via a Difference of Convex Functions. *Proceedings of the 37th International Conference on Machine Learning*. Ed. by H. D. III, A. Singh. Vol. 119. Proceedings of Machine Learning Research. PMLR, 8895–8904. URL: https://proceedings.mlr.press/v119/siahkamari20a.html (cit. on pp. 5, 31).

Sill, J. (1997): Monotonic Networks. *Advances in Neural Information Processing Systems 10 (NIPS 1997)*. URL: https://papers.nips.cc/paper_files/paper/1997/hash/83adc9225e4deb67d7ce42d58fe5157c-Paper.pdf (cit. on pp. 1, 5, 9, 83).

Simonyan, K., A. Vedaldi, A. Zisserman (2014): Deep Inside Convolutional Networks: Visualising Image Classification Models and Saliency Maps. *2nd International Conference on Learning Representations, ICLR 2014, Banff, AB, Canada, April 14-16, 2014, Workshop Track Proceedings*. Ed. by Y. Bengio, Y. LeCun. URL: http://arxiv.org/abs/1312.6034 (cit. on pp. 5, 73).

Simonyan, K., A. Zisserman (2015): Very Deep Convolutional Networks for Large-Scale Image Recognition. *3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7-9, 2015, Conference Track Proceedings*. Ed. by Y. Bengio, Y. LeCun. URL: http://arxiv.org/abs/1409.1556 (cit. on pp. 4, 23, 50, 78, 90).

Sivaprasad, S., A. Singh, N. Manwani, V. Gandhi (2021): The Curious Case of Convex Neural Networks. *Machine Learning and Knowledge Discovery in Databases. Research Track*. Ed. by N. Oliver, F. Pérez-Cruz,

S. Kramer, J. Read, J. A. Lozano. Cham: Springer International Publishing, 738–754. ISBN: 978-3-030-86486-6 (cit. on p. 5).

Sivaraman, A., G. Farnadi, T. Millstein, G. V. den Broeck (2020): Counterexample-Guided Learning of Monotonic Neural Networks. *Advances in Neural Information Processing Systems 33 (NeurIPS 2020)*. Method/tool: COMET. URL: https://proceedings.neurips.cc/paper/2020/hash/8ab70731b1553f17c11a3bbc87e0b605-Abstract.html (cit. on p. 5).

Sixt, L., M. Granz, T. Landgraf (2020): When Explanations Lie: Why Many Modified BP Attributions Fail. *ICML*. Vol. 119. Proceedings of Machine Learning Research. PMLR, 9046–9057 (cit. on pp. 6, 79, 94, 100, 106).

Smilkov, D., N. Thorat, B. Kim, F. Viégas, M. Wattenberg (2017): *SmoothGrad: removing noise by adding noise*. arXiv: 1706.03825 [cs.LG]. URL: https://arxiv.org/abs/1706.03825 (cit. on pp. 5, 73).

Springenberg, J. T., A. Dosovitskiy, T. Brox, M. Riedmiller (2015): Striving for Simplicity: The All Convolutional Net. *International Conference on Learning Representations (Workshop)* (cit. on pp. 5, 73).

Sriperumbudur, B. K., G. R. G. Lanckriet (2009): On the Convergence of the Concave-Convex Procedure. *Advances in Neural Information Processing Systems (NeurIPS)*. URL: https://papers.nips.cc/paper/3646-on-the-convergence-of-the-concave-convex-procedure (cit. on p. 5).

Sundararajan, M., A. Taly, Q. Yan (2017): Axiomatic Attribution for Deep Networks. *Proceedings of the 34th International Conference on Machine Learning*. Ed. by D. Precup, Y. W. Teh. Vol. 70. Proceedings of Machine Learning Research. PMLR, 3319–3328. URL: https://proceedings.mlr.press/v70/sundararajan17a.html (cit. on pp. 5, 73, 74).

Sutton, R. S., A. G. Barto (2018): *Reinforcement Learning: An Introduction*. 2nd ed. MIT Press (cit. on pp. 11, 12).

Syed, A., C. Rager, A. Conmy (2024): Attribution Patching Outperforms Automated Circuit Discovery. *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*. Association for Computational Linguistics, 407–416. DOI: 10.18653/v1/2024.blackboxnlp-1.25. URL: http://dx.doi.org/10.18653/v1/2024.blackboxnlp-1.25 (cit. on p. 61).

Tao, P. D., L. T. H. An (1997): Convex Analysis Approach to D.C. Programming: Theory, Algorithms and Applications. *Acta Mathematica Vietnamica* 22 (1), 289–355 (cit. on p. 5).

Theiner, J., E. Müller-Budack, R. Ewerth (2021): Interpretable Semantic Photo Geolocation. *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 1474–1484 (cit. on p. 112).

Thi, H. A. L., T. P. Dinh (2018): DC programming and DCA: thirty years of developments. *Mathematical Programming* 169 (1), 5–68. DOI: 10.1007/s10107-018-1235-y (cit. on pp. 5, 109).

Vaswani, A., N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, I. Polosukhin (2017): Attention Is All You Need. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 30, 5998–6008 (cit. on pp. 61, 107).

Voita, E., D. Talbot, F. Moiseev, R. Sennrich, I. Titov (2019): Analyzing Multi-Head Self-Attention: Specialized Heads Do the Heavy Lifting, the Rest Can Be Pruned. *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 5797–5808 (cit. on p. 61).

Vukadin, D., P. Afrić, M. Šilić, G. Delač (2024): Advancing Attribution-Based Neural Network Explainability through Relative Absolute Magnitude Layer-Wise Relevance Propagation and Multi-Component Evaluation. *ACM Transactions on Intelligent Systems and Technology* 15 (3), 1–30. ISSN: 2157-6912. DOI: 10.1145/3649458. URL: http://dx.doi.org/10.1145/3649458 (cit. on pp. 54, 61).

Wang, S., M. Gupta (2020): Deontological Ethics By Monotonicity Shape Constraints. *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*. Ed. by S. Chiappa, R. Calandra. Vol. 108. Proceedings of Machine Learning Research. PMLR, 2043–2054. URL: https://proceedings.mlr.press/v108/wang20e.html (cit. on p. 1).

Wróbel, A., S. Gairola, J. Tabor, B. Schiele, B. Zieliński, D. Rymarczyk (2026): *DAVE: Distribution-aware Attribution via ViT Gradient Decomposition*. arXiv: 2602.06613 [cs.CV]. URL: https://arxiv.org/abs/2602.06613 (cit. on pp. 6, 55, 56, 61, 64, 68, 69, 71, 72, 85, 91).

Xiong, P., T. Schnake, G. Montavon, K.-R. Müller, S. Nakajima (2026): Normalized Relevance Measure as a Unifying Framework to Explain Neural Network Latent Structures. *arXiv preprint arXiv:2606.00557*. DOI: 10.48550/arXiv.2606.00557. arXiv: 2606.00557 [cs.LG]. URL: https://arxiv.org/abs/2606.00557 (cit. on pp. 6, 54, 93, 106, 109).

Yeh, C.-K., C.-Y. Hsieh, A. Suggala, D. I. Inouye, P. K. Ravikumar (2019): On the (In)fidelity and Sensitivity of Explanations. *Advances in Neural Information Processing Systems (NeurIPS)*. Vol. 32, 10965–10976 (cit. on pp. 6, 113).

Zalgaller, V. A. (2000): Representation of functions of several variables by differences of convex functions. *Journal of Mathematical Sciences* 100, 2209–2227. DOI: 10.1007/s10958-000-0006-4. URL: https://link.springer.com/article/10.1007/s10958-000-0006-4 (cit. on pp. 5, 23).

Zech, J. R., M. A. Badgeley, M. Liu, A. B. Costa, J. J. Titano, E. K. Oermann (2018): Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLoS Medicine* 15 (11), e1002683. DOI: 10.1371/journal.pmed.1002683. URL: https://doi.org/10.1371/journal.pmed.1002683 (cit. on p. 1).

Zeiler, M. D., R. Fergus (2014): Visualizing and Understanding Convolutional Networks. *Computer Vision – ECCV 2014*. Ed. by D. Fleet, T. Pajdla, B. Schiele, T. Tuytelaars. Cham: Springer International Publishing, 818–833. ISBN: 978-3-319-10590-1 (cit. on pp. 5, 73).

Zhang, J., S. A. Bargal, Z. Lin, J. Brandt, X. Shen, S. Sclaroff (2018a): Top-down Neural Attention by Excitation Backprop. *European Conference on Computer Vision (ECCV)*. Springer, 543–559. DOI: https://doi.org/10.1007/s11263-017-1059-x (cit. on pp. 6, 54, 55, 112).

Zhang, L., G. Naitzat, L.-H. Lim (2018b): Tropical Geometry of Deep Neural Networks. *Proceedings of the 35th International Conference on Machine Learning*. Ed. by J. Dy, A. Krause. Vol. 80. Proceedings of Machine Learning Research. PMLR, 5824–5832. URL: https://proceedings.mlr.press/v80/zhang18i.html (cit. on pp. 4, 5, 9, 17–19, 21, 31, 107).

Zhou, B., A. Khosla, A. Lapedriza, A. Oliva, A. Torralba (2016): Learning Deep Features for Discriminative Localization. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2921–2929 (cit. on pp. 73, 77).

Ziegler, G. M. (1995): *Lectures on Polytopes*. Vol. 152. Graduate Texts in Mathematics. New York: Springer (cit. on p. 17).

Zimmermann, J. P., J. Berend, G. Loho, S. Lapuschkin, W. Samek (2026): *Playing the Network Backward: A Game-Theoretic Attribution Framework*. Preprint, version 2. DOI: 10.48550/arXiv.2605.06212. arXiv: 2605.06212v2 [cs.LG]. URL: https://arxiv.org/abs/2605.06212v2 (cit. on pp. 3, 115).

Zimmermann, J. P., G. Loho (2026): Hidden Monotonicity: Explaining Deep Neural Networks via their DC Decomposition. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. Highlight, 24107–24117. arXiv: 2601.07700 [cs.CV]. URL: https://openaccess.thecvf.com/content/CVPR2026/html/Zimmermann_Hidden_Monotonicity_Explaining_Deep_Neural_Networks_via_their_DC_Decomposition_CVPR_2026_paper.html (cit. on pp. 2, 3, 19, 23, 31, 39, 40, 46, 54, 61, 62, 73, 106, 115).